

Cuda Lattice Gauge Document

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1 Data

1.1 Index of lattice

1.1.1 UINT Index of lattice

Generally, in CLG, we have three kinds of indexes:

- **site index**
- **link index**
- **fat index**

Let the lattice have $V = L_x \times L_y \times L_z \times L_t$ sites.

Note: for $D = 3$, we assume $L_x = 1, L_{y,z,t} > 1$; for $D = 2$, we assume $L_x = L_y = 1, L_{z,t} > 1$.

For a site at (x, y, z, t)

$$siteIndex = x \times L_y \times L_z \times L_t + y \times L_z \times L_t + z \times L_t + t \quad (1)$$

For a link at direction dir , link with site at (x, y, z, t) , and on a lattice with number of directions of links is $dirCount$,

$$linkIndex = siteIndex \times dirCount + dir \quad (2)$$

Note: we do NOT assume dimension equal number of links. For example for $D = 2$ triangle lattice, number of directions of links is 6, for $D = 2$ hexagon number of directions of links is 3. Only for square lattice, number of links equal dimension.

For a link at direction dir , link with site at (x, y, z, t) , and on a lattice with number of directions of links is $dirCount$,

$$fatIndex = \begin{cases} siteIndex \times (dirCount + 1); & \text{for site.} \\ siteIndex \times (dirCount + 1) + (dir + 1); & \text{for link} \end{cases} \quad (3)$$

1.1.2 SIndex of lattice

1.1.3 Index and boundary condition, a int2 or a uint2 structure

In CLGLib, sometimes, the index function return a uint2 structure.

1.1.4 Index walking

1.2 CParemers

2 Update scheme

2.1 HMC

HMC is abbreviation for hybrid Monte Carlo.

2.1.1 The Fermion action

Cooperating with HMC, the fermion is usually the 'Pseudofermions'.

We begin with Eq. (1.85) and Eq. (1.86) of Ref. [1].

$$Z = \int \mathcal{D}[U] \prod_{f=1}^{N_f} \mathcal{D}[\bar{\psi}_f] \mathcal{D}[\psi_f] \exp \left(-S_G[U] - \sum_{f=1}^{N_f} \bar{\psi}_f \left(\hat{D}_f \right) \psi_f \right) \quad (4)$$

where $\hat{D}_f = D + m_f$. (Note that, there seems a typo in Eq. (1.85) of Ref. [1] and Eq. (8.31) of Ref. [2], we have $S_F = +\bar{\psi}D\psi$, see also Eqs. (2.5) and (8.39) of Ref. [2] and Eq. (7.6) of Ref. [3], Eq. (3.75) of Ref. [1], etc.)

It can be evaluated as Eq. (1.86) of Ref. [1] (or Eq. (4.19) of Ref. [4]) (Note, there is another minus sign in Eq. (5.28) of Ref. [2])

$$\begin{aligned} \int \mathcal{D}\bar{\psi}\psi \exp(-\bar{\psi}A\psi) &= \det(A), \\ Z &= \prod_{f=1}^{N_f} \det(\hat{D}_f) \int \mathcal{D}[U] \exp(-S_G[U]). \end{aligned} \quad (5)$$

On the other hand, with the help of Gaussian integral of complex vectors Eq. (3.17) of Ref. [4]

$$\int d\mathbf{v}^\dagger d\mathbf{v} \exp(-\mathbf{v}^\dagger \mathbf{A} \mathbf{v}) = \pi^N (\det \mathbf{A})^{-1} \quad (6)$$

which is (3.31) of Ref. [1]

$$\frac{1}{\det(\mathbf{A})} = \int \mathcal{D}[\eta] \exp(-\eta^\dagger \mathbf{A} \eta) \quad (7)$$

where η now is a complex Bosonic field, and the normalization

$$\mathcal{D}[\eta] = \prod \frac{d\text{Re}(\eta_i) d\text{Im}(\eta_i)}{\pi}, \quad 1 = \int \mathcal{D}[\eta] \exp(-\eta^\dagger \eta) \quad (8)$$

is assumed. With the condition such that

$$\lambda(\mathbf{A} + \mathbf{A}^\dagger) > 0. \quad (9)$$

where $\lambda(\mathbf{M})$ denoted as eigen-values of $\mathbf{M}$.

We now, concentrate on two degenerate fermion flavours. i.e. considering

$$S_F = \bar{\psi}_u \hat{D} \psi_u + \bar{\psi}_d \hat{D} \psi_d. \quad (10)$$

Using $\det(DD^\dagger) = \det(D) \det(D^\dagger)$ and $\det(M^{-1}) = (\det(M))^{-1}$ and **$\det(D) = \det(D^\dagger)$ (Only for Wilson Fermions or γ_5 -hermiticity fermions, $\hat{D}^\dagger = \gamma_5 D \gamma_5 + m = \gamma_5 (D + m) \gamma_5 = \gamma_5 \hat{D} \gamma_5$, and $\det(\hat{D}^\dagger) = \det(\gamma_5) \det(\hat{D}) \det(\gamma_5) = \det(\hat{D})$. See also Ref. [5].), one can show Eq. (8.9) of Ref. [2] (Eq. (2.77) of Ref. [6])**

$$\int \mathcal{D}[\bar{\psi}] \mathcal{D}[\psi] \exp \left(-\bar{\psi}_u \hat{D} \psi_u - \bar{\psi}_d \hat{D} \psi_d \right) = \det(\hat{D} \hat{D}^\dagger) = \int \mathcal{D}[\phi] \exp(-\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi) \quad (11)$$

where ϕ now is a complex Bosonic field. (Note that, there is a sign typo in Eq. (8.31) of Ref. [2], see also Eqs. (8.38) and (8.39) of Ref. [2])

So, generally, we are using HMC to evaluate the action with 'Pseudofermions', or in other words, we are working with an action including only gauge and bosons.

$$S = S_G + S_{pf} = S_G + \phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \quad (12)$$

where pf is short for pseudofermion.

2.1.2 Basic idea, force from gauge field

The basic idea is to use a molecular dynamics simulation, i.e, it is a integration of Langevin equation.

Treating $SU(N)$ matrix U on links as coordinate, HMC will generate a pair of configurations, (P, U) , where P is momentum and $P \in \mathfrak{su}(N)$.

One can:

1. Create a random $P = i \sum_a \omega_a T_a$, where $\omega_a \in \mathbb{R}$.
2. Obtain $\dot{P}$, $\dot{U}$. Note that, dot is $d/d\tau$, where τ is ‘Markov time’.
3. Numerically evaluate the differential equation, and use a Metropolis accept / reject to update.

- About the randomized P

The randomized P is chosen according to normal distribution $\exp(-P^2/2)$

Note that, here P corresponds to Q , not U , for $U = \exp(i \sum_a q_a T_a)$, there are 8 **real** variables denoting as ω_i .

Using $P = \sum_a \omega_a T_a$, $\text{tr}((T^a) \cdot (T^b)) = \frac{1}{2} \delta_{ab}$. So one have $\frac{1}{2} \sum_a \omega_a^2 = \text{tr}[P^2]$.

It is usually written as distribution $\exp(-\text{tr}(P^2))$ (where P is a matrix, and $\text{tr}[P^2] = \frac{1}{2} p^2$ where $p = (\omega_1, \omega_2, \dots, \omega_8)$).

Using the property of normal distribution

$$\begin{aligned} \text{if } \{X\} &\sim N(\mu_X, \sigma_X^2), \quad \{Y\} \sim N(\mu_Y, \sigma_Y^2), \\ \{X + Y\} &\sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2). \end{aligned} \quad (13)$$

One can randomize ω_a using $\exp(-\omega_a \omega_a)$. Then using $P = \frac{1}{\sqrt{8N}} \sum \omega_a T^i$, where N is the number of links.

Note: Here is a difference between Refs. [2] and [1] and Bridge++ [7]

Note, by Eq. (8.16) of Ref. [2], $P^2 = \sum_{n \in \Lambda} P^2(n)$, so when the lattice is large, P become very small. See also the definition of $\langle P, P \rangle$ below Eq. (2.42) of Ref. [1].

However, in Bridge++, it uses distribution $\exp(-\text{tr}(P^2)/DOF)$, where ‘DOF’ is the degrees of freedom, i.e., number of links.

We use the distribution same as in Bridge++. Imagining that for a very small (hot) $\beta \rightarrow 0$, the force is also almost 0 so momentum is unchanged when evolution. Considering a very large lattice such that the momentum is very small when using distribution $\exp(-\text{tr}(P^2))$, the gauge field will stay near the initial value rather than becoming hot (randomized). So we think it should be $\exp(-\text{tr}(P^2)/DOF)$.

- Force

Defined by Newton, dp/dt is a force, so $\dot{P}$ is called ‘force’. See Eqs. (2.53), (2.56) and (2.57) of Ref. [1], for $SU(N)$,

$$\begin{aligned} S_G[U_\mu(n)] &= -\frac{\beta}{N} \text{Retr} [U_\mu(n) \Sigma_\mu^\dagger(n)] \\ \Sigma_\mu(n) &= \sum_{\mu \neq \nu} (U_\nu(n) U_\mu(n + a\nu) U_\nu^{-1}(n + a\mu) + U_\nu^{-1}(n - a\nu) U_\mu(n - a\nu) U_\nu(n - a\nu + a\mu)) \end{aligned} \quad (14)$$

Note that $S_G \neq \sum_{\mu, n} S_G[U_\mu(n)]$. $S_G[U_\mu(n)]$ is convenient for derivate which collecting all terms related to the specified bond. For plaquettes with 4 edges, $S_G = \frac{1}{4} \sum_{\mu, n} S_G[U_\mu(n)]$.

S_G the action for a particular $U_\mu(n)$. Σ is the ‘staple’(see Eq. (359)). The staple for $U_\mu(n)$ is independent of $U_\mu(n)$, denoting

$$U_\mu(n) = \exp \left(i \sum_a \omega_a(\mu, n) T_a \right) U_\mu^0(n) \quad (15)$$

so

$$\begin{aligned} \frac{\partial}{\partial \omega_a(\mu, n)} S_G &= -\frac{\beta}{N} \text{Retr} \left[\frac{\partial}{\partial \omega_a} U_\mu(n) \Sigma_\mu^\dagger(n) \right] = -\frac{\beta}{2N} \text{tr} \left[\frac{\partial}{\partial \omega_a} (U_\mu(n) \Sigma_\mu^\dagger(n) + \Sigma_\mu(n) U_\mu^\dagger(n)) \right] \\ &= -i \frac{\beta}{2N} \text{tr} [T_a U_\mu(n) \Sigma_\mu^\dagger(n) - \Sigma_\mu(n) T_a^\dagger U_\mu^\dagger(n)] = -i \frac{\beta}{2N} \text{tr} [T_a (U_\mu(n) \Sigma_\mu^\dagger(n) - \Sigma_\mu(n) U_\mu^\dagger(n))] \\ &= \frac{\beta}{N} \text{Im tr} [T_a U_\mu(n) \Sigma_\mu^\dagger(n)] \end{aligned} \quad (16)$$

This is the Eq. (8.41) of Ref. [2].

Using (Checked by Mathematica that Eq. (8.42) of Ref. [2] is incompatible with our notation, but replacing the $UA - A^\dagger U^\dagger$ of Eq. (8.42) with $\{UA\}_{TA}$ is correct. Also, Eq. (2.58) of Ref. [1] is different from ours, in our formalism, it is correct by replacing $2T_a \text{Re}[tr[T_a \cdot W]]$ of Eq. (2.58) with $2iT_a \text{Im}[tr[T_a \cdot W]]$)

$$\sum_a \text{tr} [T_a (U_\mu(n) \Sigma_\mu^\dagger(n) - \Sigma_\mu(n) U_\mu^\dagger(n))] T_a = 2i \sum_a \text{Im} [T_a U_\mu(n) \Sigma_\mu^\dagger(n)] T_a = \{U_\mu(n) \Sigma_\mu^\dagger(n)\}_{TA}$$

$$\{W\}_{TA} = \frac{W - W^\dagger}{2} - \text{tr} \left(\frac{W - W^\dagger}{2N} \right) \mathbb{I} \quad (17)$$

where $\mathbb{I}$ is identity matrix. Therefor

$$\dot{\omega}_a = -\frac{\partial}{\partial \omega_a(\mu, n)} S_G$$

$$F_\mu(x) = \dot{P}_\mu(x) = i \sum \dot{\omega}_a T_a = -i \frac{\partial}{\partial \omega_a(\mu, n)} S_G T_a = -\frac{\beta}{2N} \{U_\mu(n) \Sigma_\mu^\dagger(n)\}_{TA} \quad (18)$$

Note that, $\dot{\omega}_a = \frac{\beta}{N} \text{Im}[tr[T_a \cdot W]]$ is still a **real** number.

Eq. (18) is same as Eqs. (2.53), (2.56) and (2.57) of Ref. [1].

- Integrator

Knowing $\dot{P}$, and $\dot{U}$, to obtain U and P is simply

$$U(\tau + d\tau) \approx \dot{U} d\tau + U(\tau), \quad P(\tau + d\tau) \approx \dot{P} d\tau + P(\tau) \quad (19)$$

A more accurate calculation is done by integrator, for example, the leap frog integrator, the M step leap frog integral is described in Ref. [2],

$$\epsilon = \frac{\tau}{M} \quad (20a)$$

$$U_\mu(x, (n+1)\epsilon) = U_\mu(x, n\epsilon) + \epsilon P_\mu(x, n\epsilon) + \frac{1}{2} F_\mu(x, n\epsilon) \epsilon^2 \quad (20b)$$

$$P_\mu(x, (n+1)\epsilon) = P_\mu(x, n\epsilon) + \frac{1}{2} (F_\mu(x, (n+1)\epsilon) + F_\mu(x, n\epsilon)) \epsilon \quad (20c)$$

So, knowing $U(n\epsilon)$ we can calculate $F(n\epsilon)$ using Eq. (18). Knowing $U(n\epsilon), P(n\epsilon), F(n\epsilon)$, we can calculate $U((n+1)\epsilon)$ using Eq. (20).b. Then we are able to calculate $F((n+1)\epsilon)$ again using Eq. (18). Then we can calculate $P((n+1)\epsilon)$ using Eq. (20).c.

2.1.3 Force of pseudofermions

For important sampling, one can generate both U and ϕ by e^{-S} . In molecular dynamics simulation, it can be simplified as:

1. Evaluate U use force of U and ϕ on U .
2. Evaluate ϕ use force of U and ϕ on ϕ .

The second step can be simplified as, generating random complex numbers ϕ according to $\exp(-\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi) = \exp(-\phi^\dagger (\hat{D}^\dagger)^{-1} \hat{D}^{-1} \phi)$. $D[U]$ is a function of U .

How to get randomized ϕ ? Let χ be random **complex** numbers according to $\exp(-\chi^\dagger \chi)$. Let $\hat{D}^{-1} \phi = \chi$, ϕ is the random **complex** number satisfying distribution we want ($\exp(-\phi^\dagger (\hat{D}^\dagger)^{-1} \hat{D}^{-1} \phi)$). So, first get χ and then let $\phi = D\chi$.

Using the Wilson Fermion action

$$\hat{D} = C(D + 1)$$

$$D = -\kappa \sum_\mu ((1 - \gamma_\mu) U_\mu(x_L) \delta_{x_L, (x+\mu)_R} + (1 + \gamma_\mu) U_\mu^{-1}(x_L - \mu) \delta_{x_L, (x-\mu)_R}) \quad (21)$$

with $C = m_f + (4/a) = 1/2a\kappa$ and $\kappa = 1/(2am_f + 8)$. One can rescale the field and set $C = 1$.

The force of ϕ on U is obtained as $\partial_{\omega_a} S_{pf}$. The result for Wilson Fermion action is shown in Eqs. (8.39), (8.44) and (8.45) of Ref. [2] as

$$\begin{aligned}
 F &= i \sum_a \dot{\omega}_a T_a = i \sum_a (-\partial_{\omega_a} (S_G[U_\mu(n)] + S_{pf}[U_\mu(n)])) T_a = F_G + F_{pf}. \\
 F_{pf} &= i \sum_a (-\partial_{\omega_a} S_{pf}[U_\mu(n)]) T_a = -i \sum_a T^a \frac{\partial}{\partial \omega_a} \left(\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \right). \\
 \frac{\partial}{\partial \omega_a} \left(\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \right) &= - \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right)^\dagger \left(\frac{\partial \hat{D}}{\partial \omega_a} \hat{D}^\dagger + \hat{D} \frac{\partial \hat{D}^\dagger}{\partial \omega_a} \right) \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right). \\
 \frac{\partial \hat{D}}{\partial \omega_\mu^a} &= \left(\frac{\partial D}{\partial \omega_\mu^a} \right)_{x_L, x_R} = -i\kappa \{ (1 - \gamma_\mu) T^a U_\mu(x) \delta_{x, x_L} \delta_{x, (x+\mu)_R} - (1 + \gamma_\mu) U_\mu^{-1}(x) T^a \delta_{x, (x+\mu)_L} \delta_{x, x_R} \} \\
 \hat{D}^\dagger &= \gamma_5 \hat{D} \gamma_5, \quad \frac{\partial \hat{D}^\dagger}{\partial \omega_\mu^a} = \gamma_5 \frac{\partial D}{\partial \omega_\mu^a} \gamma_5
 \end{aligned} \tag{22}$$

where F_G is force from U introduced in Sec. 2.1.2, T^a are $SU(3)$ generators. x_L, x_R are coordinate index of the left and right pseudofermion field. And

$$\begin{aligned}
 U_\mu &= \exp(i \sum_a \omega_\mu^a T^a) U_0, \quad \frac{\partial U_\mu}{\partial \omega_\mu^a} = i T^a U_\mu, \quad \frac{\partial U_\mu^\dagger}{\partial \omega_\mu^a} = -i U_\mu^\dagger T^a, \\
 (T^a)^\dagger &= T^a, \quad \frac{\partial M^{-1}}{\partial \omega_\mu^a} = -M^{-1} \frac{\partial M}{\partial \omega_\mu^a} M^{-1}
 \end{aligned} \tag{23}$$

are used. (Note that, Eq. (8.45) of Ref. [2] has a sign typo, see also Eq. (2.82) of Ref. [6])

We can simplify it further by $(\hat{D}^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi)^\dagger = ((\hat{D} \hat{D}^\dagger)^{-1} \phi)^\dagger \hat{D}$, so

$$\begin{aligned}
 \phi_1 &= \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right), \quad \phi_2 = \hat{D}^\dagger \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right) = D^{-1} \phi, \quad \phi_1^\dagger D = \phi_2^\dagger, \\
 \frac{\partial}{\partial \omega_a} \left(\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \right) &= - \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right)^\dagger \left(\frac{\partial \hat{D}}{\partial \omega_a} \hat{D}^\dagger + \hat{D} \frac{\partial \hat{D}^\dagger}{\partial \omega_a} \right) \left((\hat{D} \hat{D}^\dagger)^{-1} \phi \right) \\
 &= - \left(\phi_1^\dagger \frac{\partial D}{\partial \omega_\mu^a} \phi_2 + \phi_2^\dagger \frac{\partial D^\dagger}{\partial \omega_\mu^a} \phi_1 \right) = -2\text{Re} \left[\left(\phi_1^\dagger \frac{\partial D}{\partial \omega_\mu^a} \phi_2 \right) \right]
 \end{aligned} \tag{24}$$

and

$$\begin{aligned}
 \frac{\partial D}{\partial \omega_\mu^a} &= -i\kappa M_a, \\
 (M_a)_{x_L, x_R} &= \{ (1 - \gamma_\mu) T^a U_\mu \delta_{x_L, (x+\mu)_R} - (1 + \gamma_\mu) U_\mu^{-1} T^a \delta_{(x+\mu)_L, x_R} \} \\
 \frac{\partial}{\partial \omega_a} \left(\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \right) &= -2\kappa \text{Im} \left[\left(\phi_1^\dagger M \phi_2 \right) \right]
 \end{aligned} \tag{25}$$

Again, $\dot{\omega}$ is a **real** number, and

$$F_{pf} = -i \sum_a T^a \frac{\partial}{\partial \omega_a} \left(\phi^\dagger (\hat{D} \hat{D}^\dagger)^{-1} \phi \right) = 2i\kappa \sum_a \text{Im} \left[\left(\phi_1^\dagger M_a \phi_2 \right) \right] T_a \tag{26}$$

So we can calculate ϕ_1 first, then $\phi_2 = \hat{D}^\dagger \phi_1$. Then contract the spinor and color space with $\partial D / \partial \omega$.

Note that, D is changing when integrating the Langevin equation.

The last part is how to calculate $(\hat{D} \hat{D}^\dagger)^{-1}$.

- Anti-Hermitian traceless of the force

See from Eq. (18), the force from the gauge field is an anti-Hermitian traceless matrix.

The result above can be further simplified. Note that

$$\begin{aligned}
 \phi_{L1}(n) &= \phi_1(n), \quad \phi_{R1}(n) = (1 - \gamma_\mu) \phi_2(n + \mu), \\
 \phi_{L2}(n) &= \phi_1(n + \mu), \quad \phi_{R2}(n) = (1 + \gamma_\mu) \phi_2(n),
 \end{aligned} \tag{27}$$

One have

$$\begin{aligned}
 \text{Im} \left[\phi_1^\dagger M \phi_2 \right]_\mu^a(n) &= \text{Im} \left[\phi_{L1}^\dagger T^a U_\mu(n) \phi_{R1} \right] - \text{Im} \left[\phi_{L2}^\dagger U_\mu^\dagger(n) T^a \phi_{R2} \right] \\
 &= \text{Im} \left[\phi_{L1}^\dagger T^a U_\mu(n) \phi_{R1} \right] + \text{Im} \left[\phi_{R2}^\dagger T^a U_\mu(n) \phi_{L2} \right]
 \end{aligned} \tag{28}$$

For any vector

$$\text{Im} [L^\dagger T U R] = \text{Im} \left[\sum_{\alpha, \beta, \rho} L_\alpha^* T_{\alpha\beta} U_{\beta\rho} R_\rho \right] = \text{Im} \left[\sum_{\alpha, \beta, \rho} T_{\alpha\beta} U_{\beta\rho} R_\rho L_\alpha^* \right] = \text{Im} [\text{tr} [T U (R L^\dagger)]] \quad (29)$$

So

$$\begin{aligned} F_\mu^{pf}(n) &= 2i\kappa \text{Im} \left[\phi_1^\dagger M \phi_2 \right]_\mu (n) = \kappa \left(2i \sum_a \text{Imtr} \left[T^a U_\mu(n) \left(\phi_{R1} \phi_{L1}^\dagger + \phi_{R2} \phi_{L2}^\dagger \right) \right] T^a \right) \\ &= i\kappa \left\{ U_\mu(n) \left(\phi_{R1} \phi_{L1}^\dagger + \phi_{R2} \phi_{L2}^\dagger \right) \right\} \Big|_{TA} \end{aligned} \quad (30)$$

which is also an anti-Hermitian traceless matrix.

So, the momentum is always anti-Hermitian traceless..

For anti-Hermitian traceless matrix M , the $\exp(M)$ can be simplified as Appendix. A of Ref. [8].

2.1.4 Solver in HMC

To calculate $(\hat{D}\hat{D}^\dagger)^{-1}$, we need a solver. The detail of solvers will be introduced in Sec. 3. Here we establish a simple introduction.

Let M be a matrix operating on a vector, for example, $M = (\hat{D}\hat{D}^\dagger)$, the goal of the solver is to find x such $b = M \cdot x$, and therefor $x = (\hat{D}\hat{D}^\dagger)^{-1}b$.

We first introduce the CG algorithm for real vector and real matrix, define

$$Q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \cdot A \cdot \mathbf{x} - \mathbf{x}^T \mathbf{b}. \quad (31)$$

so that one can try to find the minimum of Q , and at the minimum

$$\frac{\partial}{\partial \mathbf{x}} Q(\mathbf{x}) = 0 = A \cdot \mathbf{x} - \mathbf{b}. \quad (32)$$

To find the minimum, one can use gradient. Starting from a random point on a curve, calculate the falling speed and move it until it is stable.

For complex vector, one can use BiCGStab in Table. 6.2 in Ref. [2]. It can be described as

Algorithm 1 BiCGStab, note that, the numbers are **complex** number.

```

x = b ▷ Use b as trail solution and start.
for  $i = 0$  to  $r$  do
    r = b -  $A\mathbf{x}$  ▷ Restart  $r$  times
     $\mathbf{r}_h = \mathbf{r}^*$  (Note, that we use  $r^*$  as in [9] which is tested to be better.)
    for  $j = 0$  to  $itera$  do
         $\rho = \mathbf{r}_h^* \cdot \mathbf{r}_j$ 
        if  $j = 0$  then
            p = r
        else
             $\beta = \alpha \times \rho / (\omega \times \rho_p)$ 
            p = r +  $\beta(\mathbf{p} - \omega\mathbf{v})$ 
        end if
        v =  $A\mathbf{p}$ 
         $\alpha = \rho / (\mathbf{r}_h^* \cdot \mathbf{v})$ 
        s = r -  $\alpha\mathbf{v}$ 
        if  $0 \neq j$  and  $0 = \text{mod}(j, 5)$  then
             $er = \|\mathbf{s}\|$  ▷ Check deviation every 5 steps
            if  $er < \epsilon$  then
                return x
            end if
        end if
        t =  $A\mathbf{s}$ 
         $\omega = \mathbf{s}^* \cdot \mathbf{t} / \|\mathbf{t}\|$ 
        r = s -  $\omega\mathbf{t}$ 
        x = x +  $\alpha\mathbf{p} + \omega\mathbf{s}$ 
         $\rho_p = \rho$  ▷ Preserve the last calculated  $\rho$  because we still need it
    end for
end for

```

2.1.5 Leap frog integrator

In Sec. 2.1.2, the basic idea is introduced. However, the implementation is slightly different.

$$U_\mu(0, x) = gauge(x), \quad P_\mu(0, x) = \sum_a r_a(\mu, x) T_a \quad (33a)$$

$$F_\mu(n\epsilon, x) = -\frac{\beta}{2N} \{U_\mu(n\epsilon, x) \Sigma_\mu(n\epsilon, x)\}_{TA} \quad (33b)$$

$$P_\mu(\frac{1}{2}\epsilon, x) = P_\mu(0, x) + \frac{\epsilon}{2} F_\mu(0, x) \quad (33c)$$

$$U_\mu((n+1)\epsilon, x) = \exp\left(i\epsilon P_\mu((n+\frac{1}{2})\epsilon, x)\right) U_\mu(n\epsilon, x) \quad (33d)$$

$$P_\mu((n+\frac{1}{2})\epsilon, x) = P_\mu((n-\frac{1}{2})\epsilon, x) + \epsilon F_\mu(n\epsilon, x) \quad (33e)$$

Note that, the sign of F is ‘+’ here which is different from Ref. [2], because in Ref. [2], $F = \partial_{\mu,n} S = -\dot{P}$. Here we define $F = \dot{P} = -\partial_{\mu,n} S$.

Or simply written as

$$P_\epsilon \circ U_\epsilon \circ P_{\frac{1}{2}\epsilon}(P_0, U_0) \quad (34)$$

The pseudo code can be written as

Algorithm 2 leap-frog integration

```

f = CalculateForce(actions, U)
p = p + 0.5 × εf
for i = 1 to n do
    U = exp(εp)U
    f = CalculateForce(actions, U)
    if i = n then
        p = p + 0.5 × εf
    else
        p = p + εf
    end if
end for

```

▷ We still need to update **p** for the Metropolis step.

2.1.6 A summary of HMC with pseudofermions

Now, every part is ready. We summary the HMC following the Sec.8.2.3 in Ref. [2]. The HMC with fermions can be divided into 6 steps.

1. Generate a complex Bosonic field with $\chi \sim \exp(-\chi^\dagger \chi)$, and $\phi = \hat{D}\chi$.
2. Generate a momentum field P by $\exp(-tr(P^2))$.
3. Calculate $E = tr(P^2) + S_G(U) + S_{pf}(U, \phi)$.
4. Use U_0 to calculate F , evaluate P and U using integrator. Here, ϕ is treated as a constant field.
5. Finally, use P', U' to calculate $E' = tr(P'^2) + S_G(U') + S_{pf}(U', \phi)$. Use a Metropolis to accept or reject the result (configurations) **Note, by Refs. [2] and [6] ‘reject’ means add a duplicated old configuration..**
6. Iterate from 1 to 5, until the number of configurations generated is sufficient.

- More on Metropolis step:

If the hybrid Monte Carlo can be implemented exactly, then, when equilibrium is reached, H should be unchanged, so, in some implementation, the Metropolis step can be ignored to archive a better accept rate. The parameter **Metropolis** of parameter **Updater** can be set to 1 if Metropolis step is enabled and 0 otherwise.

2.2 Optimization of HMC**2.2.1 Omelyan integrator**

The Omelyan integrator can be simply written as (c.f. Eq. (2.80) of Ref. [1])

$$P_{\lambda\epsilon} \circ U_{\frac{1}{2}\epsilon} \circ P_{(1-2\lambda)\epsilon} \circ U_{\frac{1}{2}\epsilon} \circ P_{\lambda\epsilon} (P_0, U_0) \quad (35)$$

with

$$\lambda = \frac{1}{2} - \frac{(2\sqrt{326} + 36)^{\frac{1}{3}}}{12} + \frac{1}{6(2\sqrt{326} + 36)^{\frac{1}{3}}} \approx 0.19318332750378364 \quad (36)$$

In practical, the λ is a tunable parameter, and usually, $2\lambda = 0.3 \sim 0.5$ [6]. The **Omelyan2Lambda** parameter of **Updater** is a input parameter to set 2λ , which if left blank is set to be 0.38636665500756728 by default.

Usually, for each sub-step, it is 2 times slower then leap-frog, and for one trajectory, it is 1.5 time faster [6], implying the number of sub-step needed is about 1/3 of leap-frog.

2.2.2 Omelyan force-gradient integrator

Start from the approximation

$$\log \left(\exp\left(\frac{\epsilon S}{6}\right) \exp\left(\frac{\epsilon T}{2}\right) \exp\left(\frac{2}{3}\epsilon S + \frac{\epsilon^3}{72}[S, [S, T]]\right) \exp\left(\frac{\epsilon T}{2}\right) \exp\left(\frac{\epsilon S}{6}\right) \right) = S + T + \mathcal{O}(\epsilon^4) + \mathcal{O}(\epsilon^6) \quad (37)$$

with [10]

$$\mathcal{O}(\epsilon^4) \sim 10^{-4}\epsilon^4 \quad (38)$$

The other 4 steps are the usual ones, except for $\exp\left(\frac{2}{3}\epsilon S + \frac{\epsilon^3}{72}[S, [S, T]]\right)$ which correspond to

$$\begin{aligned} \omega_i &\rightarrow \omega_i - \frac{2}{3}\tau \frac{\partial}{\partial \omega_i} S + \frac{1}{36}\tau^3 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j} \frac{\partial}{\partial \omega_i} S \\ p_i &= \sum_a \omega_i^a T^a \end{aligned} \quad (39)$$

Use the approximation [11]

$$\begin{aligned} &\frac{2}{3}\tau \frac{\partial}{\partial \omega_i} S - \frac{1}{36}\tau^3 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j} \frac{\partial}{\partial \omega_i} S \\ &= \frac{2}{3}\tau \exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) \frac{\partial}{\partial \omega_i} S + \mathcal{O}(\tau^5) \end{aligned} \quad (40)$$

Let U' be a function of U , solving

$$\begin{aligned} &\frac{2}{3}\tau \exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) \frac{\partial}{\partial \omega_i} S(U) = \frac{2}{3}\tau \frac{\partial}{\partial \omega_i} S(U') \\ &\exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) \frac{\partial}{\partial \omega_i} U \frac{\partial S(U)}{\partial U} = \frac{\partial}{\partial \omega_i} U' \frac{\partial S(U')}{\partial U'} \\ &\frac{\partial}{\partial \omega_i} \left(\exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) U \right) = \frac{\partial}{\partial \omega_i} U' \\ &\exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) U = U', \\ &U' = \exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) \frac{\partial}{\partial \omega_j}\right) U = \exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) T_i\right) U \end{aligned} \quad (41)$$

This approximation can be divided into 3 steps:

1. Calculate $U' = \exp\left(-\frac{1}{24}\tau^2 \sum_j \left(\frac{\partial}{\partial \omega_j} S \right) T_i\right) U$.
2. Use $S[U']$ and $\frac{2}{3}\tau$ to update P .
3. Restore U .

It is easy to implement, we do not need to calculate second derivative, and $\sum_j \left(\frac{\partial}{\partial \omega_j} S \right) T_i$ is already implemented, it is nothing but the **force**.

It is almost as accurate as force-gradient, see the compare in Ref. [12]

2.2.3 Multi-rate integrator (nested integrator)

Following Ref. [12]

Assuming the action is $S = S_F + S_G$ with $S_G \gg S_F$, one can evaluate S_G more often than S_F . In the case of lattice QCD, often, the S_G is the cheap gauge force, and S_F is the expensive fermion force.

The nested scheme is different for leap-frog Omelyan and force-gradient integrator, but they are similar

- Nested leap-frog

$$\begin{aligned}\Delta(h) &= \exp(\frac{h}{2}S_F)\Delta_m(h)\exp(\frac{h}{2}S_F), \\ \Delta_m(h) &= \left(\exp(\frac{h}{2m}S_G)\exp(\frac{h}{m}T)\exp(\frac{h}{2m}S_G)\right)^m.\end{aligned}\quad (42)$$

- Nested Omelyan

$$\begin{aligned}\Delta(h) &= \exp(\lambda h S_F)\Delta_m(\frac{h}{2})\exp((1-2\lambda)h S_F)\Delta_m(\frac{h}{2})\exp(\epsilon h S_F), \\ \Delta_m(\textcolor{red}{h}) &= \left(\exp(\frac{\lambda h}{m}S_G)\exp(\frac{h}{2m}T)\exp(\frac{1-2\lambda}{m}h S_G)\exp(\frac{h}{2m}T)\exp(\frac{\lambda h}{m}S_G)\right)^m.\end{aligned}\quad (43)$$

Note, it is $\Delta_m(\frac{h}{2})$ in the first line.

- Nested force-gradient

$$\begin{aligned}\Delta(h) &= \exp(\frac{h}{6}S_F)\Delta_m(\frac{h}{2})\exp(\frac{2}{3}h S_F + \frac{1}{72}h^3 C_F)\Delta_m(\frac{h}{2})\exp(\frac{h}{6}S_F), \\ \Delta_m(\textcolor{red}{h}) &= \left(\exp(\frac{h}{6m}S_G)\exp(\frac{h}{2m}T)\exp\left(\frac{2}{3}\frac{h}{m}S_G + \frac{1}{72}\left(\frac{h}{m}\right)^3 C_G\right)\exp(\frac{h}{2m}T)\exp(\frac{h}{6m}S_G)\right)^m.\end{aligned}\quad (44)$$

Note about the integrator: when analyzing the error of the integrators, it is assumed e^T , e^{S_G} and e^{S_F} can be calculated accurately. It is almost true for e^{S_G} , and almost true for e^T as long as ϵ is not too large, but it is not true for e^{S_F} . Typically, using an optimized integrator, it needs more accurate criterion for solvers.

2.2.4 Cached solution

The pseudo fermion field is generate only once for a trajectory and is not changed. Also, the gauge field is changing slowly in one trajectory, this make the solutions for $\mathbf{x}_1 = D^{-1}\mathbf{b}$ or $\mathbf{x}_2 = (DD^\dagger)^{-1}\mathbf{b}$, where D depends on U and $\mathbf{b}$ is the pseudo fermion field, only change slowly.

So, once $\mathbf{x}_{1,2}$ is obtained, in the same trajectory, $\mathbf{x}_{1,2}$ can be set as the initial trail solution for the solver.

2.3 Staggered Fermion

2.3.1 The D=1 staggered fermion and Jordan-Wigner transformation

One way to introduce fermion is to use the Ising action (which is particularly useful in quantum computer simulation of $\mathbb{Z}_2$ lattice gauge [13]). The Jordan-Wigner transformation of $D = 1$ Ising model is famous, now considering $D = 1$ XY model

$$H = \frac{1}{4a} \sum_n (\sigma_x(n)\sigma_x(n+1) + \sigma_y(n)\sigma_y(n+1)) \quad (45)$$

with

$$\phi^\dagger(n) = \sigma^+(n) \left\{ \prod_{m < n} (-\sigma_z(m)) \right\}, \quad \phi(n) = \left\{ \prod_{m < n} (-\sigma_z(m)) \right\} \sigma^-(n), \quad \sigma^\pm = \frac{\sigma_x \pm i\sigma_y}{2} \quad (46)$$

one can verify that ϕ is fermion

$$\{\phi(x), \phi^\dagger(y)\} = \delta_{xy}, \quad \{\phi^\dagger(x), \phi^\dagger(y)\} = \{\phi(x), \phi(y)\} = 0 \quad (47)$$

and

$$H = \frac{1}{2a} \sum_n (\phi^\dagger(n)\phi(n+1) - \phi^\dagger(n+1)\phi(n)) \quad (48)$$

with EoM same as naive discretized fermion

$$\dot{\phi}(n) = -i[\phi(n), H] = \frac{1}{2a} (\phi(n+1) - \phi(n-1)) \quad (49)$$

2.3.2 The relationship between the naive fermion and staggered fermion

Consider the naive discretized **massless** fermion

$$S_F = a^4 \sum_n \sum_\mu \frac{\bar{\psi}(n) \gamma_\mu U_\mu(n) \psi(n+\mu) + \bar{\psi}(n) \gamma_\mu U_{-\mu}(n) \psi(n-\mu)}{2a} \quad (50)$$

Although γ matrix can mix different component of the spinor, one can however, reorder the component, for example

$$\bar{\psi}(n) \gamma_x \psi(n+x) = \begin{pmatrix} \phi_x(n) \\ \phi_y(n) \\ \phi_z(n) \\ \phi_t(n) \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \phi_t(n+x) \\ \phi_z(n+x) \\ \phi_y(n+x) \\ \phi_x(n+x) \end{pmatrix} \quad (51)$$

One can see ϕ_x only couple to ϕ_x . **It can be shown that, such reorder is independent of path from any given starting site.**

To show that, just start from $\phi_i(n)$, reorder $\phi_i(n+\mu)$ based on $\phi_i(n)$, then $\phi_i(n+\mu+\nu)$ based on $\phi_i(n+\mu)$, then $\phi_i(n+\nu)$ based on $\phi_i(n+\mu+\nu)$ then $\phi_i(n)$ based on $\phi_i(n+\nu)$ (a loop), use the fact that there are always even number of γ_μ for any μ in a loop, and $\gamma_\mu^2 = 1$, so **a loop-reorder will not change the order of component**, which result in **reorder is independent of path**, and therefore **$\psi(x)$ on each site has an ambiguous reorder.**

- An easy reorder

Since it has been proved that reorder is independent of path, the reorder only depend on the coordinate (x, y, z, t) . An easy way is to define $\psi(n) = \gamma_1^x \gamma_2^y \gamma_3^z \gamma_4^t \chi(n)$. Note that, for $i \neq j$, $\gamma_i \gamma_j = -\gamma_j \gamma_i$, **so there is a sign due to commutate gamma matrix**

$$\begin{aligned} \chi(n) &= \gamma_4^t \gamma_3^z \gamma_2^y \gamma_1^x \psi(n) \\ \bar{\chi}(n) \chi(n+y) &= \bar{\psi}(n) \gamma_1^x \gamma_2^y \gamma_3^z \gamma_4^t \gamma_4^t \gamma_3^z \gamma_2^y \gamma_1^x \psi(n+y) = (-1)^x \bar{\psi}(n) \gamma_y \psi(n+y) \\ \bar{\chi}(n) \chi(n+z) &= \bar{\psi}(n) \gamma_1^x \gamma_2^y \gamma_3^z \gamma_4^t \gamma_4^t \gamma_3^z \gamma_2^y \gamma_1^x \psi(n+z) = (-1)^{x+y} \bar{\psi}(n) \gamma_z \psi(n+z) \end{aligned} \quad (52)$$

Defining $\eta_\mu(x)$ so that **(Note that μ in $\eta_\mu(x)$ can be $\mu = 5$)**

$$\begin{aligned} \eta_\mu(x) &= (-1)^{\sum_{\nu < \mu} x_\nu} \\ \eta_\mu(x) \bar{\chi}(n) \chi(n+\mu) &= \bar{\psi}(n) \gamma_\mu \psi(n+\mu) \end{aligned} \quad (53)$$

Note that, it hold with massive fermions that the naive discretization can be written as

$$S_F = a^4 \sum_n \left\{ \sum_\mu \eta_\mu(n) \bar{\chi}(n) \frac{U_\mu(n) \chi(n+\mu) - U_{-\mu}(n) \chi(n-\mu)}{2a} + m \bar{\chi}(n) \chi(n) \right\} \quad (54)$$

Here χ is still a 4-component spinor, which is in fact a reorder of $\psi(n)$, the components of χ do NOT mix with each other. Now by the fact each component of χ is equivalent (degenerate),

$$\begin{aligned} S_F &= a^4 \sum_n \sum_\alpha \left\{ \sum_\mu \eta_\mu(n) \bar{\chi}_\alpha(n) \frac{U_\mu(n) \chi_\alpha(n+\mu) - U_{-\mu}(n) \chi_\alpha(n-\mu)}{2a} + m \bar{\chi}_\alpha(n) \chi_\alpha(n) \right\} \\ &= 4a^4 \sum_n \left\{ \sum_\mu \eta_\mu(n) \bar{\chi}(n) \frac{U_\mu(n) \chi(n+\mu) - U_{-\mu}(n) \chi(n-\mu)}{2a} + m \bar{\chi}(n) \chi(n) \right\} \end{aligned} \quad (55)$$

In the last line, we redefine $\chi(x)$ as a scalar (1-component) field.

Kogut-Susskind staggered fermion is just naive discretized fermion.

2.3.3 Symmetries of the staggered fermions

Now we concentrate on

$$D_{st}(n|m) = m \delta_{m,n} + \sum_\mu \eta_\mu(n) \frac{U_\mu(n) \delta_{n,n+\mu} - U_{-\mu}(n) \delta_{n,n-\mu}}{2a} \quad (56)$$

- D_{st} is γ_5 -hermiticity

$$D_{st}^\dagger(n|m) = \eta_5(\textcolor{red}{n}) D_{st}^\dagger(n|m) \eta_5(\textcolor{red}{m}) \quad (57)$$

- **massless** case is **anti-hermitian traceless**

$$D_{st}^\dagger(n|m) = -D_{st}(n|m) \quad (58)$$

- Chiral symmetry

The **massless** S_F is unchanged under transformation

$$\chi(n) \rightarrow \exp(i\alpha\eta_5(n))\chi(n), \quad \bar{\chi}(n) \rightarrow \bar{\chi}(n)\exp(i\alpha\eta_5(n)) \quad (59)$$

Just note that they are somehow different, the chiral symmetry is from $\gamma_\mu \exp(i\alpha\gamma_5) = \exp(-i\alpha\gamma_5)\gamma_\mu$, and the chiral symmetry of staggered fermion is from $\eta_\mu \exp(i\alpha\eta_5(n+\mu)) = \exp(i\alpha\eta_5(n+\mu))\eta_\mu = \exp(-i\alpha\eta_5(n))\eta_\mu$.

- Sign problem of the staggered fermions

Let $D_{st}(m=0) = U^\dagger \Lambda U$ where Λ is diagonal, U is unitary. $D_{st}^\dagger(m=0) = U^\dagger \Lambda^\dagger U = -D_{st}(m=0) = -U^\dagger \Lambda U$, so Λ is a diagonal matrix with $\Lambda^\dagger = -\Lambda$, Λ should be pure imaginary number.

Then, use $D = U^\dagger \Lambda U + mU^\dagger U = U^\dagger (\Lambda + m) U$, and $\det[D] = \det[U^\dagger] \det[\Lambda + m] \det[U]$. And use $\det[U^\dagger] = \det[U^{-1}] = 1/\det[U]$, one find $\det[D_{st}] = \det[\Lambda + m]$, where Λ is pure imaginary number, and $m \geq 0$ is real number. One find **the eigenvalues of D_{st} are complex number with real part exactly the mass.**

I know the sum of eigenvalues is real number, but how to prove they are complex conjugate pairs?

It has been proved that $\det[D_{st}] \geq m$, so one do not worry about the sign problem, even with a single flavour. And It is a consequence of (i) massless case is anti-hermitian traceless; (ii) the massive case is η_5 -hermiticity.

2.4 The RHMC for staggered fermion

From now on, we consider $N_f = 2 + 1$. Two key quantities should be calculated.

- Calculate the action

$$\exp(-S) = (\det[D_{st}(m_{ud})])^{\frac{1}{2}} (\det[D_{st}(m_s)])^{\frac{1}{4}} \exp(-S_G) \quad (60)$$

The problem is how to evaluate $(\det[D])^\alpha$. Before that, we have to prepare some tools.

2.4.1 The rational approximation and Remes algorithm

The first thing to do is to find a rational approximation of a function, here we use Remes algorithm as proposed by Ref. [14]

The Remes algorithm is implemented in Mathematica, which finds

$$g(x) = c + \sum_i \frac{a_i}{x + b_i} \approx f(x), \quad (61)$$

in an interval. This have been implemented in Mathematica so we use the result of Mathematica directly. An example is

```
1 << FunctionApproximations\;  
   res1 = Apart[MiniMaxApproximation[1/Sqrt[x], {x, {0.003, 1}, 3, 3}][[2]][[1]]]  
   Join[{Part[res1, 1]}, Table[Numerator[Part[res1, n]], n, 2, 4], Table[Denominator[Part[res1, n]] /. x -> 0, {n, 2, 4}]]
```

2.4.2 The RHMC with $N_f = 2$

We mainly follow Ref. [14].

Similar as the HMC, the first step is to rewrite the action use a Boson field (pseudo-fermion field), still starting with

$$Z = \int \mathcal{D}[U] \prod_{f=1}^{N_f} \mathcal{D}[\bar{\psi}_f] \mathcal{D}[\psi_f] (\det D_{st})^{\frac{1}{4}} \exp(-S_G[U]) \quad (62)$$

Note that the 1/4 dose NOT come from the 1/4 in Eq. (55), it is to deal with the 4 doublers. With

$$D_{st}(n|m) = 2m\delta_{m,n} + \sum_{\mu} \eta_{\mu}(n) (U_{\mu}(n)\delta_{n,n+\mu} - U_{-\mu}(n)\delta_{n,n-\mu}) \quad (63)$$

It is very interesting that, although stated in many literatures, this is **not** the true fourth root method. The true fourth root method is after the ‘pseudofermion’ step. First, just like the usual HMC to evaluate

$$Z = \int \mathcal{D}[U] \prod_{f=1}^{N_f} (\det D_{st}) \exp(-S_G[U]) \quad (64)$$

For $N_f = 2$, before the 4th root, it is

$$Z = \int \mathcal{D}[U] (\det D_{st}^\dagger D_{st}) \exp(-S_G[U]) \quad (65)$$

Note that because D_{st0} for massless is anti-Hermitian, $D_{st}^\dagger D_{st} = -\frac{1}{4a^2} D_{st0}^2 + m^2$, for simplicity we denote $A = -D_{st0}^2 + 4a^2 m^2$ (with a rescale of ψ assumed), using

$$\det A = \frac{1}{\det A^{-1}} = \int \mathcal{D}[\phi] \exp(-\phi^\dagger A^{-1} \phi). \quad (66)$$

For $N_f = 2$ it is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger A^{-1} \phi) \quad (67)$$

The true fourth root method is taken here as

$$\begin{aligned} Z &= \int \mathcal{D}[U] (\det A)^{\frac{N_f}{4}} \exp(-S_G[U]) \\ Z &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger A^{-\frac{N_f}{4}} \phi) \end{aligned} \quad (68)$$

Note that, in Ref. [14] the fourth root is just to remove doublers, so for $N_f = 2$, it is $A^{1/2}$ in spite that $A = D^\dagger D$ already considered $N_f = 2$. However, it should not be correct. **According to Ref. [15], A should bring another doubler, so it should be $N_f/8$, or, one can use a even (or odd) pseudofermion to remove another doubler.**

In the following, we concentrate on

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger A^{-\frac{1}{2}} \phi) \quad (69)$$

- Pseudofermion refreshment of staggered fermion

Note! The refreshment of staggered fermion use a different action! It can be summarized as $Z_{MC} \approx Z_{MD}$, where Z_{MC} is to refresh the pseudofermion, Z_{MD} is for molecular dynamic.

$$\begin{aligned} Z_{MC} &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger (A^{-\frac{1}{4}})^2 \phi) \\ Z_{MD} &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger r_2(A) \phi) \end{aligned} \quad (70)$$

where $r_2(A) \approx A^{-\frac{1}{2}}$.

Then, the refreshment step is:

Let χ be random **complex** numbers according to $\exp(-\chi^\dagger \chi)$. Let $A^{-\frac{1}{4}} \phi = \chi$, ϕ is the random **complex** number satisfying distribution we want $(\exp(-\phi^\dagger (A^{-\frac{1}{4}})^\dagger A^{-\frac{1}{4}} \phi))$. So, first get χ and then let $\phi = A^{+\frac{1}{4}} \chi$.

- MD step of staggered fermion

Let $A^{-\frac{1}{2}} = c_0 + \sum_i \frac{\alpha_i}{A + \beta_i}$, it is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - c_0 \phi^\dagger \phi - \sum_i \phi^\dagger \frac{\alpha_i}{A + \beta_i} \phi\right) \quad (71)$$

then

$$\begin{aligned} F &= -iT_a \frac{\partial}{\partial \omega_a} \left(\sum_i \phi^\dagger \frac{\alpha_i}{A + \beta_i} \phi \right) \\ &= -iT_a \sum_i \left(\phi^\dagger \frac{\alpha_i}{(A + \beta_i)^2} \frac{\partial}{\partial \omega_a} A \phi \right) \end{aligned} \quad (72)$$

The order of matrix product is wrong, it is in fact

$$F = -iT_a \sum_i \left(\phi^\dagger \frac{\sqrt{\alpha_i}}{(A + \beta_i)} \left(\frac{\partial}{\partial \omega_a} A \right) \frac{\sqrt{\alpha_i}}{(A + \beta_i)} \phi \right) \quad (73)$$

Now, let $\phi_i = \frac{\sqrt{\alpha_i}}{(A+\beta_i)}\phi$, it is

$$F = -iT_a \sum_i \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i \quad (74)$$

Note it is not $\phi' = \sum_i \frac{\sqrt{\alpha_i}}{(M+\beta_i)}\phi$ **because generally,** $\phi_i^\dagger A' \phi_j \neq 0$

To calculate the A' we use

$$\begin{aligned} \frac{\partial}{\partial \omega_a} A &= \left(\frac{\partial}{\partial \omega_a} D_{st0}^\dagger \right) D_{st0} + D_{st0}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \\ \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i &= \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0}^\dagger \right) \phi_{i,d} + \phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \phi_i \\ &= \left(\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \phi_i \right)^\dagger + \left(\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \phi_i \right) \\ &= 2\text{Re} \left[\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \phi_i \right] \end{aligned} \quad (75)$$

with $\phi_{i,d} = D_{st0}\phi_i$.

$$\begin{aligned} \frac{\partial}{\partial \omega_a} D_{st0} &= \sum_\mu \left(\eta_\mu(n) \left(\frac{\partial}{\partial \omega_a} U_\mu(n) \right) \delta_{n,n+\mu} - \eta_\mu(n+\mu) \left(\frac{\partial}{\partial \omega_a} U_\mu^\dagger(n) \right) \delta_{n+\mu,n} \right) \\ &= i\eta_\mu(n) \left((T_a U_\mu(n)) \delta_{n,n+\mu} + (U_\mu^\dagger(n) T_a) \delta_{n+\mu,n} \right) \end{aligned} \quad (76)$$

note that the $\eta_\mu(n+\mu)$ follows $\delta_{n+\mu,n}$. The last step we use $\eta_\mu(n+\mu) = \eta_\mu(n)$. The $T_a = T_a^\dagger$, let

$$\begin{aligned} \phi_A &= \phi_i, \phi_B = T_a U_\mu(n) \phi_i \\ \phi_C &= \phi_{i,d}, \phi_D = T_a U_\mu(n) \phi_{i,d} \end{aligned} \quad (77)$$

it is

$$\begin{aligned} \phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_{st0} \right) \phi_i &= i\eta(n) \left(\phi_C^\dagger(n) \phi_B(n+\mu) + \phi_D^\dagger(n+\mu) \phi_A(n) \right) \\ \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i &= i\eta(n) \left(\phi_C^\dagger(n) \phi_B(n+\mu) + \phi_D^\dagger(n+\mu) \phi_A(n) \right) - i\eta(n) \left(\phi_B^\dagger(n+\mu) \phi_C(n) + \phi_A^\dagger(n) \phi_D(n+\mu) \right) \\ &= i\eta(n) \left\{ \left(\phi_C^\dagger(n) \phi_B(n+\mu) - \phi_B^\dagger(n+\mu) \phi_C(n) \right) + \left(\phi_D^\dagger(n+\mu) \phi_A(n) - \phi_A^\dagger(n) \phi_D(n+\mu) \right) \right\} \\ &= -2\eta(n) \text{Im} \left[\phi_C^\dagger(n) \phi_B(n+\mu) + \phi_D^\dagger(n+\mu) \phi_A(n) \right] \\ &= -2\eta(n) \text{Im} \left[\phi_{i,d}^\dagger T_a U_\mu(n) \delta_{n,n+\mu} \phi_i + \phi_{i,d}^\dagger(n+\mu) U_\mu^\dagger(n) T_a \delta_{n+\mu,n} \phi_i \right] \end{aligned} \quad (78)$$

let $M(n|m) = U_\mu(n) \delta_{n,n+\mu}$, it is

$$\phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i = -2\eta(n) \text{Imtr} \left\{ T_a \left[(U_\mu(n) \phi_i(n+\mu)) \phi_{i,d}^\dagger(n) \right] + T_a \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n+\mu))^\dagger \right] \right\} \quad (79)$$

One can verify that for any matrix

$$\begin{aligned} 2i \sum_a T_a \text{Imtr}[T_a M] &= -2i \sum_a T_a \text{Retr}[iT_a M] = \{M\}_{TA} \\ 2i \sum_a T_a \text{Imtr}[iT_a M] &= 2i \sum_a T_a \text{Retr}[T_a M] = \{iM\}_{TA} \end{aligned} \quad (80)$$

so

$$\begin{aligned} F_\mu(n) &= -iT_a \sum_i \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i \\ &= \eta_\mu(n) \sum_a \sum_i 2iT_a \text{Imtr} \left\{ T_a \left[(U_\mu(n) \phi_i(n+\mu)) \phi_{i,d}^\dagger(n) \right] + T_a \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n+\mu))^\dagger \right] \right\} \\ &= \eta_\mu(n) \sum_i \left\{ \left[(U_\mu(n) \phi_i(n+\mu)) \phi_{i,d}^\dagger(n) \right] + \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n+\mu))^\dagger \right] \right\}_{TA} \end{aligned} \quad (81)$$

with

$$\begin{aligned} \phi_i &= \frac{\sqrt{\alpha_i}}{(D_{st}^\dagger D_{st}) + \beta_i} \phi, \quad \alpha_i, \beta_i \in A^{-\frac{1}{2}} \\ \phi_{i,d} &= D_{st0} \phi_i \end{aligned} \quad (82)$$

Here is the summary of $N_f = 2$ RHMC for staggered fermion

1. Refresh ϕ using $\phi = (D_{st}^\dagger D_{st})^{\frac{1}{4}} \chi$, where χ is random Gaussian complex field.
2. Calculate force using Eq. (108).
3. Metropolis step.

Some tips on RHMC of staggered fermions:

- Let $r_1(M) \approx M^a$, $r_2(M) \approx M^b$, $r_1(M)$ is generally less accurate than $(r_2(M))^{1/b}$ when $b > 1$.
- For $N_f = 2 + 1$, there are other optimizations.

2.4.3 The RHMC with $N_f = 2 + 1$

Before we continue, it is necessary to extend the nested integrator, here we use Force gradient because it is the best one, a two level integrator is introduced in Sec. 2.2.3, which is modified as

$$\begin{aligned} \Delta(h) &= \exp\left(\frac{h}{6}S_F\right)\Delta_m\left(\frac{h}{2m}\right)\exp\left(\frac{2}{3}hS_F + \frac{1}{72}h^3C_F\right)\Delta_m\left(\frac{h}{2m}\right)\exp\left(\frac{h}{6}S_F\right), \\ \Delta_m(\bar{h}) &= \left(\exp\left(\frac{h}{6}S_G\right)\exp\left(\frac{h}{2}T\right)\exp\left(\frac{2}{3}hS_G + \frac{1}{72}h^3C_G\right)\exp\left(\frac{h}{2}T\right)\exp\left(\frac{h}{6}S_G\right)\right)^m. \end{aligned} \quad (83)$$

The Lagrangian of $N_f = 2 + 1$ is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi_1^\dagger A_{ud}^{-\frac{1}{2}} \phi_1 - \phi_2^\dagger A_s^{-\frac{1}{4}} \phi_2\right) \quad (84)$$

where $A = D^\dagger D$ (Note that it is always $A^{-\frac{N_f}{4}}$). Which is also

$$\begin{aligned} Z &= (\det[A_{ud}])^{\frac{1}{2}} (\det[A_s])^{\frac{1}{4}} \int \mathcal{D}[U] \exp(-S_G[U]) \\ &= \frac{(\det[A_{ud}])^{\frac{1}{2}}}{(\det[A_{ud}])^{\frac{1}{2}}} (\det[A_s])^{\frac{3}{4}} \int \mathcal{D}[U] \exp(-S_G[U]) \\ &= \left(\det\left[\frac{A_{ud}}{A_{ud}}\right]\right)^{\frac{1}{2}} (\det[A_s])^{\frac{3}{4}} \int \mathcal{D}[U] \exp(-S_G[U]) \\ &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi_1^\dagger \left(\frac{A_s}{A_{ud}}\right)^{\frac{1}{2}} \phi_1 - \phi_2^\dagger A_s^{-\frac{3}{4}} \phi_2\right) \end{aligned} \quad (85)$$

using $A_{ud,s} = -D_{st0}^2 + 4a^2 m_{ud,s}^2$, so $A_s = A_{ud} + 4a^2(m_s^2 - m_{ud}^2)$. Defining $\delta m^2 = m_s^2 - m_{ud}^2$, it is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi_1^\dagger \left(\frac{A_{ud} + 4a^2 \delta m^2}{A_{ud}}\right)^{\frac{1}{2}} \phi_1 - \phi_2^\dagger A_s^{-\frac{3}{4}} \phi_2\right) \quad (86)$$

This is called ‘mass precondition’, the force from the D_{ud} is much smaller and one can have a three level nested integrator. Defining

$$\begin{aligned} r_{1,MC}(x) &= \left(\frac{x + 4a^2 \delta m^2}{x}\right)^{-\frac{1}{4}}, \quad r_{2,MC}(x) = x^{\frac{3}{8}} \\ r_{1,MD}(x) &= \left(\frac{x + 4a^2 \delta m^2}{x}\right)^{\frac{1}{2}}, \quad r_{2,MD}(x) = x^{-\frac{3}{4}} \end{aligned} \quad (87)$$

All the other steps are same as $N_f = 2$.

Possible choices are listed in Table. 1

2.4.4 Multi field Optimization

The multi-level nested integrator is useless compared with multi field optimization. Instead of evaluating $\bar{\psi} A \psi$, one use $\sum_i^\alpha \bar{\psi}_i A^{\frac{1}{\alpha}} \psi_i$. Typically, the force from the s quark is much larger than the mass preconditioned u, d quarks, so we can use $\sum_{i=1,2,3} \bar{\psi}_i A^{-\frac{1}{4}} \psi_i$ instead of $\bar{\psi} A^{-\frac{3}{4}} \psi$. And then use a nested integrator.

I don't know why but it is tested that Omelyan nested with leapfrog is the best choice, better than approximate force gradient.

schemes	MC	MD	MC	MD	MC	MD	MC	MD
$N_f = 1$	$x^{\frac{1}{8}}$	$x^{-\frac{1}{4}}$						
$N_f = 2$	$x^{\frac{1}{4}}$	$x^{-\frac{1}{2}}$						
$N_f = 2 + 1$	$x^{\frac{1}{4}}$	$x^{-\frac{1}{2}}$	$x^{\frac{1}{8}}$	$x^{-\frac{1}{4}}$				
$N_f = 2 + 1$	$\left(\frac{x+4a^2\delta m^2}{x}\right)^{-\frac{1}{4}}$	$\left(\frac{x+4a^2\delta m^2}{x}\right)^{\frac{1}{2}}$	$x^{\frac{3}{8}}$	$x^{-\frac{3}{4}}$				
$N_f = 2 + 1$	$\left(\frac{x+4a^2\delta m^2}{x}\right)^{-\frac{1}{4}}$	$\left(\frac{x+4a^2\delta m^2}{x}\right)^{\frac{1}{2}}$	$x^{\frac{1}{8}}$	$x^{-\frac{1}{4}}$	$x^{\frac{1}{8}}$	$x^{-\frac{1}{4}}$	$x^{\frac{1}{8}}$	$x^{-\frac{1}{4}}$

表 1: The Fermions for $N_f = 2 + 1$.

2.4.5 Tip to implement higher orders staggered fermion

As required by anti-Hermiticity and gauge invariance, for higher orders, we always have

$$\bar{\chi}(n_1)U(n_1, n_2)\chi(n_2) - \bar{\chi}(n_2)U^\dagger(n_1, n_2)\chi(n_1) \quad (88)$$

Let $U(n_1, n_2) = U(n_1, n_3)U_\mu(n_3)$

2.5 The RHMC for staggered fermion with diagonal mass term

In this section, we consider that

$$D(m) = D_{st} + M(m) \quad (89)$$

where M is not a number but a diagonal matrix.

- Calculate the action

$$\exp(-S) = (\det[D(m_{ud})])^{\frac{1}{2}} (\det[D])^{\frac{1}{4}} \exp(-S_G) \quad (90)$$

The problem is how to evaluate $(\det[D])^\alpha$. Before that, we have to prepare some tools.

2.5.1 The RHMC with $N_f = 2$ and with diagonal mass term

We mainly follow Ref. [14].

Similar as the HMC, the first step is to rewrite the action use a Boson field (pseudo-fermion field), still starting with

$$Z = \int \mathcal{D}[U] \prod_{f=1}^{N_f} \mathcal{D}[\bar{\psi}_f] \mathcal{D}[\psi_f] (\det D)^{\frac{1}{4}} \exp(-S_G[U]) \quad (91)$$

with

$$D(n|m) = 2M(n|m)\delta_{m,n} + \sum_{\mu} \eta_{\mu}(n) (U_{\mu}(n)\delta_{n,n+\mu} - U_{-\mu}(n)\delta_{n,n-\mu}) \quad (92)$$

Note that

$$\begin{pmatrix} ia & b \\ -b & -ia \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \begin{pmatrix} iaA & bB \\ -bA & -iaB \end{pmatrix}.$$

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} ia & b \\ -b & -ia \end{pmatrix} = \begin{pmatrix} iaA & bA \\ -bB & -iaB \end{pmatrix}. \quad (93)$$

$$[D_0, M] \neq 0$$

Still we use $A = 4a^2 D^\dagger D \neq -D_0^2 + 4a^2 m^2$, and $A^\dagger = A$.

$$\det A = \frac{1}{\det A^{-1}} = \int \mathcal{D}[\phi] \exp(-\phi^\dagger A^{-1} \phi). \quad (94)$$

For $N_f = 2$ it is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp(-S_G[U] - \phi^\dagger A^{-1} \phi) \quad (95)$$

The true fourth root method is taken here as

$$\begin{aligned} Z &= \int \mathcal{D}[U] (\det A)^{\frac{N_f}{4}} \exp(-S_G[U]) \\ Z &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi^\dagger A^{-\frac{N_f}{4}} \phi\right) \end{aligned} \quad (96)$$

In the following, we concentrate on

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi^\dagger A^{-\frac{1}{2}} \phi\right) \quad (97)$$

- Pseudofermion refreshment of staggered fermion

Similarly it is always,

$$\begin{aligned} Z_{MC} &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi^\dagger \left(A^{-\frac{1}{4}}\right)^2 \phi\right) \\ Z_{MD} &= \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - \phi^\dagger r_2(A) \phi\right) \end{aligned} \quad (98)$$

where $r_2(A) \approx A^{-\frac{1}{2}}$.

Then, the refreshment step is:

Let χ be random **complex** numbers according to $\exp(-\chi^\dagger \chi)$. Let $A^{-\frac{1}{4}} \phi = \chi$, ϕ is the random **complex** number satisfying distribution we want $(\exp(-\phi^\dagger \left(A^{-\frac{1}{4}}\right)^\dagger A^{-\frac{1}{4}} \phi))$. So, first get χ and then let $\phi = A^{+\frac{1}{4}} \chi$.

- MD step of staggered fermion

Let $A^{-\frac{1}{2}} = c_0 + \sum_i \frac{\alpha_i}{A + \beta_i}$, it is

$$Z = \int \mathcal{D}[U] \mathcal{D}[\phi] \exp\left(-S_G[U] - c_0 \phi^\dagger \phi - \sum_i \phi^\dagger \frac{\alpha_i}{A + \beta_i} \phi\right) \quad (99)$$

then (for any matrix, $\frac{\partial M^{-1}}{\partial x} = M^{-1} \frac{\partial M}{\partial x} M^{-1}$)

$$\begin{aligned} F &= -iT_a \frac{\partial}{\partial \omega_a} \left(\sum_i \phi^\dagger \frac{\alpha_i}{A + \beta_i} \phi \right) \\ &= -iT_a \sum_i \left(\phi^\dagger \frac{\sqrt{\alpha_i}}{(A + \beta_i)} \left(\frac{\partial}{\partial \omega_a} A \right) \frac{\sqrt{\alpha_i}}{(A + \beta_i)} \phi \right) \end{aligned} \quad (100)$$

Now, let $\phi_i = \frac{\sqrt{\alpha_i}}{(A + \beta_i)} \phi$, **it is required and satisfied that, $\left(\frac{\sqrt{\alpha_i}}{(A + \beta_i)}\right)^\dagger = \frac{\sqrt{\alpha_i}}{(A + \beta_i)}$ when A is Hermitian**, it is

$$F = -iT_a \sum_i \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i \quad (101)$$

We need to calculate $\frac{\partial A}{\partial \omega_a}$, which is

$$\begin{aligned} A &= D^\dagger D \\ \frac{\partial}{\partial \omega_a} A &= \left(\frac{\partial}{\partial \omega_a} D_0^\dagger \right) D + D^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \\ \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i &= \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} D_0^\dagger \right) \phi_{i,d} + \phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \phi_i \\ &= \left(\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \phi_i \right)^\dagger + \left(\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \phi_i \right) \\ &= 2\text{Re} \left[\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \phi_i \right] \end{aligned} \quad (102)$$

with $\phi_{i,d} = D \phi_i$.

$$\begin{aligned} \frac{\partial}{\partial \omega_a} D_0 &= \sum_\mu \left(\eta_\mu(n) \left(\frac{\partial}{\partial \omega_a} U_\mu(n) \right) \delta_{n,n+\mu} - \eta_\mu(n + \mu) \left(\frac{\partial}{\partial \omega_a} U_\mu^\dagger(n) \right) \delta_{n+\mu,n} \right) \\ &= i\eta_\mu(n) \left((T_a U_\mu(n)) \delta_{n,n+\mu} + (U_\mu^\dagger(n) T_a) \delta_{n+\mu,n} \right) \end{aligned} \quad (103)$$

note that the $\eta_\mu(n + \mu)$ follows $\delta_{\mathbf{n}+\boldsymbol{\mu},\mathbf{n}}$. The last step we use $\eta_\mu(n + \mu) = \eta_\mu(n)$. The $T_a = T_a^\dagger$, let

$$\begin{aligned}\phi_A &= \phi_i, \phi_B = T_a U_\mu(n) \phi_i \\ \phi_C &= \phi_{i,d}, \phi_D = T_a U_\mu(n) \phi_{i,d}\end{aligned}\tag{104}$$

it is

$$\begin{aligned}\phi_{i,d}^\dagger \left(\frac{\partial}{\partial \omega_a} D_0 \right) \phi_i &= i\eta(n) \left(\phi_C^\dagger(n) \phi_B(n + \mu) + \phi_D^\dagger(n + \mu) \phi_A(n) \right) \\ \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i &= i\eta(n) \left(\phi_C^\dagger(n) \phi_B(n + \mu) + \phi_D^\dagger(n + \mu) \phi_A(n) \right) - i\eta(n) \left(\phi_B^\dagger(n + \mu) \phi_C(n) + \phi_A^\dagger(n) \phi_D(n + \mu) \right) \\ &= i\eta(n) \left\{ \left(\phi_C^\dagger(n) \phi_B(n + \mu) - \phi_B^\dagger(n + \mu) \phi_C(n) \right) + \left(\phi_D^\dagger(n + \mu) \phi_A(n) - \phi_A^\dagger(n) \phi_D(n + \mu) \right) \right\} \\ &= -2\eta(n) \text{Im} \left[\phi_C^\dagger(n) \phi_B(n + \mu) + \phi_D^\dagger(n + \mu) \phi_A(n) \right] \\ &= -2\eta(n) \text{Im} \left[\phi_{i,d}^\dagger T_a U_\mu(n) \delta_{n,n+\mu} \phi_i + \phi_{i,d}^\dagger(n + \mu) U_\mu^\dagger(n) T_a \delta_{n+\mu,n} \phi_i \right]\end{aligned}\tag{105}$$

let $M(n|m) = U_\mu(n) \delta_{n,n+\mu}$, it is

$$\phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i = -2\eta(n) \text{Imtr} \left\{ T_a \left[(U_\mu(n) \phi_i(n + \mu)) \phi_{i,d}^\dagger(n) \right] + T_a \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n + \mu))^\dagger \right] \right\}\tag{106}$$

One can verify that for any matrix

$$\begin{aligned}2i \sum_a T_a \text{Imtr}[T_a M] &= -2i \sum_a T_a \text{Retr}[iT_a M] = \{M\}_{TA} \\ 2i \sum_a T_a \text{Imtr}[iT_a M] &= 2i \sum_a T_a \text{Retr}[T_a M] = \{iM\}_{TA}\end{aligned}\tag{107}$$

so

$$\begin{aligned}F_\mu(n) &= -iT_a \sum_i \phi_i^\dagger \left(\frac{\partial}{\partial \omega_a} A \right) \phi_i \\ &= \eta_\mu(n) \sum_a \sum_i 2iT_a \text{Imtr} \left\{ T_a \left[(U_\mu(n) \phi_i(n + \mu)) \phi_{i,d}^\dagger(n) \right] + T_a \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n + \mu))^\dagger \right] \right\} \\ &= \eta_\mu(n) \sum_i \left\{ \left[(U_\mu(n) \phi_i(n + \mu)) \phi_{i,d}^\dagger(n) \right] + \left[\phi_i(n) (U_\mu(n) \phi_{i,d}(n + \mu))^\dagger \right] \right\}_{TA}\end{aligned}\tag{108}$$

with

$$\begin{aligned}\phi_i &= \frac{\sqrt{\alpha_i}}{(D^\dagger D) + \beta_i} \phi, \quad \alpha_i, \beta_i \in A^{-\frac{1}{2}} \\ \phi_{i,d} &= \textcolor{red}{D} \phi_i\end{aligned}\tag{109}$$

2.6 The update of boson field and gauge field

2.6.1 action

Consider

$$S = D_\mu \phi D_\mu \phi + m\phi^2 + \lambda\phi^4\tag{110}$$

where

$$D_\mu \phi = \frac{U_\mu(n) \phi(n + \mu) - \phi(n)}{a}\tag{111}$$

so

$$\begin{aligned}S &= \sum_n \sum_\mu \frac{\phi^\dagger(n + \mu) U_\mu^\dagger(n) - \phi^\dagger(n)}{a} \frac{U_\mu(n) \phi(n + \mu) - \phi(n)}{a} \\ &\quad + m \sum_n \phi^\dagger(n) \phi(n) + \lambda \sum_n (\phi^\dagger(n) \phi(n))^2 \\ &= - \sum_n \sum_\mu \frac{\phi^\dagger(n + \mu) U_\mu^\dagger(n) \phi(n) + \phi^\dagger(n) U_\mu(n) \phi(n + \mu)}{a^2} + \sum_n \sum_\mu \frac{\phi^\dagger(n + \mu) \phi(n + \mu) + \phi^\dagger(n) \phi(n)}{a^2} \\ &\quad + m \sum_n \phi^\dagger(n) \phi(n) + \lambda \sum_n (\phi^\dagger(n) \phi(n))^2 \\ &= \sum_n \left\{ -\frac{1}{a^2} \phi^\dagger(n) \sum_\mu (U_{-\mu}(n) \phi(n - \mu) + U_\mu(n) \phi(n + \mu)) + \left(\frac{2D}{a^2} + m \right) \phi^\dagger(n) \phi(n) + \lambda (\phi^\dagger(n) \phi(n))^2 \right\}\end{aligned}\tag{112}$$

2.6.2 boson force

Take ϕ and $\phi^\dagger$ as two different functions of ϕ_r and ϕ_i ,

$$\begin{aligned}\frac{\partial S}{\partial \phi_r(n)} &= \frac{\partial S}{\partial \phi(\phi_r, \phi_i, n)} + \frac{\partial S}{\partial \phi^\dagger(\phi_r, \phi_i, n)} \\ -i \frac{\partial S}{\partial \phi_i(n)} &= \frac{\partial S}{\partial \phi(\phi_r, \phi_i, n)} - \frac{\partial S}{\partial \phi^\dagger(\phi_r, \phi_i, n)}\end{aligned}\quad (113)$$

$$\begin{aligned}\frac{\partial S}{\partial \phi^\dagger(\phi_r, \phi_i, n)} &= \left(\frac{2D}{a^2} + m \right) \phi(n) + 2\lambda (\phi^\dagger(n) \phi(n)) \phi(n) - \frac{1}{a^2} \sum_\mu (U_{-\mu}(n) \phi(n - \mu) + U_\mu(n) \phi(n + \mu)) \\ \frac{\partial S}{\partial \phi(\phi_r, \phi_i, n)} &= \left(\frac{\partial S}{\partial \phi^\dagger(n)} \right)^\dagger\end{aligned}\quad (114)$$

therefor,

$$\frac{\partial S}{\partial \phi(n)} = 2 \left\{ \left(\frac{2D}{a^2} + m \right) \phi(n) + 2\lambda (\phi^\dagger(n) \phi(n)) \phi(n) - \frac{1}{a^2} \sum_\mu (U_{-\mu}(n) \phi(n - \mu) + U_\mu(n) \phi(n + \mu)) \right\} \quad (115)$$

2.6.3 gauge force

For one link, there are two contributions,

$$S_{link} = -\frac{1}{a^2} \sum_n \sum_\mu (\phi^\dagger(n + \mu) U_\mu^\dagger(n) \phi(n) + \phi^\dagger(n) U_\mu(n) \phi(n + \mu)) \quad (116)$$

taking

$$\begin{aligned}U_\mu &= \exp(i \sum_a \omega_\mu^a T^a) U_0, \quad \frac{\partial U_\mu}{\partial \omega_\mu^a} = iT^a U_\mu, \quad \frac{\partial U_\mu^\dagger}{\partial \omega_\mu^a} = -i U_\mu^\dagger T^a, \\ (T^a)^\dagger &= T^a, \quad \frac{\partial M^{-1}}{\partial \omega_\mu^a} = -M^{-1} \frac{\partial M}{\partial \omega_\mu^a} M^{-1}\end{aligned}\quad (117)$$

so

$$\begin{aligned}\frac{\partial S_{link}}{\partial \omega_a(\mu, n)} &= -\frac{1}{a^2} (-i \phi^\dagger(n + \mu) U_\mu^\dagger(n) T^a \phi(n) + i \phi^\dagger(n) T^a U_\mu(n) \phi(n + \mu)) \\ &= -i \frac{1}{a^2} \left(-\text{Tr} [\phi(n) (T^a U_\mu(n) \phi(n + \mu))^\dagger] + \text{Tr} [T^a U_\mu(n) \phi(n + \mu) \phi^\dagger(n)] \right) \\ &= 2 \frac{1}{a^2} \text{ImTr} [T^a U_\mu(n) \phi(n + \mu) \phi^\dagger(n)]\end{aligned}\quad (118)$$

and using

$$\begin{aligned}2i \sum_a T_a \text{Imtr}[T_a M] &= -2i \sum_a T_a \text{Retr}[iT_a M] = \{M\}_{TA} \\ 2i \sum_a T_a \text{Imtr}[iT_a M] &= 2i \sum_a T_a \text{Retr}[T_a M] = \{iM\}_{TA}\end{aligned}\quad (119)$$

so

$$\begin{aligned}\dot{\omega}_a &= -\frac{\partial}{\partial \omega_a(\mu, n)} S_{link} \\ F_\mu(x) = \dot{P}_\mu(x) &= i \sum_a \dot{\omega}_a T_a = -i \frac{\partial}{\partial \omega_a(\mu, n)} S_{link} T_a = -\frac{1}{a^2} \{U_\mu(n) \phi(n + \mu) \phi^\dagger(n)\}_{TA}\end{aligned}\quad (120)$$

2.6.4 other situation for boson

Consider again,

$$D_\mu \phi D_\nu \phi \quad (121)$$

this time, we use

$$D_\mu \phi = \frac{U_\mu(n) \phi(n + \mu) - U_{-\mu}(n) \phi(n - \mu)}{2a} \quad (122)$$

so

$$\begin{aligned}
S &= \sum_n \sum_\mu \frac{\phi^\dagger(n+\mu)U_\mu^\dagger(n) - \phi^\dagger(n)}{a} \frac{U_\mu(n)\phi(n+\mu) - \phi(n)}{a} \\
&+ m \sum_n \phi^\dagger(n)\phi(n) + \lambda \sum_n (\phi^\dagger(n)\phi(n))^2 \\
&= - \sum_n \sum_\mu \frac{\phi^\dagger(n+\mu)U_\mu^\dagger(n)\phi(n) + \phi^\dagger(n)U_\mu(n)\phi(n+\mu)}{a^2} + \sum_n \sum_\mu \frac{\phi^\dagger(n+\mu)\phi(n+\mu) + \phi^\dagger(n)\phi(n)}{a^2} \\
&+ m \sum_n \phi^\dagger(n)\phi(n) + \lambda \sum_n (\phi^\dagger(n)\phi(n))^2 \\
&= \sum_n \left\{ -\frac{1}{a^2} \phi^\dagger(n) \sum_\mu (U_{-\mu}(n)\phi(n-\mu) + U_\mu(n)\phi(n+\mu)) + \left(\frac{2D}{a^2} + m \right) \phi^\dagger(n)\phi(n) + \lambda (\phi^\dagger(n)\phi(n))^2 \right\}
\end{aligned} \tag{123}$$

2.7 HISQ Fermion

2.7.1 Calculate force using contract

Using

$$\{W^\dagger\}_{TA} = -\{W\}_{TA} \tag{124}$$

$$F_\mu(n) = \eta_\mu(n) \sum_i \left\{ \left[(U_\mu(n)\phi_i(n+\mu)) \phi_{i,d}^\dagger(n) \right] + [\phi_i(n)(U_\mu(n)\phi_{i,d}(n+\mu))^\dagger] \right\}_{TA} \tag{125}$$

with

$$\begin{aligned}
\phi_i &= \frac{\sqrt{\alpha_i}}{(D_{st}^\dagger D_{st}) + \beta_i} \phi, \quad \alpha_i, \beta_i \in A^{-\frac{1}{2}} \\
\phi_{i,d} &= D_{st} \phi_i
\end{aligned} \tag{126}$$

can be

$$F_\mu(n) = \eta_\mu(n) \sum_i \left\{ U_\mu(n) \left[\phi_i(n+\mu) \phi_{i,d}^\dagger(n) - \phi_{i,d}(n+\mu) \phi_i^\dagger(n) \right] \right\}_{TA} \tag{127}$$

note that, $\phi_{i,d}$ can also be $D_{st0}\phi_i$, then let

$$\begin{aligned}
\phi_i &= \phi_i^e + \phi_i^o \\
\phi_{i,d} &= D_{st0}\phi_i = D_{st0}\phi_i^o + D_{st0}\phi_i^e
\end{aligned} \tag{128}$$

So, if ϕ is even field, then ϕ_i is even field, and $\phi_{i,d}$ is odd field,

$$F_\mu(n) = \begin{cases} \eta_\mu(n) \sum_i \left\{ -U_\mu(n) \left[\phi_{i,d}(n+\mu) \phi_i^\dagger(n) \right] \right\}_{TA} & n \in \text{even} \\ \eta_\mu(n) \sum_i \left\{ U_\mu(n) \left[\phi_i(n+\mu) \phi_{i,d}^\dagger(n) \right] \right\}_{TA} & n \in \text{odd} \end{cases} \tag{129}$$

If ϕ is initialed as even field **to remove doubler**, with $\phi_{i,\text{even}} = \phi_i$ and $\phi_{i,\text{odd}} = \phi_{i,d}$, then

$$F_\mu(n) = -(-1)^n \eta_\mu(n) \sum_i \left\{ U_\mu(n) \left[\phi_i(n+\mu) \phi_i^\dagger(n) \right] \right\}_{TA} \tag{130}$$

where $(-1)^n = -1$ if n is odd.

2.7.2 Derivate on effective gauge

2.7.2.1 Derivate to be calculated

Considering the term $\phi_x^\dagger U_1 U_2 U_3 \phi_y$, if the D_0 was anti-Hermitian, there must also be a $-\phi_y^\dagger U_3^\dagger U_2^\dagger U_1^\dagger \phi_x$ term. If it was $\phi^\dagger D_0^\dagger D_0 \phi$.

We need to calculate

$$f = U \frac{\partial S}{\partial U} \tag{131}$$

to make sure U is $SU(3)$, we do traceless anti-Hermitian at last,

$$f = \left\{ U \frac{\partial S}{\partial U} \right\}_{TA} \quad (132)$$

then, we consider the derivate, which is,

$$\frac{\partial}{\partial U} \left(\phi^\dagger D_0^\dagger D_0 \phi \right) \quad (133)$$

2.7.2.2 Some notations

The derivate on matrix is a tensor, denote,

$$(AB)^i{}_k \rightarrow A^i{}_j B^j{}_k \quad (134)$$

using matrix element, the derivate is defined as,

$$\begin{aligned} \left(\frac{\partial f}{\partial A} \right)^i{}_l &= \frac{\partial f}{\partial A^l{}_i} \\ \left(\frac{\partial B}{\partial A} \right)^i{}_j{}^k{}_l &= \frac{\partial B^i{}_l}{\partial A^j{}_k} \end{aligned} \quad (135)$$

Taking the case of $\phi_x^\dagger U \phi_y - \phi_y^\dagger U^\dagger \phi_x$ as an example, it is, both D and $D^\dagger$ contains U , and they are anti-Hermitian,

$$\frac{\partial}{\partial U^i{}_l} \left((\phi^\dagger)_m \left(c + \sum_i \frac{a_i}{(D^\dagger)D + b_i} \right)^m{}_k \phi^k \right) \quad (136)$$

Before move on, we need to check the derivate on matrix, it is known,

$$\frac{\partial A^{-1}}{\partial a} = -A^{-1} \frac{\partial A}{\partial a} A^{-1}. \quad (137)$$

so,

$$\frac{\partial M^{-1}}{\partial A^j{}_k} = -M^{-1} \frac{\partial M}{\partial A^j{}_k} M^{-1}. \quad (138)$$

compare the matrix element of the left and right (treat A as a number now), it is,

$$\frac{\partial M^{-1}{}^i{}_l}{\partial A^j{}_k} = -M^{-1}{}^i{}_m \frac{\partial M^m{}_n}{\partial A^j{}_k} M^{-1}{}^n{}_l. \quad (139)$$

and can be written as (now A is a matrix),

$$\left(\frac{\partial M^{-1}}{\partial A} \right)^i{}_j{}^k{}_l = -(M^{-1})^i{}_m \left(\frac{\partial M}{\partial A} \right)^m{}_j{}^k{}_n (M^{-1})^n{}_l. \quad (140)$$

now, assume $M = D^\dagger D + b$, so

$$\begin{aligned} \left(\frac{\partial M}{\partial U} \right)^m{}_j{}^k{}_n &= \left(\frac{\partial (D^\dagger D)^m{}_n}{\partial U^j{}_k} \right) = \left(\frac{\partial (D^{\dagger m}{}_h D^h{}_n)}{\partial U^j{}_k} \right) \\ &= \frac{\partial D^{\dagger m}{}_h}{\partial U^j{}_k} D^h{}_n + D^{\dagger m}{}_h \frac{\partial D^h{}_n}{\partial U^j{}_k} \end{aligned} \quad (141)$$

$$\begin{aligned} \left(\frac{\partial M^{-1}}{\partial A} \right)^m{}_i{}^l{}_k &= -(M^{-1})^m{}_p \left(\frac{\partial M}{\partial A} \right)^p{}_i{}^l{}_q (M^{-1})^q{}_k. \\ \left(\frac{\partial M}{\partial A} \right)^p{}_i{}^l{}_q &= \frac{\partial D^{\dagger p}{}_h}{\partial U^i{}_l} D^h{}_q + D^{\dagger p}{}_h \frac{\partial D^h{}_q}{\partial U^i{}_l} \end{aligned} \quad (142)$$

so,

$$\begin{aligned} &\frac{\partial}{\partial U^i{}_l} \left((\phi^\dagger)_m \left(c + \sum_s \frac{a_s}{(D^\dagger)D + b_s} \right)^m{}_k \phi^k \right) \\ &= \sum_s a_s \left((\phi^\dagger)_m \left(\frac{\partial \frac{1}{(D^\dagger)D + b_s}}{\partial U} \right)^m{}_i{}^l{}_k \phi^k \right) \\ &= - \sum_s a_s \phi^\dagger{}_m \left(\frac{1}{D^\dagger D + b_s} \right)^m{}_p \left(\frac{\partial D^{\dagger p}{}_h}{\partial U^i{}_l} D^h{}_q + D^{\dagger p}{}_h \frac{\partial D^h{}_q}{\partial U^i{}_l} \right) \left(\frac{1}{D^\dagger D + b_s} \right)^q{}_k \phi^k \\ &= - \sum_s a_s \left(\phi^\dagger \frac{1}{D^\dagger D + b_s} \right)_p \left(\frac{\partial D^{\dagger p}{}_h}{\partial U^i{}_l} D^h{}_q + D^{\dagger p}{}_h \frac{\partial D^h{}_q}{\partial U^i{}_l} \right) \left(\frac{1}{D^\dagger D + b_s} \phi \right)^q \end{aligned} \quad (143)$$

it is,

$$\begin{aligned} & - \sum_s a_s \left\{ \left(\phi^\dagger \frac{1}{D^\dagger D + b_s} \right)_p \left(\frac{\partial D^{\dagger p}_h}{\partial U^{i_l}} D^{h_q} \right) \left(\frac{1}{D^\dagger D + b_s} \phi \right)^q + \left(\phi^\dagger \frac{1}{D^\dagger D + b_s} \right)_p \left(D^{\dagger p}_h \frac{\partial D^{h_q}}{\partial U^{i_l}} \right) \left(\frac{1}{D^\dagger D + b_s} \phi \right)^q \right\} \\ & = - \sum_s a_s \left\{ \left(\phi^\dagger \frac{1}{D^\dagger D + b_s} \right)_p \left(\frac{\partial D^{\dagger p}_h}{\partial U^{i_l}} \right) \left(D \frac{1}{D^\dagger D + b_s} \phi \right)^h + \left(\phi^\dagger \frac{1}{D^\dagger D + b_s} D^\dagger \right)_h \left(\frac{\partial D^{h_q}}{\partial U^{i_l}} \right) \left(\frac{1}{D^\dagger D + b_s} \phi \right)^q \right\} \end{aligned} \quad (144)$$

let $\phi = \frac{1}{D^\dagger D + b_s} \phi$, $\phi_d = D \frac{1}{D^\dagger D + b_s} \phi$, it is,

$$= - \sum_s a_s \left\{ \phi^\dagger_p \left(\frac{\partial D^{\dagger p}_h}{\partial U^{i_l}} \right) \phi_d^h + \phi_d^\dagger_h \left(\frac{\partial D^{h_q}}{\partial U^{i_l}} \right) \phi^q \right\} \quad (145)$$

so we can always start from,

$$\frac{\partial}{\partial U^{i_l}} S = - \sum_s a_s \left\{ \phi^\dagger_p \left(\frac{\partial D^{\dagger p}_h}{\partial U^{i_l}} \right) \phi_d^h + \phi_d^\dagger_h \left(\frac{\partial D^{h_q}}{\partial U^{i_l}} \right) \phi^q \right\} \quad (146)$$

$$\begin{aligned} D &= \eta_\mu (U^{eff} \delta_{n,n+\mu} - U^{eff,\dagger} \delta_{n+\mu,n}) \\ D^\dagger &= \eta_\mu (U^{eff,\dagger} \delta_{n+\mu,n} - U^{eff} \delta_{n,n+\mu}) \end{aligned} \quad (147)$$

so,

$$\begin{aligned} \frac{\partial}{\partial U^{i_l}(n,\mu)} S &= - \sum_s a_s \left\{ \phi^\dagger(n+\mu)_p \left(\eta_\mu \frac{\partial U^{eff,\dagger p}_h}{\partial U^{i_l}} \right) \phi_d(n)^h + \phi^\dagger(n)_p \left(-\eta_\mu \frac{\partial U^{eff p}_h}{\partial U^{i_l}} \right) \phi_d(n+\mu)^h \right. \\ & \quad \left. + \eta_\mu \phi_d^\dagger(n)_h \left(\frac{\partial U^{eff h}}{\partial U^{i_l}} \right) \phi(n+\mu)^q + \phi_d^\dagger(n+\mu)_p \left(-\eta_\mu \frac{\partial U^{eff,\dagger p}_h}{\partial U^{i_l}} \right) \phi(n)^h \right\} \end{aligned} \quad (148)$$

So, with even field, it is,

$$\begin{aligned} \frac{\partial}{\partial U^{i_l}(n,\mu)} S &= \frac{\partial U^{eff p}_h}{\partial U^{i_l}} f_0^h_p + \frac{\partial U^{eff,\dagger p}_h}{\partial U^{i_l}} (f_0^\dagger)^h_p \\ f_0 &= \begin{cases} -\eta_\mu(n) \phi_d(n+\mu)^h \phi^\dagger(n)_p & n \in \text{even}, \\ \eta_\mu(n) \phi(n+\mu)^h \phi_d^\dagger(n)_p & n \in \text{odd} \end{cases} \end{aligned} \quad (149)$$

2.7.2.3 Derivate on a link

Now consider $\phi^\dagger(n) U_1 U_2 U_3^\dagger \phi_d(n+\mu) - \phi_d^\dagger(n+\mu) U_3 U_2^\dagger U_1^\dagger \phi(n)$ as an example, ($\eta_\mu(n)$ is ignored), when n is even, the term contains U_2 is,

$$\begin{aligned} & (\phi_d^\dagger(n))_m \frac{\partial D^m_n}{\partial U^{i_l}} \phi(n+\mu)^n \\ &= \phi_d^\dagger(n)_m U_1^m_j \frac{\partial (U_2)^j_k}{\partial U^{i_l}} U_3^{\dagger k}_n \phi(n+\mu)^n \\ &= \phi_d^\dagger(n)_m U_1^m_j \delta^j_i \delta^l_k U_3^{\dagger k}_n \phi(n+\mu)^n \\ &= (\phi_d^\dagger(n) U_1)_i (U_3^\dagger \phi(n+\mu))^l \\ &= (U_3^\dagger \phi(n+\mu) \phi_d^\dagger(n) U_1)^l_i \end{aligned} \quad (150)$$

We need to compare with the old method to check whether this is really correct.

For $\phi_x^\dagger U_1^\dagger U_2 U_3 \phi_y$, to calculate force on U_2 , the path is divided into four sets,

$$\begin{aligned} p_{left}^\dagger &= 1, & p_{right} &= U_1^\dagger U_2 U_3 \\ p_{left}^\dagger &= U_1^\dagger, & p_{right} &= U_2 U_3 \\ p_{left}^\dagger &= U_1^\dagger U_2, & p_{right} &= U_3 \\ p_{left}^\dagger &= U_1^\dagger U_2 U_3, & p_{right} &= 1 \end{aligned} \quad (151)$$

If the first link of p_{left} is U_1 , the result is

$$\left\{ - (p_{right} \phi_y) (p_{left} \phi_{x,d})^\dagger - (p_{left} \phi_x) (p_{right} \phi_{y,d})^\dagger \right\}_{TA} \quad (152)$$

so, there are three forces,

$$\left\{ - (U_2 U_3 \phi_y) (U_1 \phi_{x,d})^\dagger - (U_1 \phi_x) (U_2 U_3 \phi_{y,d})^\dagger \right\}_{TA} \quad (153)$$

which is

$$\left\{ -U_2 U_3 \phi_y \phi_{x,d}^\dagger U_1^\dagger - U_1 \phi_x \phi_{y,d}^\dagger U_3^\dagger U_2^\dagger \right\}_{TA} \quad (154)$$

Using $\{\dots\}_{TA}$ is anti-Hermitian,

$$\{W^\dagger\}_{TA} = -\{W\}_{TA} \quad (155)$$

so,

$$\left\{ -U_2 U_3 \phi_y \phi_{x,d}^\dagger U_1^\dagger - U_1 \phi_x \phi_{y,d}^\dagger U_3^\dagger U_2^\dagger \right\}_{TA} \quad (156)$$

which is,

$$\left\{ U_1 \phi_{x,d} \phi_y^\dagger U_3^\dagger U_2^\dagger - U_1 \phi_x \phi_{y,d}^\dagger U_3^\dagger U_2^\dagger \right\}_{TA} \quad (157)$$

if the first link of p_{right} is U_2 , the force is,

$$\left\{ (p_{right} \phi_y) (p_{left} \phi_{x,d})^\dagger + (p_{left} \phi_x) (p_{right} \phi_{y,d})^\dagger \right\}_{TA} \quad (158)$$

the force is the same as U_1 but with a minus sign, the result is,

$$\begin{aligned} & - \left\{ U_1 \phi_{x,d} \phi_y^\dagger U_3^\dagger U_2^\dagger - U_1 \phi_x \phi_{y,d}^\dagger U_3^\dagger U_2^\dagger \right\}_{TA} \\ & = \left\{ U_2 U_3 \phi_y \phi_{x,d}^\dagger U_1^\dagger - U_2 U_3 \phi_{y,d} \phi_x^\dagger U_1^\dagger \right\}_{TA} \end{aligned} \quad (159)$$

if the first link of p_{right} is U_2 , the force is,

$$\left\{ (p_{right} \phi_y) (p_{left} \phi_{x,d})^\dagger + (p_{left} \phi_x) (p_{right} \phi_{y,d})^\dagger \right\}_{TA} \quad (160)$$

which is

$$\begin{aligned} & \left\{ (U_3 \phi_y) (U_2^\dagger U_1 \phi_{x,d})^\dagger + (U_2^\dagger U_1 \phi_x) (U_3 \phi_{y,d})^\dagger \right\}_{TA} \\ & = \left\{ U_3 \phi_y \phi_{x,d}^\dagger U_1^\dagger U_2 - U_3 \phi_{y,d} \phi_x^\dagger U_1^\dagger U_2 \right\}_{TA} \end{aligned} \quad (161)$$

2.7.2.4 Projection

Note that the order of the derivate. Let $f(M) = \bar{\phi} A M A \phi$, $g(M) = B M B$, then

$$\begin{aligned} f(g(M)) &= A B M B A \\ \frac{\partial}{\partial M^n_m} \bar{\phi}_i (A B M B A)^i_l \phi^l &= \bar{\phi}_i (A B)^i_j \frac{\partial M^j_k}{\partial M^n_m} (B A)^k_l \phi^l \\ &= (\bar{\phi} A B)_j \delta^j_m \delta^n_k (B A \phi)^k \\ &= (\bar{\phi} A B)_m (B A \phi)^n \\ &= (B A \phi \bar{\phi} A B)^n_m \end{aligned} \quad (162)$$

Note that, it is like $g(f(M))$.

2.7.3 Projection of effective gauge

The next step is to project the effective gauge on to $U(3)$ or $SU(3)$. It calculates,

$$\begin{aligned} U^R &= \frac{U}{\sqrt{U^\dagger U}} \\ \tilde{U}^R &= \frac{1}{\det[U^R]^{\frac{1}{N_C}}} U^R \end{aligned} \quad (163)$$

So, the main problem is to calculate $(U^\dagger U)^{-1/2}$, which can be calculated using rational approximation,

$$(U^\dagger U)^{-1/2} = c_0 + \sum_i \frac{a_i}{U^\dagger U + b_i} \quad (164)$$

using,

$$\begin{aligned} \left(\frac{\partial A \left(c + \frac{1}{A^\dagger A + b} \right)}{\partial A} \right)_{i_j k_l} &= \delta_{ij}^k \left(c + \frac{1}{A^\dagger A + b} \right)_{k_l} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)_{i_j} \left(\frac{1}{A^\dagger A + b} \right)_{k_l} \\ \left(\frac{\partial \left(c + \frac{1}{A^\dagger A + b} \right) A^\dagger}{\partial A} \right)_{i_j k_l} &= - \left(\frac{1}{A^\dagger A + b} A^\dagger \right)_{i_j} \left(\frac{1}{A^\dagger A + b} A^\dagger \right)_{k_l} \end{aligned} \quad (165)$$

so, let $M = A \left(c + \frac{1}{A^\dagger A + b} \right)$ if we want to calculate, let $\phi = \phi(n)$, $\phi_d = \phi_d(n + \mu)$

$$\begin{aligned} f &= \frac{\partial}{\partial A^{i_l}} \left(\phi^\dagger M^\dagger \phi_d + \phi_d^\dagger M \phi \right) \\ &= \left(\phi^\dagger_m \frac{\partial (M^\dagger)^m_n}{\partial A^{i_l}} \phi_d^n \right) + \left(\phi_d^\dagger_n \frac{\partial M^n_k}{\partial A^{i_l}} \phi^k \right) \\ &= - \left(\phi^\dagger_m \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^m_i \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^l_n \phi_d^n \right) \\ &\quad + \left(\phi_d^\dagger_n \left(\delta^n_i \left(c + \frac{1}{A^\dagger A + b} \right)_{k_l} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)^n_i \left(\frac{1}{A^\dagger A + b} \right)_{k_l} \right) \phi^k \right) \\ &= - \left(\left(\phi^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right)_i \left(\frac{1}{A^\dagger A + b} A^\dagger \phi_d \right)^l \right) \\ &\quad + \left(\phi_d^\dagger \left(c + \frac{1}{A^\dagger A + b} \right) \phi \right)^l - \left(\phi_d^\dagger A \frac{1}{A^\dagger A + b} A^\dagger \right)_i \left(\frac{1}{A^\dagger A + b} \phi \right)^l \\ &= \left(\left(c + \frac{1}{A^\dagger A + b} \right) \phi \phi_d^\dagger \right)^l_i - \left(\frac{1}{A^\dagger A + b} \phi \phi_d^\dagger A \frac{1}{A^\dagger A + b} A^\dagger \right)^l_i - \left(\frac{1}{A^\dagger A + b} A^\dagger \phi_d \phi^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right)^l_i \\ &= \left(\left(c + \frac{1}{A^\dagger A + b} \right) f_0 \right)^l_i - \left(\frac{1}{A^\dagger A + b} f_0 A \frac{1}{A^\dagger A + b} A^\dagger \right)^l_i - \left(\frac{1}{A^\dagger A + b} A^\dagger f_0^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right)^l_i \end{aligned} \quad (166)$$

$$\begin{aligned} F &= f^\dagger \\ &= \left(f_0^\dagger \left(c + \frac{1}{A^\dagger A + b} \right) \right)^l_i - \left(A \frac{1}{A^\dagger A + b} A^\dagger f_0^\dagger \frac{1}{A^\dagger A + b} \right)^l_i - \left(A \frac{1}{A^\dagger A + b} f_0 A \frac{1}{A^\dagger A + b} \right)^l_i \\ &= \left(F_0 \left(c + \frac{1}{A^\dagger A + b} \right) \right)^l_i - \left(A \frac{1}{A^\dagger A + b} A^\dagger F_0 \frac{1}{A^\dagger A + b} \right)^l_i - \left(A \frac{1}{A^\dagger A + b} F_0^\dagger A \frac{1}{A^\dagger A + b} \right)^l_i \end{aligned} \quad (167)$$

Note that, the force of projection is calculated in front of effective gauge.

Before move on, we need to check the derivate on determinant, it is known,

$$\frac{\partial \det(M)}{\partial M^{i_j}} = \det(M) (M^{-1})^j_i \quad (168)$$

so,

$$\begin{aligned} \frac{\partial \det(M)^{-\frac{1}{3}}}{\partial M^{i_j}} &= -\frac{1}{3} \det(M)^{-\frac{1}{3}} (M^{-1})^j_i \\ \frac{\partial (\det(M))^{-\frac{1}{3}}}{\partial U^{i_j}} &= \frac{\partial (\det(M))^{-\frac{1}{3}}}{\partial M^{k_l}} \frac{\partial M^{k_l}}{\partial U^{i_j}} = -\frac{1}{3} \det(M)^{-\frac{1}{3}} (M^{-1})^l_k \left(\frac{\partial M}{\partial U} \right)^{k_i j_l} \end{aligned} \quad (169)$$

also,

$$\frac{\partial (\det(M)^*)^{-\frac{1}{3}}}{\partial U^{i_j}} = -\frac{1}{3} (\det(M)^*)^{-\frac{1}{3}} (M^{-1\dagger})^l_k \left(\frac{\partial M^\dagger}{\partial U} \right)^{k_i j_l} \quad (170)$$

we calculate the two terms separately,

$$\begin{aligned}
& \frac{\partial}{\partial A^i_j} \phi_d^\dagger M \det(M)^{-\frac{1}{3}} \phi \\
&= \det(M)^{-\frac{1}{3}} \frac{\partial}{\partial A^i_j} \phi_d^\dagger M \phi + \phi_d^\dagger M \phi \frac{\partial \det(M)^{-\frac{1}{3}}}{\partial A^i_j} \\
&= \det(M)^{-\frac{1}{3}} f_1 + \text{tr}(M f_0) \left(-\frac{1}{3} \det(M)^{-\frac{1}{3}} (M^{-1})^l_k \left(\frac{\partial M}{\partial U} \right)^{k_i j_l} \right) \\
&= \det(M)^{-\frac{1}{3}} \left(f_1 - \frac{1}{3} \text{tr}(M f_0) \left((M^{-1})^l_k \left(\delta^k_i \left(c + \frac{1}{A^\dagger A + b} \right) \right)^{j_l} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)^{k_i} \left(\frac{1}{A^\dagger A + b} \right)^{j_l} \right) \right) \\
&= \det(M)^{-\frac{1}{3}} \left(f_1 - \frac{1}{3} \text{tr}(M f_0) \left((M^{-1})^l_i \left(c + \frac{1}{A^\dagger A + b} \right) \right)^{j_l} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)^{k_i} \left(\frac{1}{A^\dagger A + b} M^{-1} \right)^{j_k} \right) \\
&= \det(M)^{-\frac{1}{3}} \left(f_1 - \frac{1}{3} \text{tr}(M f_0) \left(\left(c + \frac{1}{A^\dagger A + b} \right) M^{-1} - \left(\frac{1}{A^\dagger A + b} M^{-1} A \frac{1}{A^\dagger A + b} A^\dagger \right) \right) \right)
\end{aligned} \tag{171}$$

and

$$\begin{aligned}
& \frac{\partial}{\partial A^i_j} \phi^\dagger M^\dagger (\det(M)^*)^{-\frac{1}{3}} \phi_d \\
&= (\det(M)^*)^{-\frac{1}{3}} \frac{\partial}{\partial A^i_j} \phi^\dagger M^\dagger \phi_d + \phi^\dagger M^\dagger \phi_d \frac{\partial (\det(M)^*)^{-\frac{1}{3}}}{\partial A^i_j} \\
&= (\det(M)^*)^{-\frac{1}{3}} f_2 + \phi^\dagger M^\dagger \phi_d \frac{\partial (\det(M)^*)^{-\frac{1}{3}}}{\partial A^i_j} \\
&= (\det(M)^*)^{-\frac{1}{3}} \left(f_2 - \frac{1}{3} \text{tr}[M^\dagger f_0^\dagger] (M^{-1})^{\dagger l}_k \left(\frac{\partial M^\dagger}{\partial A} \right)^{k_i j_l} \right) \\
&= (\det(M)^*)^{-\frac{1}{3}} \left(f_2 + \frac{1}{3} \text{tr}[M^\dagger f_0^\dagger] (M^{-1})^{\dagger l}_k \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{k_i} \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{j_l} \right) \\
&= (\det(M)^*)^{-\frac{1}{3}} \left(f_2 + \frac{1}{3} \text{tr}[M^\dagger f_0^\dagger] \left(\frac{1}{A^\dagger A + b} A^\dagger (M^{-1})^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right) \right)
\end{aligned} \tag{172}$$

so, let $R = \frac{1}{3} \det(M)^{-\frac{1}{3}} \text{tr}(M f_0) M^{-1}$, it is

$$\begin{aligned}
& \frac{\partial}{\partial A^i_j} \phi_d^\dagger M \det(M)^{-\frac{1}{3}} \phi = \det(M)^{-\frac{1}{3}} f_1 - \left(c + \frac{1}{A^\dagger A + b} \right) R + \left(\frac{1}{A^\dagger A + b} R A \frac{1}{A^\dagger A + b} A^\dagger \right) \\
& \frac{\partial}{\partial A^i_j} \phi^\dagger M^\dagger (\det(M)^*)^{-\frac{1}{3}} \phi_d = (\det(M)^*)^{-\frac{1}{3}} f_2 + \left(\frac{1}{A^\dagger A + b} A^\dagger R^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right)
\end{aligned} \tag{173}$$

so,

$$\begin{aligned}
F_1 &= \left(F_0 \left(c + \frac{1}{A^\dagger A + b} \right) \right) - \left(A \frac{1}{A^\dagger A + b} A^\dagger F_0 \frac{1}{A^\dagger A + b} \right) \\
F_2 &= - \left(A \frac{1}{A^\dagger A + b} F_0^\dagger A \frac{1}{A^\dagger A + b} \right) \\
F &= (\det(M)^*)^{-\frac{1}{3}} F_1 + \det(M)^{-\frac{1}{3}} F_2 - R^\dagger \left(c + \frac{1}{A^\dagger A + b} \right) + A \frac{1}{A^\dagger A + b} (R A + A^\dagger R^\dagger) \frac{1}{A^\dagger A + b}
\end{aligned} \tag{174}$$

Note $R = \frac{1}{3} \det(M)^{-\frac{1}{3}} \text{tr}(M f_0) M^{-1} = \frac{1}{3} \det(M)^{-\frac{1}{3}} \text{tr}(\hat{M} f_0) \hat{M}^{-1}$, with $\hat{M} = M \det(M)^{-\frac{1}{3}}$. Let $J_0 = (\det(M)^*)^{-\frac{1}{3}} F_0$, it is

$$\begin{aligned}
F_1 &= \left(J_0 \left(c + \frac{1}{A^\dagger A + b} \right) \right) - \left(A \frac{1}{A^\dagger A + b} A^\dagger J_0 \frac{1}{A^\dagger A + b} \right) \\
F_2 &= - \left(A \frac{1}{A^\dagger A + b} J_0^\dagger A \frac{1}{A^\dagger A + b} \right) \\
F &= F_1 + F_2 - R^\dagger \left(c + \frac{1}{A^\dagger A + b} \right) + A \frac{1}{A^\dagger A + b} (R A + A^\dagger R^\dagger) \frac{1}{A^\dagger A + b} \\
R &= \frac{1}{3} \text{tr}(\hat{M} J_0^\dagger) \hat{M}^{-1}
\end{aligned} \tag{175}$$

which is,

$$\begin{aligned}
F &= (J_0 - R^\dagger) \left(c + \frac{1}{A^\dagger A + b} \right) - A \frac{1}{A^\dagger A + b} (A^\dagger J_0 + J_0^\dagger A - R A - A^\dagger R^\dagger) \frac{1}{A^\dagger A + b} \\
R &= \frac{1}{3} \text{tr}(\hat{M} J_0^\dagger) \hat{M}^{-1}
\end{aligned} \tag{176}$$

which is,

$$F = K_0 \left(c + \frac{1}{A^\dagger A + b} \right) - A \frac{1}{A^\dagger A + b} (A^\dagger K_0 + K_0^\dagger A) \frac{1}{A^\dagger A + b} \quad (177)$$

$$K_0 = J_0 - R^\dagger$$

We can do it in another way, consider

$$\phi_d^\dagger A \det(A)^{-\frac{1}{3}} \phi \quad (178)$$

so,

$$\begin{aligned} & \frac{\partial}{\partial A^{i_j}} \left(\phi_d^\dagger A \det(A)^{-\frac{1}{3}} \phi \right) \\ &= \det(A)^{-\frac{1}{3}} \frac{\partial}{\partial A^{i_j}} \left(\phi_d^\dagger A \phi \right) + \left(\phi_d^\dagger A \phi \right) \frac{\partial}{\partial A^{i_j}} \det(A)^{-\frac{1}{3}} \\ &= \det(A)^{-\frac{1}{3}} \left(\phi \phi_d^\dagger \right) - \frac{1}{3} \det(A)^{-\frac{1}{3}} \left(\phi_d^\dagger A \phi \right) (A^{-1}) \\ &= \det(A)^{-\frac{1}{3}} \left(\phi \phi_d^\dagger \right) - \frac{1}{3} \text{tr} \left(\det(A)^{-\frac{1}{3}} \phi \phi_d^\dagger A \right) (A^{-1}) \end{aligned} \quad (179)$$

this has been checked numerically, by using.

$$\frac{\partial f}{\partial A^{i_j}} = \left\{ \frac{f(M + \epsilon E^j_i) - f(M)}{\epsilon} \right\} \quad (180)$$

so,

$$\begin{aligned} j_0 &= \det(A)^{-\frac{1}{3}} f_0 \\ f &= j_0 - \frac{1}{3} \text{tr} (j_0 A) (A^{-1}) \\ F &= j_0^\dagger - \frac{1}{3} \text{tr} \left(A^\dagger j_0^\dagger \right) (A^{-1})^\dagger \\ &= J_0 - \frac{1}{3} \text{tr} \left(A^\dagger J_0 \right) A \\ J_0 &= (\det(A)^{-\frac{1}{3}})^* F_0 \end{aligned} \quad (181)$$

This result is misleading! Consider the full expression,

$$F^i_k = U^i_j \frac{\partial S}{\partial U^k_j} \quad (182)$$

it is

2.7.4 Projection of effective gauge

From arXiv:hep-lat/0403019.

Derivate of a scalar function and a matrix function,

$$\begin{aligned} \left(\frac{\partial f(A)}{\partial A} \right)^i_j &\equiv \frac{\partial f(A)}{\partial A^j_i} \\ \left(\frac{\partial M(A)}{\partial A} \right)^i_{j^k_l} &\equiv \frac{\partial M(A)^i_l}{\partial A^j_k} \end{aligned} \quad (183)$$

The chain rule is, **Eq. (28) of arXiv:hep-lat/0403019 seem wrong!**

$$\left(\frac{\partial f}{\partial B} \right)^i_l = \left(\frac{\partial f}{\partial A} \right)^j_k \left(\frac{\partial A}{\partial B} \right)^k_{l^i_j} \quad (184)$$

The reason of the first chain rule,

$$\left(\frac{\partial f}{\partial B} \right)^i_l = \left(\frac{\partial f}{\partial B^l_i} \right) = \left(\frac{\partial f}{\partial A^k_j} \right) \left(\frac{\partial A^k_j}{\partial B^l_i} \right) = \left(\frac{\partial f}{\partial A} \right)^j_k \left(\frac{\partial A}{\partial B} \right)^k_{l^i_j} \quad (185)$$

From arXiv:0710.0737. (and part from arXiv:hep-lat/0403019)

Given f^n

$$(f^{n-1})^i_l = (f^n)^j_k \left(\frac{\partial U^n}{\partial U^{n-1}} \right)^k_l{}^i_j + (f^{n\dagger})^j_k \left(\frac{\partial U^{n\dagger}}{\partial U^{n-1}} \right)^k_l{}^i_j \quad (186)$$

The reason, let $S(U^n, U^{n\dagger})$ be a **scalar and real** function (the action) of matrix U . And,

$$(f^{n-1})^i_l = \left(\frac{\partial S}{\partial U^{n-1}} \right)^i_l = \left(\frac{\partial S}{\partial U^n} \right)^j_k \left(\frac{\partial U^n}{\partial U^{n-1}} \right)^k_l{}^i_j + \left(\frac{\partial S}{\partial U^{n\dagger}} \right)^j_k \left(\frac{\partial U^{n\dagger}}{\partial U^{n-1}} \right)^k_l{}^i_j \quad (187)$$

with $f^n = \frac{\partial S}{\partial U^n}$, $f^{n\dagger} = \frac{\partial S^*}{\partial U^{n\dagger}} = \frac{\partial S}{\partial U^{n\dagger}}$.

2.7.4.1 f0

I find it difficult to consider objects like $\partial U^\dagger / \partial U$, I think the correct way should be considering $U^\dagger$ and U as independent variables, but all as functions of ω_a . **The force is then $F = i\dot{\omega}_a T_a$, where $\dot{\omega}_a = -\partial S / \partial \omega_a$.**

By using,

$$\frac{\partial U_\mu}{\partial \omega_\mu^a} = iT^a U_\mu, \quad \frac{\partial U_\mu^\dagger}{\partial \omega_\mu^a} = -iU_\mu^\dagger T^a, \quad (188)$$

it is,

$$\frac{\partial S}{\partial \omega_a} = \text{tr} \left[\frac{\partial S}{\partial U} \frac{\partial U}{\partial \omega_a} + \frac{\partial S}{\partial U^\dagger} \frac{\partial U^\dagger}{\partial \omega_a} \right] \quad (189)$$

note that $\partial S / \partial \omega_a$ is a scalar, S, ω_a are both scalars, and both $\frac{\partial S}{\partial U}$ and $\frac{\partial U}{\partial \omega_a}$ are matrices.

So,

$$\begin{aligned} F &= -i \frac{\partial S}{\partial \omega_a} T_a = -i \text{tr} \left[\frac{\partial S}{\partial U} iT^a U - \frac{\partial S}{\partial U^\dagger} iU^\dagger T^a \right] T_a \\ &= -i \text{tr} \left[i \frac{\partial S}{\partial U} T^a U - i \frac{\partial S}{\partial U^\dagger} U^\dagger T^a \right] T_a \\ &= -i \text{tr} \left[i \frac{\partial S}{\partial U} T^a U + \left(i \frac{\partial S}{\partial U} T^a U \right)^\dagger \right] T_a \\ &= -2i \text{tr} \left[\text{Re} \left[i \frac{\partial S}{\partial U} T^a U \right] \right] T_a \end{aligned} \quad (190)$$

for any matrix

$$2i \sum_a T_a \text{Imtr}[T_a M] = -2i \sum_a T_a \text{Retr}[iT_a M] = \{M\}_{TA} \quad (191)$$

So, as long as $\partial S / \partial U^\dagger = (\partial S / \partial U)^\dagger$,

$$F = \{U^i{}_j \frac{\partial S}{\partial U^k{}_j}\}_{TA} \quad (192)$$

here, when calculating $\partial S / \partial U$, the $U^\dagger$ is considered as independent of U .

Before move on, we need to check the derivate on matrix, it is known,

$$\frac{\partial A^{-1}}{\partial a} = -A^{-1} \frac{\partial A}{\partial a} A^{-1}. \quad (193)$$

so,

$$\frac{\partial M^{-1}}{\partial A^j{}_k} = -M^{-1} \frac{\partial M}{\partial A^j{}_k} M^{-1}. \quad (194)$$

compare the matrix element of the left and right (treat A as a number now), it is,

$$\frac{\partial M^{-1}{}^i{}_l}{\partial A^j{}_k} = -M^{-1}{}^i{}_m \frac{\partial M^m{}_n}{\partial A^j{}_k} M^{-1}{}^n{}_l. \quad (195)$$

and can be written as (now A is a matrix),

$$\left(\frac{\partial M^{-1}}{\partial A} \right)^i{}_j{}^k{}_l = -(M^{-1})^i{}_m \left(\frac{\partial M}{\partial A} \right)^m{}_j{}^k{}_n (M^{-1})^n{}_l. \quad (196)$$

$$\begin{aligned}
S &= (\psi^\dagger)_m \left(c + \sum_i \frac{a_i}{DD^\dagger + b_i} \right) m_k \psi^k \\
\left(\frac{\partial \frac{1}{DD^\dagger + b_s}}{\partial U} \right)_{j^k l}^i &= - \left(\frac{1}{DD^\dagger + b_s} \right)^i_m \frac{\partial (DD^\dagger)}{\partial U^{j_k}} \left(\frac{1}{DD^\dagger + b_s} \right)^{n_l} \\
\phi &= \frac{1}{DD^\dagger + b_s} \psi, \quad \phi_d = D^\dagger \frac{1}{DD^\dagger + b_s} \psi, \\
\frac{\partial}{\partial U^{i_l}} S &= - \sum_s a_s \left\{ \phi^\dagger_p \left(\frac{\partial D^p_h}{\partial U^{i_l}} \right) \phi_d^h + \phi_d^\dagger_h \left(\frac{\partial D^{\dagger h}_q}{\partial U^{i_l}} \right) \phi^q \right\} \\
\frac{\partial}{\partial U^{i_l}} S &= - \sum_s a_s \left\{ \phi^\dagger_p \left(\frac{\partial D^p_h}{\partial U^{i_l}} \right) \phi_d^h + \left(\phi^\dagger_h \left(\frac{\partial D^h_q}{\partial U^{\dagger i_l}} \right) \phi_d^q \right)^\dagger \right\} \\
\frac{\partial}{\partial U_{\mu(n)^{i_l}}} S &= - \sum_s a_s \left\{ \eta_\mu(n) \phi(n)^\dagger_p \delta^p_i \delta^l_h \phi_d(n + \mu)^h - (\eta_\mu(n) \phi^\dagger(n + \mu)_h \delta^p_i \delta^l_h \phi_d(n)^q)^\dagger \right\} \\
(f_0)^l_i &= -\eta_\mu(n) \sum_s a_s \left\{ \phi_d(n + \mu)^l \phi(n)^\dagger_i - \phi(n + \mu)^l \phi_d^\dagger(n)_i \right\}
\end{aligned} \tag{197}$$

If pseudofermion is even, then if n is even, $\phi(n + \mu) = \phi_d(n) = 0$, if n is odd, $\phi(n) = \phi_d(n + \mu) = 0$.

$$(f_0)^l_i = \begin{cases} -\eta_\mu(n) \phi_d(n + \mu)^l \phi(n)^\dagger_i & \text{even} \\ \eta_\mu(n) \phi(n + \mu)^l \phi_d^\dagger(n)_i & \text{odd} \end{cases} \tag{198}$$

This is Eq. (2.7) of arXiv:0710.0737.

2.7.4.2 U3 to SU3

$$\begin{aligned}
U_1 &= \det(U_0)^{-\frac{1}{3}} U_0 \\
\left(\frac{\partial \det(A)}{\partial A} \right)^i_j &= \det(A) (A^{-1})^i_j \\
(f_1)^i_l &= (f_0)^j_k \left(\frac{\partial U_1}{\partial U_0} \right)^k_{l^i_j} \\
&= (f_0)^j_k \left(\frac{\partial \det(U_0)^{-\frac{1}{3}} U_0}{\partial U_0} \right)^k_{l^i_j} \\
&= (f_0)^j_k \left(\frac{\partial \det(U_0)^{-\frac{1}{3}}}{\partial (U_0)^l_i} (U_0)^k_j + \det(U_0)^{-\frac{1}{3}} \left(\frac{\partial U_0}{\partial U_0} \right)^k_{l^i_j} \right) \\
&= (f_0)^j_k \left(-\frac{1}{3} \det(U_0)^{-\frac{4}{3}} \frac{\partial \det(U_0)}{\partial (U_0)^l_i} (U_0)^k_j + \det(U_0)^{-\frac{1}{3}} \delta^i_j \delta^k_l \right) \\
&= (f_0)^j_k \left(-\frac{1}{3} \det(U_0)^{-\frac{4}{3}} \det(U_0) (U_0^{-1})^i_l (U_0)^k_j + \det(U_0)^{-\frac{1}{3}} \delta^i_j \delta^k_l \right) \\
&= \det(U_0)^{-\frac{1}{3}} (f_0)^i_l - \frac{1}{3} \det(U_0)^{-\frac{1}{3}} \text{tr} [U_0 f_0] (U_0^{-1})^i_l \\
&= \det(U_0)^{-\frac{1}{3}} \left(f_0 - \frac{1}{3} \det(U_0)^{-\frac{1}{3}} \text{tr} [U_0 f_0] U_0^{-1} \right)
\end{aligned} \tag{199}$$

so

$$\begin{aligned}
f_1 &= \det(U_0)^{-\frac{1}{3}} \left(f_0 - \frac{1}{3} \text{tr} [U_0 f_0] U_0^{-1} \right) \\
f_1^\dagger &= \left(\det(U_0)^{-\frac{1}{3}} \right)^* \left(f_0^\dagger - \frac{1}{3} \text{tr} [U_0^\dagger f_0^\dagger] (U_0^{-1})^\dagger \right)
\end{aligned} \tag{200}$$

Note that, U_0 is unitary, so $U_0^{-1} = U_0^\dagger$, and $U_1 = \det(U_0)^{-\frac{1}{3}} U_0$, so $\text{tr} [U_0^\dagger f_0^\dagger] (U_0^{-1})^\dagger = \text{tr} [U_1^\dagger f_0^\dagger] (U_1^{-1})^\dagger$.

$$\begin{aligned}
j_0 &= \left(\det(U_0)^{-\frac{1}{3}} \right)^* f_0^\dagger \\
f_1^\dagger &= \left(j_0 - \frac{1}{3} \text{tr} [U_1^\dagger j_0] U_1 \right)
\end{aligned} \tag{201}$$

2.7.4.3 Rational approximation

$$\begin{aligned}
\left(U_0^\dagger U_0\right)^{-1/2} &= c_0 + \sum_i \frac{a_i}{U_0^\dagger U_0 + b_i} \\
U_1 &= U_0 \left(c_0 + \sum_i \frac{a_i}{U_0^\dagger U_0 + b_i} \right) \\
(f_1)^i_l &= (f_0)^j_k \left(\frac{\partial U_1}{\partial U_0} \right)^{k\ i}_l + (f_0)^{\dagger j}_k \left(\frac{\partial U_1^\dagger}{\partial U_0} \right)^{k\ i}_l
\end{aligned} \tag{202}$$

Before move on, we need to check the derivate on matrix, it is known,

$$\left(\frac{\partial M^{-1}}{\partial A} \right)^{i\ j\ k}_l = -(M^{-1})^i_m \left(\frac{\partial M}{\partial A} \right)^{m\ j\ k}_n (M^{-1})^n_l. \tag{203}$$

now, assume $M = A^\dagger A + b$, so

$$\begin{aligned}
\left(\frac{\partial M}{\partial A} \right)^{m\ j\ k}_n &= \left(\frac{\partial (A^\dagger A)^{m\ n}}{\partial A^{j\ k}} \right) = \left(\frac{\partial (A^{\dagger m}_h A^h_n)}{\partial A^{j\ k}} \right) \\
&= A^{\dagger m}_h \left(\frac{\partial (A^h_n)}{\partial A^{j\ k}} \right) = A^{\dagger m}_h \delta^h_j \delta^k_n = A^{\dagger m}_j \delta^k_n
\end{aligned} \tag{204}$$

so,

$$\left(\frac{\partial (A^\dagger A + b)^{-1}}{\partial A} \right)^{i\ j\ k}_l = - \left(\frac{1}{A^\dagger A + b} \right)^i_m (A^{\dagger m}_j \delta^k_n) \left(\frac{1}{A^\dagger A + b} \right)^n_l = - \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{i\ j} \left(\frac{1}{A^\dagger A + b} \right)^{k\ l} \tag{205}$$

Now, consider,

$$\begin{aligned}
\left(\frac{\partial A \left(c + \frac{1}{A^\dagger A + b} \right)}{\partial A} \right)^{i\ j\ k}_l &= \left(\frac{\partial A^i_m \left(c + \frac{1}{A^\dagger A + b} \right)^{m\ l}}{\partial A^{j\ k}} \right) \\
&= \frac{\partial A^i_m}{\partial A^{j\ k}} \left(c + \frac{1}{A^\dagger A + b} \right)^{m\ l} + A^i_m \frac{\partial \left(c + \frac{1}{A^\dagger A + b} \right)^{m\ l}}{\partial A^{j\ k}} \\
&= \delta^i_j \delta^k_m \left(c + \frac{1}{A^\dagger A + b} \right)^{m\ l} + A^i_m \left(\frac{\partial (A^\dagger A + b)^{-1}}{\partial A} \right)^{m\ j\ k}_l \\
&= \delta^i_j \left(c + \frac{1}{A^\dagger A + b} \right)^{k\ l} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)^{i\ j} \left(\frac{1}{A^\dagger A + b} \right)^{k\ l}
\end{aligned} \tag{206}$$

similarly,

$$\left(\frac{\partial \left(c + \frac{1}{A^\dagger A + b} A^\dagger \right)}{\partial A} \right)^{i\ j\ k}_l = \left(\frac{\partial (A^\dagger A + b)^{-1}}{\partial A} \right)^{i\ j\ k}_m A^{\dagger m}_l = - \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{i\ j} \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{k\ l} \tag{207}$$

$$\begin{aligned}
(f_1)^i_l &= (f_0)^j_k \left(\delta^k_l \left(c + \frac{1}{A^\dagger A + b} \right)^{i\ j} - \left(A \frac{1}{A^\dagger A + b} A^\dagger \right)^{k\ l} \left(\frac{1}{A^\dagger A + b} \right)^{i\ j} \right) + (f_0)^{\dagger j}_k \left(- \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{k\ l} \left(\frac{1}{A^\dagger A + b} A^\dagger \right)^{i\ j} \right) \\
&= \left(c + \frac{1}{A^\dagger A + b} \right) f_0 - \left(\frac{1}{A^\dagger A + b} f_0 A \frac{1}{A^\dagger A + b} A^\dagger \right) - \left(\frac{1}{A^\dagger A + b} A^\dagger f_0^\dagger \frac{1}{A^\dagger A + b} A^\dagger \right) \\
&= \left(c + \frac{1}{A^\dagger A + b} \right) f_0 - \left(\frac{1}{A^\dagger A + b} \left(f_0 A + A^\dagger f_0^\dagger \right) \frac{1}{A^\dagger A + b} \right) A^\dagger
\end{aligned} \tag{208}$$

2.7.4.4 Caylay Hamilton

$$\begin{aligned}
U_1 &= U_0 \left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \\
(f_1)^i_l &= (f_0)^j_k \left(\frac{\partial U_1}{\partial U_0} \right)^{k_i}_{l_j} + (f_0)^{\dagger j}_k \left(\frac{\partial U_1^\dagger}{\partial U_0} \right)^{k_i}_{l_j} \\
&= (f_0)^j_k \left(\frac{\partial U_0^k_m \left(\left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \right)^{m_j}}{\partial U_0^{l_i}} \right) + (f_0)^{\dagger j}_k \left(\frac{\partial \left(\left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \right)^{k_m} (U_0^\dagger)^{m_j}}{\partial U_0^{l_i}} \right) \\
&= (f_0)^j_k \left(\frac{\partial U_0^k_m}{\partial U_0^{l_i}} \left(\left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \right)^{m_j} + U_0^k_m \frac{\partial \left(\left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \right)^{m_j}}{\partial U_0^{l_i}} \right) + (f_0)^{\dagger j}_k \left(\frac{\partial \left(\left(U_0^\dagger U_0 \right)^{-\frac{1}{2}} \right)^{k_m}}{\partial U_0^{l_i}} \right) U_0^{\dagger m_j}
\end{aligned} \tag{209}$$

let $Q = U_0^\dagger U_0$,

$$\begin{aligned}
(f_1)^i_l &= (f_0)^j_k \left(\delta^k_l \delta^i_m \left(Q^{-\frac{1}{2}} \right)^{m_j} + U_0^k_m \frac{\partial \left(Q^{-\frac{1}{2}} \right)^{m_j}}{\partial U_0^{l_i}} \right) + (f_0)^{\dagger j}_k \left(\frac{\partial \left(Q^{-\frac{1}{2}} \right)^{k_m}}{\partial U_0^{l_i}} \right) U_0^{\dagger m_j} \\
&= \left(Q^{-\frac{1}{2}} f_0 \right)^i_l + (f_0 U_0)^j_m \frac{\partial \left(Q^{-\frac{1}{2}} \right)^{m_j}}{\partial U_0^{l_i}} + \left(U_0^\dagger f_0^\dagger \right)^m_k \frac{\partial \left(Q^{-\frac{1}{2}} \right)^k_m}{\partial U_0^{l_i}}
\end{aligned} \tag{210}$$

Chain rule of matrix function,

$$\left(\frac{\partial M(A)}{\partial A} \right)^{i_j}_{k_l} \equiv \frac{\partial M(A)^i_l}{\partial A^j_k} \tag{211}$$

$$\begin{aligned}
\left(\frac{\partial M}{\partial B} \right)^{i_j}_{k_l} &= \frac{\partial M^i_l}{\partial B^j_k} = \frac{\partial M^i_l}{\partial A^{m_n}} \frac{\partial A^{m_n}}{\partial B^j_k} \\
&= \left(\frac{\partial M}{\partial A} \right)^{i_m}_{n_l} \left(\frac{\partial A}{\partial B} \right)^{m_j}_{k_n}
\end{aligned} \tag{212}$$

$$\begin{aligned}
(f_1)^i_l &= \left(Q^{-\frac{1}{2}} f_0 \right)^i_l + \left(f_0 U_0 + U_0^\dagger f_0^\dagger \right)^j_m \left(\frac{\partial \left(Q^{-\frac{1}{2}} \right)}{\partial U_0} \right)^{m_l}_{i_j} \\
&= \left(Q^{-\frac{1}{2}} f_0 \right)^i_l + \left(f_0 U_0 + U_0^\dagger f_0^\dagger \right)^j_m \left(\frac{\partial \left(Q^{-\frac{1}{2}} \right)}{\partial Q} \right)^{m_p}_{q_j} \left(\frac{\partial Q}{\partial U_0} \right)^{p_l}_{i_q}
\end{aligned} \tag{213}$$

Using,

$$\left(\frac{\partial Q}{\partial U_0} \right)^{p_l}_{i_q} = \left(\frac{\partial U_0^{\dagger p}_t U_0^t_q}{\partial U_0^{l_i}} \right) = U_0^{\dagger p}_t \delta^t_l \delta^i_q = U_0^{\dagger p}_l \delta^i_q \tag{214}$$

The remaining thing is,

$$\begin{aligned}
&\left(\frac{\partial (F_0 + F_1 Q + F_2 Q^2)}{\partial Q} \right)^{m_p}_{q_j} \\
&= \frac{\partial F_0}{\partial Q^p_q} \delta^m_j + \frac{\partial F_1}{\partial Q^p_q} Q^m_j + \frac{\partial F_2}{\partial Q^p_q} (Q^2)^m_j + F_1 \delta^m_p \delta^q_j + F_2 \frac{\partial Q^m_t Q^t_j}{\partial Q^p_q} \\
&= \frac{\partial F_0}{\partial Q^p_q} \delta^m_j + \frac{\partial F_1}{\partial Q^p_q} Q^m_j + \frac{\partial F_2}{\partial Q^p_q} (Q^2)^m_j + F_1 \delta^m_p \delta^q_j + F_2 (\delta^m_p Q^q_j + Q^m_p \delta^q_j)
\end{aligned} \tag{215}$$

Then, we need to know,

$$\frac{\partial F_i}{\partial Q^p_q} = \sum_j \frac{\partial F_i}{\partial c_j} \frac{\partial c_j}{\partial Q^p_q} \tag{216}$$

From definition, $\frac{\partial \text{tr}[M]}{\partial M^{k_j}} = \delta^j_k$, this leads to,

$$\frac{\partial \text{tr}[\frac{1}{n} Q^n]}{\partial Q} = Q^{n-1} \tag{217}$$

Since $c_j = \frac{1}{j+1} \text{tr}[Q^{j+1}]$, and define $b_{ij} = \frac{\partial F_i}{\partial c_j}$

$$\frac{\partial F_i}{\partial Q^p_q} = \sum_j \frac{\partial F_i}{\partial c_j} \frac{\partial c_j}{\partial Q^p_q} = \sum_j b_{ij} (Q^j)^q_p \quad (218)$$

And,

$$\begin{aligned} & \left(\frac{\partial (F_0 + F_1 Q + F_2 Q^2)}{\partial Q} \right)^{m_p q_j} \\ &= \sum_s b_{0s} (Q^s)^q_p \delta^m_j + \sum_s b_{1s} (Q^s)^q_p Q^m_j + \sum_s b_{2s} (Q^s)^q_p (Q^2)^m_j + F_1 \delta^m_p \delta^q_j + F_2 (\delta^m_p Q^q_j + Q^m_p \delta^q_j) \end{aligned} \quad (219)$$

Define B matrix $B_n = b_{n0} + b_{n1}Q + b_{n2}Q^2$,

$$\begin{aligned} & \left(\frac{\partial (F_0 + F_1 Q + F_2 Q^2)}{\partial Q} \right)^{m_p q_j} = B_0^q_p \delta^m_j + B_1^q_p Q^m_j + B_2^q_p (Q^2)^m_j + F_1 \delta^m_p \delta^q_j + F_2 (\delta^m_p Q^q_j + Q^m_p \delta^q_j) \\ & \left(\frac{\partial Q}{\partial U_0} \right)^{p_l i_q} = U_0^\dagger p_l \delta^i_q \\ & \left(\frac{\partial (Q^{-\frac{1}{2}})}{\partial Q} \right)^{m_p q_j} \left(\frac{\partial Q}{\partial U_0} \right)^{p_l i_q} = (B_0 U_0^\dagger)^i_l \delta^m_j + (B_1 U_0^\dagger)^i_l Q^m_j + (B_2 U_0^\dagger)^i_l (Q^2)^m_j + F_1 U_0^\dagger m_l \delta^i_j + F_2 (U_0^\dagger m_l Q^i_j + (Q U_0^\dagger)^m_l \delta^i_j) \end{aligned} \quad (220)$$

go back to,

$$(f_1)^i_l = \left((Q^{-\frac{1}{2}} f_0)^i_l + (f_0 U_0 + U_0^\dagger f_0^\dagger)^j_m \left(\frac{\partial (Q^{-\frac{1}{2}})}{\partial Q} \right)^{m_p q_j} \left(\frac{\partial Q}{\partial U_0} \right)^{p_l i_q} \right) \quad (221)$$

Note, $B_n^\dagger = B_n$, define,

$$\begin{aligned} M &= f_0 U_0 + U_0^\dagger f_0^\dagger \\ a_n &= \text{tr}[M Q^n], \end{aligned} \quad (222)$$

$$(f_1)^i_l = Q^{-\frac{1}{2}} f_0 + \sum a_n B_n U_0^\dagger + F_1 M U_0^\dagger + F_2 Q M U_0^\dagger + F_2 M Q U_0^\dagger \quad (223)$$

Note: M , and Q^n are Hermitian, and therefore $\text{tr}[M Q^n]$ is real.

$$f_1^\dagger = f_0^\dagger (Q^{-\frac{1}{2}})^\dagger + U_0 (\sum a_n B_n) + F_1 U_0 M + F_2 U_0 (Q M + M Q) \quad (224)$$

The final task is to calculate, $b_{ij} = \frac{\partial f_i}{\partial c_j}$. To do this, we need to calculate $\frac{\partial u, v, w}{\partial c_j}$. (denote $u_{0,1,2} = u, v, w$). To do this, we need to calculate $\frac{\partial u_i}{\partial g_j}$ and $\frac{\partial g_i}{\partial c_j}$.

$$\begin{aligned} c_0 &= g_0 + g_1 + g_2 \\ c_1 &= \frac{1}{2}(g_0^2 + g_1^2 + g_2^2) \\ c_2 &= \frac{1}{3}(g_0^3 + g_1^3 + g_2^3) \\ u &= \sqrt{g_0} + \sqrt{g_1} + \sqrt{g_2} \\ v &= \sqrt{g_0 g_1} + \sqrt{g_0 g_2} + \sqrt{g_1 g_2} \\ w &= \sqrt{g_0 g_1 g_2} \end{aligned} \quad (225)$$

It can be verified,

$$\begin{aligned} c_0 &= u^2 - 2v \\ c_1 &= \frac{1}{2}(u^4 - 4u^2 v + 4uw + 2v^2) \\ c_2 &= \frac{1}{3}(u^6 - 6u^4 v + 6u^3 w + 9u^2 v^2 - 12uvw - 2v^3 + 3w^2) \\ \frac{\partial C}{\partial U} &= \begin{pmatrix} 2u & -2 & 0 \\ 2u^3 - 4uv + 2w & -2u^2 + 2v & 2u \\ 2u^5 - 8u^3 v + 6u^2 w + 6uv^2 - 4vw & -2u^4 + 6u^2 v - 4uw - 2v^2 & 2u^3 - 4uv + 2w \end{pmatrix} \end{aligned} \quad (226)$$

And, use f_i in Eq. (30) of hep-lat/0702028,

$$\frac{\partial f}{\partial c} = \frac{\partial f}{\partial U} \left(\frac{\partial c}{\partial U} \right)^{-1}$$

$$= \begin{pmatrix} \frac{1}{2} \left(\frac{4u^3}{(w-uv)^2} + \frac{u^4(u^2+v)}{(w-uv)^3} + \frac{3uv}{w^2} + \frac{2}{w-uv} - \frac{v^3}{w^3} - \frac{3}{w} \right) & \frac{w^3(-5u^4+10u^2v+2v^2)-u^2v^2w(u^4+u^2v-6v^2)+u^3v^4(u^2-2v)-uw^2(u^6-6u^4v+6u^2v^2+6v^3)-3uw^4}{2w^3(uv-w)^3} & \frac{w^3(-5u^4+10u^2v+2v^2)-u^2v^2w(u^4+u^2v-6v^2)+u^3v^4(u^2-2v)-uw^2(u^6-6u^4v+6u^2v^2+6v^3)-3uw^4}{2w^3(uv-w)^3} \\ \frac{w^3(-5u^4+10u^2v+2v^2)-u^2v^2w(u^4+u^2v-6v^2)+u^3v^4(u^2-2v)-uw^2(u^6-6u^4v+6u^2v^2+6v^3)-3uw^4}{2w^3(uv-w)^3} & \frac{-u^3v^4+u^2v^2w(u^2+3v)+w^3(4u^2+v)+uw^2(u^4-4u^2v-3v^2)}{2w^3(uv-w)^3} & \frac{w^3(-5u^4+10u^2v+2v^2)-u^2v^2w(u^4+u^2v-6v^2)+u^3v^4(u^2-2v)-uw^2(u^6-6u^4v+6u^2v^2+6v^3)-3uw^4}{2w^3(uv-w)^3} \\ \frac{-u^3v^4+u^2v^2w(u^2+3v)+w^3(4u^2+v)+uw^2(u^4-4u^2v-3v^2)}{2w^3(uv-w)^3} & \frac{-u^3v^4+u^2v^2w(u^2+3v)+w^3(4u^2+v)+uw^2(u^4-4u^2v-3v^2)}{2w^3(uv-w)^3} & \frac{-u^3v^4+u^2v^2w(u^2+3v)+w^3(4u^2+v)+uw^2(u^4-4u^2v-3v^2)}{2w^3(uv-w)^3} \end{pmatrix} \quad (227)$$

The equation in hep-lat/0702028 is checked and OK.

2.8 The update for clover Wilson Dirac fermion

In the previous sections there are some mistakes, so I will try to sort out everything in this section. But I'll keep the mistakes in the previous sections until I'm sure I understand everything and I'm sure I get things correct now.

2.8.1 Revisit of force of HMC and RHMC

I find it difficult to consider objects like $\partial U^\dagger/\partial U$, I think the correct way should be considering $U^\dagger$ and U as independent variables, but all as functions of ω_a . **The force is then $F = i\dot{\omega}_a T_a$, where $\dot{\omega}_a = -\partial S/\partial \omega_a$.**

By using,

$$\frac{\partial U_\mu}{\partial \omega_\mu^a} = iT^a U_\mu, \quad \frac{\partial U_\mu^\dagger}{\partial \omega_\mu^a} = -iU_\mu^\dagger T^a, \quad (228)$$

it is,

$$\frac{\partial S}{\partial \omega_a} = \text{tr} \left[\frac{\partial S}{\partial U} \frac{\partial U}{\partial \omega_a} + \frac{\partial S}{\partial U^\dagger} \frac{\partial U^\dagger}{\partial \omega_a} \right] \quad (229)$$

note that $\partial S/\partial \omega_a$ is a scalar, S, ω_a are both scalars, and both $\frac{\partial S}{\partial U}$ and $\frac{\partial U}{\partial \omega_a}$ are matrices.

So,

$$\begin{aligned} F &= -i \frac{\partial S}{\partial \omega_a} T_a = -i \text{tr} \left[\frac{\partial S}{\partial U} iT^a U - \frac{\partial S}{\partial U^\dagger} iU^\dagger T^a \right] T_a \\ &= -i \text{tr} \left[i \frac{\partial S}{\partial U} T^a U - i \frac{\partial S}{\partial U^\dagger} U^\dagger T^a \right] T_a \\ &= -i \text{tr} \left[i \frac{\partial S}{\partial U} T^a U + \left(i \frac{\partial S}{\partial U} T^a U \right)^\dagger \right] T_a \\ &= -2i \text{tr} \left[\text{Re} \left[i \frac{\partial S}{\partial U} T^a U \right] \right] T_a \end{aligned} \quad (230)$$

for any matrix

$$2i \sum_a T_a \text{Imtr}[T_a M] = -2i \sum_a T_a \text{Retr}[iT_a M] = \{M\}_{TA} \quad (231)$$

So, as long as $\partial S/\partial U^\dagger = (\partial S/\partial U)^\dagger$,

$$F = \{U^i_j \frac{\partial S}{\partial U^k_j}\}_{TA} \quad (232)$$

here, when calculating $\partial S/\partial U$, the $U^\dagger$ is considered as independent of U .

Then we can use the result in Sec. 2.7.2.2, **Note that, different from the case of staggered fermion, $DD^\dagger \neq D^\dagger D$ in the case of Wilson-Dirac fermion, so in the following, we use $DD^\dagger$ because $\psi^\dagger(D^\dagger)^{-1}D^{-1}\psi = \psi^\dagger(DD^\dagger)^{-1}\psi$.**

$$\begin{aligned} S &= (\psi^\dagger)_m \left(c + \sum_i \frac{a_i}{DD^\dagger + b_i} \right) m_k \psi^k \\ \phi &= \frac{1}{DD^\dagger + b_s} \psi, \quad \phi_d = D^\dagger \frac{1}{DD^\dagger + b_s} \psi, \\ \frac{\partial}{\partial U^i_l} S &= - \sum_s a_s \left\{ \phi^\dagger_p \left(\frac{\partial D^p_h}{\partial U^i_l} \right) \phi_d^h + \phi_d^\dagger_h \left(\frac{\partial D^{\dagger h}_q}{\partial U^i_l} \right) \phi^q \right\} \end{aligned} \quad (233)$$

For HMC,

$$\begin{aligned} S &= (\psi^\dagger)_m ((DD^\dagger)^{-1})^m_k \psi^k \\ \phi &= \frac{1}{DD^\dagger} \psi, \quad \phi_d = D^\dagger \frac{1}{DD^\dagger} \psi, \\ \frac{\partial}{\partial U^i_l} S &= - \left\{ \phi^\dagger_p \left(\frac{\partial D^p_h}{\partial U^i_l} \right) \phi_d^h + \phi_d^\dagger \left(\frac{\partial D^{\dagger h}_q}{\partial U^i_l} \right) \phi^q \right\} \end{aligned} \quad (234)$$

2.8.2 The force of RHMC of Wilson-Dirac fermion

Using $\kappa = 1/(2am_f + 8)$, the Wilson-Dirac fermion operator is,

$$D = 1 - \kappa \sum_{\mu} ((1 - \gamma_{\mu}) U_{\mu}(x_L) \delta_{x_L, (x+\mu)_R} + (1 + \gamma_{\mu}) U_{\mu}^{-1}(x_L - \mu) \delta_{x_L, (x-\mu)_R}) \quad (235)$$

$$\begin{aligned} \frac{\partial}{\partial U^i_l(n, \mu)} S &= \kappa \sum_s a_s \left\{ \phi^\dagger(n + \mu)_p (1 + \gamma_{\mu}^{\dagger}) \delta^p_i \delta^h_l \phi_d(n)^h + \phi_d^\dagger(n + \mu)_p (1 - \gamma_{\mu}) \delta^p_i \delta^h_l \phi(n)^h \right\} \\ &= \kappa \sum_s a_s \left\{ \phi^\dagger(n + \mu)_i (1 + \gamma_{\mu}) \phi_d(n)^i + \phi_d^\dagger(n + \mu)_i (1 - \gamma_{\mu}) \phi(n)^i \right\} \\ &= \kappa \sum_s a_s \left\{ (1 + \gamma_{\mu}) \phi_d(n)^i \phi^\dagger(n + \mu)_i + (1 - \gamma_{\mu}) \phi(n)^i \phi_d^\dagger(n + \mu)_i \right\} \end{aligned} \quad (236)$$

For HMC, just remove $\sum_s a_s$ and use the correct definition of ϕ, ϕ_d .

2.8.3 The force of RHMC of clover Wilson-Dirac fermion

Start from the clover term, which is defined using,

$$F_{\mu\nu} = \frac{1}{8i} (P_{\mu\nu} - P_{\mu\nu}^\dagger) \quad (237)$$

There are different definitions of $P_{\mu\nu}$, the original one was,

$$P_{\mu\nu} = U_{\mu, \nu} + U_{\nu, -\mu} - U_{\mu, -\nu} - U_{-\nu, -\mu} \quad (238)$$

Using $U_{a,b} = U_{b,a}^\dagger$, it is equivalent to use,

$$P_{\mu\nu}^a = U_{\mu, \nu} - U_{\mu, -\nu} - U_{\mu, -\nu} + U_{-\mu, -\nu} \quad (239)$$

or

$$P_{\mu\nu}^b = U_{\mu, \nu} + U_{\nu, -\mu} + U_{-\nu, \mu} + U_{-\mu, -\nu} \quad (240)$$

The first definition ($P_{\mu\nu}^a$), μ is always in front of ν .

The second definition, all signs are sum, and all are $U_{\mu\nu}$ plaquette (but disconnect at different positions).

The force come from each plaquette disconnected at four corners.

Denoting, $L_{\mu, \nu, k}$ as the plaquette disconnected at the position k , i.e.,

$$\begin{aligned} L_{\mu, \nu, 0}(n) &= \bar{\psi}_l(n) U_{\mu, \nu, -\mu, -\nu}(n) \sigma_{\mu\nu} \psi_r(n) \\ L_{\mu, \nu, 1}(n) &= \bar{\psi}_l(n + \mu) U_{\nu, -\mu, -\nu, \mu}(n + \mu) \sigma_{\mu\nu} \psi_r(n + \mu) \\ L_{\mu, \nu, 2}(n) &= \bar{\psi}_l(n + \mu + \nu) U_{-\mu, -\nu, \mu, \nu}(n + \mu + \nu) \sigma_{\mu\nu} \psi_r(n + \mu + \nu) \\ L_{\mu, \nu, 3}(n) &= \bar{\psi}_l(n + \nu) U_{-\nu, \mu, \nu, -\mu}(n + \nu) \sigma_{\mu\nu} \psi_r(n + \nu) \\ \sum_n \bar{\psi}_l \sigma_{\mu\nu} P_{\mu\nu}^b \psi_r &= \sum_{n, k} L_{\mu, \nu, k}(n) \\ \sum_n \bar{\psi}_l \sigma_{\mu\nu} (P_{\mu\nu}^b)^\dagger \psi_r &= \sum_{n, k} L_{\mu, -\nu, k}(n) \end{aligned} \quad (241)$$

Note, denote the staple as $St_{\mu,\nu,k}$, where k is the insertion of $\sigma_{\mu\nu}\psi_r\bar{\psi}_l$

$$\begin{aligned}\frac{\partial}{\partial U_\mu} L_{\mu,\nu,k}(n) &= St_{\mu,\nu,k} \\ \frac{\partial}{\partial U_\nu} L_{\mu,\nu,k}(n) &= St_{\nu,-\mu,k} \\ \frac{\partial}{\partial U_\mu} L_{\mu,-\nu,k}(n) &= St_{\mu,-\nu,k} \\ \frac{\partial}{\partial U_\nu} L_{\mu,-\nu,k}(n) &= St_{\nu,\mu,k}\end{aligned}\tag{242}$$

The actual minus sign of ν -link comes from the fact that it should be $\sigma_{\nu\mu}$. So for all $\mu > \nu$ in staple, there is an extra minus sign.

so,

$$\frac{\partial}{\partial U_\mu} \bar{\psi}_l \sigma_{\mu\nu} F_{\mu\nu} \psi_r = \sum_{k, \pm\nu \neq \mu} (-1)^{if \nu < \mu} \times (\pm 1) \times St_{\mu, \pm\nu, k}\tag{243}$$

And,

$$F_{\mu\nu}^\dagger = F_{\mu\nu}, \sigma_{\mu\nu, E}^\dagger = \sigma_{\mu\nu, E}, (\gamma_4 \sigma_{\mu\nu}^M)^\dagger = (\gamma_4 \sigma_{\mu\nu}^M)\tag{244}$$

Although both $\sigma_{\mu\nu}^{E,M}$ are γ_5 -hermitian, only $\bar{\psi} \sigma_{\mu\nu}^M \psi$ is hermitian.

2.8.4 The force of RHMC of Wilson-Dirac fermion with even pseudofermion

we do not use even-pseudofermion because I'm not sure whether the κ term can be decoupled in $D^\dagger D$.

2.8.5 STOUT link smearing

The definition of Stout smearing is,

$$\begin{aligned}U_\nu^{(n+1)}(x) &= \exp(i\rho Q_\mu^{(n)}(x)) U_\nu^{(n)}(x) \\ Q_\mu^{(n)}(x) &= \frac{1}{2i} \left(W_\mu^{(n)}(x) - \frac{1}{N_c} \text{tr} [W_\mu^{(n)}(x)] \right) \\ W_\mu^{(n)}(x) &= S_\mu^{(n)}(x) U_\mu^{(n)\dagger}(x) - U_\mu^{(n)}(x) S_\mu^{(n)\dagger}(x) \\ S_\mu^{(n)}(x) &= \sum_{\nu \neq \mu} (U_\nu^{(n)}(x) U_\mu^{(n)}(x + \nu) U_\nu^{(n)\dagger}(x + \mu) + U_\nu^{(n)\dagger}(x - \nu) U_\mu^{(n)}(x - \nu) U_\nu^{(n)}(x + \mu - \nu))\end{aligned}\tag{245}$$

2.8.6 STOUT link smearing force

2.9 HMC diagnostics

The HMC diagnostic suite contains two tests: molecular-dynamics reversibility and a finite-difference check of the production force. The implementation is in `Code/CLGTest/Tests/TestHmcDiagnostics.cpp`; its test configurations are in `Bin/Debug/TestSuit_HmcDiagnostics.yaml`.

2.9.1 Gauge-dependent state

Let U denote the thin gauge field supplied to an action. Before evaluating the force or energy at a particular value of U , the diagnostic rebuilds all objects derived from that same U in the production order:

1. update the registered cache before gauge updating;
2. update the registered cache before smearing;
3. construct the registered smeared or effective gauge field, if the smearing object is enabled during an update;
4. update the registered cache after smearing.

Thus an energy evaluated at a perturbed gauge field never uses a smeared field, staple, plaquette, or $F_{\mu\nu}$ cache made from a different gauge field.

For a fermion action, the diagnostic first performs this refresh at the unperturbed field U_0 and then calls `PrepareForHMC` once. Denote the resulting pseudofermion field by ϕ_0 . The same ϕ_0 is used for the production-force calculation and for every positive and negative perturbation defined below. In particular, ϕ_0 is not regenerated when the link, direction, or finite-difference step size changes.

2.9.2 Production-force convention

For each link l , `CalculateForce` adds a raw matrix Y_l to the raw force field. The force actually used by the integrator is

$$F_l = \text{TA}\left(U_{0,l}Y_l^\dagger\right), \quad (246)$$

where $U_{0,l}$ is link l of the unperturbed gauge field and

$$\text{TA}(M) = \frac{M - M^\dagger}{2} - \frac{1}{3}\text{tr}\left(\frac{M - M^\dagger}{2}\right)\mathbb{I}. \quad (247)$$

Here $\mathbb{I}$ is the 3×3 identity matrix and tr is the color-matrix trace. In the code, Eq. (246) is implemented by `LeftMul(U0,FALSE,TRUE)` followed by `TA()`. Directions fixed by `BindDir` are then set to zero in F .

For two gauge-shaped matrix fields A and B , define the real Frobenius inner product

$$\langle A, B \rangle = \text{Re} \sum_l \text{tr}\left(A_l^\dagger B_l\right). \quad (248)$$

This is the real part returned by `A.Dot(B)`. The CLGLib force normalization predicts the action derivative

$$D_{\text{force}} = -2\langle F, X \rangle. \quad (249)$$

The perturbation field X used below is nonzero on only one link l . Consequently Eq. (249) reduces to

$$D_{\text{force}} = -2 \text{Re tr}\left(F_l^\dagger X_l\right), \quad (250)$$

which is calculated by `-2.0*pForce->Dot(pX).x`.

2.9.3 Finite-difference force test

The diagnostic tests each continuous action separately. Fix one such action, its unperturbed thin gauge field U_0 , and, for a fermion action, the pseudofermion field ϕ_0 defined above. The action value written below is therefore

$$S[U] \equiv S[U; \phi_0] \quad (251)$$

for a fermion action; for a gauge action there is no pseudofermion argument.

Choose a link l which is neither Dirichlet nor fixed by `BindDir`. Choose a 3×3 matrix X_l satisfying

$$X_l^\dagger = -X_l, \quad \text{tr}(X_l) = 0, \quad \|X_l\|_F = 1, \quad (252)$$

where

$$\|X_l\|_F^2 = \text{tr}(X_l^\dagger X_l). \quad (253)$$

Define the gauge-shaped field X by setting its value to X_l on link l and to zero on every other link. This definition makes $\langle F, X \rangle$ in Eq. (249) depend only on the selected link.

Let $\Delta > 0$ be the YAML parameter `Delta`, and let N_h be the integer YAML parameter `DeltaLevels`. For $k = 0, 1, \dots, N_h - 1$, define

$$h_k = \frac{\Delta}{2^k}. \quad (254)$$

For each k , make two independent copies of U_0 . Change only link l and define

$$U_{+,k}(l) = \exp(h_k X_l) U_0(l), \quad U_{+,k}(l') = U_0(l') \quad (l' \neq l), \quad (255)$$

$$U_{-,k}(l) = \exp(-h_k X_l) U_0(l), \quad U_{-,k}(l') = U_0(l') \quad (l' \neq l). \quad (256)$$

The multiplication is on the left. In the implementation it is performed by `pX->ExpMult(+h,pWorkGauge)` and `pX->ExpMult(-h,pWorkGauge)`. Because X_l is traceless and anti-Hermitian, both exponentials are in $SU(3)$, so the perturbed link remains in $SU(3)$.

After rebuilding all gauge-dependent data separately from $U_{+,k}$ and $U_{-,k}$, calculate

$$S_{+,k} = S[U_{+,k}], \quad S_{-,k} = S[U_{-,k}]. \quad (257)$$

The central finite-difference estimate of the directional derivative at U_0 is

$$D_{\text{FD},k} = \frac{S_{+,k} - S_{-,k}}{2h_k}. \quad (258)$$

The production-force prediction D_{force} in Eq. (250) is calculated once at U_0 and is the same for all k .

For every level define the absolute disagreement

$$e_k = |D_{\text{FD},k} - D_{\text{force}}|, \quad (259)$$

the comparison scale

$$q_k = \max(|D_{\text{FD},k}|, |D_{\text{force}}|), \quad (260)$$

and the allowed error

$$b_k = \epsilon_{\text{abs}} + \epsilon_{\text{rel}} q_k, \quad (261)$$

where ϵ_{abs} and ϵ_{rel} are the YAML parameters `TolAbs` and `TolRel`, respectively.

A link-and-direction trial fails if any required value $S_{+,k}$, $S_{-,k}$, $D_{\text{FD},k}$, D_{force} , e_k , or b_k is not finite. Otherwise the trial passes exactly when

$$\text{there exists } k \in \{0, \dots, N_h - 1\} \text{ such that } e_k \leq b_k. \quad (262)$$

There is no additional plateau, monotonic-decrease, convergence-rate, or adjacent-level requirement. The code reports all levels and identifies the level with the smallest e_k , but the smallest-error level has no separate acceptance rule beyond Eq. (262). Every sampled direction on every sampled link must pass, and every selected continuous action is tested independently.

The central difference in Eq. (258) has an $O(h_k^2)$ truncation error in exact arithmetic. Several values of h_k are used because decreasing h_k reduces this truncation error but can amplify subtraction, round-off, and fermion-solver errors. This observation explains the step-size sweep; it is not an extra pass condition.

2.9.4 Reversibility

Reversibility is tested using one prepared Hamiltonian state. The test calls `Prepare(FALSE,0)` exactly once, which prepares the action state and generates the initial momentum. Let the saved integrator working gauge and momentum immediately after that call be U_0 and P_0 . The test performs

$$(U_0, P_0) \longrightarrow (U_1, P_1), \quad (263)$$

negates the momentum, and then performs a second evolution without another call to `Prepare`:

$$(U_1, -P_1) \longrightarrow (U_2, P_2). \quad (264)$$

The same integrator, step size, step count, action state, and boundary conditions are therefore used in both evolutions. Reversibility requires

$$U_2 \simeq U_0, \quad P_2 \simeq -P_0. \quad (265)$$

The diagnostic reports average and maximum Frobenius norms of $U_2 - U_0$ and $P_2 + P_0$ and compares them with the configured tolerances.

The two calls to `Evaluate()` modify the integrator working fields but do not copy them to the lattice gauge field. Reversibility checks the complete molecular-dynamics map, but it does not prove that the force is correct: an incorrect force can still be used reversibly in both evolutions.

3 Sparse linear algebra solver

Given a matrix A and a vector $\mathbf{b}$. The solver works out the solution

$$\mathbf{b} = A\mathbf{x}, \quad \mathbf{x} = A^{-1}\mathbf{b}. \quad (266)$$

3.1 Krylov subspace

In short, the Krylov subspace methods assumes

$$\mathbf{x} \approx \sum_{l=0}^{k-1} C_l A^l \mathbf{b} \in K_k = \text{span} \{ \mathbf{b}, A\mathbf{b}, \dots, A^{k-1}\mathbf{b} \}. \quad (267)$$

with finite k , where C_k are coefficients. The equation $0 = \mathbf{b} - A\mathbf{x}$ becomes

$$\|\mathbf{b} - \sum_{l=0}^{k-1} C_l A^{l+1} \mathbf{b}\| = 0 \quad (268)$$

This is a problem in $k+1$ dimension, where k is independent of the dimension of $\mathbf{b}$, and usually significantly smaller than the dimension of $\mathbf{b}$. The Eq. (268) can be understood that, if $\mathbf{x}_k \approx A^{-1}\mathbf{b}$ is approximation of the solution in k dimension, in the $k+1$ dimension

$$(\mathbf{b} - A\mathbf{x})_{k+1} \perp K_k \quad (269)$$

That is, if we have a multi-dimension vector v_n , and its projection in 3-dimension ($D = k+1$) is a vector $\mathbf{v}_3$, if we want to find a plane ($D = k$) such that the projection of v_n in the plane is minimized, the plane is chosen to be the one orthogonal to $\mathbf{v}_3$.

3.2 GMRES

This section we follow Refs. [16] and [9].

Assume a set of basis has been found. For example, if the subspace is found by using modified Gram-Schmidt as

Algorithm 3 Arnoldi with modified Gram-Schmidt

```

 $\mathbf{v}^{(0)} = \mathbf{x}_0 / \|\mathbf{x}_0\|$ 
for  $i = 0$  to  $k-1$  do
   $\mathbf{w} = A\mathbf{v}^{(i)}$ 
  for  $j = 0$  to  $i$  do
     $c = \mathbf{v}^{(j)*} \cdot \mathbf{w}$ 
     $\mathbf{w} = \mathbf{w} - c\mathbf{v}^{(j)}$ 
     $h[j, i] = c$ 
  end for
   $h[i+1, i] = \|\mathbf{w}\|$ 
   $\mathbf{v}^{(i+1)} = \mathbf{w} / \|\mathbf{w}\|$ 
end for

```

Note that $(\mathbf{w} - (\mathbf{v}_i^* \cdot \mathbf{w}) \mathbf{v}_i)^* \cdot \mathbf{v}_i = 0$, and $\mathbf{x}_0$ is a trial solution, which can be set to be $\mathbf{b}$ at first. Now we obtain $k+1$ unitary orthogonal vectors, such that

$$\mathbf{v}_i^* \cdot \mathbf{v}_j = \delta_{ij}, \quad A\mathbf{v}_{i-1} = \sum_{j=0}^i h[j, i-1] \mathbf{v}_j, \quad (270)$$

That is

$$\begin{pmatrix} Av_0 \\ Av_1 \\ Av_2 \\ \dots \\ Av_{k-1} \end{pmatrix} = (v_0, v_1, \dots, v_{k-1}, v_k) \begin{pmatrix} h[0,0] & h[0,1] & \dots & h[0,k-2] & h[0,k-1] \\ h[1,0] & h[1,1] & \dots & h[1,k-2] & h[1,k-1] \\ 0 & h[2,1] & \dots & h[2,k-2] & h[2,k-1] \\ 0 & 0 & \dots & \dots & \dots \\ \dots & \dots & \dots & h[k-1,k-2] & h[k-1,k-1] \\ 0 & 0 & \dots & 0 & h[k,k-1] \end{pmatrix} \quad (271)$$

which can be written as

$$(Av)_k = v_{k+1}H \quad (272)$$

Assume the solution is

$$\mathbf{x} = \mathbf{x}_0 + \sum_{i=0}^{k-1} y_i \mathbf{v}_i = \mathbf{x}_0 + \mathbf{y} = \mathbf{x}_0 + v_k y, \quad (273)$$

Using $\mathbf{r}_0 = \mathbf{b} - A\mathbf{x}_0$, to minimize $\|\mathbf{b} - A\mathbf{x}\|$ is to minimize $\|\mathbf{r}_0 - A\mathbf{y}\|$. We always choose $\mathbf{v}_0 = \mathbf{r}_0/\|\mathbf{r}_0\|$, denote $\beta = \|\mathbf{r}_0\|$, it is to minimize

$$\operatorname{argmin} \|\beta \mathbf{e}_0 - Hy\|. \quad (274)$$

Or, to solve an equation in k dimension

$$\beta \mathbf{e}_0 - Hy = 0, \quad y = H^{-1}\beta \mathbf{e}_0 = H^{-1}g \quad (275)$$

Now, we need to solve H^{-1} , we can do this by applying rotation matrix, defining (This is also called **Givens rotation**)

$$J_0 = \begin{pmatrix} R & 0 \\ 0 & \mathbb{I}_{k-2} \end{pmatrix}_{D=k} = \begin{pmatrix} c_0^* & s_0^* & 0 & \dots & 0 \\ -s_0 & c_0 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}_{D=k} \quad (276)$$

Note that c_0^* and s_0^* is necessary to keep unitary. (s_0^* seems not necessary? we only need to keep the length of g unchanged) So that

$$0 = g - Hy \rightarrow 0 = J_0 g - J_0 Hy \quad (277)$$

with (Note that the first 2 lines are changed entirely)

$$H' = \begin{pmatrix} h'_{0,0} & h'_{0,1} & h'_{0,2} & \dots & h'_{0,k-1} \\ 0 & h'_{1,1} & h'_{1,2} & \dots & h'_{1,k-1} \\ 0 & h_{2,1} & h_{2,2} & \dots & h_{2,k-1} \\ 0 & 0 & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & h_{k,k-1} \end{pmatrix} \quad (278)$$

$$g' = (c_0^* \beta, -s_0 \beta, 0, \dots)$$

$$c_0 = \frac{h_{00}}{\sqrt{h_{00}^2 + h_{10}^2}}, \quad s_0 = \frac{h_{10}}{\sqrt{h_{00}^2 + h_{10}^2}}$$

where $\mathbb{I}_l$ is dimension l identity matrix. Similarly, after this, one can rotation matrices

$$J_1 = \begin{pmatrix} \mathbb{I}_1 & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & \mathbb{I}_{k-3} \end{pmatrix}_{D=k}, J_2 = \begin{pmatrix} \mathbb{I}_2 & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & \mathbb{I}_{k-4} \end{pmatrix}_{D=k}, \dots \quad (279)$$

To make H triangular.

The algorithm is

Algorithm 4 Rotate H

```

 $g[0] = \beta$ 
for  $i = 0$  to  $k - 1$  do
   $d = 1/\sqrt{|h[i, i]|^2 + |h[i + 1, i]|^2}$ 
   $cs = h[i, i] \times d, sn = h[i + 1, i] \times d$ 
  for  $j = i$  to  $k - 1$  do
     $h_{ij} = h[i, j]$ 
     $h[i, j] = cs^* \times h_{ij} + sn^* \times h[i + 1, j]$ 
     $h[i + 1, j] = cs \times h[i + 1, j] - sn \times h_{ij}$ 
  end for
   $minus_g = -g[i]$ 
   $g[i] = cs^* \times g[i]$ 
   $g[i + 1] = sn \times minus_g$ 
end for

```

After the rotation, $g[k]$ is the residue. If it is small enough, the last step is to solve $y = H^{-1}g$, where H is a upper triangular matrix. It can be iterated as

$$y[k - 1] = \frac{g[k - 1]}{h[k - 1, k - 1]}, y[k - 2] = \frac{1}{h[k - 2, k - 2]} (g[k - 2] - h[k - 2, k - 1]y[k - 1]), \dots \quad (280)$$

The algorithm is backward substitution

Algorithm 5 Solve Y

```

for  $i = k - 1$  to  $0$  do
  for  $j = i + 1$  to  $k - 1$  do
     $g[i] - = h[i, j] \times y[j]$ 
  end for
   $y[i] = g[i]/h[i, i]$ 
end for
return  $\mathbf{x}_0 + \sum_{i=0}^{k-1} y[i]\mathbf{v}^{(i)}$ 

```

Note that, the first step, the modified Gram-Schmidt step will produce more and more unitary normalized vectors, so the GMRES usually has a restart step. Let r denote the restart times, for example, the full algorithm with k is (GMRES(m) means GMRES with modified Gram-Schmidt, there is also GMRES with Household, etc)

Algorithm 6 GMRES(m)

```

x0 = b ▷ Use b as trail and start
for  $i = 1$  to  $r$  do
  r0 = b − Ax0
   $\beta = \|\mathbf{r}_0\|$ 
  v(0) = r0/ $\beta$ 
  for  $i = 0$  to  $k - 1$  do
    w = Av(i)
    for  $j = 0$  to  $i$  do
       $c = \mathbf{v}^{(j)*} \cdot \mathbf{w}$ 
      w − =  $c\mathbf{v}^{(j)}$ 
       $h[j, i] = c$ 
    end for
     $h[i + 1, i] = \|\mathbf{w}\|$ 
    v(i+1) = w/‖w‖
  end for
  RotateH( $k$ )
  x = SolveY( $k$ )
  if  $|g[k]| < \epsilon$  then
    return x ▷ Succeed, with the solution
  end if
  x0 = x ▷ Use the last solution as trail and restart
end for
return  $x$  ▷ Failed, with the last best solution

```

where *RotateH*(k) and **x** = *SolveY*(k) is described in Algorithms. 4 and 5.

3.3 GCR

This section we follow Ref. [9].

The GCR solver is similar to GMRES in Sec. 3.2, but the orthogonal basis are obtained in a different way. If one have a set of orthogonal basis such that

$$A\mathbf{p}_i^* \cdot A\mathbf{p}_j = \delta_{ij}, \quad (281)$$

The solution **x** is the residue projected into this basis (Note, here we do NOT assume the basis are normalized)

$$\begin{aligned} \mathbf{r}_0 &= \mathbf{b} - A\mathbf{x}_0 \\ \mathbf{x} &= \mathbf{x}_0 + \sum_{i=0}^{\infty} \frac{\mathbf{r}_0^* \cdot A\mathbf{p}_i}{\|A\mathbf{p}_i\|} \mathbf{p}_i \end{aligned} \quad (282)$$

So, the iteration is

$$\mathbf{x} \approx \mathbf{x}_k = \mathbf{x}_0 + \sum_{i=0}^k \frac{\mathbf{r}_0^* \cdot A\mathbf{p}_i}{\|A\mathbf{p}_i\|} \mathbf{p}_i \quad (283)$$

which can be obtained order by order as

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \frac{(\mathbf{b} - A\mathbf{x}_{k-1})^* \cdot A\mathbf{p}_{k-1}}{\|A\mathbf{p}_{k-1}\|} \mathbf{p}_{k-1} \quad (284)$$

There are GCR, ORTHOMIN, ORTHODIR. Both GCR and ORTHOMIN have oscillation (tested with random Gaussian pseudo fermion field and random gauge field), when iterating, sometimes, $\|\mathbf{p}^i\| \gg \|\mathbf{p}^{i-1}\|$, and $\|\mathbf{p}^{i+1}\| \ll \|\mathbf{p}^i\|$ and $\|\mathbf{p}^{i+2}\| \gg \|\mathbf{p}^{i+1}\|$, so we use ORTHODIR. The algorithm is

Algorithm 7 incomplete GCR with restart

```

x = b ▷ Use b as trail and start
for  $i = 0$  to  $r$  do ▷ restart  $r$  times
  r = b −  $A\mathbf{x}$ ,  $\mathbf{p}_0 = \mathbf{r}$ 
  for  $j = 0$  to  $k - 1$  do
     $\alpha = (A\mathbf{p}_j)^* \cdot \mathbf{r} / \|A\mathbf{p}_j\|^2$ 
    x = x +  $\alpha\mathbf{p}_j$ 
    r = r −  $\alpha A\mathbf{p}_j$ 
    if  $\|\mathbf{r}\| < \epsilon$  then return x ▷ Success
    end if
     $\mathbf{p}_{j+1} = A\mathbf{p}_j$ 
    for  $k = j - l + 1$  to  $j$  do
       $\beta = (A\mathbf{p}_k)^* \cdot A^2\mathbf{p}_j / \|A\mathbf{p}_k\|^2$ 
       $\mathbf{p}_{j+1} = \mathbf{p}_{j+1} + \beta\mathbf{p}_k$ 
    end for
  end for
end for
return x ▷ Failed with the closest result

```

Note that, GCR is much slower than GMRES and BiCGStab. A strategy to improve the speed is to restart quickly.

3.4 TFQMR

The QMR and TFQMR is based on Lanczos process [9]

Algorithm 8 Lanczos process

```

Prepare two vectors  $\mathbf{v}_1^\dagger \mathbf{w}_1 = 1$ 
 $\beta = \delta = 0$ ,  $w_0 = v_0 = 1$ 
for  $i = 0$  to  $m$  do
  u =  $A\mathbf{v}_i$ 
   $\alpha = \mathbf{u}^\dagger \mathbf{w}_i$ 
   $\mathbf{v}_{i+1} = \mathbf{u} - \alpha\mathbf{v}_i - \beta\mathbf{v}_{i-1}$ 
   $\mathbf{w}_{i+1} = A^\dagger \mathbf{w}_i - \alpha\mathbf{w}_i - \delta\mathbf{w}_{i-1}$ 
   $l = \mathbf{v}_{i+1}^\dagger \mathbf{w}_{i+1}$ 
   $\delta = \sqrt{|l|}$ ,  $\beta = l/\delta$ 
   $\mathbf{w}_{i+1} = \mathbf{w}_{i+1}/\beta$ ,  $\mathbf{v}_{i+1} = \mathbf{v}_{i+1}/\delta$ 
   $T[i+1, i+1] = \alpha$ ,  $T[i+2, i+1] = \beta$ ,  $T[i+1, i+2] = \delta$ 
end for
 $T[m+1, m+1] = (A\mathbf{v}_{m+1})^\dagger \mathbf{w}_{m+1}$ 

```

Let T_m be the $m \times m$ part of $T_{m+1, m+1}$, one have

$$\begin{aligned}
 AV_m &= V_m T_m + T[m, m+1] \mathbf{e}_m^T \mathbf{v}_{m+1} \\
 A^\dagger W_m &= W_m T_m^\dagger + T[m+1, m] \mathbf{e}_m^T \mathbf{w}_{m+1} \\
 W_m^T AV_m &= T_m
 \end{aligned} \tag{285}$$

Use $AV_m = V_m T_m + T[m, m+1] \mathbf{e}_m^T \mathbf{v}_{m+1}$, one can implement a GMRES method, which is called QMR, or one can implement a FOM method, which is called two sided Lanczos method.

In the TFQMR, one no need to calculate $A^\dagger$ (There are also methods making use of the γ_5 -hermiticity to simplify this.), which can be summarized as

Algorithm 9 TFQMR

```

x = x0 is the initial guess
for i = 0 to r do                                     ▷ Restart
    u = w = b − Ax0, d = 0, r = u*
    v = Au, uA = v
     $\theta = 0, \tau = |\mathbf{u}|$                                      ▷  $\theta, \tau \in \mathbb{R}$ 
     $\rho = \mathbf{r}^\dagger \mathbf{u}, \alpha = \eta = 0$                        ▷  $\rho, \alpha, \eta \in \mathbb{C}$ 
    for m = 0 to ... do
        if m is even then
             $\alpha = \frac{\rho}{\mathbf{r}^\dagger \mathbf{v}}$ 
        end if
        w = w −  $\alpha \mathbf{u}_A$ 
        d = u +  $\frac{\eta \theta}{\alpha} \mathbf{d}$ 
         $\theta = \frac{|\mathbf{w}|^2}{\tau^2}, c = \frac{1}{1+\theta}, \tau = \tau \times \sqrt{\theta c}, \eta = c\alpha$    ▷  $c \in \mathbb{R}$ 
        x = x +  $\eta \mathbf{d}$ 
        if m is odd then
             $\rho' = \mathbf{r}^\dagger \mathbf{w}, \beta = \frac{\rho'}{\rho}, \rho = \rho'$            ▷ If  $\rho$  is too small, stop
            u = w +  $\beta \mathbf{u}$ 
            v =  $\beta(\mathbf{u}_A + \beta \mathbf{v})$ 
            uA = Au
            v = uA + v
        else
            u = u −  $\alpha \mathbf{v}$ 
            uA = Au
        end if
        Use  $\mathbf{r}^\dagger \mathbf{v} < \epsilon$  to check whether to stop
    end for
    x0 = x
end for

```

3.5 GCRO-DR and GMRES-MDR

‘A comparison with the methods seen in the previous chapter indicates that in many cases, GMRES will be faster if the problem is well conditioned, resulting in a moderate number of steps required to converge. If many steps (say, in the hundreds) are required, then BICGSTAB and TFQMR may perform better. If memory is not an issue, GMRES or DQGMRES, with a large number of directions, is often the most reliable choice. The issue then is one of trading robustness for memory usage. In general, a sound strategy is to focus on finding a good preconditioner rather than the best accelerator’. [9].

That might be because the Krylov space will converge to the domain eigen-vector.

From Fig. 1. 9 of Ref. [17], the low mode is the most critical problem, so CLGLib first implement low mode deflation preconditioner.

In the following, we follow Ref. [18].

3.5.1 Brief introduction to deflation preconditioner

In short, the preconditioner means, one solve

$$M^{-1}Ax = M^{-1}b, \quad (286)$$

or

$$\begin{cases} AM^{-1}u = b \\ x = M^{-1}u \end{cases} \quad (287)$$

instead of $Ax = b$. If M is chosen carefully, it is usually faster.

Now, considering $A \in \mathbb{C}^{n \times n}$ and a matrix $Z \in \mathbb{C}^{n \times k}$ such that $Z = (v_1, v_2, \dots, v_k)$ and each row is a vector $v_i \in \mathbb{C}^n$ such that $v_i^\dagger v_j = \delta_{ij}$. So Z acts like a Unitary matrix $Z^\dagger Z = \mathbb{I}^{k \times k}$. Then we can use Z to project A on a subspace, as

$$T = Z^\dagger A Z, \quad Z^\dagger A = T Z^\dagger \quad (288)$$

so

$$\begin{aligned} Ax = b &\Rightarrow (\mathbb{I} + Z(T^{-1} - \mathbb{I})Z^\dagger)Ax = (\mathbb{I} + Z(T^{-1} - \mathbb{I})Z^\dagger)b \\ &\Rightarrow (A - ZZ^\dagger A)x + ZZ^{-1}Z^\dagger Ax = b - ZZ^\dagger b + ZZ^{-1}Z^\dagger b \end{aligned} \quad (289)$$

Note that $ZZ^\dagger \in \mathbb{C}^{n \times n}$ is not identity matrix (**also not a unitary matrix, but is an Hermitian matrix**). $T \in \mathbb{C}^{k \times k}$ is a small matrix. And then, one can solve

$$\begin{aligned} ZT^{-1}Z^\dagger Ax &= ZT^{-1}Z^\dagger b, \quad Z^\dagger A = TZ^\dagger \\ ZT^{-1}TZ^\dagger x &= ZT^{-1}Z^\dagger b \\ x &= ZT^{-1}Z^\dagger b \end{aligned} \quad (290)$$

exactly, while solving $(A - ZZ^\dagger A)x = b - ZZ^\dagger b$ by iteration methods such as GMRES.

This is the so-called **subspace deflation**.

3.5.2 Brief intro to GCRO-DR

Start from Eq. (272). Assume after the first-step GMRES, we have the orthogonal-normal basis v_i which can be written as a matrix $V_m \in \mathbb{C}^{n \times m}$, $V_{m+1} \in \mathbb{C}^{n \times (m+1)}$, $H \in \mathbb{C}^{(m+1) \times m}$. On the other hand, will be introduced later, we have a set of deflation vectors, or a matrix $P_k \in \mathbb{C}^{n \times k}$, such that

$$\begin{aligned} AV_m P_k &= V_{m+1} H P_k \\ \tilde{Y}_k &\equiv V_m P_k \in \mathbb{C}^{n \times k} \end{aligned} \quad (291)$$

Then $\tilde{Y}_k$ is the deflation matrix.

Consider the matrix $HP_k = QR$, where QR is the QR factorization, with $Q \in \mathbb{C}^{(m+1) \times k}$ and $R \in \mathbb{C}^{k \times k}$. And define

$$C_k \equiv V_{m+1} Q \in \mathbb{C}^{n \times k}. \quad (292)$$

So, if R which is a small upper triangular matrix such that R^{-1} can be easily calculated, it is

$$\begin{aligned} AV_m P_k &= A\tilde{Y}_k = V_{m+1} H P_k = V_{m+1} Q R = C_k R \\ C_k &= A\tilde{Y}_k R^{-1} = AU, \quad U \equiv \tilde{Y}_k R^{-1} \in \mathbb{C}^{n \times k} \end{aligned} \quad (293)$$

Finally, the problem in GMRES Eq. (272) is changed as

$$\begin{aligned} \tilde{U}_k &= U_k D_k = U_k \begin{pmatrix} \frac{1}{\|u_1\|} & 0 & 0 & 0 \\ 0 & \frac{1}{\|u_2\|} & 0 & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \frac{1}{\|u_k\|} \end{pmatrix} \in \mathbb{C}^{n \times k} \\ V_m^{(1)} &= (U_k, V_{m-k}) \in \mathbb{C}^{n \times m} \\ V_{m+1}^{(2)} &= (C_k, V_{m-k+1}) \in \mathbb{C}^{n \times (m+1)} \\ H' &= \begin{pmatrix} D_k & B_{m-k} \\ 0 & H_{m-k} \end{pmatrix} \in \mathbb{C}^{(m+1) \times m} \\ AV_m^{(1)} &= V_{m+1}^{(2)} H' \end{aligned} \quad (294)$$

where V are orthogonal-normal basis obtained in GMRES, and $B_{m-k} = AV_{m-k}$. Note that $B_{m-k} \in \mathbb{C}^{(m-k) \times k}$ but $H_{m-k} \in \mathbb{C}^{(m-k+1) \times (m-k)}$.

From Eq. (294), we find

- The subspace is k dimension subspace of m dimension Krylov space.
- With H , V and P known, we are able to calculate $QR = HP$, $U = V_m P R^{-1}$, $C = V_{m+1} Q$.

3.5.3 The choice of deflation subspace

In the above, we have assumed $P_k \in \mathbb{C}^{m \times k}$ is already known. Now we concentrate on this part.

Let $A \in \mathbb{C}^{n \times n}$, $V \in \mathbb{C}^{n \times k}$, **If V is formed as orthogonal normal basis of subspace S** , then, if $(\lambda, w \in \mathbb{C}^m)$ is eigen-pair of $V^\dagger AV$, $(\lambda, u = V^\dagger w \in \mathbb{C}^n)$ is eigen-pair of A .

$$\begin{aligned} H_m p_i &= \lambda_i p_i, \quad H_m = V_m^\dagger A V_m \\ A(V_m p_i) &= V_m H_m p_i = \lambda_i (V_m p_i) \end{aligned} \quad (295)$$

Therefor, P_k is a matrix with k rows, and each row is a eigen-vector of H_m (H_m denoting the first m row of H_{m+1}), then, $V P_k$ is a matrix with k rows such that each row is a eigen-vector of A (approximately since $AV \approx VH \Rightarrow H \approx V^\dagger AV$).

The first GMRES cycle will generate H_m (denoting the first m row of H_{m+1}), and $H_m \omega = \theta \omega$ is solved. However, starting from the second cycle of GCRO-DR, it is not $AV_m = V_{m+1} H_{m+1}$ but $AV_m^{(1)} = V_{m+1}^{(2)} H'_{m+1}$, such that $V_{m+1}^{(2)}$ are orthogonal basis but $V_m^{(1)}$ are not orthogonal basis! (Therefor $V^\dagger AV$ does not hold!). In this case, it is another eigen-problem which should be solved. This will be listed below without explain.

By Ref. [18], there are three strategies, Ritz eigen-vector (REV), harmonic Ritz eigen vector (HEV) and singular value decomposition (SVD). Either it is $REV > HEV > SVD$ or $SVD > HEV > REV$, so we only list REV and SVD here.

Note that $\tilde{U}_k$ is the normalized U_k , H_{m+1} means the H_{m+1} of GMRES procedure, and H'_{m+1} means H'_{m+1} obtained in GCRO-DR procedure, H_m and H'_m means the upper m rows of H_{m+1} and H'_{m+1} .

- REV

The k small eigen value of m , such that m is

$$\begin{cases} H_m \omega = \theta \omega, \\ \begin{pmatrix} \tilde{U}_k^\dagger C_k & \tilde{U}_k^\dagger V_{m-k+1} \\ 0 & (I_{m-k}, 0) \end{pmatrix} H'_{m+1} \omega = \theta \begin{pmatrix} \tilde{U}_k^\dagger \tilde{U}_k & \tilde{U}_k^\dagger V_{m-k} \\ V_{m-k}^\dagger \tilde{U}_k & I_{m-k} \end{pmatrix} \omega, \end{cases} \quad (296)$$

- HEV

The k larger eigen value of m , such that m is

$$\begin{cases} H_m^\dagger \omega = \theta H_{m+1}^\dagger H_{m+1} \omega, \\ H_{m+1}^\dagger \begin{pmatrix} C_k^\dagger \tilde{U}_k & 0 \\ V_{m-k+1}^\dagger \tilde{U}_k & \begin{pmatrix} I_{m-k} \\ 0 \end{pmatrix} \end{pmatrix} \omega = \theta H_{m+1}^\dagger H'_{m+1} \omega, \end{cases} \quad (297)$$

- SVD

The k small eigen value of m , such that m is

$$\begin{cases} H_m^\dagger H_m \omega = \theta \omega, \\ H_{m+1}^\dagger H'_{m+1} \omega = \theta \begin{pmatrix} \tilde{U}_k^\dagger \tilde{U}_k & 0 \\ 0 & I_{m-k} \end{pmatrix} \omega, \end{cases} \quad (298)$$

Although H_m is usually a small matrix, we still need to know how to calculate the eigen-value and eigen-vectors.

Note that the second line of REV and SVD, and both line of HEV, that is a **generalized eigen-value problem (GEV)**.

3.5.4 Eigen solver

The eigen solver is implemented following Ref. [19].

There are many strategies. The most common algorithm is to transform a matrix to a **Hessenberg matrix**.

- Householder reflection

Tested that householder reduction is faster than symmetric or unsymmetric Lanczos method when the matrix is large. On the other hand, for a Hermitian matrix, Householder can also produce Hermitian tri-diagonal matrix.

Note that this might be not true when the matrix is huge, and Hessenberg reduction is not a full reduction. In our case, we concentrate on matrix with $5 < m < 50$. Tested when about $7 < m < 30$ ($30 \times 30 \approx 1024$ is the maximum thread count on test machine), Householder is faster.

Also, as tested, the quality of QR factorization affects the QR iteration very much. At the same time, compared with QR iteration, the QR factorization is relatively cheap, so we also use Householder to do the QR factorization.

The householder reduction is to insert zeros into a vector, which can be briefly written as

$$\mathbf{v} = \begin{pmatrix} x_1 + e^{i \arg x_1} |\mathbf{x}| \\ x_2 \\ \dots \\ x_n \end{pmatrix}, \quad U = \mathbb{I} - \frac{2\mathbf{v}\mathbf{v}^\dagger}{\mathbf{v}^\dagger\mathbf{v}}, \quad U \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} = \begin{pmatrix} \frac{|x_1|}{x_1} |\mathbf{x}| \\ 0 \\ \dots \\ 0 \end{pmatrix}, \quad (299)$$

Note that U is at the same time unitary and Hermitian. Since it is unitary, it can be used as QR factorization, $A = QR$ where Q is unitary and R is upper triangular, and to transform a matrix to Henssenberg matrix $A = U^\dagger H U$, where U is unitary and H is upper Henssenberg matrix.

The algorithm is not listed, the procedure can be written as

$$\begin{aligned} A_0 &= U_0^\dagger U_0 A_0 = U_0^\dagger A_1, \quad U_0 A_0 = \left(\mathbb{I} - \frac{2\mathbf{v}\mathbf{v}^\dagger}{\mathbf{v}^\dagger\mathbf{v}} \right) A_0 = A_1 = \begin{pmatrix} + & + & + & + \\ 0 & + & + & + \\ 0 & + & + & + \\ 0 & + & + & + \end{pmatrix} \\ A_0 &= U_0^\dagger U_1^\dagger U_1 A_1, \quad U_1 A_1 = \begin{pmatrix} \mathbb{I}_1 & 0 \\ 0 & \mathbb{I} - \frac{2\mathbf{v}\mathbf{v}^\dagger}{\mathbf{v}^\dagger\mathbf{v}} \end{pmatrix} A_1 = A_2 = \begin{pmatrix} + & + & + & + \\ 0 & + & + & + \\ 0 & 0 & + & + \\ 0 & 0 & + & + \end{pmatrix} \\ A_0 &= U_0^\dagger U_1^\dagger U_2^\dagger R, \quad R = U_2 A_2 = \begin{pmatrix} \mathbb{I}_2 & 0 \\ 0 & \mathbb{I} - \frac{2\mathbf{v}\mathbf{v}^\dagger}{\mathbf{v}^\dagger\mathbf{v}} \end{pmatrix} A_2 = R = \begin{pmatrix} + & + & + & + \\ 0 & + & + & + \\ 0 & 0 & + & + \\ 0 & 0 & 0 & + \end{pmatrix} \end{aligned} \quad (300)$$

Similarly, note that if only insert zeroes from the second row

$$U A = \begin{pmatrix} \mathbb{I}_1 & 0 \\ 0 & \mathbb{I} - \frac{2\mathbf{v}\mathbf{v}^\dagger}{\mathbf{v}^\dagger\mathbf{v}} \end{pmatrix} A = \begin{pmatrix} + & + & + & + \\ + & + & + & + \\ 0 & + & + & + \\ 0 & + & + & + \end{pmatrix} \quad (301)$$

then

$$U A U^\dagger = U A \begin{pmatrix} \mathbb{I}_1 & 0 \\ 0 & @ \end{pmatrix} = \begin{pmatrix} + & + & + & + \\ + & + & + & + \\ 0 & + & + & + \\ 0 & + & + & + \end{pmatrix} \begin{pmatrix} \mathbb{I}_1 & 0 \\ 0 & @ \end{pmatrix} = \begin{pmatrix} + & +@ & +@ & +@ \\ + & +@ & +@ & +@ \\ 0 & +@ & +@ & +@ \\ 0 & +@ & +@ & +@ \end{pmatrix} \quad (302)$$

So that it is kept Henssenberg.

- Shifted QR iteration

Let $A = U_0^\dagger H_0 U_0$ where H is a Henssenberg matrix, then, let $H_0 = U_1 R$, it can be shown that $H_1 = R U_1 = U_1^\dagger (U_1 R) U_1$ is still a Henssenberg.

Also, $H = U_1 H_1 U_1^\dagger$, so $A = (U_0^\dagger U_1) H_1 (U_1^\dagger U_0)$.

So, H_1 has same eigen-value as A .

Apart from that, H_i can approach a upper triangular matrix. It is noted that, if the QR factorization is performed to a shifted matrix $H - \sigma I$, where σ is an approximate eigen-value of H , it will converge much fast.

In CLGLib, we use Wilkinson shift, which is the eigen-value of the right-bottom 2×2 irreducible matrix, and is the one closer to the right-bottom corner element. It can be written as

Algorithm 10 Shifted QR Iteration

```

for  $H$  is not a triangular do
   $\sigma$  be the eigen-value of the  $2 \times 2$  matrix of right-bottom matrix which is closer to the right bottom element.
   $QR = H - \sigma I$ 
   $H' = RQ + \sigma I$ 
  if  $H_{n-1,n} \approx 0$  then
    Reduce to a  $n - 1$  Henssenberg matrix problem.
  end if
end for

```

Once the upper triangular matrix is obtained, the eigen-values are just the diagonal elements.

- Implicit shifted QR iteration

The **Implicit shifted QR iteration** sometimes also called **Double shifted QR iteration** or **Double shifted QR iteration**.

The details are not listed here, it uses Householder to chase the zero to the bottom and right, it is a little bit better convergent, and is said to be more stable. It can be written as

Algorithm 11 Implicit shifted QR iteration

```

for  $T$  a Hessenberg matrix with  $n \geq 3$ . (In the case of  $n = 2$ , the eigen-value can be directly obtained.) do
   $H$  a irreducible Hessenberg matrix with  $n \geq 3$ .  $T = \begin{pmatrix} + & + & + \\ 0 & H & + \\ 0 & 0 & + \end{pmatrix}$   $\triangleright$  In the case of  $n = 2$ , the eigen-value can be directly
  obtained.
   $H_{2 \times 2}$  be the  $2 \times 2$  matrix of right-bottom matrix.  $s = \text{tr}(H_{2 \times 2})$  and  $t = \det(H_{2 \times 2})$ 
   $x = H_{1,1}(H_{1,1} - s) + H_{1,2}H_{2,1} + t$ 
   $y = H_{2,1}(H_{1,1} + H_{2,2} - s)$ 
   $z = H_{2,1}H_{3,2}$ 
   $H' = RQ + \sigma I$ 
  for  $k = 0$  to  $n - 3$  do
     $h$  be Householder matrix to zero  $\mathbf{v} = (x, y, z)^T \rightarrow (|\mathbf{v}|, 0, 0)^T$ .
     $q = \max(1, k)$ ,  $H(k+1 : k+3, q : n) = hH(k+1 : k+3, q : n)$ .
     $r = \min(k+4, n)$ ,  $H(1 : 4, k+1 : k+3) = H(1 : 4, k+1 : k+3)h^\dagger$ .  $\triangleright$  Note that  $h^\dagger = h$ 
     $x = H(k+2, k+1)$ ,  $y = H(k+3, k+1)$ 
    if  $k < n - 3$  then
       $z = H(k+4, k+1)$ 
    end if
  end for
   $h$  be Householder matrix to zero  $\mathbf{v} = (x, y)^T \rightarrow (|\mathbf{v}|, 0)^T$ .
   $H(n-1 : n, n-2 : n) = hH(n-1 : n, n-2 : n)$ ,  $H(1 : n, n-1 : n) = H(1 : n, n-1 : n)h^\dagger$ .
end for

```

- Inverse power iteration

Once the eigen-values are obtained, one can calculate the approximate eigen-vector correspond the the eigen-value using inverse power iteration. The inverse power iteration performs well with the original matrix A .

Algorithm 12 Inverse power Iteration

$\mathbf{v}$ is a normalized vector.
for $\|(A - \sigma I)\mathbf{v}\| > \epsilon$ **do**
 $QR = (A - \sigma I)$
 $\mathbf{v} = R^{-1}Q^\dagger \mathbf{v}$
 $\mathbf{v} = \mathbf{v}/\|\mathbf{v}\|$
end for

The R is upper triangular, so R^{-1} is just a modification of Algorithm. 5.

Algorithm 13 Backward substitution

for $i = k - 1$ to 0 **do**
 for $j = i + 1$ to $k - 1$ **do**
 $\mathbf{y}[i] -= r[i, j]\mathbf{y}[j]$
 end for
 $\mathbf{y}[i] = \mathbf{y}[i]/r[i, i]$
end for
return $\mathbf{u}[k] = \mathbf{y}[k]$.

- Eigen vector of upper triangular matrix

The inverse power iteration is incompatible with upper triangular matrix, because $R - \lambda I$ is singular, for the inverse power iteration, $R - \lambda I$ is only nearly singular, however, for a upper triangular, it is almost exactly a singular. Although one can shift the eigen value a little bit, but one can also obtain eigen vector exactly. by the procedure below.

Suppose

$$\begin{pmatrix} r_{1,1} - \lambda_k & \dots & r_{1,k-1} \\ 0 & \dots & \dots \\ 0 & 0 & r_{k-1,k-1} - \lambda_k \end{pmatrix} \begin{pmatrix} x_1 \\ \dots \\ x_{k-1} \end{pmatrix} = \begin{pmatrix} y_1 \\ \dots \\ y_{k-1} \end{pmatrix} \quad (303)$$

$(R - \lambda_k \mathbb{I})\mathbf{x} = 0$ can be written as

$$\begin{pmatrix} r_{1,1} - \lambda_k & \dots & r_{1,k-1} & r_{1,k} & \dots \\ 0 & \dots & \dots & \dots & \dots \\ 0 & 0 & r_{k-1,k-1} - \lambda_k & r_{k-1,k} & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \end{pmatrix} \begin{pmatrix} x_1 \\ \dots \\ x_{k-1} \\ 1 \\ 0 \\ \dots \end{pmatrix} = \begin{pmatrix} y_1 + r_{1,k} \\ \dots \\ y_{k-1} + r_{k-1,k} \\ 0 \\ \dots \end{pmatrix} = 0 \quad (304)$$

leads to the equation

$$\begin{pmatrix} r_{1,1} - \lambda_k & \dots & r_{1,k-1} \\ 0 & \dots & \dots \\ 0 & 0 & r_{k-1,k-1} - \lambda_k \end{pmatrix} \begin{pmatrix} x_1 \\ \dots \\ x_{k-1} \end{pmatrix} = \begin{pmatrix} -r_{1,k} \\ \dots \\ -r_{k-1,k} \end{pmatrix} \quad (305)$$

which can be solved using backward shift, i.e. Algorithm. 13.

- Generalized eigen-value problem

The generalized eigen-value problem can be transformed to a eigen-value problem

$$A\mathbf{v} = \lambda B\mathbf{v} \Rightarrow B = QR \Rightarrow R^{-1}Q^\dagger A\mathbf{v} = \lambda \mathbf{v} \quad (306)$$

3.5.5 Implementation of GCRO-DR

Now, we concentrate on the implementation of GCRO-DR. First of all, we need to know how to apply $\mathbf{x} - AB^\dagger \mathbf{v}$, where $A, B \in \mathbb{C}^{n \times k}$ and $\mathbf{v} \in \mathbb{C}^n$.

Algorithm 14 $\mathbf{x} = \mathbf{x} - AB^\dagger \mathbf{v}$

```

for  $i = 0$  to  $k - 1$  do
   $\mathbf{x} = \mathbf{x} - (\mathbf{b}_k^\dagger \mathbf{x}) \mathbf{a}_k$ 
end for

return  $\mathbf{x}$ 

```

The second thing is QR decompose of $\mathbb{C}^{n \times k}$ and $\mathbb{C}^{(m+1) \times k}$ matrix. For the $\mathbb{C}^{n \times k}$ matrix, the usually Arnoldi with modified Gram-Schmidt, i.e. Algorithm. 3 can be used.

Algorithm 15 modified Gram-Schmidt for QR factorization decompose of $A\tilde{Y}_k$

```

for  $i = 0$  to  $k - 1$  do
   $y_i = Ay_i$ 
end for
 $\mathbf{v}^{(0)} = y_0 / \|\mathbf{y}_0\|$ 
for  $i = 0$  to  $k - 1$  do
   $\mathbf{w} = \mathbf{y}_{i+1}$ 
  for  $j = i + 1$  to  $k - 1$  do
     $c = \mathbf{v}^{(j)*} \cdot \mathbf{w}$ 
     $\mathbf{w} = \mathbf{w} - c\mathbf{v}^{(j)}$ 
     $r[j, i] = c$ 
  end for
   $\mathbf{v}^{(i+1)} = \mathbf{w} / r[i + 1, i + 1]$ 
end for

return  $Q = (\mathbf{v}_0, \dots, \mathbf{v}_{k-1}), R = r[i, j]$ .

```

Finally we have to calculate YR^{-1} . This is a forward substitution.

$$U = YR^{-1}, \quad UR = Y, \quad R^T U^T = Y^T \quad U^T = (R^T)^{-1} Y^T. \quad (307)$$

3.5.6 Implement of GCRO-DR

We present pseudo-code of GCRO-DR can be found in Ref. [18]. The only difference is that we always make sure C_k and V_{m-k+1} are orthogonal to each other. It can be written as

Algorithm 16 GCRO-DR

```

if  $U_k$  is defined from solving a previous linear system then
    Let  $[Q, R] = AU_k$  be QR decomposition of  $AU_k$ .
     $C_k = Q$ .
     $U_k = U_k R^{-1}$ .
     $\mathbf{r}^{(0)} = A\mathbf{x}^{(0)} - \mathbf{b}$ 
else
    Perform GMRES to get  $W_{m+1} = (C_k, V_{m-k+1})$ 
    Update  $\mathbf{x}^{(0)}, \mathbf{r}^{(0)}$  as  $\mathbf{x}^{(0)} = \mathbf{x}^{(0)} + V_m y, \mathbf{r}^{(0)} = V_{m+1}(\beta \mathbf{e}_1 - H_{m+1} y)$ , which is in fact part of GMRES.
    Compute eigen-vector problem and obtain  $P_k \in \mathbb{C}^{m \times k}$ .
     $U_k = V_m P_k$ 
    Let  $[Q, R] = H_{m+1} P_k$  be QR decomposition.
     $C_k = V_{m+1} Q$ 
     $U_k = U_k R^{-1}$ 
end if
for  $\hat{i} = 1$  to  $r$  do ▷ restart  $r$  times.
     $\mathbf{x}^{(i-1)} = \mathbf{x}^{(i-1)} + U_k C_k^\dagger \mathbf{r}^{(i-1)}$ 
     $\mathbf{r}^{(i-1)} = \mathbf{r}^{(i-1)} - C_k C_k^\dagger \mathbf{r}^{(i-1)}$ 
    Reset  $H_{m+1} = 0$ .  $W_{m+1}(k) = V_{m-k+1}(0) = \mathbf{r}^{(i)} / \|\mathbf{r}^{(i)}\|$ .
     $H_{m+1}(k, k) = 1 / \|U_k\|$ , normalize  $U_k$ .
    Perform Arnoldi procedure on matrix  $(1 - CC^\dagger)A$ , to obtain  $V_{m-k+1}$ , and set  $H_{k:m+1, k:m}$ . And  $H_{0:k, m} = C_k^\dagger A V_{m-k}$ .
     $\hat{V} = (U_k, V_{m-k})$  and  $W_{m+1} = (C_k, V_{m-k+1})$ .
    Solve  $\arg \min \|r\| \|\mathbf{e}_k - H_{m+1} y\|$ .
     $\mathbf{x}^{(i)} = \mathbf{x}^{(i-1)} + \hat{V}_m y, \mathbf{r}^{(i)} = \mathbf{r}^{(i-1)} - W_{m+1} H_{m+1} y$ . ▷ Check the error here. If reach the criterion, return.
    Compute eigen-vector problem and obtain  $P_k \in \mathbb{C}^{m \times k}$ .
     $U_k = V_m P_k$ 
    Let  $[Q, R] = H_{m+1} P_k$  be QR decomposition.
     $C_k = V_{m+1} Q$ 
     $U_k = U_k R^{-1}$ 
end for

```

3.5.7 Implement of GMRES-MDR

The GMRES-MDR is almost the same as GCRO-DR, except for 3 things.

1. It set a threshold on eigen-values to decrease k if a larger k is not necessary.
2. It check the speed of convergence to switch between REV and SVD.
3. At first iteration, if U_k is defined, it use another algorithm to obtain U_k and C_k .

Algorithm 17 First iteration of GMRES-MDR if U_k is defined.

```

 $[Q, R] = U_k$ 
if REV then
     $Q^\dagger A Q \omega = \theta \omega$ 
end if
if HEV then
     $Q^\dagger A^\dagger A Q \omega = \theta Q^\dagger A^\dagger Q \omega$ 
end if
if SVD then
     $Q^\dagger A^\dagger A Q \omega = \theta^2 \omega$ 
end if
 $U_k = Q \omega_k$ 

```

We only implement the third because the first two can be tunable by parameters.

3.5.8 Test of GCRO-DR and GMRES-MDR

It is tested that, for both GCRO-DR and GMRES-MDR are suitable for the low-mode case.

We run with unitary gauge and $\kappa = 0.1249$. ($\kappa_c = 0.125$). GCRO-DR with $m = 16$ and $k = 4$ will run even faster than $dim = 50$ GMRES.

However, if it is not the low-mode case, GMRES is faster.

3.6 Even-odd preconditioner

The equation for the D operator can be written as

$$\begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} \begin{pmatrix} z_e \\ z_o \end{pmatrix} = \begin{pmatrix} \phi_e \\ \phi_o \end{pmatrix} \quad (308)$$

Where D_{oe} means $D_{o \leftarrow e}$. Often, D_{oo}^{-1} and D_{ee}^{-1} is very easy to calculate. In the case of Wilson-Dirac fermion without $\mathcal{O}(a)$ improvement, it is $\mathbb{I}$.

It can be divided into 3 step to solve this equation

1. Calculate $\tilde{\phi}_e = \phi_e - D_{eo}D_{oo}^{-1}\phi_o$.
2. Solve $(D_{ee} - D_{eo}D_{oo}^{-1}D_{oe})z_e = \tilde{\phi}_e$.
3. $z_o = D_{oo}^{-1}(\phi_o - D_{oe}z_e)$.

This is because (Note the order of D 's should not change.)

$$\begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} \begin{pmatrix} z_e \\ z_o \end{pmatrix} = \begin{pmatrix} 1 & D_{eo}D_{oo}^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} D_{ee} - D_{eo}D_{oo}^{-1}D_{oe} & 0 \\ D_{oe} & D_{oo} \end{pmatrix} \begin{pmatrix} z_e \\ z_o \end{pmatrix} = \begin{pmatrix} \phi_e \\ \phi_o \end{pmatrix} \quad (309)$$

Using

$$\begin{pmatrix} 1 & -D_{eo}D_{oo}^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & D_{eo}D_{oo}^{-1} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (310)$$

So that

$$\begin{pmatrix} D_{ee} - D_{eo}D_{oo}^{-1}D_{oe} & 0 \\ D_{oe} & D_{oo} \end{pmatrix} \begin{pmatrix} z_e \\ z_o \end{pmatrix} = \begin{pmatrix} 1 & -D_{eo}D_{oo}^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \phi_e \\ \phi_o \end{pmatrix} = \begin{pmatrix} \tilde{\phi}_e \\ \phi_o \end{pmatrix} \quad (311)$$

Which leads to two equations

$$\begin{aligned} (D_{ee} - D_{eo}D_{oo}^{-1}D_{oe})z_e &= \tilde{\phi}_e, \\ D_{oe}z_e + D_{oo}z_o &= \phi_o, \end{aligned} \quad (312)$$

which leads to step 1,2,3.

Because the D matrix is sparse, For n sites, $D\phi$ is not $O(n^2)$ but $O((c+1)n)$ ($c = 2D$ for neighbour sites), so in the even-odd preconditioner, D_{eo} is not $O(n^2/4)$ but $O(cn/2)$. Solving $(D_{ee} - D_{eo}D_{oo}^{-1}D_{oe})z_e = \tilde{\phi}_e$, one need to calculate D_{eo} and D_{oe} , so it is not $O(n^2/2)$ but $O(cn)$, as a result the effect of even-odd preconditioner is limited.

There is one more problem: for HMC, one need to solve mainly $(DD^\dagger)^{-1}$. Which can not be even-odd decomposed, so one need to at first solve D^{-1} then $(D^\dagger)^{-1}$, the solution in the last step cannot be set as an initial guess.

Note, for some reason I do not know, the GCRO-DR and GMRES-MDR not yet support even-odd preconditioner.

Note, for periodic boundary condition, and odd extent lattice, the $D_{ee} \neq \mathbb{I}$ and $D_{oo} \neq \mathbb{I}$. For example, for $4 \times 4 \times 3 \times 3$, $(0,0,0,1)$ is connected with $(0,0,2,1)$. Note that, D_{oo} is neither a diagonal matrix, so D_{oo}^{-1} need to be solved, which makes the even-odd inefficient, so we do NOT support such case.

3.7 The multi-shift solver

The multi-shift solver are used to solve $(A + a_n)x = b$ for different a_n at once.

Note that, it is only possible when A is on the numerator, for example, when using Arnoldi or Lanczos algorithm.

3.7.1 The multi-shift version of GMRES

We still use Algorithm 3 to build $\mathbf{v}_i$, such that

$$\mathbf{v}_i^* \cdot \mathbf{v}_j = \delta_{ij}, \quad A\mathbf{v}_{i-1} = \sum_{j=0}^i h[j, i-1] \mathbf{v}_j, \quad (313)$$

holds.

However, now we want to solve

$$((A + a_n)v)_k = v_{k+1}H \quad (314)$$

For that to happen, we need

$$A\mathbf{v}_{i-1} = \sum_{j=0}^i h[j, i-1] \mathbf{v}_j \rightarrow (A + a_n)\mathbf{v}_{i-1} = \sum_{j=0}^i (h[j, i-1] + a_n \delta_{j,i-1}) \mathbf{v}_j \quad (315)$$

Note that the sum over j will contain $\mathbf{v}_{i-1}$, so simply use $\hat{H} = h[j, i-1] + a_n \delta_{j,i-1}$ instead of $H = h[j, i-1]$ one will arrive at Eq. 314.

When considering the restart, one should assume the residue of the shifted system is collinear to the seed system, $\hat{r}_n = \beta_n r$ which can be guaranteed when $x_0 = 0$ and for the first iteration, $\hat{r}_n^{k=0} = r = b$, where k is index of restart, r is the residue of seed system.

For the second iteration (after the first restart), to make sure again $\hat{r}_n = \beta_n r$, one need to solve

$$\begin{aligned} \beta &= |\mathbf{r}^{k-1}| \\ \mathbf{z}_{m+1} &= \beta \mathbf{e}_1 - H_{(m+1) \times m} \cdot \mathbf{y}_m \\ \left(\hat{H}_{(m+1) \times m} \mathbf{z}_{m+1} \right)_{(m+1) \times (m+1)} \cdot \begin{pmatrix} \hat{y}_m \\ \beta_n^k \end{pmatrix} &= \beta_n^{k-1} \beta \mathbf{e}_1 \end{aligned} \quad (316)$$

where k is the index of restart, $\mathbf{r}$ is the residue of the seed system (zero shift system), $\hat{y}_m$ is the solution of $\argmin(\beta^{k-1} \mathbf{e}_1 - H y_m)$ (GMRES of seed system).

Then, $x_n = x_n + \sum_j \hat{y}_j v_j$.

It can be summarized as [20]. Let the seed system be $Ax = b$, than

Algorithm 18 shifted GMRES

$\mathbf{x} = 0, \mathbf{x}_n = 0, \mathbf{r} = \mathbf{b}, \beta_n = 1$

for $i = 0$ to $k - 1$ **do**

$\mathbf{r} = \mathbf{b} - A\mathbf{x}, \beta = |\mathbf{r}|$

 Solve $A\mathbf{x} = \mathbf{b}$, obtain $\mathbf{v}_m, H_{m+1,m}$ and y_m

 Update $\mathbf{x}$ use $\mathbf{v}_m$ and y_m as $\mathbf{x} = \mathbf{x} + \sum \mathbf{v}_j y_j$

$\mathbf{z}_{m+1} = \beta \mathbf{e}_1 - H_{m+1,m} y_m$

for $n = 0$ to n **do**

 The left $(m+1) \times m$ of $\hat{H}_{m+1,m+1}$ is $H_{m+1,m} + a_n \mathbb{I}_{m,m}$

 The right column of $\hat{H}_{m+1,m+1}$ is $\mathbf{z}_{m+1}$

 Solve $\hat{H}_{m+1,m+1} \hat{y}_{m+1} = \beta_n \beta \mathbf{e}_1$

 ▷ use QR factorization or Givens rotation

 Update $\beta_n = \hat{y}_{m+1}[m+1]$

 ▷ β_n is complex number

 Update $\mathbf{x}_n$ use $\mathbf{v}_m$ and first m element of $\hat{y}_{m+1}$ as $\mathbf{x}_n = \mathbf{x}_n + \sum \mathbf{v}_j \hat{y}_j$

end for

end for

3.7.2 The multi-shift FOM

FOM is very like GMRES, but [for the shifted system, FOM is tested to be much faster](#). It can be summarized as [\[21\]](#)

Algorithm 19 shifted FOM

```

 $\mathbf{x}_n = 0, \mathbf{r}^{(k=0)} = \mathbf{b}, \beta_n = |\mathbf{b}|$ 
for  $i = 0$  to  $k - 1$  do
  Solve Krylov space for  $A\mathbf{x} = \mathbf{b}$ , obtain  $\mathbf{v}_m, \mathbf{v}_{m+1}$  and  $H_{m,m+1}$  ▷ Not solve  $\beta \mathbf{e}_1 - Hy$ 
   $\mathbf{r}^{(k)} = \mathbf{v}_{m+1}$ 
  for  $n = 0$  to  $n$  do
    if  $|\beta_n| > \epsilon$  then ▷  $\beta_n$  is complex number
      Solve  $(H_{m,m} + a_n \mathbb{I}_m) y_m = \beta_n \mathbf{e}_1$  ▷ use Givens rotation
       $\beta_n = y_m[m] - h[m, m+1]$ 
      Update  $\mathbf{x}_n$  use  $\mathbf{v}_m$  and  $y_m$  as  $\mathbf{x}_n = \mathbf{x}_n + \sum \mathbf{v}_j y_j$ 
    end if
  end for
end for

```

where $H_{m,m}$ is the upper $m \times m$ square matrix of $H_{m,m+1}$, and $h[m, m+1]$ is the last element of $H_{m,m+1}$.

3.7.3 The multi-shift BiCGStab

The speed of BiCGStab-m is similar as multi-shift GMRES. However, for a fixed number of shift, the memory is fixed (the accuracy is independent of memory)

The algorithm is from Ref. [\[22\]](#), which can be written as

Algorithm 20 multi-shift BiCGStab

$\mathbf{x}^{(n)} = 0, \mathbf{s} = \mathbf{s}^{(n)} = \mathbf{r} = \mathbf{b}, \mathbf{w}_0 = \mathbf{w} = \mathbf{r}^\dagger, \mathbf{s}_A = A\mathbf{s}$
 $\beta = \alpha = 0, \delta = \mathbf{w}_0^\dagger \mathbf{r}, \phi = \frac{\mathbf{w}_0^\dagger \mathbf{s}_A}{\delta} \quad \triangleright \beta, \alpha, \delta, \phi \in \mathbb{C}$
 $\beta^{(n)} = \zeta^{(n)} = \zeta_p^{(n)} = \rho^{(n)} = \chi^{(n)} = s^{(n)} = 1 \quad \triangleright \beta^{(n)}, \zeta^{(n)}, \zeta_p^{(n)}, \rho^{(n)}, \chi^{(n)} \in \mathbb{C}, s^{(n)} \in \mathbb{R}$
for $i = 0$ to m **do**
 $\beta_1 = -\frac{1}{\phi}$
 for $n = 0$ to n with $s^{(n)} > \epsilon$ **do**
 if $i = 0$ **then**
 $\zeta^{(n)} = \frac{1}{1 - a_n \beta_1}$
 $\beta^{(n)} = \zeta^{(n)} \beta_1$
 else
 $\zeta_p = \zeta_p^{(n)}, \zeta_p^{(n)} = \zeta^{(n)}, d = \zeta_p \beta$
 $\zeta^{(n)} = \frac{d \zeta_p^{(n)}}{\alpha \beta_1 (\zeta_p - \zeta_p^{(n)}) + d(1 - a_n \beta_1)}$
 $\beta^{(n)} = \frac{\beta_1 \zeta^{(n)}}{\zeta_p^{(n)}}$
 end if
 end for
 $\beta = \beta_1$
 $\mathbf{w} = \mathbf{r} - \beta \mathbf{s}_A, \mathbf{w}_A = A\mathbf{w}$
 $\chi = \frac{\mathbf{w}_A^\dagger \mathbf{w}}{\mathbf{w}_A^\dagger \mathbf{w}_A}$
 for $n = 0$ to n with $s^{(n)} > \epsilon$ **do**
 $d = \frac{1}{1 + a_n \chi}, \chi^{(n)} = \frac{\chi}{d}$
 $d_1 = \chi^{(n)} \rho^{(n)}, d_2 = d_1 \zeta^{(n)}$
 $\mathbf{x}^{(n)} = \mathbf{x}^{(n)} - \beta^{(n)} \mathbf{s}^{(n)} + d_2 \mathbf{w}$
 $\mathbf{s}^{(n)} = \mathbf{s}^{(n)} - \frac{d_2}{\beta^{(n)}} \mathbf{w} + \frac{d_1 \zeta_p^{(n)}}{\beta^{(n)}} \mathbf{r}$
 $\rho^{(n)} = \frac{\rho^{(n)}}{d}$
 end for
 $\mathbf{r} = \mathbf{w} - \chi \mathbf{w}_A$
 $\delta_1 = \mathbf{w}_0^\dagger \mathbf{r}, \alpha = -\frac{\beta \delta_1}{\delta \chi}, \delta = \delta_1$
 for $n = 0$ to n with $s^{(n)} > \epsilon$ **do**
 $\alpha^{(n)} = \frac{\alpha \beta^{(n)} \zeta^{(n)}}{\beta \zeta_p^{(n)}}$
 $\mathbf{s}^{(n)} = \zeta^{(n)} \rho^{(n)} \mathbf{r} + \alpha^{(n)} \mathbf{s}^{(n)}$
 $s^{(n)} = |\mathbf{s}^{(n)}|$
 end for
 $\mathbf{s} = \mathbf{r} + \alpha(\mathbf{s} - \chi \mathbf{s}_A) \quad \triangleright \text{If } |\mathbf{s}| \text{ is small, quit.}$
 $\mathbf{s}_A = A\mathbf{s}$
 $\phi = \frac{\mathbf{w}_0^\dagger \mathbf{s}_A}{\delta}$
end for

It is tested that multi-shift FOM is faster than BiCGStab and GMRES. However, the memory usage of FOM and GMRES grow with m , the dimension of Krylov space, and grow as n vectors, the number of shifted a_n as $m + n$. The memory of BiCGStab does not grow with accuracy but grow with $2n$ vectors.

4 Miscellaneous topics

4.1 Gauge Fixing

4.1.1 Introduction of FFT before start

A brief introduction of FFT.

FFT is to calculate Discrete Fourier Transform (DFT), in 1D, it is

$$\begin{aligned}\tilde{x}_m &= \sum_n x_n W_N^{mn} \\ W_N^j &\equiv \exp(-i \frac{2\pi j}{N})\end{aligned}\tag{317}$$

- Cooley-Tukey mapping

Let $N = N_1 \times N_2$, we first calculate DFT of subset $I_{n_1} = \{n_2 N_1 + n_1\}$, such that

$$\tilde{x}_m = \sum_{n_1} S_{n_1}, \quad S_{n_1} = \sum_{n_2} x_{n_2 N_1 + n_1} W_N^{m(n_2 N_1 + n_1)}\tag{318}$$

Note that, S_{n_1} can be further factorized as

$$\tilde{x}_m = \sum_{n_1} W_N^{mn_1} S'_{n_1}, \quad S'_{n_1} = \sum_{n_2} x_{n_2 N_1 + n_1} W_N^{mn_2 N_1}\tag{319}$$

then, note that, N can be divided by N_1 (the result is N_2), so $W_N^{mn_2 N_1} = W_{N_2}^{mn_2}$, so S' is just DFT of subset I_{n_1} . Then, we can also decompose

$$m = m_1 N_2 + m_2\tag{320}$$

to write

$$\tilde{x}_{m_1 N_2 + m_2} = \sum_{n_1} W_N^{n_1(m_1 N_2 + m_2)} S'_{n_1} = \sum_{n_1} W_{N_1}^{n_1 m_1} W_N^{n_1 m_2} S'_{n_1}\tag{321}$$

The $W_N^{n_1 m_2}$ is twiddle factor, after ‘twiddle’, $S''_{n_1} = W_N^{n_1 m_2} S'_{n_1}$, the result is again a DFT with size N_1

$$\tilde{x}_{m_1 N_2 + m_2} = \sum_{n_1} W_{N_1}^{n_1 m_1} S''_{n_1}\tag{322}$$

The FFT is implemented in cuFFT, [We may use a batched 3D cuFFT and a batched 1D cuFFT to implement 4D FFT.](#)

4.1.2 Cornell Gauge Fixing and FFT accelerated

The Cornell Gauge Fixing is the steepest descend gauge fixing. The **Landau Gauge** for example. The Landau gauge needs $\partial_\mu A_\mu = 0$. One finds that if

$$F(A) = \sum_n \text{tr} [A_\mu^2(n)]\tag{323}$$

is minimized, which means $\partial_\mu F(A) = 0$, and leads to $\partial_\mu A_\mu = 0$. In other words, the Landau gauge fixing is to find the minimum of $F(A)$ (using steepest descend method).

The steepest descend method can be simply described as

$$x_{n+1} \rightarrow x_n - \alpha \left. \frac{df(x)}{dx} \right|_{x=x_n}\tag{324}$$

where x is a vector, and x_n means iteration for n-times, α is a tunable parameter.

Using the **Cornell gauge fixing**, we follow Ref. [\[23\]](#)

The Cornell gauge fixing is a steepest descend algorithm, which can be described as

Algorithm 21 Cornell gauge fixing

```

for  $i = 0$  to max iteration do
   $A_\mu(n) = U_\mu(n).TA()$ 
   $\Gamma(n) = \sum_\mu (A_\mu(n - \mu) - A_\mu(n))$ 
  if  $\sum_n \Gamma(n)\Gamma^\dagger(n) < \epsilon$  then return ▷ Succeed.
  end if
   $G(n) = \exp(-\alpha_0 \Gamma(n))$ 
   $U_\mu(n) = G(n)U_\mu(n)G^\dagger(n + \mu)$ 
end for

```

Note:

- TA means traceless anti-Hermitian.
- For a traceless anti-Hermitian matrix, $M^\dagger M = 2(|m_{11} + m_{22}|^2 + |m_{12}|^2 + |m_{13}|^2 + |m_{23}|^2)$, where m_{11} and m_{22} are pure imaginary numbers.
- For a traceless anti-Hermitian matrix, $\exp(M)$ can be calculated as Appendix. A of Ref. [8].
- α_0 is a tunable parameter usually set to $0.05 - 0.1$.

The Cornell gauge fixing can be Fourier accelerated. At first, prepare the table such that

$$f_p(n) = \begin{cases} \frac{4N_d}{2V(N_d - \sum_\mu \cos(\frac{2\pi n_\mu}{L_\mu}))}, & N_d \neq \sum_\mu \cos(\frac{2\pi n_\mu}{L_\mu}); \\ \frac{4N_d}{V}, & N_d = \sum_\mu \cos(\frac{2\pi n_\mu}{L_\mu}). \end{cases} \quad (325)$$

where $N_d = 4$ is the number of dimension (Note, for Coulomb gauge, it is not 4), and V is the volume of the FFT transform (just the volume of the lattice, for the case of Coulomb gauge, it is the spatial volume.). L_μ is the extend of the direction.

Then, the step to generate gauge transform is modified by insert a FFT and an inverse FFT such that

$$G(n) = \exp(-\alpha_0 \Gamma(n)) \rightarrow G(n) = \exp(-\alpha_0 \hat{F} f_p(n) F \Gamma(n)) \quad (326)$$

where the FFT of a matrix is the FFT of each matrix element.

4.1.3 Los Alamos Gauge Fixing and over relaxation

Using the **Los Alamos gauge fixing**, we follow Ref. [24]

The idea is to maximize $F(U) = \sum_n \sum_\mu \text{ReTr}[U_\mu(n)]$.

By rewrite

$$F(U) = \sum_n \sum_\mu \text{ReTr}[U_\mu(n)] = \frac{1}{2} \sum_n \sum_\mu \text{ReTr}[U_\mu(n) + U_\mu^\dagger(n - \mu)] = \frac{1}{2} \sum_n \text{ReTr}[\omega(n)], \quad (327)$$

$$\omega(n) \equiv \sum_\mu (U_\mu(n) + U_\mu^\dagger(n - \mu)).$$

Note that, if for the gauge transform such that only even sites or odd sites are non-unity, the transform of ω is $G(n)\omega(n)$ or $\omega(n)G^\dagger(n)$, and $\text{ReTr}[G(n)\omega(n)] = \text{ReTr}[\omega(n)G^\dagger(n)]$. So we just need to find a $G(n)$ such that $\text{ReTr}[G(n)\omega(n)] \geq \text{ReTr}[\omega(n)]$, which is known as **Cabibbo-Marinari trick**, which is

$$G(n) = ABC$$

$$a_{11} = \frac{1}{\sqrt{|a_{11}|^2 + |a_{12}|^2}}(m_{11}^* + m_{22}), \quad a_{12} = \frac{1}{\sqrt{|a_{11}|^2 + |a_{12}|^2}}(m_{21}^* - m_{12})$$

$$b_{11} = \frac{1}{\sqrt{|b_{11}|^2 + |b_{13}|^2}}(m_{11}^* + m_{33}), \quad b_{13} = \frac{1}{\sqrt{|b_{11}|^2 + |b_{13}|^2}}(m_{31}^* - m_{13})$$

$$c_{22} = \frac{1}{\sqrt{|c_{22}|^2 + |c_{23}|^2}}(m_{22}^* + m_{33}), \quad c_{23} = \frac{1}{\sqrt{|c_{22}|^2 + |c_{23}|^2}}(m_{32}^* - m_{23}) \quad (328)$$

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ -a_{12}^* & a_{11}^* & 0 \\ 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} b_{11} & 0 & b_{13} \\ 0 & 1 & 0 \\ -b_{31}^* & 0 & b_{11}^* \end{pmatrix}, C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{22} & c_{23} \\ 0 & -c_{23}^* & c_{22}^* \end{pmatrix}$$

The Los Alamos gauge fixing with over relaxation ω can be summarized as

Algorithm 22 Los Alamos gauge fixing with over relaxation ω

```

for  $i = 0$  to max iteration do
  if  $\sum_n \Gamma(n) \Gamma^\dagger(n) < \epsilon$  then return                                 $\triangleright$  Succeed.  $\Gamma$  is defined in the above.
  end if
  for all odd sites
     $G(n) = \sum_\mu (U_\mu(n) + U_\mu^\dagger(n - \mu))$ 
     $G(n) = (1 - \omega) \mathbb{I}_{3 \times 3} + \omega G(n)$ 
     $G(n) = \text{CabibboMarinariProjection}(G(n))$ 
    if  $n$  is odd then
       $U_\mu(n) = G(n) U_\mu(n)$ 
    else
       $U_\mu(n) = U_\mu(n) G^\dagger(n + \mu)$ 
    end if
  for all even sites, do the same thing.
end for

```

Note:

- There is no need to check convergence every iteration.
- When $\omega = 1$, there is no over relaxation, $\omega = \frac{2}{1+\frac{3}{L}}$ is often used.

4.1.4 Coulomb Gauge

The Coulomb gauge is very similar to Landau gauge, however, note that the gauge transform can be performed time slice by time slice. Usually, the count of iteration to convergence is different for each time slice.

4.1.5 Logarithm definition

The usual definition of U_μ is $U_\mu = e^{iaA_\mu}$, there for the real definition of A should be $iaA_\mu = \log(U_\mu)$.

To calculate take an example of $U^{\frac{1}{k}}$ first, let

$$A = P^{-1} D P \quad (329)$$

where D is a diagonal matrix, let

$$B = P^{-1} D^{\frac{1}{k}} P, \quad B^k = \left(P^{-1} D^{\frac{1}{k}} P \right)^k = P^{-1} \left(D^{\frac{1}{k}} \right)^k P = P^{-1} D P = A \quad (330)$$

The problem reduce to a eigne system problem.

- Eigenvalue of 3×3 matrix

The eigenvalues of a 3×3 matrix can be obtained by solve the equation

$$A = \det [\alpha \mathbb{I}_3 - A] = 0 = \alpha^3 - \alpha^2 \text{tr} [A] - \alpha \frac{1}{2} (\text{tr} [A^2] - \text{tr}^2 [A]) - \det [A] \quad (331)$$

Using the fact that, if $A = aB + b\mathbb{I}_3$, the eigenvalue of A is $\lambda_A = a\lambda_B + b$, we can make a traceless matrix

$$B = \frac{1}{\sqrt{\frac{\text{tr} [(A - \frac{\text{tr} [A]}{3})^2]}{6}}} \left(A - \frac{\text{tr} [A]}{3} \right) \quad (332)$$

such that

$$\text{tr} [B] = 0, \quad \text{tr} [B^2] = 6 \quad (333)$$

and the eigenvalue equation for B is

$$\lambda_B^3 - 3\lambda_B - \det [B] = 0 \quad (334)$$

which can be solved analytically (for example, by using Mathematica).

- Eigenvector of 3×3 matrix

If we assume there is no degeneracy due to float point precision, then any column of the matrix $(A - \lambda_2 \mathbb{I}_3)(A - \lambda_3 \mathbb{I}_3)$ is an eigenvector correspond to λ_1 , because $(A - \lambda_1 \mathbb{I}_3)(A - \lambda_2 \mathbb{I}_3)(A - \lambda_3 \mathbb{I}_3) = 0$.

- Power, logarithm, and exponential

Similar as the power,

$$\begin{aligned} A &= P^{-1}DP \\ \log(A) &= P \log(D)P^{-1} \\ \exp(A) &= P \exp(D)P^{-1} \end{aligned} \quad (335)$$

where the exponential and logarithm of a diagonal matrix is easy to do.

4.2 Gamma matrix for KS Staggered fermion

4.2.1 Gamma 1234

Defining

$$link_\mu(2h) = \sum_{\mathbf{A}=n \setminus \mu \times 0,1} \eta_\mu(2h + \mathbf{A}) \bar{\chi}(2h + \mathbf{A} + \mu) \chi(2h + \mathbf{A}) + \bar{\chi}(2h + \mathbf{A}) \chi(2h + \mathbf{A} + \mu) \quad (336)$$

as a pair of line **in** one grid of 2^4 hypercube, sum over other directions. Here $n \setminus \mu = (y, z, t)$ for $\mu = x$.

It is easy to verify

$$\begin{aligned} \frac{1}{16} link_\mu(2h) &= \bar{q} \gamma_\mu \otimes I q \\ \frac{1}{16} link_\mu(2h + \mu) &= \frac{1}{2} ((\bar{q}(2h + 2\mu) \gamma_\mu \otimes I q(2h) + \bar{q}(2h + 2\mu) \gamma_5 \otimes \tau_5 \tau_\mu q(2h)) \\ &\quad + (\bar{q}(2h\mu) \gamma_\mu \otimes I q(2h + 2\mu) - \bar{q}(2h) \gamma_5 \otimes \tau_5 \tau_\mu q(2h + 2\mu))) \end{aligned} \quad (337)$$

and

$$\begin{aligned} \frac{1}{16} \left(link_\mu(2h) + \frac{1}{2} link_\mu(2h + \mu) + \frac{1}{2} link_\mu(2h - \mu) \right) \\ = 2\bar{q} \gamma_\mu \otimes I q + a^2 [\bar{q} \gamma_\mu \otimes I \Delta_\mu q]_+ - a [\bar{q} \gamma_5 \otimes \tau_5 \tau_\mu \partial_\mu q]_- \end{aligned} \quad (338)$$

where

$$\begin{aligned} [\hat{O}_1 \bar{q} \Gamma \hat{O}_2 q]_\pm &= \hat{O}_1 \bar{q} \Gamma \hat{O}_2 q \pm \hat{O}_2 \bar{q} \Gamma \hat{O}_1 q \\ \partial_i q &= \frac{q(2h+2) - q(2h-2)}{4a} \quad \Delta_i q = \frac{q(2h+2) + q(2h-2) - 2q(2h)}{4a^2} \end{aligned} \quad (339)$$

After shift the first line, it is

$$\begin{aligned} (2a)^4 \sum_h \bar{q} \gamma_\mu \otimes I q + \mathcal{O}(a) &= a^4 \frac{1}{2} \sum_n \sum_{s=\pm 1} (\eta_\mu(n) \bar{\chi}(n\mu) \chi(n + s\mu)) \\ \mathcal{O}(a) &= \frac{1}{2} (2a)^4 \sum_h (a^2 [\bar{q} \gamma_\mu \otimes I \Delta_\mu q]_+ - a [\bar{q} \gamma_5 \otimes \tau_5 \tau_\mu \partial_\mu q]_-) \end{aligned} \quad (340)$$

4.2.2 Gamma 12, 13, 14, 23, 24

Note, we use $\sigma_{12}^E = \frac{i}{2} (\gamma_1^E \gamma_2^E - \gamma_2^E \gamma_1^E)$ in this subsection.

Defining

$$plane_{ij}(2h) = \frac{1}{16} \sum_{ijkl=0,1} \eta_i(2h + \vec{v}_{ijkl}) \eta_j(2h + \vec{v}_{ijkl}) \bar{\chi}(2h + \vec{v}_{ijkl}) \chi(2h + \vec{1}_{ij} - \vec{v}_{ij} + \vec{v}_{kl}) \quad (341)$$

where (taking $i, j = x, y$ for example) $\vec{v}_{ijkl} = (n_i, n_j, n_k, n_l)$, $\vec{v}_{ij} = (n_i, n_j, 0, 0)$, $\vec{v}_{kl} = (0, 0, n_k, n_l)$, $\vec{1}_{ij} = (1, 1, 0, 0)$. This is the diagonal lines of square i, j , sum over k, l .

Then, for $i < j$, $\delta_{i,j} = \pm 1$

$$\begin{aligned}
 plane_{ij}(2h) &= \bar{q}(2h)i\sigma_{ij}q(2h) \\
 plane_{ij}(2h + \vec{\delta}_i) &= \frac{1}{2}\bar{q}(2h + 2\vec{\delta}_i)(i\sigma_{ij} - \delta_i\gamma_5\gamma_j \otimes \tau_5\tau_i)q(2h) \\
 &\quad + \frac{1}{2}\bar{q}(2h)(i\sigma_{ij} + \delta_i\gamma_5\gamma_j \otimes \tau_5\tau_i)q(2h + 2\vec{\delta}_i) \\
 plane_{ij}(2h + \vec{\delta}_j) &= \frac{1}{2}\bar{q}(2h + 2\vec{\delta}_j)(i\sigma_{ij} + \delta_j\gamma_5\gamma_i \otimes \tau_5\tau_j)q(2h) \\
 &\quad + \frac{1}{2}\bar{q}(2h)(i\sigma_{ij} - \delta_j\gamma_5\gamma_i \otimes \tau_5\tau_j)q(2h + 2\vec{\delta}_j) \\
 plane_{ij}(2h + \vec{\delta}_{ij}) &= \frac{1}{4} \sum_{s_{i,j}=\pm 1} \left\{ \bar{q}(2h + \vec{\delta}_{ij} - s_i s_j \vec{s}_{ij}) \right. \\
 &\quad \times \left(i\sigma_{ij} - s_i\gamma_5\gamma_i \otimes \tau_5\tau_j + s_j\gamma_5\gamma_j \otimes \tau_5\tau_i + s_i s_j \frac{\tau_i\tau_j - \tau_j\tau_i}{2} \right) q(2h + \vec{\delta}_{ij} + s_i s_j \vec{s}_{ij}) \Big\}
 \end{aligned} \tag{342}$$

So, for $i < j$,

$$\begin{aligned}
 (2a)^4 \sum_h \bar{q}i\sigma_{ij} \otimes Iq + \mathcal{O}(a) &= a^4 \frac{1}{4} \sum_n \sum_{s_{i,j}=\pm 1} (\eta_i(n)\eta_j(n)\bar{\chi}(n)\chi(n + \vec{s}_{ij})) \\
 \mathcal{O}(a) &= (2a)^4 \sum_h P_1 + (2a)^4 \sum_h P_2 \\
 P_1 &= -a [(\partial_i \bar{q}\gamma_5\gamma_j \otimes \tau_5\tau_i - \partial_j \bar{q}\gamma_5\gamma_i \otimes \tau_5\tau_j)q]_- + a^2 [\bar{q}i\sigma_{ij}(\Delta_i + \Delta_j)q]_+ \\
 P_2 &= [m_{ij} - m_{ji}]_- + [\bar{q}i\sigma_{ij}(a^4\Delta_i\Delta_j + a^2(\Delta_i + \Delta_j))q + a^4\Delta_j\bar{q}i\sigma_{ij}\Delta_i q]_+ + a^2 [\bar{q}\tau_{ij}\partial_i\partial_j q - \partial_i\bar{q}\tau_{ij}\partial_j q]_+ \\
 m_{ij} &= \left(\frac{a}{2} + a^3\Delta_i\right) \bar{q}\gamma_5\gamma_i \otimes \tau_5\tau_j\partial_j q + \bar{q}\gamma_5\gamma_i \otimes \tau_5\tau_j \left(a^3\partial_j\Delta_i + \frac{a}{2}\partial_j\right) q \\
 \tau_{ij} &= \frac{\tau_i\tau_j - \tau_j\tau_i}{2}
 \end{aligned} \tag{343}$$

4.2.3 Gamma 51234

After calculate, the result happens to be **opposite** to

$$\begin{aligned}
 \gamma_5\gamma_1 &= -\gamma_2\gamma_3\gamma_4 \\
 \gamma_5\gamma_2 &= \gamma_1\gamma_3\gamma_4 \\
 \gamma_5\gamma_3 &= -\gamma_1\gamma_2\gamma_4 \\
 \gamma_5\gamma_4 &= \gamma_1\gamma_2\gamma_3
 \end{aligned} \tag{344}$$

Defining

$$\begin{aligned}
 \eta_{51}(n) &= \eta_2(n)\eta_3(n)\eta_4(n) \\
 \eta_{52}(n) &= -\eta_1(n)\eta_3(n)\eta_4(n) \\
 \eta_{53}(n) &= \eta_1(n)\eta_2(n)\eta_4(n) \\
 \eta_{54}(n) &= -\eta_1(n)\eta_2(n)\eta_3(n)
 \end{aligned} \tag{345}$$

Then, define the cube as

$$cube_i(2h) = \frac{1}{16} \sum_{ijkl=0,1} \eta_{5i}(2h + \vec{v}_{ijkl})\bar{\chi}(2h + \vec{v}_{ijkl})\chi(2h + \vec{1}_{jkl} - \vec{v}_{jkl} + \vec{v}_i) \tag{346}$$

as the links connecting diagonal of cube $n \setminus i$, and then sum over i .

$$\begin{aligned}
\Gamma &= \gamma_5 \gamma_\mu \otimes I, \quad G = \gamma_5 \otimes \tau_\mu, \quad \Gamma_i = \gamma_\mu \gamma_i \otimes \tau_5 \tau_i, \quad \Gamma_{ij} = \gamma_5 \gamma_\mu \gamma_i \gamma_j \otimes \tau_j \tau_i, \\
CubeA_\mu(h_1, h_2) &= \bar{q}(h_1) \Gamma q(h_2) \\
CubeB_\mu(h_1, h_2, i \neq \mu, s = \pm 1) &= \frac{1}{2} [\bar{q}(h_1) \Gamma q(h_2) + s \bar{q}(h_1) \Gamma_i q(h_2)] \\
CubeC_\mu(h_1, h_2, i, j, s_i = \pm 1, s_j = \pm 1) &= \frac{1}{4} \left[\bar{q}(h_1) \Gamma q(h_2) + \sum_{k=i,j} s_k \bar{q}(h_1) \Gamma_k q(h_2) + s_i s_j \bar{q}(h_1) \Gamma_{ij} q(h_2) \right] \\
CubeD_\mu(h_1, h_2, s_{ijk \neq \mu} = \pm 1) &= \frac{1}{8} \left[\bar{q}(h_1) \Gamma q(h_2) + \sum_{l \neq \mu} s_l \bar{q}(h_1) \Gamma_l q(h_2) - \sum_{a < b, a \neq \mu, b \neq \mu} s_a s_b \bar{q}(h_1) \Gamma_{ab} q(h_2) \right. \\
&\quad \left. - s_i s_j s_k \bar{q}(h_2) G q(h_2) \right]
\end{aligned} \tag{347}$$

It can be verified $\delta = \pm 1$

$$\begin{aligned}
cube_\mu(2h) &= CubeA_\mu(2h, 2h) \\
cube_\mu(2h + \vec{\delta}_i) &= \sum_{s=\pm 1} \left(CubeB_\mu(2h + 2\vec{\delta}_i, 2h, i, s) + CubeB_\mu(2h, 2h + 2\vec{\delta}_i, i, s) \right) \\
cube_\mu(2h + \vec{\delta}_{ij}) &= \sum_{s_{ij}=\pm 1} CubeC_\mu(2h + L_c(\delta_{ij}, s_{ij}), 2h + R_c(\delta_{ij}, s_{ij}), i, j, s_i, s_j) \\
cube_\mu(2h + \vec{\delta}_{ijk}) &= \sum_{s_{ijk}=\pm 1} CubeD_\mu(2h + L_d(\delta_{ijk}, s_{ijk}), 2h + R_d(\delta_{ijk}, s_{ijk}), s_{ijk}) \\
L_{c,d}(\delta, s) &= \vec{\delta} + \vec{s}, \quad R_{c,d}(\delta, s) = \vec{\delta} - \vec{s}
\end{aligned} \tag{348}$$

Finally

$$\begin{aligned}
(2a)^4 \sum_h \bar{q} \gamma_5 \gamma_\mu \otimes I q + \mathcal{O}(a) &= a^4 \frac{1}{8} \sum_n \sum_{s_{i,j,k \neq \mu} = \pm 1} (\eta_{5\mu}(n) \bar{\chi}(n) \chi(n + \vec{s}_{ijk})) \\
\mathcal{O}(a) &= (2a)^4 \sum_h \left(\frac{1}{2} \sum_{i \neq \mu} B_i + \frac{1}{4} \sum_{i \neq j \neq \mu} \sum_{P(i,j)} C_{ij} + \frac{1}{8} \sum_{P(i,j,k)} D_{ijk} \right)
\end{aligned} \tag{349}$$

where $\sum_{P(i,j)} C_{ij} = C_{ij} + C_{ji}$, $\sum_{P(i,j,k)} D_{ijk} = D_{ijk} + D_{jki} + D_{kij}$ and

$$\begin{aligned}
B_i &= 2a [\partial_i \bar{q} \Gamma_i q]_- + 2a^2 [\Delta_i \bar{q} \Gamma q]_+ \\
C_{ij} &= 4a [\partial_i \bar{q} \Gamma_i q]_- + 2a^2 [2\Delta_i \bar{q} \Gamma q + \partial_i \bar{q} \Gamma_{ij} \partial_j q - \partial_i \partial_j \bar{q} \Gamma_{ij} q]_+ \\
&\quad + a^3 [\partial_i \bar{q} \Gamma_i \Delta_j q + \partial_i \Delta_j \bar{q} \Gamma_i q]_- + a^4 [\Delta_i \bar{q} \Gamma \Delta_j q + \Delta_i \Delta_j \bar{q} \Gamma q]_+ \\
D_{ijk} &= \left[8a^6 \Delta_i \Delta_j \bar{q} \Gamma \Delta_k q + 8a^4 \Delta_i \bar{q} \Gamma \Delta_j q + \left(\frac{8a^6}{3} \Delta_i \Delta_j \Delta_k + 8a^4 \Delta_i \Delta_j + 8a^2 \Delta_i \right) \bar{q} \Gamma q \right]_+ \\
&\quad + [8a^5 (\partial_i \Delta_j \bar{q} \Gamma_i \Delta_k q + \partial_i \Delta_k \bar{q} \Gamma_i \Delta_j q + \partial_i \bar{q} \Gamma_i \Delta_j \Delta_k q) \\
&\quad + 8a^3 \partial_i \bar{q} (\Gamma_i \Delta_j q + \Gamma_i \Delta_k q) + 8a \partial_i (a^4 \Delta_j \Delta_k + a^2 (\Delta_j + \Delta_k) + 1) \bar{q} \Gamma_i q]_- \\
&\quad + [8a^4 (\partial_j \Delta_k \bar{q} \Gamma_{ij} \partial_j q + \partial_i \Delta_k \bar{q} \Gamma_{ij} \partial_j q - \partial_i \partial_j \bar{q} \Gamma_{ij} \Delta_k q) \\
&\quad + 8a^2 \partial_i \bar{q} \Gamma_{ij} \partial_j q - 8a^2 (a^2 \partial_j \partial_k \Delta_i + \partial_i \partial_j) \bar{q} \Gamma_{ij} q]_+ + \left[8a^3 \partial_i \partial_j \bar{q} G \partial_k q - \frac{8a^3}{3} \partial_i \partial_j \partial_k \bar{q} G q \right]_-
\end{aligned} \tag{350}$$

here

$$\begin{aligned}
[\hat{O}_1 \bar{q} \Gamma \hat{O}_2 q]_+ &= \hat{O}_1 \bar{q} \Gamma \hat{O}_2 q + \hat{O}_2 \bar{q} \Gamma \hat{O}_1 q \\
[\hat{O}_1 \bar{q} \Gamma \hat{O}_2 q]_- &= \hat{O}_1 \bar{q} \Gamma \hat{O}_2 q - \hat{O}_2 \bar{q} \Gamma \hat{O}_1 q
\end{aligned} \tag{351}$$

4.2.4 Gamma5

I will give up on details.

Note that, $\eta_1(n) = 1$, so

$$\eta_{51}(n) = \eta_2 \eta_3 \eta_4 = \eta_1 \eta_2 \eta_3 \eta_4 \tag{352}$$

It can be found that

$$\bar{q}(2h)\gamma_5 q(2h) = \frac{1}{16} \sum_{s_{ijkl}=\pm 1} \eta_{51}(2h + \vec{s}_{ijkl}) \bar{\chi}(2h + \vec{s}_{ijkl}) \chi(2h + \vec{1}_{ijkl} - \vec{s}_{ijkl}) \quad (353)$$

which are diagonals of a hypercube.

I will assume

$$(2a)^4 \sum_h \bar{q}\gamma_5 \otimes Iq + \mathcal{O}(a) = a^4 \frac{1}{16} \sum_n \sum_{s_{i,j,k,l} \neq \pm 1} (\eta_{51}(n) \bar{\chi}(n) \chi(n + \vec{s}_{ijkl})) \quad (354)$$

4.2.5 Dimensions

Let $S_F = a^4 \sum (\bar{\psi} D\psi + 2am\bar{\psi}\psi)$.

$$\begin{aligned} S_F &\sim a^0 \\ \psi &\sim a^{-2} \\ q \sim \chi &\sim a^{-\frac{3}{2}} \\ \langle \bar{\chi}\chi \rangle &= 2a\langle \bar{\psi}\psi \rangle \sim a^{-3} \end{aligned} \quad (355)$$

4.2.6 Simulations

$$\begin{aligned} iC\bar{q}\gamma_\mu q &\rightarrow iaC \sum_{n,\delta} \bar{\chi}(n) \chi(n+\delta) \\ iC\bar{q}\sigma_\mu q &\rightarrow aC \sum_{n,\delta} \bar{\chi}(n) \chi(n+\delta) \\ C\bar{q}\gamma_5\gamma_\mu q &\rightarrow \frac{aC}{4} \sum_{n,\delta} \bar{\chi}(n) \chi(n+\delta) \\ iC\bar{q}\gamma_5 q &\rightarrow \frac{iaC}{8} \sum_{n,\delta} \bar{\chi}(n) \chi(n+\delta) \\ \frac{i\Omega}{2} \bar{q}\gamma_4\sigma_{12}q &= \frac{\Omega}{2} \bar{q}\gamma_5\gamma_3q \rightarrow \frac{a\Omega}{8} \sum_{n,\delta} \bar{\chi}(n) \chi(n+\delta) \end{aligned} \quad (356)$$

When simulating, there are cases that **pure imaginary coefficients** are used. The difference is that, for **anti-hermitian** D operator, the link should be

$$\begin{aligned} C\bar{\chi}(n)L\chi(n+\delta) - C\bar{\chi}(n+\delta)L^\dagger\chi(n) \\ iC\bar{\chi}(n)L\chi(n+\delta) + iC\bar{\chi}(n+\delta)L^\dagger\chi(n) \end{aligned} \quad (357)$$

Therefore, for the pure imaginary case,

$$\begin{aligned} \frac{\partial}{\partial\omega_a} D_{st0} &= iC \left(\frac{\partial}{\partial\omega_a} L \right) \delta_{n,n+\mu} + iC \left(\frac{\partial}{\partial\omega_a} L^\dagger \right) \delta_{n+\mu,n} \\ &= iC (T^a L) \delta_{n,n+\mu} - iC (L^\dagger T^a) \delta_{n+\mu,n} \end{aligned} \quad (358)$$

Using ‘one link’ function in the program, for σ^{ij} and $\gamma_5\gamma_\mu$, it is usual ‘+’, for γ_μ, γ_5 , it is a ‘-’.

5 Measurement

5.1 Plaquette Energy

For $SU(N)$, for square lattice, the gauge action can be written as

$$\begin{aligned} S_G &= \beta \frac{1}{N} \sum_n \sum_{\mu > \nu} (N - \text{tr} [U_\mu(n) U_\nu(n + a\mu) U_\mu^{-1}(n + a\nu) U_\nu^{-1}(n)]) \\ S_G &= \frac{1}{4} \beta \frac{1}{N} \sum_n ((2(D-1))N - \text{tr} [U_\mu(n) \Sigma_\mu(n)]) , \\ \Sigma_\mu(n) &= \sum_{\mu \neq \nu} (U_\nu(n) U_\mu(n + a\nu) U_\nu^{-1}(n + a\mu) + U_\nu^{-1}(n - a\nu) U_\mu(n - a\nu) U_\nu(n - a\nu + a\mu)) \end{aligned} \quad (359)$$

The plaquette energy is defined as

$$\begin{aligned} \langle S \rangle &= \frac{1}{N\Lambda} \sum_n \sum_{\mu > \nu} (\text{tr} [U_\mu(n) U_\nu(n + a\mu) U_\mu^{-1}(n + a\nu) U_\nu^{-1}(n)]) . \\ &= \frac{1}{N\Lambda} \sum_{n, \mu} \text{tr} [U_\mu(n) \Sigma_\mu(n)] . \end{aligned} \quad (360)$$

which is the average (average according to configurations) energy of plaquettes per plaquette (average according to plaquettes).

This is also $\langle W^{1 \times 1} \rangle$.

5.2 Meson Correlator

5.2.1 Meson Wave Function

We need at first construct an observable which is a bound state of two fermions and **has the same quantum number** as mesons. In short, we want to know

$$O(x) = \bar{\psi}(x) \Gamma \psi(x) \quad (361)$$

where Γ is a (product of) gamma matrix.

5.2.2 Meson Correlator

The correlator is defined as

$$C(x, y) = \langle \bar{O}(x) O(y) \rangle \quad (362)$$

where

$$\begin{aligned} \langle W \rangle &= \frac{1}{Z} \int \mathcal{D}[U, \bar{\psi}, \psi] W \exp(-S) \\ Z &= \int \mathcal{D}[U, \bar{\psi}, \psi] \exp(-S), \quad S = S_G + S_{pf} \end{aligned} \quad (363)$$

- iso-triplet

Denote the variables as C_T and O_T .

We need to calculate (green variables are constant)

$$\begin{aligned} C_T(n, m) &= \langle \bar{\psi}^{f_1}(n) \Gamma \psi^{f_2}(n) \bar{\psi}^{f_2}(m) \Gamma \psi^{f_1}(m) \rangle \\ &= \sum_{a, b, c_i} \Gamma_{a_1, b_1} \Gamma_{a_2, b_2} \langle \bar{\psi}_{a_1, c_1}^{f_1}(n) \psi_{b_1, c_1}^{f_2}(n) \bar{\psi}_{a_2, c_2}^{f_2}(m) \psi_{b_2, c_2}^{f_1}(m) \rangle \end{aligned} \quad (364)$$

Note that, they are all Grassman numbers (exchange three times will introduce a minus sign), and they can be averaged according to different fields, so

$$C_T(n, m) = - \sum_{a, b, c_i} \Gamma_{a_1, b_1} \Gamma_{a_2, b_2} \langle \psi_{b_1, c_1}^{f_2}(n) \bar{\psi}_{a_2, c_2}^{f_2}(m) \rangle_{f_1} \langle \psi_{b_2, c_2}^{f_1}(m) \bar{\psi}_{a_1, c_1}^{f_1}(n) \rangle_{f_2} \quad (365)$$

Using the Wick theorem for Grassman numbers (f is flavour index, c is color index, a, b are spinor index).

$$\begin{aligned}\langle \dots \rangle &= \frac{1}{Z_f} \int \mathcal{D}[\psi] \dots \exp \left(- \sum_{l,m} \bar{\psi}_l M_{lm} \psi_m \right). \\ \langle \psi_{i_1} \dots \psi_{i_n} \bar{\psi}_{j_1} \dots \bar{\psi}_{j_n} \rangle &= \sum_P \text{sign}(P) \prod_n^N (M^{-1})_{i_n, j_{P_n}}. \\ \langle \psi^f(n)_{a,c_1} \bar{\psi}_{b,c_2}^f(m) \rangle &= -D_{f,a,b,c_1,c_2}^{-1}(n, m).\end{aligned}\tag{366}$$

Then we can multiply gamma matrix back

$$C_T(n, m) = -\text{tr}_{c,s} [\Gamma D_{f_1}^{-1}(n, m) \Gamma D_{f_2}^{-1}(m, n)]\tag{367}$$

The trace is for both color and spinor space.

- iso-singlet

Denote the variables as C_S and O_S .

5.2.3 Sources

- Fourier transform

Usually, one need to know the observable in momentum space, which is

$$\tilde{C}(\mathbf{p}, n_t; \mathbf{0}, 0) \equiv \frac{1}{\sqrt{\Lambda_3}} \sum_{\mathbf{n} \in \Lambda_3} \exp(-i\mathbf{a}\mathbf{n} \cdot \mathbf{p}) C(\mathbf{n}, n_t; \mathbf{0}, 0)\tag{368}$$

where Λ_3 denotes the spatial lattice.

For hadron spectroscopy,

$$\tilde{C}(\mathbf{p}, n_t; \mathbf{0}, 0) \propto \exp(-an_t E_0(\mathbf{p})) \times (1 + \mathcal{O}(e^{-an_t \Delta E}))\tag{369}$$

where $E_0(\mathbf{p})$ is the ground state energy (dissipative relation?) and ΔE is the energy gap between ground state and the lowest excitation, and

$$E_0(\mathbf{p}) = \sqrt{m_H^2 + |\mathbf{p}|^2} \times (1 + \mathcal{O}(a|\mathbf{p}|))\tag{370}$$

For zero momentum, we find m_H . That is why the lattice at t -dir is usually larger than the spatial directions.

From Eq. (368), we only need to calculate $C(n, 0)$ for all n . That is a **point source**.

Using $\{\gamma_\mu, \gamma_5\} = 0$ and $\gamma_5^2 = 1$, $\{\gamma_\mu, \gamma_5\} = 0$, so

$$\begin{aligned}(\Gamma D^{-1}(n, m) \Gamma D^{-1}(m, n)) &= (\Gamma D^{-1}(n, m) \Gamma \gamma_5 (D^{-1}(n, m))^\dagger \gamma_5) \\ \text{tr}_{c,s} [\Gamma D^{-1}(n, m) \Gamma \gamma_5 (D^{-1}(n, m))^\dagger \gamma_5] &= \text{tr}_{c,s} [\gamma_5 \Gamma D^{-1}(n, m) \Gamma \gamma_5 (D^{-1}(n, m))^\dagger] \\ &= \pm \text{tr}_{c,s} [\Gamma' D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger] \\ &= \pm \text{tr}_{c,s} [\Gamma'^\dagger D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger]\end{aligned}\tag{371}$$

where $\Gamma' = \Gamma \gamma_5$ and $\pm$ come from $\gamma_5 \Gamma = \pm \Gamma \gamma_5$, and $\pm$ come from both $\gamma_5 \Gamma = \pm \Gamma \gamma_5$ and $\Gamma^\dagger = \pm \Gamma^\dagger$. Note that it is in fact a **real** number because

$$\begin{aligned}\text{tr}_{c,s} [\Gamma'^\dagger D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger] &= \text{tr}_{c,s} [D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger \Gamma'^\dagger] \\ \left(\text{tr}_{c,s} [\Gamma'^\dagger D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger] \right)^* &= \text{tr}_{c,s} \left[\left(D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger \Gamma'^\dagger \right)^\dagger \right] \\ &= \text{tr}_{c,s} [\Gamma' D^{-1}(n, m) \Gamma'^\dagger (D^{-1}(n, m))^\dagger] = \text{tr}_{c,s} [\Gamma'^\dagger D^{-1}(n, m) \Gamma' (D^{-1}(n, m))^\dagger]\end{aligned}\tag{372}$$

With point source, we need only to calculate $D^{-1}(n, 0)$, which is a $12 \times 12 = 144$ elements matrix field on each site, with the matrix element

$$\begin{aligned}D^{-1}(n, m_0)_{c_1, c_2, s_1, s_2} &= \sum_{m, c_3, s_3} D^{-1}(n, m)_{c_1, c_3, s_1, s_3} (S(m_0, c_2, s_2; m, c_3, s_3)) \\ D^{-1}(n, m_0)_{:, c_2, :, s_2} &= D^{-1} \phi_{m_0, c_2, s_2}^S\end{aligned}\tag{373}$$

In the last line, $\cdot, c_2, \cdot, s_2$ denote one column of the 12×12 matrix, and ϕ_{m_0, c_2, s_2}^S is pseudo-fermion field with only one non-zero element (the **point source** at m_0 , in our case, $m_0 = (\mathbf{0}, 0)$)

$$\phi_{m_0, c_2, s_2}^S(m)_{c, s} = \delta(m - m_0) \delta(c - c_2) \delta(s - s_2) \quad (374)$$

In matrix form it is

$$\begin{pmatrix} D_{1,cs}^{-1} \\ D_{2,cs}^{-1} \\ D_{3,cs}^{-1} \\ \dots \\ D_{10,cs}^{-1} \\ D_{11,cs}^{-1} \\ D_{12,cs_2}^{-1} \end{pmatrix} = \begin{pmatrix} D_{1,1}^{-1} & D_{1,2}^{-1} & \dots & D_{1,cs}^{-1} & \dots & D_{1,11}^{-1} & D_{1,12}^{-1} \\ D_{2,1}^{-1} & D_{2,2}^{-1} & \dots & D_{2,cs}^{-1} & \dots & D_{2,11}^{-1} & D_{2,12}^{-1} \\ D_{3,1}^{-1} & D_{3,2}^{-1} & \dots & D_{3,cs}^{-1} & \dots & D_{3,11}^{-1} & D_{3,12}^{-1} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ D_{10,1}^{-1} & D_{10,2}^{-1} & \dots & D_{10,cs}^{-1} & \dots & D_{10,11}^{-1} & D_{10,12}^{-1} \\ D_{11,1}^{-1} & D_{11,2}^{-1} & \dots & D_{11,cs}^{-1} & \dots & D_{11,11}^{-1} & D_{11,12}^{-1} \\ D_{12,1}^{-1} & D_{12,2}^{-1} & \dots & D_{12,cs}^{-1} & \dots & D_{12,11}^{-1} & D_{12,12}^{-1} \end{pmatrix} \begin{pmatrix} 0 \\ \dots \\ 0 \\ 1_{idx=cs} \\ 0 \\ \dots \\ 0 \end{pmatrix} \quad (375)$$

So, we need 12 point sources to fill the 12×12 matrix. Now, for each site, we can calculate the trace, and obtain a **real** field defined on sites

$$\Phi(n) = \text{tr}_{c,s}(\Gamma D^{-1} \Gamma D^{-1}) \quad (376)$$

The final step is to sum the spatial lattice with the weight $e^{-ian \cdot \mathbf{p}}$ for each n_t , note that, the result should be calculated for each (assume periodic boundary condition for spatial directions)

$$\mathbf{p} \in \left\{ (p_1, p_2, p_3) | p_i = \frac{2\pi}{aN_i} k_i, k_i = -\frac{N_i}{2} - 1, \dots, \frac{N_i}{2} \right\} \quad (377)$$

where N_i is the length of the lattice at i direction.

Therefor, $\tilde{C}(\mathbf{p})$ is a complex field defined on spatial **reciprocal space**.

For spectroscopy, we need only the data for $\mathbf{p} = 0$, because when $\Delta E \ll 1$

$$\tilde{C}(n_t) \equiv \tilde{C}(\mathbf{0}, n_t; \mathbf{0}, 0) \propto \exp(-an_t m_H). \quad (378)$$

Note that, for $k_i = 0$, we need **even** number of length at spatial directions.

- Detail of implementation

We can write D^{-1} in the form of 4×4 matrices, with elements as 3×3 matrices, as

$$D^{-1} = \begin{pmatrix} U_{11} & U_{12} & U_{13} & U_{14} \\ U_{21} & U_{22} & U_{23} & U_{24} \\ U_{31} & U_{32} & U_{33} & U_{34} \\ U_{41} & U_{42} & U_{43} & U_{44} \end{pmatrix} \quad (379)$$

If our pseudo-fermion field is organized as

$$D^{-1} \phi^S \equiv \phi^{s \times 3 + c}(n) = (d_0, d_1, d_2, d_3), d_s = (v_0, v_1, v_2) \quad (380)$$

Using Eq. (375), one have

$$\begin{aligned} U_{ij} &= \begin{pmatrix} \phi^{j \times 3 + 0}(d_i, v_0) & \phi^{j \times 3 + 1}(d_i, v_0) & \phi^{j \times 3 + 2}(d_i, v_0) \\ \phi^{j \times 3 + 0}(d_i, v_1) & \phi^{j \times 3 + 1}(d_i, v_1) & \phi^{j \times 3 + 2}(d_i, v_1) \\ \phi^{j \times 3 + 0}(d_i, v_2) & \phi^{j \times 3 + 1}(d_i, v_2) & \phi^{j \times 3 + 2}(d_i, v_2) \end{pmatrix}, \\ U_{ij}^T &= \begin{pmatrix} \phi^{j \times 3 + 0}(d_i, v_0) & \phi^{j \times 3 + 0}(d_i, v_1) & \phi^{j \times 3 + 0}(d_i, v_2) \\ \phi^{j \times 3 + 1}(d_i, v_0) & \phi^{j \times 3 + 1}(d_i, v_1) & \phi^{j \times 3 + 1}(d_i, v_2) \\ \phi^{j \times 3 + 2}(d_i, v_0) & \phi^{j \times 3 + 2}(d_i, v_1) & \phi^{j \times 3 + 2}(d_i, v_2) \end{pmatrix} \end{aligned} \quad (381)$$

The Γ intersect between D^{-1} and $(D^{-1})^\dagger$ is a permutation of rows in spinor space, for example

$$\gamma_1 = \begin{pmatrix} 0 & 0 & 0 & c_1 \\ 0 & 0 & c_2 & 0 \\ 0 & c_3 & 0 & 0 \\ c_4 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{aligned} p(1) &= 4, & p(2) &= 3, & p(3) &= 2, & p(4) &= 1 \\ c_1 &= -i, & c_2 &= -i, & c_3 &= i, & c_4 &= i \end{aligned} \quad (382)$$

Mason	correlator	Γ	Γ'	mass
Scalar (0^{++})	a_0	$1, \gamma_4$	$\gamma_5, \gamma_4 \gamma_5$	
Pseudoscalar (0^{-+})	$\pi^+ = \bar{d} \gamma_5 u$	γ_5	1	139.57061(24)MeV
Vector (1^{--})	$\rho^+ = d \gamma_i \bar{u}$	γ_i	$\gamma_i \gamma_5$	775.11(34)MeV
Axial vector (1^{+-})				
Tensor (1^{+-})				

表 2

where c_i are coefficients and $c_i \in \mathbb{Z}_4$ group. $p(i) = j$ denote that the none-zero of the i -th row is j -element, (also, the none-zero of the i -th column is j -element, because $\gamma_\mu^\dagger = \gamma_\mu$).

So one have such that

$$\Gamma' D^{-1} = \begin{pmatrix} c_1 U_{p(1)1} & c_1 U_{p(1)2} & c_1 U_{p(1)3} & c_1 U_{p(1)4} \\ c_2 U_{p(2)1} & c_2 U_{p(2)2} & c_2 U_{p(2)3} & c_2 U_{p(2)4} \\ c_3 U_{p(3)1} & c_3 U_{p(3)2} & c_3 U_{p(3)3} & c_3 U_{p(3)4} \\ c_4 U_{p(4)1} & c_4 U_{p(4)2} & c_4 U_{p(4)3} & c_4 U_{p(4)4} \end{pmatrix}, \quad (383)$$

$$\Gamma' D^{-1} \Gamma'^\dagger = \begin{pmatrix} c_1 c_1^* U_{p(1)p(1)} & c_1 c_2^* U_{p(1)p(2)} & c_1 c_3^* U_{p(1)p(3)} & c_1 c_4^* U_{p(1)p(4)} \\ c_2 c_1^* U_{p(2)p(1)} & c_2 c_2^* U_{p(2)p(2)} & c_2 c_3^* U_{p(2)p(3)} & c_2 c_4^* U_{p(2)p(4)} \\ c_3 c_1^* U_{p(3)p(1)} & c_3 c_2^* U_{p(3)p(2)} & c_3 c_3^* U_{p(3)p(3)} & c_3 c_4^* U_{p(3)p(4)} \\ c_4 c_1^* U_{p(4)p(1)} & c_4 c_2^* U_{p(4)p(2)} & c_4 c_3^* U_{p(4)p(3)} & c_4 c_4^* U_{p(4)p(4)} \end{pmatrix}$$

Finally we have (Note U is not a $SU(3)$ matrix)

$$\text{tr}_{c,s} [\Gamma' D^{-1} \Gamma'^\dagger (D^{-1})^\dagger] = \sum_{ij} c_i c_j^* \text{tr}_c [U_{p(i)p(j)} U_{ij}^\dagger] = \sum_{ij} c_i c_j^* \text{tr}_c [U_{ij}^\dagger U_{p(i)p(j)}] \quad (384)$$

This can be further simplified, note that the result should be a real number, so for $i \neq j$, if $\text{tr} [U_{p(i)p(j)}^\dagger U_{ij}]$ is present, so must be $\text{tr} [U_{ij}^\dagger U_{p(i)p(j)}]$ with the same sign. This is guaranteed by symmetric matrix, i.e. if $p(i) = a$, one must have $p(a) = i$, and also $c_i c_j^* = c_{p(i)} c_{p(j)}^*$ as shown below.

To prove $c_i c_j^* = c_{p(i)} c_{p(j)}^*$, we need to consider:

1. $\Gamma^\dagger = \pm \Gamma$ and $p(i) = i$. In this case, $c_i = c_{p(i)}$, and $c_i c_j^* = c_{p(i)} c_{p(j)}^*$ is straightforward.
2. $\Gamma^\dagger = \Gamma$ and $p(i) \neq i$. In this case, $c_i = c_{p(i)}^*$, so $c_i c_j^* = c_{p(i)}^* c_{p(j)}$. Note that, c_i are either all real or all imaginary, so $c_i c_j^* = c_{p(i)}^* c_{p(j)} = c_{p(i)} c_{p(j)}^*$.
3. $\Gamma^\dagger = -\Gamma$ and $p(i) \neq i$. In this case, $c_i = -c_{p(i)}^*$, so $c_i c_j^* = c_{p(i)}^* c_{p(j)} = c_{p(i)} c_{p(j)}^*$.

So, we have two cases, one for $p(1) = 1$, and one for $p(1) \neq 1$

$$\begin{aligned} & \text{tr}_{c,s} [\Gamma' D^{-1} \Gamma'^\dagger (D^{-1})^\dagger] \\ &= \begin{cases} 2 \sum_{i>1, j>i} c_i c_j^* \text{Retr}_c [U_{ij}^\dagger U_{p(i)p(j)}] + \sum_{i=1,2,3,4} \text{tr}_c [U_{ii}^\dagger U_{ii}] & p(i) = i \\ 2 \sum_{i>1, j>i} c_i c_j^* \text{Retr}_c [U_{ij}^\dagger U_{p(i)p(j)}] + 2 \sum_{i=1,k} \text{Retr}_c [U_{ii}^\dagger U_{p(i)p(i)}] & p(1) \neq 1, k \end{cases} \end{aligned} \quad (385)$$

5.2.4 Summary of parameters

The input gamma matrix is the Γ' , here is a summary of gamma matrices.

5.2.5 Gauge smearing

Note: I think I did NOT understand gauge smearing correctly, will go back to this after staggered fermion being implemented.

In HMC with fermions, the computer power is consumed mainly in solving the D^{-1} . The small eigenvalues of the D operator is the main reason to slow down the solver, which is the so called **low mode** or **exceptional configurations**.

Gauge smearing (gauge smoothing) is one of the method to ease the problem by replacing the original configuration with a gauge equivalent but easier configuration. There are several different smearing methods. In CLGLib, only two are implemented.

- APE

It use

$$U'_\mu = \mathcal{P} \left((1 - \alpha)U_\mu + \frac{\alpha}{6}\Sigma_\mu \right) \quad (386)$$

where Σ_μ is the staple, (see Eq. (359)). In CLGLib, staples are cached. After smoothing, $\mathcal{P}$ is a projection project to result to $SU(3)$ and can be approximated as

Algorithm 23 $\mathcal{P}(U)$ approximately

```

 $U = U / \sqrt{\text{tr}(U^\dagger U / 3)}$ 
for  $i = 0$  to  $r$  do ▷ iterate r times
   $x = U \left( \frac{3}{2} - \frac{1}{2}U^\dagger U \right)$ 
   $U = \left( 1 - \frac{i}{3}\text{Im}(\det(x)) \right) x$ 
end for
return  $U$ 

```

Usually, iterate for 4 times, it can archive α accuracy.

- APE stout

In this approach, it construct a $SU(3)$ candidate directly by the staples. (Therefor, no need to project). Using

$$\Omega_\mu = \rho_\mu \Sigma_\mu U_\mu^\dagger, \quad Q_\mu = \{\Omega_\mu\}_{TA}, \quad U'_\mu = \exp(Q_\mu) U_\mu \quad (387)$$

where ρ_μ is usually set to be $\rho_{1,2,3} = \rho, \rho_4 = 0$. Note that, there is no sum over μ in the above equation. Also, note that, exp is not accurate unless ρ is small enough, however, one can iterate the smearing for a few sub-steps.

5.3 Chiral Condensate of Wilson fermion

5.3.1 Wall source

5.3.2 Decay constant

5.3.3 Effective Quark Mass

The quark mass can be measured using

$$m_q = \frac{c_\delta}{2} \frac{\langle 0 | \nabla_4 A_4 | \pi \rangle}{\langle 0 | P_5(n) | \pi \rangle} \quad (388)$$

where

$$\begin{aligned} \nabla_4 A_4(t) &= A_4(t+1) - A_4(t-1) \rightarrow c_\delta = \frac{2m_\pi}{e^{+m_\pi} - e^{-m_\pi}}; \\ \nabla_4 A_4(t) &= A_4(t+1) - A_4(t) \rightarrow c_\delta = \frac{m_\pi}{1 - e^{-m_\pi}}; \end{aligned} \quad (389)$$

5.4 Stochastic Methods

The stochastic methods is based on

$$\frac{1}{K} \sum_k \phi_{c,s}^{k*}(x) \phi_{c',s'}^k(y) \approx \delta_{xy} \delta_{cc'} \delta_{ss'} \quad (390)$$

where ϕ is a Gaussian distributed pseud-fermion, or $\mathbb{Z}_4$ random distributed pseud-fermion (the elements are 1, -1, i , $-i$ average distributed).

Therefor, for any matrix,

$$\text{tr}[M] \approx \frac{1}{K} \sum_k \phi^{k\dagger} M \psi^k \quad (391)$$

This is useful to measure some objects easily.

5.4.1 For condensations

For example, to measure $\langle \bar{\psi} O \psi \rangle$, where $\Gamma = 1, \gamma_5$ or γ_i etc.

For Grassman field, and for $S_F = -\bar{\psi} D \psi$

$$\langle \psi_i \bar{\psi}_j \rangle = (D^{-1})_{ji} \quad (392)$$

which means

$$\langle \psi_{c',s'}(n) \bar{\psi}_{c,s}(m) \rangle = (D^{-1})_{c',s',s}(n|m) \quad (393)$$

Then consider $\langle \bar{\psi} O_{cs,c's'} \psi \rangle$ which is

$$\begin{aligned} \langle \bar{\psi} O_{cs,c's'} \psi \rangle &= \langle \sum_{cs,c's',mn} \bar{\psi}_{cs}(m) O_{cs,c's'}(m|n) \psi_{c's'}(n) \rangle = \sum_{cs,c's',mn} \langle \bar{\psi}_{cs}(m) \psi_{c's'}(n) \rangle O_{cs,c's'}(m|n) \\ &= - \sum_{cs,c's',mn} \langle \psi_{c's'}(n) \bar{\psi}_{cs}(m) \rangle O_{cs,c's'}(m|n) = - \sum_{cs,c's',mn} (D^{-1})_{c',c,s's}(n|m) O_{cs,c's'}(m|n) \end{aligned} \quad (394)$$

The sum over m is matrix multiply, and sum over n is a trace, so

$$\langle \bar{\psi} O_{cs,c's'} \psi \rangle = -\text{tr}_{cs,m} [(D^{-1})_{c',c,s's}(n|m) O_{cs,c's'}(m|n)] \quad (395)$$

Note that, put D^{-1} right most is more convenient to calculate, use $\text{tr}[ABC] = \text{tr}[BCA]$, it is

$$\langle \bar{\psi} O_{cs,c's'} \psi \rangle = -\text{tr}_{cs,m} [O_{cs,c's'}(m|n) (D^{-1})_{c',c,s's}(n|m)] \quad (396)$$

For any operator, one can always calculate $D^{-1} \phi^k$ first.

5.4.2 For densities

Similarly, the observables defined as $O(x) = \langle \bar{\psi}(x) O \psi(x) \rangle$ can be calculated, the discretized version is

$$O(n) = \langle \sum_{cs} \bar{\psi}_{cs}(n) (O\psi)_{cs}(n) \rangle = \langle \sum_{cs} \bar{\psi}_{cs}(n) \left(\sum_{c's',m} O_{cs,c's'}(n|m) \psi_{c's'}(m) \right) \rangle \quad (397)$$

All the same as above, it becomes

$$O(n) = - \sum_{cs,c's',m} (D^{-1})_{c's',cs}(m|n) O_{cs,c's'}(n|m) \quad (398)$$

On the other hand, the n -component of element product of two vectors $\frac{1}{K} \sum_k (\phi^\dagger, O D^{-1} \phi)$ (denoted as $X(n)$) is

$$\begin{aligned} X(n) &= \frac{1}{K} \sum_k \sum_{cs} \phi_{cs}^*(n) \sum_{c_a s_a, c_b s_b, a, b} O_{cs, c_a s_a}(n|a) (D^{-1})_{c_a s_a, c_b s_b}(a|b) \phi_{c_b s_b}(b) \\ &\approx \sum_{cs} \delta_{nb} \delta_{cc_b} \delta_{ss_b} \sum_{c_a s_a, c_b s_b, a, b} O_{cs, c_a s_a}(n|a) (D^{-1})_{c_a s_a, c_b s_b}(a|b) \\ &= \sum_{cs} \sum_{c_a s_a, a} O_{cs, c_a s_a}(n|a) (D^{-1})_{c_a s_a, cs}(a|n) \end{aligned} \quad (399)$$

Do the variable rename $a \rightarrow m, c_a \rightarrow c', s_a \rightarrow s'$, it is

$$\begin{aligned} X(n) &\approx \sum_{cs} \sum_{c's',m} O_{cs,c's'}(n|m) (D^{-1})_{c',c,s's}(m|n) = \sum_{cs,c's',m} (D^{-1})_{c's',cs}(m|n) O_{cs,c's'}(n|m) \\ O(n) &\approx -X(n) \end{aligned} \quad (400)$$

Note, the $\sum_{cs}$ can be written as tr_{cs} , and the sum $\sum_{c's',m}$ is just matrix product, so it is the trace of n -component of element product of two vectors $\frac{1}{K} \sum_k \text{tr}_{cs} [(\phi^\dagger, O D^{-1} \phi)]$.

It is in fact the median result to calculate the condensation.

5.5 Staggered Fermion Spectrum

5.5.1 Interpretation Of Staggered Fermion

Start with the Lagrangian

$$S_F = \frac{1}{2a} \sum_{\mu} \eta_{\mu}(n) (\bar{\chi}(n) U_{\mu}(n) \chi(n + \mu) - \bar{\chi}(n) U_{-\mu}(n) \chi(n - \mu)) + m \bar{\chi} \chi \quad (401)$$

which is also ($\eta_{\mu}(n) = \eta_{\mu}(n + \mu)$)

$$S_F = \frac{1}{2a} \sum_{\mu} \eta_{\mu}(n) (\bar{\chi}(n) U_{\mu}(n) \chi(n + \mu) - \bar{\chi}(n + \mu) U_{\mu}^{\dagger}(n) \chi(n)) + m \bar{\chi} \chi \quad (402)$$

Following Ref. [25], the quark is defined on a 2^4 hypercube on a even lattice, defining

$$n_{\mu} = 2h_{\mu} + s_{\mu}, \quad h_{\mu} = 0, 1, 2, \dots, N_{\mu}/2 - 1, \quad s_{\mu} = 0, 1 \quad (403)$$

Without gauge field, the transform is **NOTE: there are actually two conventions! We are not using Ref. [25] here, but the other one for example used in Ref. [26], see also Eq. (30) of Ref. [27].**

$$q_{\alpha,a}(h) = \frac{1}{C\sqrt{C}} \sum_{s_{\mu}} (\Gamma_s)_{\alpha,a} V_s(2h) \chi(2h + s), \quad \bar{q}_{\alpha,a}(h) = \frac{1}{C\sqrt{C}} \sum_s \bar{\chi}(2h + s) V_s^{\dagger}(2h) (\Gamma_s^{\dagger})_{\alpha,a} \quad (404)$$

$$\Gamma_s = \gamma_1^{s_x} \gamma_2^{s_y} \gamma_3^{s_z} \gamma_4^{s_t}$$

where $C = 2^{d-2}$, Γ_s is a $C^2 \times C^2$ matrix, and we write the case $d = 4$, and

$$V_s(2h) = (U_x(2h))^{s_x} (U_y(2h + s_x \mathbf{x}))^{s_y} (U_z(2h + s_x \mathbf{x} + s_y \mathbf{y}))^{s_z} (U_t(2h + s_x \mathbf{x} + s_y \mathbf{y} + s_z \mathbf{z}))^{s_t} \quad (405)$$

$V_s(2h)$ is the Wilson line link $2h$ to $2h + s$. But, there are in fact many Wilson lines can link $2h$ to $2h + s$, do we have a free choice here?

Note that $V_s^{\dagger}(2h) V_s = 1$.

5.5.2 Go Back To Staggered Fermion

Consider $\sum_a \bar{q}_a q_a$, which is $\sum_{\alpha,a} \bar{q} q$. One can verify that

$$\sum_{\alpha,a} \bar{q}(2h) q(2h) = \frac{1}{16} \sum_s \bar{\chi}(2h + s) \chi(2h + s) \quad (406)$$

If one don't care about the $\mathcal{O}(a)$ terms, then the

$$\sum_{\alpha_1, \alpha_2, a} \bar{q}_{\alpha_1, a}(2h) (\gamma_{\mu})_{\alpha_1, \alpha_2} q_{\alpha_2, a}(2h) = \frac{1}{16} \eta_{\mu}(s) \sum_t (\bar{\chi}(2h + t + \mu) \chi(2h + t) + \bar{\chi}(2h + t) \chi(2h + t + \mu)) \quad (407)$$

where t is $\{s_i\} \setminus s_{\mu}$, for example, when $\mu = x$, $t = (0, t_y, t_z, t_t)$ with $t_{y,z,t} = 0, 1$.

$$\begin{aligned} & \bar{q}(2h) \gamma_{\mu} \otimes \mathbb{I} (q(2h + 2\mu) - q(2h - 2\mu)) \\ &= \eta_{\mu}(s) \frac{1}{16} \sum_t \bar{\chi}(2h + t + \mu) \chi(2h + t + 2\mu) + \eta_{\mu}(s) \frac{1}{16} \sum_t \bar{\chi}(2h + t) \chi(2h + t + 3\mu) \\ & - \eta_{\mu}(s) \frac{1}{16} \sum_t \bar{\chi}(2h + t + \mu) \chi(2h + t - 2\mu) - \eta_{\mu}(s) \frac{1}{16} \sum_t \bar{\chi}(2h + t) \chi(2h + t - \mu) \end{aligned} \quad (408)$$

The 3μ term is at higher order, note that

$$\partial^2 \rightarrow \frac{f(n + \mu) + f(n - \mu) - 2f(n)}{2a^2} \quad (409)$$

To see it clearly, we use

$$\begin{aligned} & \sum_{\alpha_1, \alpha_2, a_1, a_2} \bar{q}_{\alpha_1, a_1}(2h) (\gamma_5)_{\alpha_1, \alpha_2} (\gamma_5^* \gamma_{\mu}^*)_{a_1, a_2} q_{\alpha_2, a_2}(2h) \\ &= \frac{1}{16} \eta_{\mu}(s) \sum_t (-\bar{\chi}(2h + t + \mu) \chi(2h + t) + \bar{\chi}(2h + t) \chi(2h + t + \mu)) \end{aligned} \quad (410)$$

The gamma matrices in the taste space is denoted as τ , so we have

$$\begin{aligned}
 & \bar{q}(2h)\gamma_\mu \otimes \mathbb{I}_\tau q(2h+2\mu) - \bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h+2\mu) \\
 &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) \\
 & \bar{q}(2h)\gamma_\mu \otimes \mathbb{I}_\tau q(2h-2\mu) + \bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h-2\mu) \\
 &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t) \chi(2h+t-\mu) \\
 &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu)
 \end{aligned} \tag{411}$$

So

$$\begin{aligned}
 & \bar{q}(2h)\gamma_\mu \otimes \mathbb{I}_\tau q(2h+2\mu) - \bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h+2\mu) \\
 & - (\bar{q}(2h)\gamma_\mu \otimes \mathbb{I}_\tau q(2h-2\mu) + \bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h-2\mu)) + 2\bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h) \\
 &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) - \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu) \\
 & + \frac{1}{8} \eta_\mu(s) \sum_t (-\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu))
 \end{aligned} \tag{412}$$

which is

$$\begin{aligned}
 & 4a\bar{q}(2h)\gamma_\mu \otimes \mathbb{I}_\tau \nabla_\mu q(2h) - 4a^2 \bar{q}(2h)\gamma_5 \otimes \tau_5^* \tau_\mu^* \Delta_\mu q(2h) \\
 &= \frac{1}{8} \eta_\mu(s) \sum_t \{ \bar{\chi}(2h+t) \chi(2h+t+\mu) - \bar{\chi}(2h+t+\mu) \chi(2h+t) \\
 & + \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) - \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu) \}
 \end{aligned} \tag{413}$$

Omitting the $\mathcal{O}(a)$ term, it go back to the original action.

5.5.3 Staggered Meson Operators

The meson operators come from group theory. The meson operator can be written compactly as

$$\bar{\chi}(x) S(x, x') \chi(x') \tag{414}$$

where S is a weight function, or a phase function, or a sign function.

In the following, we follow Ref. [28].

For the reason of interpolation of quarks, we allow the spatial coordinate to be different, but require them to be in a unit cube. Also, usually, we require t be the same. That is something like

$$M(t) = \sum_{\mathbf{x}, \mathbf{x}' \in \square} \bar{\chi}_c(\mathbf{x}, t) S(\mathbf{x}, \mathbf{x}') \chi_c(\mathbf{x}', t) \tag{415}$$

where c is color index.

With the definition (**Note that D here is not Dirac operator but merely a shift operator**)

$$D_{\mathbf{A}} \phi(\mathbf{x}) = \sum_{\delta_1=\pm 1} \sum_{\delta_2=\pm 1} \sum_{\delta_3=\pm 1} \phi(\mathbf{x} + \sum_{x,y,z} A_i \delta_i \mathbf{e}_i) \tag{416}$$

Usually, one can write

$$\begin{aligned}
 M(t) &= \sum_{\mathbf{A} \in \mathbb{Z}_2^3} S(\mathbf{A}) M_{\mathbf{A}, \mathbf{A}+\delta}(t) \\
 M_{\mathbf{A}, \mathbf{B}} &= \sum_{\mathbf{x} \in \mathbb{Z}^3} D_{\mathbf{A}} \bar{\chi}_c(2\mathbf{x}) D_{\mathbf{B}} \chi_c(2\mathbf{x})
 \end{aligned} \tag{417}$$

Note that, sum of $\mathbf{A}$ vector run over all 2^3 corners of the cube, the $\mathbf{B} = \mathbf{A} + \delta$ vector is mod 2, i.e. also in the unit cube.

This $M(t)$ correspond to a sum of two terms

$$M(t) = \sum_{\mathbf{y} \in \mathbb{Z}^3/2} (\bar{q} \gamma_S \otimes \tau_F q + \bar{q} \gamma_S \gamma_4 \gamma_5 \otimes \tau_F \tau_4 \tau_5 q) \tag{418}$$

idx	$S(\mathbf{x})$	δ	$P = +\sigma_s$			$P = -\sigma_s$		
			M	J_R^{PC}	$\gamma_S \otimes \tau_F$	M	J_R^{PC}	$\gamma_S \otimes \tau_F$
0	1	0	f_0	0_S^{++}	$\mathbb{I} \otimes \mathbb{I}$	π	0_A^{+-}	$\gamma_4 \gamma_5 \otimes \tau_4 \tau_5$
1	$\eta_4 \xi_4$	0	?	0_A^{+-}	$\gamma_4 \otimes \tau_4$	π	0_A^{+-}	$\gamma_5 \otimes \tau_5$
2	$\eta_k \xi_k \epsilon$	0	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_k \tau_5$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_k \tau_4$
3	$\eta_4 \xi_4 \eta_k \xi_k \epsilon$	0	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_l \tau_m$	ρ	1_A^{--}	$\gamma_k \otimes \tau_k$
4	η_k	$\mathbf{e}_k$	ω	1_S^{--}	$\gamma_k \otimes \mathbb{I}$	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_4 \tau_5$
5	$\eta_4 \xi_4 \eta_k$	$\mathbf{e}_k$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_4$	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_5$
6	$\xi_k \epsilon$	$\mathbf{e}_k$	π	0_A^{+-}	$\gamma_5 \otimes \tau_k \tau_5$	?	0_A^{+-}	$\gamma_4 \otimes \tau_k \tau_4$
7	$\eta_4 \xi_4 \xi_k \epsilon$	$\mathbf{e}_k$	π	0_A^{+-}	$\gamma_4 \gamma_5 \otimes \tau_l \tau_m$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_k$
8	$\eta_k \xi_k \eta_l \epsilon$	$\mathbf{e}_l$	ρ	1_A^{--}	$\gamma_m \gamma_4 \otimes \tau_k \tau_5$	a_1	1_A^{++}	$\gamma_m \gamma_5 \otimes \tau_k \tau_4$
9	$\eta_4 \xi_4 \eta_k \xi_k \eta_l \epsilon$	$\mathbf{e}_l$	ρ	1_A^{--}	$\gamma_m \otimes \tau_l \tau_m$	b_1	1_A^{+-}	$\gamma_k \gamma_l \otimes \tau_k$
10	$\eta_k \eta_l \epsilon_{klm}$	$\mathbf{e}_k + \mathbf{e}_l$	h_1	1_S^{+-}	$\gamma_k \gamma_l \otimes \mathbb{I}$	ρ	1_A^{--}	$\gamma_m \otimes \tau_4 \tau_5$
11	$\eta_4 \xi_4 \eta_k \eta_l \epsilon_{klm}$	$\mathbf{e}_k + \mathbf{e}_l$	a_1	1_A^{++}	$\gamma_m \gamma_5 \otimes \tau_4$	ρ	1_A^{--}	$\gamma_m \gamma_4 \otimes \tau_5$
12	$\xi_k \xi_l \epsilon_{klm}$	$\mathbf{e}_k + \mathbf{e}_l$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_k \tau_l$	π	0_A^{+-}	$\gamma_4 \gamma_5 \otimes \tau_m$
13	$\eta_4 \xi_4 \xi_k \xi_l \epsilon_{klm}$	$\mathbf{e}_k + \mathbf{e}_l$	?	0_A^{+-}	$\gamma_4 \otimes \tau_m \tau_5$	π	0_A^{+-}	$\gamma_5 \otimes \tau_m \tau_4$
14	$\eta_m \xi_m \eta_k \xi_l$	$\mathbf{e}_k + \mathbf{e}_l$	b_1	1_A^{+-}	$\gamma_k \gamma_m \otimes \tau_l \tau_m$	ρ	1_A^{--}	$\gamma_l \otimes \tau_k$
15	$\eta_4 \xi_4 \eta_m \xi_m \eta_k \xi_l$	$\mathbf{e}_k + \mathbf{e}_l$	a_1	1_A^{++}	$\gamma_l \gamma_5 \otimes \tau_k \tau_5$	ρ	1_A^{--}	$\gamma_l \gamma_4 \otimes \tau_k \tau_4$
16	$\eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	η	0_S^{+-}	$\gamma_4 \gamma_5 \otimes \mathbb{I}$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_4 \tau_5$
17	$\eta_4 \xi_4 \eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	π	0_A^{+-}	$\gamma_5 \otimes \tau_4$	?	0_A^{+-}	$\gamma_4 \otimes \tau_5$
18	$\eta_k \xi_k \epsilon \eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_k \tau_5$	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_k \tau_4$
19	$\eta_4 \xi_4 \eta_k \xi_k \epsilon \eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	ρ	1_A^{--}	$\gamma_k \otimes \tau_l \tau_m$	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_k$

表 3: $R = S, A$ means singlet or adjoint. For $P = -\sigma_s$, spin matrix is $\gamma_S \gamma_4 \gamma_5$, taste matrix is $\tau_F \tau_4 \tau_5$. $k, l, m = 1, 2, 3$ or $k, l, m = x, y, z$ with $k \neq l \neq m$. **If we do not have s quark, how can we have η, ω, f_0 and h_1 ?**

where τ are taste gamma matrices. (Note that, in the previous subsection, we use $\tau = \gamma$, in Eqs. (410 - 413), it is always γ^* , so here we directly define $\tau_\mu = \gamma_\mu^*$)

γ_S and τ_F are related to δ as

$$\gamma_S = \gamma_1^{S_x} \gamma_2^{S_y} \gamma_3^{S_z} \gamma_4^{S_t}, \quad \tau_F = \tau_1^{F_x} \tau_2^{F_y} \tau_3^{F_z} \tau_4^{F_t}, \quad \delta = S + F \quad (419)$$

The pion and rhon are listed in Table. 3 (This table is copied from Ref. [28])

The η, ξ, ϵ in Table 3 are

$$\begin{aligned} \eta_\mu(n) &= (-1)^{\sum_{i<\mu} n_i}, \quad \xi_\mu(n) = (-1)^{\sum_{i>\mu} n_i}, \quad \eta_\mu(\nu) = \xi_\nu(\mu), \\ \epsilon(n) &= (-1)^{\sum_{i=1}^4 n_i} = \eta_5, \end{aligned} \quad (420)$$

and ϵ_{ijk} is the Levi-Civita tensor.

5.5.4 Coulomb Wall Source

For staggered fermion, to simplify the calculation, one uses the wall source, which in fact correspond to a modified meson operator defined as

$$\begin{aligned} \tilde{M}_{\mathbf{A}, \mathbf{B}} &= \sum_{\mathbf{x}} \bar{\chi}_c(2\mathbf{x} + \mathbf{A}) \sum_{\mathbf{y}} \chi_c(2\mathbf{y} + \mathbf{B}) \\ \tilde{M}(t) &= \sum_{\mathbf{A}} S(\mathbf{A}) \tilde{M}_{\mathbf{A}, \mathbf{A}+\delta} \end{aligned} \quad (421)$$

$\tilde{M}(t)$ carries same quantum number as $M(t)$ [28]. This is only a function of δ and t

Since it is a wall source, to make sure **gauge invariance**, this correlator should be measured in **Coulomb gauge**. How to fix the gauge is introduced in Sec. 4.1.4.

The expression of the sources are

$$s_{\mathbf{A}, \mathbf{c}}(\mathbf{n}, t) = \sum_{\mathbf{x}} \delta_{c, c'} \delta_{2\mathbf{x} + \mathbf{A}, \mathbf{n}} \delta_{t, n_4} \quad (422)$$

idx	$S(\mathbf{x})$	δ	$P = +\sigma_s$			$P = -\sigma_s$		
			M	J_R^{PC}	$\gamma_S \otimes \tau_F$	M	J_R^{PC}	$\gamma_S \otimes \tau_F$
0	1	0	f_0	0_S^{++}	$\mathbb{I} \otimes \mathbb{I}$	π	0_A^{-+}	$\gamma_4 \gamma_5 \otimes \tau_4 \tau_5$
1	ϵ	0	?	0_A^{+-}	$\gamma_4 \otimes \tau_4$	π	0_A^{-+}	$\gamma_5 \otimes \tau_5$
2	$\eta_k \xi_k \epsilon$	0	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_k \tau_5$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_k \tau_4$
3	$\eta_k \xi_k$	0	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_l \tau_m$	ρ	1_A^{--}	$\gamma_k \otimes \tau_k$
4	η_k	$\mathbf{e}_k$	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_4 \tau_5$	ω	1_S^{--}	$\gamma_k \otimes \mathbb{I}$
5	$\eta_k \epsilon$	$\mathbf{e}_k$	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_5$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_4$
6	$\xi_k \epsilon$	$\mathbf{e}_k$	?	0_A^{+-}	$\gamma_4 \otimes \tau_k \tau_4$	π	0_A^{-+}	$\gamma_5 \otimes \tau_k \tau_5$
7	ξ_k	$\mathbf{e}_k$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_k$	π	0_A^{-+}	$\gamma_4 \gamma_5 \otimes \tau_l \tau_m$
8	$\eta_k \xi_k \eta_l \epsilon$	$\mathbf{e}_l$	a_1	1_A^{++}	$\gamma_m \gamma_5 \otimes \tau_k \tau_4$	ρ	1_A^{--}	$\gamma_m \gamma_4 \otimes \tau_k \tau_5$
9	$\eta_k \xi_k \eta_l$	$\mathbf{e}_l$	b_1	1_A^{+-}	$\gamma_k \gamma_l \otimes \tau_k$	ρ	1_A^{--}	$\gamma_m \otimes \tau_l \tau_m$
10	$\eta_k \eta_l \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	h_1	1_S^{+-}	$\gamma_k \gamma_l \otimes \mathbb{I}$	ρ	1_A^{--}	$\gamma_m \otimes \tau_4 \tau_5$
11	$\eta_k \eta_l \epsilon \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	a_1	1_A^{++}	$\gamma_m \gamma_5 \otimes \tau_4$	ρ	1_A^{--}	$\gamma_m \gamma_4 \otimes \tau_5$
12	$\xi_k \xi_l \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_k \tau_l$	π	0_A^{-+}	$\gamma_4 \gamma_5 \otimes \tau_m$
13	$\xi_k \xi_l \epsilon \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	?	0_A^{+-}	$\gamma_4 \otimes \tau_m \tau_5$	π	0_A^{-+}	$\gamma_5 \otimes \tau_m \tau_4$
14	$\eta_m \xi_m \eta_k \xi_l \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	b_1	1_A^{+-}	$\gamma_k \gamma_m \otimes \tau_l \tau_m$	ρ	1_A^{--}	$\gamma_l \otimes \tau_k$
15	$\eta_m \xi_m \eta_k \xi_l \epsilon \epsilon_{23}$	$\mathbf{e}_k + \mathbf{e}_l$	a_1	1_A^{++}	$\gamma_l \gamma_5 \otimes \tau_k \tau_5$	ρ	1_A^{--}	$\gamma_l \gamma_4 \otimes \tau_k \tau_4$
16	$\eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	a_0	0_A^{++}	$\mathbb{I} \otimes \tau_4 \tau_5$	η	0_S^{-+}	$\gamma_4 \gamma_5 \otimes \mathbb{I}$
17	$\eta_1 \eta_2 \eta_3 \epsilon$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	?	0_A^{+-}	$\gamma_4 \otimes \tau_5$	π	0_A^{-+}	$\gamma_5 \otimes \tau_4$
18	$\eta_k \xi_k \epsilon \eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	a_1	1_A^{++}	$\gamma_k \gamma_5 \otimes \tau_k \tau_4$	ρ	1_A^{--}	$\gamma_k \gamma_4 \otimes \tau_k \tau_5$
19	$\eta_k \xi_k \eta_1 \eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$	b_1	1_A^{+-}	$\gamma_l \gamma_m \otimes \tau_k$	ρ	1_A^{--}	$\gamma_k \otimes \tau_l \tau_m$

表 4: In $t - x - y - z$ order. In this case, $\eta_i = (-1)^{\sum_{j < i} n_j}$, $\xi_i = (-1)^{\sum_{j < i} n_j}$, $\epsilon = (-1)^{\sum_i n_i}$, where the sum is over $n_{x,y,z}$. Note that, for some k , there is a sign difference between $k = 3, l = 1, m = 2$ and others, which cannot be characteristic using ϵ_{klm} , we use a ϵ_{13} instead, for $k = 1, l = 2, m = 3$ and $k = 2, l = 3, m = 1$, $\epsilon_{23} = +1$, for $k = 3, m = 1, l = 2$, $\epsilon_{23} = -1$. Note that, sometimes there is a -1 factor on $+\sigma$ or $-\sigma$ which can be absorbed in to $A^\pm$ (where $A^\pm$ is coefficient of correlators $C^\pm(t) = A^\pm \left(e^{-m^\pm t} + e^{-m^\pm (L_t - t)} \right)$), therefore omitted.

idx	$S(\mathbf{x})$	$S(\mathbf{x})$	$S(\mathbf{x})$	δ
0	1	1	1	0
1	$\eta_4 \xi_4$	η_4	η_4	0
2	$\eta_k \xi_k \epsilon$	$\xi_1 \eta_4$	η_2	0
3	$\eta_4 \xi_4 \eta_k \xi_k \epsilon$	ξ_1	ξ_1	0
4	η_k	1	1	$\mathbf{e}_x$
5	$\eta_4 \xi_4 \eta_k$	η_4	η_4	$\mathbf{e}_x$
6	$\xi_k \epsilon$	$\xi_1 \eta_4$	η_2	$\mathbf{e}_x$
7	$\eta_4 \xi_4 \xi_k \epsilon$	ξ_1	ξ_1	$\mathbf{e}_x$
8	$\eta_k \xi_k \eta_l \epsilon$	$\xi_1 \eta_2 \eta_4$	1	$\mathbf{e}_y$
9	$\eta_4 \xi_4 \eta_k \xi_k \eta_l \epsilon$	$\xi_1 \eta_2$	η_4	$\mathbf{e}_y$
10	$\eta_k \eta_l \epsilon_{klm}$	η_2	η_2	$\mathbf{e}_x + \mathbf{e}_y$
11	$\eta_4 \xi_4 \eta_k \eta_l \epsilon_{klm}$	$\eta_4 \eta_2$	ξ_1	$\mathbf{e}_x + \mathbf{e}_y$
12	$\xi_k \xi_l \epsilon_{klm}$	$\xi_1 \xi_2$	$\xi_1 \xi_2$	$\mathbf{e}_x + \mathbf{e}_y$
13	$\eta_4 \xi_4 \xi_k \xi_l \epsilon_{klm}$	$\eta_4 \xi_1 \xi_2$	$\eta_2 \xi_2$	$\mathbf{e}_x + \mathbf{e}_y$
14	$\eta_m \xi_m \eta_k \xi_l$	$\eta_3 \xi_2$	η_4	$\mathbf{e}_x + \mathbf{e}_y$
15	$\eta_4 \xi_4 \eta_m \xi_m \eta_k \xi_l$	$\eta_4 \eta_3 \xi_2$	1	$\mathbf{e}_x + \mathbf{e}_y$
16	$\eta_1 \eta_2 \eta_3$	$\eta_2 \eta_3$	$\eta_2 \eta_3$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$
17	$\eta_4 \xi_4 \eta_1 \eta_2 \eta_3$	$\eta_4 \eta_2 \eta_3$	$\eta_2 \xi_2$	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$
18	$\eta_k \xi_k \epsilon \eta_1 \eta_2 \eta_3$	$\xi_1 \eta_4 \eta_2 \eta_3$	η_3	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$
19	$\eta_4 \xi_4 \eta_k \xi_k \epsilon \eta_1 \eta_2 \eta_3$	$\xi_1 \eta_2 \eta_3$	ξ_2	$\mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z$

表 5: Simplified version, using the fact that $A_4 = 0$, so $\eta_1 = \xi_3 = \xi_4 = 1$, $\epsilon = \eta_4$. Then, we take explicitly $k = x$

5.5.5 Modified operator

The correlation function is defined as overlap between sources.

Then

$$\begin{aligned}
C(\delta, t) &= \langle \tilde{M}_\delta(t) \overline{\tilde{M}_\delta(0)} \rangle \\
&= \sum_{c_1, c_2, \mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \left(\sum_{\mathbf{x}_1} \bar{\chi}_{c_1}(2\mathbf{x}_1 + \mathbf{A}_1, t) \sum_{\mathbf{y}_1} \chi_{c_1}(2\mathbf{x}_1 + \mathbf{A}_1 + \delta, t) \right. \\
&\quad \left. \sum_{\mathbf{y}_2} \bar{\chi}_{c_2}(2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0) \sum_{\mathbf{x}_2} \chi_{c_2}(2\mathbf{x}_2 + \mathbf{A}_2, 0) \right) \\
&= \sum_{c_1, c_2, \mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \\
&\quad \times \left\langle \sum_{\mathbf{y}_1} \chi_{c_1}(2\mathbf{x}_1 + \mathbf{A}_1 + \delta, t) \sum_{\mathbf{x}_2} \chi_{c_2}(2\mathbf{x}_2 + \mathbf{A}_2, 0) \sum_{\mathbf{y}_2} \bar{\chi}_{c_2}(2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0) \sum_{\mathbf{x}_1} \bar{\chi}_{c_1}(2\mathbf{x}_1 + \mathbf{A}_1, t) \right\rangle
\end{aligned} \tag{423}$$

Then use

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = (M^{-1})_{1,3} ((M^\dagger)^{-1})_{4,2}^* - (M^{-1})_{1,4} ((M^\dagger)^{-1})_{3,2}^*. \tag{424}$$

The second term is the ‘disconnected pieces’ (propagator within same time slice) which is expensive to calculate because one need inverses of sources at different time, the first term is the ‘connected pieces’ (propagator connecting different time slices). Connected pieces are cheaper because we only need inverses of sources at $t = 0$. I don’t know why but we drop the disconnected pieces, so that

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = (M^{-1})_{1,3} ((M^\dagger)^{-1})_{4,2}^* = (M^{-1})_{1,3} (M^{-1})_{2,4}. \tag{425}$$

So

$$\begin{aligned}
C(\delta, t) &= \sum_{c_1, c_2, \mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \\
&\quad \times \sum_{\mathbf{x}_1, \mathbf{y}_1, \mathbf{x}_2, \mathbf{y}_2} (M^{-1})_{2\mathbf{x}_1 + \mathbf{A}_1 + \delta, t, c_1; 2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0, c_2} \left(M^{\dagger^{-1}} \right)_{2\mathbf{x}_1 + \mathbf{A}_1, t, c_1; 2\mathbf{x}_2 + \mathbf{A}_2, 0, c_2}^*
\end{aligned} \tag{426}$$

π			ρ		
1-	$\gamma_5 \otimes \tau_5 (\pi\text{PS})$	1.18	3-	$\gamma_k \otimes \tau_k (\text{VT})$	1.70
0-	$\gamma_4 \gamma_5 \otimes \tau_4 \tau_5 (\tilde{\pi}\text{SC})$	2.13	2-	$\gamma_k \gamma_4 \otimes \tau_k \tau_4 (\text{PV})$	2.00
6+	$\gamma_5 \otimes \tau_k \tau_5 (\pi_3)$	2.22	5+	$\gamma_k \gamma_4 \otimes \tau_4 (\rho_3^A)$	2.26
7+	$\gamma_4 \gamma_5 \otimes \tau_l \tau_m (\tilde{\pi}_3)$	2.20	9+	$\gamma_m \otimes \tau_l \tau_m (\rho_6^C)$	2.46
13-	$\gamma_5 \otimes \tau_m \tau_4$	2.72	15-	$\gamma_l \gamma_4 \otimes \tau_k \tau_4 (\rho_6^B)$	2.97
12-	$\gamma_4 \gamma_5 \otimes \tau_m$	2.90	8+	$\gamma_m \gamma_4 \otimes \tau_k \tau_5 (\rho_6^D)$	2.57
17+	$\gamma_5 \otimes \tau_4$	3.07	11-	$\gamma_m \gamma_4 \otimes \tau_5$	3.27
16+	$\gamma_4 \gamma_5 \otimes \mathbb{I}(\eta)$	2.93	14-	$\gamma_l \otimes \tau_k (\rho_6^A)$	3.05
			10-	$\gamma_m \otimes \tau_4 \tau_5$	3.22
			19+	$\gamma_k \otimes \tau_l \tau_m (\rho_3^C)$	2.20
			18+	$\gamma_k \gamma_4 \otimes \tau_k \tau_5 (\rho_3^D)$	2.28
			4+	$\gamma_k \otimes \mathbb{I}(\omega)$	2.20

表 6: Tested result with $\beta = 5.0, N_f = 2, am = 0.25, 4^3 \times 8$ with 20 + 100 configurations. The order of mass of pions are same as most of the Refs. In Ref, they find $\pi(\tau_5) < \pi(\tau_4 \tau_5) < \pi(\tau_k \tau_5) < \pi(\tau_l \tau_m) < \pi(\tau_m \tau_4) < \pi(\tau_m) < \pi(\tau_4) < \omega$. In Ref, they find $PS < VT < PV < SC$ which is only partly consistent. In Ref. they find $\pi < \tilde{\pi} < \pi_3 < \tilde{\pi}_3 < \rho_6^A$ which is consistent. In Ref they find $\pi(\tau_5) < \pi(\tau_k \tau_5) < \pi(\tau_4 \tau_5) < \pi(\tau_m \tau_4) < \pi(\tau_l \tau_m) < \pi(\tau_m) < \pi(\tau_4) < \omega$.

Which is

$$C(\delta, t) = \sum_{c_1, c_2, \mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \phi(\mathbf{A}_1 + \delta, c_1, t) D^{-1} \phi(\mathbf{A}_2 + \delta, c_2, 0) \left(\phi(\mathbf{A}_1, c_1, t) D^{\dagger -1} \phi(\mathbf{A}_2, c_2, 0) \right)^* \quad (427)$$

It seems we have compute 24 inverses rather than 8.

The results:

5.5.6 The propagators of staggered meson

The correlation function is defined as overlap between **operator** and **source** (**It is fuzzy in literatures, it is my guess, one can measure it this way**).

Then

$$\begin{aligned} C(t) &= \langle M_\delta(t) \overline{\tilde{M}_\delta(0)} \rangle \\ &= \sum_{\mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \langle \sum_{\mathbf{x}_1} D_{\mathbf{A}_1} \bar{\chi}_{c_1}(2\mathbf{x}_1, t) D_{\mathbf{A}_1 + \delta} \chi_{c_1}(2\mathbf{x}_1, t) \\ &\quad \sum_{\mathbf{y}_2} \bar{\chi}_{c_2}(2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0) \sum_{\mathbf{x}_2} \chi_{c_2}(2\mathbf{x}_2 + \mathbf{A}_2, 0) \rangle \end{aligned} \quad (428)$$

Eq. (366) in the case of 4 Grassman numbers is

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = (M^{-1})_{1,3} (M^{-1})_{2,4} - (M^{-1})_{1,4} (M^{-1})_{2,3}. \quad (429)$$

Using $(A^\dagger)^{-1} = (A^{-1})^\dagger$, this is

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = (M^{-1})_{1,3} ((M^\dagger)^{-1})_{4,2}^* - (M^{-1})_{1,4} ((M^\dagger)^{-1})_{3,2}^*. \quad (430)$$

The first term is the ‘disconnected pieces’ (propagator within same time slice) which is expansive to calculate because one need inverses of sources at different time, the second term is the ‘connected pieces’ (propagator connecting different time slices). Connected pieces are cheaper because we only need inverses of sources at $t = 0$. I don’t known why but we drop the disconnected pieces, so that

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = - (M^{-1})_{1,4} ((M^\dagger)^{-1})_{3,2}^*. \quad (431)$$

So

$$\begin{aligned}
 C(t) &= \langle M_\delta(t) \overline{\tilde{M}_\delta(0)} \rangle \\
 &= - \sum_{\mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \sum_{\mathbf{x}_1} \\
 &\quad \langle D_{\mathbf{A}_1 + \delta} \chi_{c_1}(2\mathbf{x}_1, t) \sum_{\mathbf{x}_2} \chi_{c_2}(2\mathbf{x}_2 + \mathbf{A}_2, 0) D_{\mathbf{A}_1} \bar{\chi}_{c_1}(2\mathbf{x}_1, t) \sum_{\mathbf{y}_2} \bar{\chi}_{c_2}(2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0) \rangle \\
 &= \sum_{\mathbf{A}_1 \mathbf{A}_2} S(\mathbf{A}_1) S(\mathbf{A}_2) \sum_{\mathbf{x}_1} (D_{\mathbf{A}_1 + \delta} \phi_{sink} D^{-1} \phi_{sour}(\mathbf{A}_2 + \delta)) (D_{\mathbf{A}_1} \phi_{sink} (D^\dagger)^{-1} \phi_{sour}(\mathbf{A}_2))^*
 \end{aligned} \tag{432}$$

Use the fact that $D = D_{st0} + m$, $D^\dagger = -D_{st0} + m$, $DD^\dagger = D^\dagger D = m^2 - D_{st0}^2$. So $D^{-1} = D^\dagger (D^\dagger D)^{-1}$ and $(D^\dagger)^{-1} = D (D^\dagger D)^{-1}$. Let the inverse of source be $v = (D^\dagger D)^{-1} \phi_{source}$, we need both $D^\dagger v$ and Dv .

5.5.7 Implementation Of Sink Operator

As $w_1 = D^\dagger v$ and $w_2 = Dv$ obtained, the effect of sink operator is to pick the element of $w_{1,2}$ vectors.

5.5.8 Fit of staggered correlator

The correlator contains two meson propagators, so the fitting is

$$C(t) = A_+ (\exp(-m_+ t) + \exp(-m_+ (L_t - t))) + (-1)^t A_- (\exp(-m_- t) + \exp(-m_- (L_t - t))) \tag{433}$$

where $m_\pm$ is the mass of $P = \pm \sigma_s$.

5.5.9 Summary of Staggered Spectrum

Before summarize, we need to clarify those things.

• Q1: What is a propagator?

The propagator is just $\langle \chi_a \bar{\chi}_b \rangle_F$. Usually, it is $\langle \chi_a(m) \bar{\chi}_b(n) \rangle_F$. It is a number (complex number) which can be calculated by using

$$\langle \psi^f(n)_{a,c_1} \bar{\psi}_{b,c_2}^f(m) \rangle = -D_{f,a,b,c_1,c_2}^{-1}(n, m). \tag{434}$$

This is just the (n, m) element of the D^{-1} matrix. The elements can be extracted (projected) out by sink and sources.

• Q2: What is average of 4 fields?

We need to calculate $\langle \bar{\chi}_a \chi_b \bar{\chi}_c \chi_d \rangle_F$. What is that?

For a single flavor, That depends on the coordinate, for example

$$\begin{aligned}
 \sum_a \sum_b \langle \bar{\chi}_a \chi_b \bar{\chi}_b \chi_a \rangle_F &= - \sum_a \langle \chi_a \left(\sum_b \chi_b \bar{\chi}_b \right) \bar{\chi}_a \rangle_F = -tr[D^{-1}(a|b) D^{-1}(b|a)] \\
 \sum_a \sum_b \langle \bar{\chi}_a \chi_a \bar{\chi}_b \chi_b \rangle_F &= \sum_a \langle \chi_a \bar{\chi}_a \rangle_F \sum_b \langle \chi_b \bar{\chi}_b \rangle_F = tr[D^{-1}(a|b)] tr[D^{-1}(a|b)]
 \end{aligned} \tag{435}$$

We are lucky that, we do NOT have the sums. It seems that we should pick the pairs according to t , that we calculate

$$\begin{aligned}
 G_1(t, 0) &= \langle \sum_{\mathbf{y}_1} \chi_{c_1}(2\mathbf{y}_1 + \mathbf{A}_1 + \delta, t) \sum_{\mathbf{y}_2} \bar{\chi}_{c_2}(2\mathbf{y}_2 + \mathbf{A}_2 + \delta, 0) \rangle \\
 G_2(t, 0) &= \langle \sum_{\mathbf{x}_2} \chi_{c_2}(2\mathbf{x}_2 + \mathbf{A}_2, 0) \sum_{\mathbf{x}_1} \bar{\chi}_{c_1}(2\mathbf{x}_1 + \mathbf{A}_1, t) \rangle
 \end{aligned} \tag{436}$$

In color space, they are both 3×3 matrices, and in $\mathbf{A}$ space, they are 8×8 matrices, we need to calculate $9 \times 64 = 576$ elements.

• Q3: What about color index?

It seems incorrect to ignore the color index, and put a $(1, 1, 1)$ in the source.

• Step1: Gauge fixing to Coulomb gauge

• Step2: Build 24 source fields using $s_{\mathbf{A},c}(\mathbf{n}, 0) = \sum_{\mathbf{x}} \delta_{2\mathbf{x}+\mathbf{A},\mathbf{n}} \delta_{0,n_4} \delta_{c,c'}$

• Step3: Calculate 3×8 inverse fields using $v_{\mathbf{A},c} = (D^\dagger D)^{-1} s_{\mathbf{A},c}(\mathbf{n}, 0)$

• Step4: Calculate $w_{1,\mathbf{A},c} = D^\dagger v_{\mathbf{A},c}$ and $w_{2,\mathbf{A},c} = Dv_{\mathbf{A},c}$.

• Step5: Use sink operator to pick elements from w_1 and w_2 to calculate $\langle D_{\mathbf{A}_1 + \delta} \chi(\sum \bar{\chi}(2\mathbf{x} + \mathbf{A}_2 + \delta)) \rangle$ and $\langle D_{\mathbf{A}_1} \chi(\sum \bar{\chi}(2\mathbf{x} + \mathbf{A}_2)) \rangle$.

Note: the result of correlation function is real number not complex number.

π		ρ	
$\gamma_5 \otimes \tau_5 (\pi\text{PS})$	1.17	$\gamma_k \otimes \tau_k (\text{VT})$	1.71
$\gamma_4 \gamma_5 \otimes \tau_4 \tau_5 (\tilde{\pi}\text{SC})$	1.75	$\gamma_k \gamma_4 \otimes \tau_k \tau_4 (\text{PV})$	1.87
$\gamma_5 \otimes \tau_k \tau_5 (\pi_3)$	2.14	$\gamma_k \gamma_4 \otimes \tau_4 (\rho_3^A)$	2.18
$\gamma_4 \gamma_5 \otimes \tau_l \tau_m (\tilde{\pi}_3)$	2.59	$\gamma_m \otimes \tau_l \tau_m (\rho_6^C)$	2.27
$\gamma_5 \otimes \tau_m \tau_4$	2.67	$\gamma_l \gamma_4 \otimes \tau_k \tau_4 (\rho_6^B)$	2.55
$\gamma_4 \gamma_5 \otimes \tau_m$	2.95	$\gamma_m \gamma_4 \otimes \tau_k \tau_5 (\rho_6^D)$	2.55
$\gamma_5 \otimes \tau_4$	3.22	$\gamma_m \gamma_4 \otimes \tau_5$	2.84
$\gamma_4 \gamma_5 \otimes \mathbb{I}(\eta)$	3.08	$\gamma_l \otimes \tau_k (\rho_6^A)$	2.91
		$\gamma_m \otimes \tau_4 \tau_5$	2.96
		$\gamma_k \otimes \tau_l \tau_m (\rho_3^C)$	3.39
		$\gamma_k \gamma_4 \otimes \tau_k \tau_5 (\rho_3^D)$	3.94
		$\gamma_k \otimes \mathbb{I}(\omega)$	2.59

表 7: Tested result with $\beta = 5.0, N_f = 2, am = 0.25, 4^3 \times 8$ with 20 + 100 configurations. The order of mass of pions are same as most of the Refs. In Ref, they find $\pi(\tau_5) < \pi(\tau_4 \tau_5) < \pi(\tau_k \tau_5) < \pi(\tau_l \tau_m) < \pi(\tau_m \tau_4) < \pi(\tau_m) < \pi(\tau_4) < \omega$. In Ref, they find $PS < VT < PV < SC$ which is only partly consistent. In Ref. they find $\pi < \tilde{\pi} < \pi_3 < \tilde{\pi}_3 < \rho_6^A$ which is consistent. In Ref they find $\pi(\tau_5) < \pi(\tau_k \tau_5) < \pi(\tau_4 \tau_5) < \pi(\tau_m \tau_4) < \pi(\tau_l \tau_m) < \pi(\tau_m) < \pi(\tau_4) < \omega$.

5.5.10 Some explanation

The meson operator can be written as

$$M(x) = \bar{q}_{f_1}(x) \Gamma_A \otimes \Gamma_B^\dagger q_{f_2}(x). \quad (437)$$

Usually,

$$M(x) = \frac{1}{16} \sum_{n \in h} s_n \bar{\chi}_{f_1}(x+n) \chi_{f_2}(x+n+\delta_n). \quad (438)$$

where n are corners of hypercube, $s_n = \pm 1$ are signs, δ_n are shifts. The sum over color index is implied in the vector dot, $\bar{\chi}\chi$, where $\chi(x)$ is a SU3-vector.

When $\Gamma_A = \Gamma_B$, $\delta_n = 0$, $M(x)$ are so called ‘local operators’.

$$M_\Gamma(x) = \frac{1}{16} \sum_{n \in h} s_n \bar{\chi}_{f_1}(x+n) \chi_{f_2}(x+n). \quad (439)$$

As an example,

$$M_{\gamma_5}(x) = \frac{1}{16} \sum_{n \in h} (-1)^{\sum_{i=1,2,3,4} n_i} \bar{\chi}_{f_1}(x+n) \chi_{f_2}(x+n) \quad (440)$$

We also need translational invariance,

$$\langle \bar{\chi}(x+1) \chi(x+1) \bar{\chi}(1) \chi(1) \rangle = \langle \bar{\chi}(x) \chi(x) \bar{\chi}(0) \chi(0) \rangle \quad (441)$$

For simplicity, we use,

$$\sum_x = \sum_{x=0,1}, \quad \sum_{xy} = \sum_{x,y=0,1}, \quad \sum_{xyz} = \sum_{x,y,z=0,1}, \quad \sum_{xyzt} = \sum_{x,y,z,t=0,1} \quad (442)$$

So

$$\begin{aligned} M_{\gamma_5}(n) &= \frac{1}{16} \sum_{xyzt} (-1)^{x+y+z+t} \bar{\chi}_{f_1}(n_x+x, n_y+y, n_z+z, n_t+t) \chi_{f_2}(n_x+x, n_y+y, n_z+z, n_t+t) \\ &= \frac{1}{16} \left(\sum_{xyz} (-1)^{x+y+z} \bar{\chi}_{f_1}(n_x+x, n_y+y, n_z+z, n_t) \chi_{f_2}(n_x+x, n_y+y, n_z+z, n_t) \right. \\ &\quad \left. - \sum_{xyz} (-1)^{x+y+z} \bar{\chi}_{f_1}(n_x+x, n_y+y, n_z+z, n_t+1) \chi_{f_2}(n_x+x, n_y+y, n_z+z, n_t+1) \right) \end{aligned} \quad (443)$$

For simplicity, we use $n + \delta$ to implicitly denote $(n_x + x, n_y + y, \dots)$
so that, for simplicity, we ignore the sum of indices

$$\begin{aligned}
& \bar{M}_{\gamma_5}(n)M_{\gamma_5}(0) \\
&= \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) + \left(\sum s_x \bar{\chi}_{f_2}(\vec{x}, n_t + 1) \chi_{f_1}(\vec{x}, n_t + 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 1) \chi_{f_2}(\vec{x}, 1) \right) \\
&- \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) - \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 1) \chi_{f_2}(\vec{x}, 1) \right) \\
&= 2 \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) \\
&- \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) - \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t - 1) \chi_{f_1}(n + \vec{x}, n_t - 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right)
\end{aligned} \tag{444}$$

We have used translational invariance.

Similarly,

$$\begin{aligned}
& \sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(n + \vec{x}, 0) \chi_{f_2}(n + \vec{x}, 0) \\
&= \left(\sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z, n_t) \chi_{f_1}(n + xy, n_z, n_t) - \sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z + 1, n_t) \chi_{f_1}(n + xy, n_z + 1, n_t) \right) \\
&\times \left(\sum s_{xy} \bar{\chi}_{f_1}(xy, 0, 0) \chi_{f_1}(xy, 0, 0) - \sum s_{xy} \bar{\chi}_{f_1}(xy, 1, 0) \chi_{f_2}(xy, 1, 0) \right) \\
&= \left(2 \sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z, n_t) \chi_{f_1}(n + xy, n_z, n_t) - \sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z - 1, n_t) \chi_{f_1}(\vec{x}, z - 1, t) - \sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z + 1, n_t) \chi_{f_1}(n + xy, n_z + 1, n_t) \right) \\
&\times \sum s_{xy} \bar{\chi}_{f_1}(xy, 0, 0) \chi_{f_2}(xy, 0, 0)
\end{aligned} \tag{445}$$

If $\bar{\chi}(\vec{x}, t_1) \chi(\vec{y}, t_2) = f(|\vec{x} - \vec{y}|, t_1 - t_2)$, than $\bar{\chi}(\vec{x}, z + 1, t) \chi(\vec{y}, 0, 0) = \bar{\chi}(\vec{x}, z - 1, t) \chi(\vec{y}, 0, 0)$

$$\begin{aligned}
& \sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(n + \vec{x}, 0) \chi_{f_2}(n + \vec{x}, 0) \\
&= 2 \left(\sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z, n_t) \chi_{f_1}(n + xy, n_z, n_t) - \sum s_{xy} \bar{\chi}_{f_2}(n + xy, n_z + 1, n_t) \chi_{f_1}(n + xy, n_z + 1, n_t) \right) \\
&\times \sum s_{xy} \bar{\chi}_{f_1}(xy, 0, 0) \chi_{f_2}(xy, 0, 0) \\
&= 2 \sum_{xyz} s_{xyz} \bar{\chi}_{f_2}(n + xyz, n_t) \chi_{f_1}(n + xyz, n_t) \times \sum_{xy} s_{xy} \bar{\chi}_{f_1}(xy, 0, 0) \chi_{f_2}(xy, 0, 0)
\end{aligned} \tag{446}$$

Using similar steps, it is

$$\begin{aligned}
& \sum_{xyz} s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum_{xyz} s_x \bar{\chi}_{f_1}(n + \vec{x}, 0) \chi_{f_2}(n + \vec{x}, 0) \\
&= 4 \sum_{xyz} s_{xyz} \bar{\chi}_{f_2}(n + xyz, n_t) \chi_{f_1}(n + xyz, n_t) \times \sum_x s_x \bar{\chi}_{f_1}(x, 0, 0, 0) \chi_{f_2}(x, 0, 0, 0) \\
&= 8 \sum_{xyz} s_{xyz} \bar{\chi}_{f_2}(n + xyz, n_t) \chi_{f_1}(n + xyz, n_t) \times \bar{\chi}_{f_1}(0, 0, 0, 0) \chi_{f_2}(0, 0, 0, 0)
\end{aligned} \tag{447}$$

We arrive at

$$\begin{aligned}
& \bar{M}_{\gamma_5}(n)M_{\gamma_5}(0) \\
&= 8 \sum_{xyz} (-1)^{x+y+z} [2 \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) \\
&- \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) - \bar{\chi}_{f_2}(n + \vec{x}, n_t - 1) \chi_{f_1}(n + \vec{x}, n_t - 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0)]
\end{aligned} \tag{448}$$

Similarly

$$\begin{aligned}
& \bar{M}_I(n)M_I(0) \\
&= 8 \sum_{xyz} [2\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) \\
&\quad + \bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) + \bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)] \\
& \bar{M}_{\gamma_4}(n)M_{\gamma_4}(0) \\
&= 8 \sum_{xyz} (-1)^{x+y+z} [2\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) \\
&\quad + \bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) + \bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)] \\
& \bar{M}_{\gamma_4\gamma_5}(n)M_{\gamma_4\gamma_5}(0) \\
&= 8 \sum_{xyz} [2\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) \\
&\quad - \bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) - \bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)]
\end{aligned} \tag{449}$$

Define $C_\Gamma(x, t) = \bar{M}_\Gamma(x, t)M_\Gamma(0)$ as correlation functions, one have

$$\begin{aligned}
(C_I(x, t) + C_{\gamma_4\gamma_5}(x, t)) &= 32 \sum_{xyz} (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) \\
(C_I(x, t) - C_{\gamma_4\gamma_5}(x, t)) &= 16 \sum_{xyz} (\bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) + 16 \sum_{xyz} (\bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0))
\end{aligned} \tag{450}$$

Which is

$$\begin{aligned}
\sum_t (C_I(x, t) + C_{\gamma_4\gamma_5}(x, t)) &= 32 \sum_{xyz} \sum_{n_t \in \text{even}} (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) \\
\sum_t (C_I(x, t) - C_{\gamma_4\gamma_5}(x, t)) &= 32 \sum_{xyz} \sum_{n_t \in \text{odd}} (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0))
\end{aligned} \tag{451}$$

I don't know why, but we remove the sum over t and denote C as $C(n_t)$., then denote

$$C_\Gamma(n, n_t) = K_\Gamma (e^{-m_\Gamma n_t} + e^{-m_\Gamma(T-n_t)}) \tag{452}$$

So

$$\begin{aligned}
& \sum_{xyz} (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) \\
& \propto K_I (e^{-m_I t} + e^{-m_I(T-n_t)}) + (-1)^{n_t} K_{\gamma_4\gamma_5} (e^{-m_{\gamma_4\gamma_5} n_t} + e^{-m_{\gamma_4\gamma_5}(T-n_t)})
\end{aligned} \tag{453}$$

similarly

$$\begin{aligned}
& \sum_{xyz} (-1)^{x+y+z} (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) \\
& \propto K_{\gamma_4} (e^{-m_{\gamma_4} t} + e^{-m_{\gamma_4}(T-n_t)}) + (-1)^{n_t} K_{\gamma_5} (e^{-m_{\gamma_5} n_t} + e^{-m_{\gamma_5}(T-n_t)})
\end{aligned} \tag{454}$$

They are the first two lines in Table 3. Using the fact that A_4 in Table 3 is always 0, $\eta_4\zeta_4 = (-1)^{x+y+z}$.

Except for that, generally,

$$\begin{aligned}
& \sum_{xyz} \phi(xyz) (\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)) \\
& \propto K_\Gamma (e^{-m_\Gamma t} + e^{-m_\Gamma(T-n_t)}) + (-1)^{n_t} K_{\Gamma\gamma_4\gamma_5} (e^{-m_{\Gamma\gamma_4\gamma_5} n_t} + e^{-m_{\Gamma\gamma_4\gamma_5}(T-n_t)})
\end{aligned} \tag{455}$$

For example,

$$\begin{aligned}
& \bar{M}_{\gamma_1}(n)M_{\gamma_1}(0) \\
&= 8 \sum_{xyz} (-1)^{y+z} [2\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) \\
&\quad - \bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) - \bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)] \\
& \bar{M}_{\gamma_3\gamma_2}(n)M_{\gamma_3\gamma_2}(0) \\
&= 8 \sum_{xyz} (-1)^{y+z} [2\bar{\chi}_{f_2}(n+\vec{x}, n_t)\chi_{f_1}(n+\vec{x}, n_t)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) \\
&\quad + \bar{\chi}_{f_2}(n+\vec{x}, n_t+1)\chi_{f_1}(n+\vec{x}, n_t+1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0) + \bar{\chi}_{f_2}(n+\vec{x}, n_t-1)\chi_{f_1}(n+\vec{x}, n_t-1)\bar{\chi}_{f_1}(0)\chi_{f_2}(0)]
\end{aligned} \tag{456}$$

	W	Part. interp. Γ	Oppo. parity Γ
PS	1	$\epsilon(1)$	$\pi(\gamma_4\gamma_5)$
VT	$(-1)^x + (-1)^y + (-1)^z$	$b_1(\gamma_i\gamma_5)$	$\rho(\gamma_i\gamma_4)$
PV	$(-1)^{x+y} + (-1)^{y+z} + (-1)^{x+z}$	$a_1(\gamma_j\gamma_k)$	$\rho_2(\gamma_i)$
S	$(-1)^{x+y+z}$	$\sigma(\gamma_4)$	$\pi_2(\gamma_5)$

表 8: Four channels for local operators, see also Ref. [29].

This is the forth line of Table 3, note that $\eta_4\zeta_4\eta_1\zeta_1\epsilon = (-1)^{x+y+z} \times 1 \times 1 \times (-1)^{y+z} \times (-1)^{x+y+z} = (-1)^{y+z}$. $\bar{M}_{\gamma_1}$, $\bar{M}_{\gamma_2}$ and $\bar{M}_{\gamma_3}$ are often combined as one channel, and the correlation to measure is

$$\sum_{xyz} ((-1)^{x+y} + (-1)^{y+z} + (-1)^{x+z}) (\bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0)) \quad (457)$$

Similarly, the $\gamma_i\gamma_5$ and $\gamma_i\gamma_4$ corresponds to the third line of Table 3,

$$\begin{aligned} & \bar{M}_{\gamma_1\gamma_5}(n) M_{\gamma_1\gamma_5}(0) \\ &= 8 \sum_{xyz} (-1)^x [2\bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) \\ & \quad + \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) + \bar{\chi}_{f_2}(n + \vec{x}, n_t - 1) \chi_{f_1}(n + \vec{x}, n_t - 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0)] \\ & \bar{M}_{\gamma_1\gamma_4}(n) M_{\gamma_1\gamma_4}(0) \\ &= 8 \sum_{xyz} (-1)^x [2\bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) \\ & \quad - \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) - \bar{\chi}_{f_2}(n + \vec{x}, n_t - 1) \chi_{f_1}(n + \vec{x}, n_t - 1) \bar{\chi}_{f_1}(0) \chi_{f_2}(0)] \end{aligned} \quad (458)$$

the correlation to measure is

$$\sum_{xyz} ((-1)^x + (-1)^y + (-1)^z) (\bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0)) \quad (459)$$

Those are the only 4 local correlators correspond to only 8 local operators.

5.5.11 Another simple approach

Form the above it can be seen that, if one only cares about local operators, one needs to calculate 4 correlators, they are

$$\sum_{\vec{A}} W(\vec{A}) \left(\bar{\chi}_{f_2}(n + \vec{A}, n_t) \chi_{f_1}(n + \vec{A}, n_t) \bar{\chi}_{f_1}(0) \chi_{f_2}(0) \right) \quad (460)$$

where $\vec{A}$ are 8 corners of a cube, and the corresponding mesons are listed in Table [?]]

Then, using

$$\langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle = (M^{-1})_{1,4} (M^{-1})_{2,3} - (M^{-1})_{1,3} (M^{-1})_{2,4}. \quad (461)$$

Neglecting the ‘connected’ part, it is (From here, we explicitly add the sum of the color index)

$$\begin{aligned} P_A &= \langle \sum_{\vec{A}, c_1, c_2} W(\vec{A}) \left(\bar{\chi}_{c_1, f_2}(n + \vec{A}, n_t) \chi_{c_1, f_1}(n + \vec{A}, n_t) \bar{\chi}_{c_2, f_1}(0) \chi_{c_2, f_2}(0) \right) \rangle \\ &= - \sum_{\vec{A}, c_1, c_2} W(\vec{A}) (D_{f_1}^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right) (D_{f_2}^{-1}) \left(0, c_2 | n + \vec{A}, n_t, c_1 \right) \end{aligned} \quad (462)$$

To calculate it efficiently, we shift the second matrix to $(n + \vec{A}, n_t, c_1 | 0, c_2)$, which means

$$\begin{aligned} P_A &= \langle \sum_{\vec{A}, c_1, c_2} W(\vec{A}) \left(\bar{\chi}_{c_1, f_2}(n + \vec{A}, n_t) \chi_{c_1, f_1}(n + \vec{A}, n_t) \bar{\chi}_{c_2, f_1}(0) \chi_{c_2, f_2}(0) \right) \rangle \\ &= - \sum_{\vec{A}, c_1, c_2} W(\vec{A}) (D_{f_1}^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right) \left((D_{f_2}^\dagger)^{-1} \right) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right)^* \end{aligned} \quad (463)$$

Absorbing the constant -1 into the factors, this is the Eq. (3) of Ref. [29]. It looks that there is no way to merge the colors, at least three different sources for different colors should be used.

The $(D_{f_1}^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right)$ can be obtained by using the point source, that

$$\begin{pmatrix} D^{-1}(0, r | 0, r) \\ D^{-1}(0, g | 0, r) \\ D^{-1}(0, b | 0, r) \\ D^{-1}(1, r | 0, r) \\ D^{-1}(1, g | 0, r) \\ D^{-1}(1, b | 0, r) \\ \dots \\ D^{-1}(n, t, r | 0, r) \\ D^{-1}(n, t, g | 0, r) \\ D^{-1}(n, t, b | 0, r) \end{pmatrix} = D^{-1} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (464)$$

If $f_1 = f_2$, it is more convenient to use $D((DD^\dagger)^{-1}s)$ and $D^\dagger((DD^\dagger)^{-1}s)$ to calculate $D^{-1}s$ and $(D^\dagger)^{-1}s$.

Why is the result real numbers?

1. Gauge fixing to Coulomb gauge, if use wall source.
2. Build 3 wall(or point) sources, $s_c(x) = \sum_{c',x} \delta_{cc'} \delta_{x,0} \delta_{t,0}$, which is 0-corner of all cubic in 0-time slice.
3. Calculate 6 propagators, $p_{f_1, c_1 c_2}(x) = (D_{f_1}^{-1}) s_{c_1}(x, c_2)$, $p_{f_2, c_1 c_2}(x) = ((D_{f_2}^\dagger)^{-1}) s_{c_1}(x, c_2)$, if $f_1 = f_2$, use $(DD^\dagger)^{-1}s$ to build $D^{-1}s$ and $(D^\dagger)^{-1}s$.
4. Now, $p_{f_1}(2n + A)$ is $(D_{f_1}^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right)$, $p_{f_2}(2n + A)$ is $((D_{f_2}^\dagger)^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right)$, linear combine them as $\sum_{\vec{A}, c_1, c_2} W(\vec{A}) (D_{f_1}^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right) ((D_{f_2}^\dagger)^{-1}) \left(n + \vec{A}, n_t, c_1 | 0, c_2 \right)^*$.
5. Now we sum over $2n$, same as Ref. [29], note that, $P_M(t) = \sum_n \sum_A \sum_{c_1, c_2} W(\vec{A}) p_{f_1, c_1 c_2}(2n + A, t) p_{f_2, c_1 c_2}^*(2n + A, t)$, is $P_M(t) = \sum_n W(n) \sum_{c_1, c_2} p_{f_1, c_1 c_2}(n, t) p_{f_2, c_1 c_2}^*(n, t)$.

Note that, since $p_{f_1} \neq p_{f_2}$, there is no guarantee that $p_{f_1} p_{f_2}^*$ is a real number. To avoid imaginary part, we always use $|p_{f_1}| |p_{f_2}|$ before sum over the phase.

5.5.12 The complicated approach

If we do not shift the correlators, then it is impossible to calculate the 8 operators using one (counting color index, it is 3) propagator.

For example, if two propagators are used, 16 states can be built, however, the usual ρ meson cannot be built [30].

To build every time-local operators, at least 8 propagators (counting color index, it is 24) are needed.

To move on, we again write the correlators which are not shifted, they are already listed in Sec. 5.5.10.

$$\begin{aligned} & \bar{M}_{\gamma_5}(n) M_{\gamma_5}(0) \\ &= 2 \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t) \chi_{f_1}(n + \vec{x}, n_t) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) \\ & - \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t + 1) \chi_{f_1}(n + \vec{x}, n_t + 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) - \left(\sum s_x \bar{\chi}_{f_2}(n + \vec{x}, n_t - 1) \chi_{f_1}(n + \vec{x}, n_t - 1) \sum s_x \bar{\chi}_{f_1}(\vec{x}, 0) \chi_{f_2}(\vec{x}, 0) \right) \end{aligned} \quad (465)$$

Denote the sign function as phase ϕ , in this approach,

$$\begin{aligned} \sum_t (C_\Gamma(x, t) + C_{\Gamma\gamma_4\gamma_5}(x, t)) &= 4 \sum_{n_t \in \text{even}} \left(\sum_A \phi(A) \bar{\chi}_{f_2}(n + \vec{A}, n_t) \chi_{f_1}(n + \vec{A}, n_t) \sum_B \phi(B) \bar{\chi}_{f_1}(\vec{B}, t = 0) \chi_{f_2}(\vec{B}, t = 0) \right) \\ \sum_t (C_\Gamma(x, t) - C_{\Gamma\gamma_4\gamma_5}(x, t)) &= 4 \sum_{n_t \in \text{odd}} \left(\sum_A \phi(A) \bar{\chi}_{f_2}(n + \vec{A}, n_t) \chi_{f_1}(n + \vec{A}, n_t) \sum_B \phi(B) \bar{\chi}_{f_1}(\vec{B}, t = 0) \chi_{f_2}(\vec{B}, t = 0) \right) \end{aligned} \quad (466)$$

So, there are $8 \times 8 = 64$ propagators,

$$\begin{aligned} G_{f, AB, t} &= \bar{\chi}_f(n + \vec{A}, n_t) \chi_f(\vec{B}, 0) \\ (C_\Gamma(x, t) + (-1)^{n_t} C_{\Gamma\gamma_4\gamma_5}(x, t)) &= 4 \sum_{AB} \phi(A) \phi(B) G_{f_1, AB, t} (G_{f_2, AB, t}) \end{aligned} \quad (467)$$

At first sight, 64 propagators are needed, however, for one B , 8 elements can be obtained for each A , that is, 1 propagator to calculate $r(B) = \sum_A s(A) \bar{\chi}(n + \vec{A}, n_t) \chi(B, 0)$, then, the step is

1. Gauge fixing to Coulomb gauge, because we will use wall source.
2. Build 3×8 wall sources, $s_{c,B}(x) = \sum_{c',x} \delta_{cc'} \delta_{x,B} \delta_{t,0}$, where B runs over 8 corners of a cubic.
3. Calculate 6×8 propagators, $p_{f_1,c_1 c_2,B}(x) = (D_{f_1}^{-1})_{s_{c_1,B}(x,c_2)}$, $p_{f_2,c_1 c_2,B}(x) = ((D_{f_2}^\dagger)^{-1})_{s_{c_1,B}(x,c_2)}$, if $f_1 = f_2$, use $(DD^\dagger)^{-1}s$ to build $D^{-1}s$ and $(D^\dagger)^{-1}s$.
4. Now, $p_{f_1,B}(2n+A)$ is $G_{f_1,AB,t}$, $p_{f_2,B}^*(2n+A)$ is $G_{f_2,AB,t}^*$, linear combine them as $\sum_{AB} \phi(A)\phi(B)G_{f_1,AB,t}(G_{f_2,AB,t}^*)$.

5.6 Staggered Fermion Charge And Current

5.6.1 Staggered Fermion Gamma matrix

The observables in this sub-sub-section we want to measure are $\langle \bar{\psi} \gamma^\mu \psi \rangle$, and we consider only the taste singlet, so that we care about $\langle \bar{\psi} \gamma^\mu \otimes I \psi \rangle$.

It has been shown that

$$\sum_{\alpha_1, \alpha_2, a} \bar{q}_{\alpha_1, a}(2h) (\gamma_\mu)_{\alpha_1, \alpha_2} q_{\alpha_2, a}(2h) = \frac{1}{16} \eta_\mu(s) \sum_t (\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu)) \quad (468)$$

and

$$\begin{aligned} & \sum_{\alpha_1, \alpha_2, a_1, a_2} \bar{q}_{\alpha_1, a_1}(2h) (\gamma_5)_{\alpha_1, \alpha_2} (\gamma_5^* \gamma_\mu^*)_{a_1, a_2} q_{\alpha_2, a_2}(2h) \\ &= \frac{1}{16} \eta_\mu(s) \sum_t (-\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu)) \end{aligned} \quad (469)$$

Therefore

$$\bar{q}(2h) \gamma_\mu \otimes I q(2h) = \frac{1}{16} \eta_\mu(s) \sum_t (\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu)) \quad (470)$$

and

$$\begin{aligned} & \bar{q}(2h) \gamma_\mu \otimes \mathbb{I}_\tau q(2h+2\mu) - \bar{q}(2h) \gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h+2\mu) \\ &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) \\ & \bar{q}(2h) \gamma_\mu \otimes \mathbb{I}_\tau q(2h-2\mu) + \bar{q}(2h) \gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h-2\mu) \\ &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu) \end{aligned} \quad (471)$$

Therefore

$$\begin{aligned} & \bar{q}(2h) \gamma_\mu \otimes \mathbb{I}_\tau q(2h+2\mu) - \bar{q}(2h) \gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h+2\mu) \\ &+ \bar{q}(2h) \gamma_\mu \otimes \mathbb{I}_\tau q(2h-2\mu) + \bar{q}(2h) \gamma_5 \otimes \tau_5^* \tau_\mu^* q(2h-2\mu) \\ &+ 4\bar{q}(2h) \gamma_\mu \otimes I q(2h) - 2\bar{q}(2h) \gamma_\mu \otimes I q(2h) \\ &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) \\ &+ \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu) \\ &+ \frac{1}{8} \eta_\mu(s) \sum_t (\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu)) \end{aligned} \quad (472)$$

which is

$$\begin{aligned} & 4\bar{q}(2h) \gamma_\mu \otimes I q(2h) + \mathcal{O}(a^4 + a^2) \\ &= \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+\mu) \chi(2h+t+2\mu) \\ &+ \eta_\mu(s) \frac{1}{8} \sum_t \bar{\chi}(2h+t+2\mu) \chi(2h+t+\mu) \\ &+ \frac{1}{8} \eta_\mu(s) \sum_t (\bar{\chi}(2h+t+\mu) \chi(2h+t) + \bar{\chi}(2h+t) \chi(2h+t+\mu)) \end{aligned} \quad (473)$$

	$q(x) \sim O[\frac{3}{2}]$	$\chi(x) \sim \frac{1}{\sqrt{2a}}q(x) \sim O[2]$	$\chi(x) \sim \frac{a^2}{\sqrt{2a}}q(x) \sim O[0]$
	$\int d^4x \bar{q}Dq \sim O[0]$	$a^4 \sum_n \bar{\chi}D\chi \sim O[0]$	$\sum_n \bar{\chi}D\chi \sim O[0]$
	$m\bar{q}q \sim O[4]$	$am\bar{\chi}\chi \sim O[4]$	$am\bar{\chi}\chi \sim O[0]$
cond.	$c = \int d^4x \langle \bar{q}q \rangle \sim O[-1]$	$2a \times a^4 \sum_n \langle \bar{\chi}\chi \rangle \sim O[-1]$	$2a \sum_n \langle \bar{\chi}\chi \rangle \sim O[-1]$
c-den.	$\frac{1}{V} \int d^4x \langle \bar{q}q \rangle \sim O[3]$	$\frac{1}{a^4 \prod N_i} 2a^5 \sum_n \langle \bar{\chi}\chi \rangle \sim O[3]$	$\frac{1}{a^4 \prod N_i} 2a \sum_n \langle \bar{\chi}\chi \rangle \sim O[3]$
			$\frac{1}{\prod N_i} \frac{2}{a^3} \text{tr}[D^{-1}] \sim O[3]$
			$\frac{N_f}{4 \prod N_i} \frac{2}{a^3} \text{tr}[D^{-1}] \sim O[3]$
susp.	$s \sim c^2 \sim O[-2]$		$\langle (2a \text{tr}[D^{-1}])^2 \rangle - \langle 2a \text{tr}[D^{-1}] \rangle^2 \sim O[-2]$
			$4a^2 (\langle (\text{tr}[D^{-1}])^2 \rangle - \langle \text{tr}[D^{-1}] \rangle^2) \sim O[-2]$
s-den.			$\frac{4a^2}{a^4 \prod N_i} (\langle (\text{tr}[D^{-1}])^2 \rangle - \langle \text{tr}[D^{-1}] \rangle^2) \sim O[2]$
			$\frac{4a^2 N_f^2}{16a^4 \prod N_i} (\langle (\text{tr}[D^{-1}])^2 \rangle - \langle \text{tr}[D^{-1}] \rangle^2) \sim O[2]$
s/T^2	$s/T^2 \sim O[0]$		$\frac{4a^2 N_f^2}{16a^4 \prod N_i} \frac{N_t^2}{a^{-2}} (\langle (\text{tr}[D^{-1}])^2 \rangle - \langle \text{tr}[D^{-1}] \rangle^2) \sim O[0]$
			$\frac{4N_f^2 N_t^2}{16 \prod N_i} (\langle (\text{tr}[D^{-1}])^2 \rangle - \langle \text{tr}[D^{-1}] \rangle^2) \sim O[0]$
			$d = \frac{\text{tr}[D^{-1}]}{\prod N_i}$
			$\frac{4N_f^2 N_t^2 \prod N_i}{16} (\langle d^2 \rangle - \langle d \rangle^2) \sim O[0]$

表 9: Dimension of condensations.

P	$\int d^3x p(x) \sim O[-3]$	$a^3 \sum_x p(x) \sim O[-3]$
L	$\frac{1}{V} \int d^3x p(x) \sim O[0]$	$L = \frac{a^3}{a^3 \prod N_i} \sum_x p(x) \sim O[0]$
χ_P	$\chi_P \sim \int d^3x p(x) \int d^3y p(y) \sim O[-6]$	$(a^3 \sum_x p(x))^2 \sim O[-6]$
χ_P/V	$\chi_P/V \sim \int d^3x p(x) \int d^3y p(y)/V \sim O[-3]$	$(a^3 \prod N_i) L^2 \sim O[-3]$

表 10: Dimension of Polyakov loop.

so

$$\begin{aligned}
4 \sum_{2h} \bar{q}(2h) \gamma_\mu \otimes Iq(2h) &= \eta_\mu(n_{left}) \frac{1}{8} \sum_n (\bar{\chi}(n+\mu)\chi(n) + \bar{\chi}(n)\chi(n+\mu)) \\
2^4 \sum_{2h} \bar{q}(2h) \gamma_\mu \otimes Iq(2h) &= \eta_\mu(n_{left}) \frac{1}{2} \sum_n (\bar{\chi}(n+\mu)\chi(n) + \bar{\chi}(n)\chi(n+\mu))
\end{aligned} \tag{474}$$

Note that they are not gauge invariant, we need to add links to connect them.

5.6.2 Staggered Fermion Gamma5

We can start from

$$\bar{q}(2h) \gamma_5 \otimes Iq(2h) = \frac{1}{16} \eta_1(s) \eta_2(s) \eta_3(s) \eta_4(s) \sum_n (\bar{\chi}(2h)\chi(2h+n) + \bar{\chi}(2h+2-n)\chi(2h+t+\mu)) \tag{475}$$

where n runs over 16 corners, and the link is for the diagonal corners.

5.6.3 Multiplication of Gamma matrix

5.6.4 Dimension of condensations

Although, not related, but the Polyakov loop is also summarized here.

5.7 Grassman Number Gaussian integration

This is the note of measurement on propagators. All start from the Grassman number field Gaussian integration, with Lagrangian $L = \psi^\dagger A \psi$,

$$\langle \psi_{j_1} \psi_{j_2} \dots \psi_{j_n} \psi_{i_n}^\dagger \psi_{i_{n-1}}^\dagger \dots \psi_1^\dagger \rangle = \sum_P \text{Sign}(P) A_{j_1, i_{P_1}}^{-1} \dots A_{j_n, i_{P_n}}^{-1} \tag{476}$$

Considering $L = \bar{\psi}D\psi$, $A = \gamma_0 D$. using $(AB)^{-1} = B^{-1}A^{-1}$ and $\gamma_0^{-1} = \gamma_0$. For bilinear,

$$\begin{aligned}\langle \psi^c \psi^\dagger_a \rangle &= \sum_d D^{-1c}_d \gamma_0^d_a, \\ \langle \bar{\psi} M \psi \rangle &= \sum_{a,b,c} \langle \psi^\dagger_a \gamma_0^a_b M^b_c \psi^c \rangle = \sum_{a,b,c,d} \gamma_0^a_b M^b_c D^{-1c}_d \gamma_0^d_a = \text{Tr}[\gamma_0 M D^{-1} \gamma_0] = \text{Tr}[M D^{-1}],\end{aligned}\quad (477)$$

For ψ^4 , let $D_0^{-1} = D^{-1} \gamma_0$

$$\langle \psi^a \psi^b \psi^\dagger_c \psi^\dagger_d \rangle = D_0^{-1b}_c D_0^{-1a}_d - D_0^{-1b}_d D_0^{-1a}_c, \quad (478)$$

so,

$$\begin{aligned}\langle \bar{\psi} M_1 \psi \bar{\psi} M_2 \psi \rangle &= \sum_{a,b,c,d} \langle \psi^\dagger_c \gamma_0 M_1^c_b \psi^b \psi^\dagger_d \gamma_0 M_2^d_a \psi^a \rangle \\ &= \sum_{a,b,c,d} \gamma_0 M_1^c_b \gamma_0 M_2^d_a \langle \psi^\dagger_c \psi^b \psi^\dagger_d \psi^a \rangle \\ &= \sum_{a,b,c,d} \gamma_0 M_1^c_b \gamma_0 M_2^d_a \langle \psi^a \psi^b \psi^\dagger_c \psi^\dagger_d \rangle \\ &= \sum_{a,b,c,d} (\gamma_0 M_1^c_b \gamma_0 M_2^d_a D_0^{-1b}_c D_0^{-1a}_d - \gamma_0 M_1^c_b \gamma_0 M_2^d_a D_0^{-1b}_d D_0^{-1a}_c) \\ &= \sum_{a,b,c,d} (D_0^{-1b}_c \gamma_0 M_1^c_b D_0^{-1a}_d \gamma_0 M_2^d_a - D_0^{-1b}_d \gamma_0 M_2^d_a D_0^{-1a}_c \gamma_0 M_1^c_b) \\ &= \text{Tr}[D_0^{-1} \gamma_0 M_1] \text{Tr}[D_0^{-1} \gamma_0 M_2] - \text{Tr}[D_0^{-1} \gamma_0 M_2 D_0^{-1} \gamma_0 M_1] \\ &= \text{Tr}[D^{-1} M_1] \text{Tr}[D^{-1} M_2] - \text{Tr}[D^{-1} M_2 D^{-1} M_1]\end{aligned}\quad (479)$$

Note that, the γ_0 comes in pairs, we do not track γ_0 any more.

$$\begin{aligned}\langle \psi^a \psi^b \psi^c \bar{\psi}_d \bar{\psi}_e \bar{\psi}_f \rangle &= D^{-1c}_d D^{-1b}_e D^{-1a}_f + D^{-1b}_d D^{-1a}_e D^{-1c}_f + D^{-1c}_d D^{-1a}_e D^{-1b}_f \\ &\quad - D^{-1b}_d D^{-1c}_e D^{-1a}_f - D^{-1a}_d D^{-1b}_e D^{-1c}_f - D^{-1c}_d D^{-1a}_e D^{-1b}_f\end{aligned}\quad (480)$$

so,

$$\begin{aligned}\langle \bar{\psi} M_1 \psi \bar{\psi} M_2 \psi \bar{\psi} M_3 \psi \rangle &= -\text{Tr}[D^{-1} M_1] \text{Tr}[D^{-1} M_2] \text{Tr}[D^{-1} M_3] - \text{Tr}[D^{-1} M_1 D^{-1} M_2 D^{-1} M_3] - \text{Tr}[D^{-1} M_1 D^{-1} M_3 D^{-1} M_2] \\ &\quad + \text{Tr}[D^{-1} M_1 D^{-1} M_2] \text{Tr}[D^{-1} M_3] + \text{Tr}[D^{-1} M_1 D^{-1} M_3] \text{Tr}[D^{-1} M_2] + \text{Tr}[D^{-1} M_2 D^{-1} M_3] \text{Tr}[D^{-1} M_1]\end{aligned}\quad (481)$$

5.7.1 Disconnected and connected contribution

The $\text{Tr}[D^{-1} M_1] \text{Tr}[D^{-1} M_2]$ is the disconnected contribution, and the $\text{Tr}[D^{-1} M_2 D^{-1} M_1]$ term is connected contribution.

If the coordinate is considered as quantum number, M is local,

$$\begin{aligned}\langle \bar{\psi}(x) M_1 \psi(x) \bar{\psi}(0) M_2 \psi(0) \rangle &= \text{Tr}[D^{-1}(0, x) M_1(x)] \text{Tr}[D^{-1}(x, 0) M_2(0)] - \text{Tr}[D^{-1}(x, 0) M_2 D^{-1}(0, x) M_1]\end{aligned}\quad (482)$$

here the trace is only applied in the spinor space. Note that,

$$\begin{aligned}\langle \bar{\psi}_{f_1}(x) M_1 \psi_{f_2}(x) \bar{\psi}_{f_2}(0) M_2 \psi_{f_1}(0) \rangle &= -\text{Tr}[D^{-1}(x, 0) M_2 D^{-1}(0, x) M_1]\end{aligned}\quad (483)$$

The disconnected contribution is much easier to calculate, because, with pseudofermion ϕ ,

$$D^{-1}(0, x) = \bar{\phi}(x) \gamma_0 D^{-1} \phi(0) \quad (484)$$

6 Programming

6.1 cuda

6.1.1 blocks and threads

6.1.2 device member function

According to <https://stackoverflow.com/questions/53781421/cuda-the-member-field-with-device-ptr-and-device-member-function-to-visit-it-i>

To call device member function, the content of the class should be on device.

- First, new a instance of the class.
- Then, create a device memory using cudaMalloc.
- Copy the content to the device memory

In other words, it will work as

```

1  __global__ void _kInitialArray(int* thearray)
2  {
3      int iX = threadIdx.x + blockDim.x * blockIdx.x;
4      int iY = threadIdx.y + blockDim.y * blockIdx.y;
5      int iZ = threadIdx.z + blockDim.z * blockIdx.z;
6      thearray[iX * 16 + iY * 4 + iZ] = iX * 16 + iY * 4 + iZ;
7  }
8
9  extern "C" {
10     void _cInitialArray(int* thearray)
11     {
12         dim3 block(1, 1, 1);
13         dim3 th(4, 4, 4);
14
15         _kInitialArray << <block, th >> > (thearray);
16         checkCudaErrors(cudaGetLastError());
17     }
18 }
19
20 class B
21 {
22 public:
23     B()
24     {
25         checkCudaErrors(cudaMalloc((void**)&m_pDevicePtr, sizeof(int) * 64));
26         _cInitialArray(m_pDevicePtr);
27     }
28     ~B()
29     {
30         cudaFree(m_pDevicePtr);
31     }
32     __device__ int GetNumber(int index)
33     {
34         m_pDevicePtr[index] = m_pDevicePtr[index] + 1;
35         return m_pDevicePtr[index];
36     }
37     int* m_pDevicePtr;
38 };
39
40 __global__ void _kAddArray(int* thearray1, B* pB)
41 {
42     int iX = threadIdx.x + blockDim.x * blockIdx.x;
43     int iY = threadIdx.y + blockDim.y * blockIdx.y;
44     int iZ = threadIdx.z + blockDim.z * blockIdx.z;
45     thearray1[iX * 16 + iY * 4 + iZ] = thearray1[iX * 16 + iY * 4 + iZ] + pB->GetNumber(iX * 16 + iY * 4 + iZ);
46 }
47
48 extern "C" {

```

```

49     void _cAddArray(int* thearray1, B* pB)
50     {
51         dim3 block(1, 1, 1);
52         dim3 th(4, 4, 4);
53         _kAddArray << <block, th >> > (thearray1, pB);
54         checkCudaErrors(cudaGetLastError());
55     }
56 }
57
58 class A
59 {
60 public:
61     A()
62     {
63         checkCudaErrors(cudaMalloc((void**)&m_pDevicePtr, sizeof(int) * 64));
64         _cInitialArray(m_pDevicePtr);
65     }
66     ~A()
67     {
68         checkCudaErrors(cudaFree(m_pDevicePtr));
69     }
70     void Add(B* toAdd/*this should be a device ptr(new on device function or created by cudaMalloc)*/)
71     {
72         _cAddArray(m_pDevicePtr, toAdd);
73     }
74     int* m_pDevicePtr;
75 };
76
77
78
79 int main(int argc, char * argv[])
80 {
81     B* pB = new B();
82     A* pA = new A();
83     B* pDeviceB;
84     checkCudaErrors(cudaMalloc((void**)&pDeviceB, sizeof(B)));
85     checkCudaErrors(cudaMemcpy(pDeviceB, pB, sizeof(B), cudaMemcpyHostToDevice));
86     pA->Add(pDeviceB);
87     int* res = (int*)malloc(sizeof(int) * 64);
88     checkCudaErrors(cudaMemcpy(res, pA->m_pDevicePtr, sizeof(int) * 64, cudaMemcpyDeviceToHost));
89     printf("-----_A=");
90     for (int i = 0; i < 8; ++i)
91     {
92         printf("\n");
93         for (int j = 0; j < 8; ++j)
94             printf("res_%d=%d_\t", i * 8 + j, res[i * 8 + j]);
95     }
96     printf("\n");
97     //NOTE: We are getting data from pB, not pDeviceB, this is OK, ONLY because m_pDevicePtr is a pointer
98     checkCudaErrors(cudaMemcpy(res, pB->m_pDevicePtr, sizeof(int) * 64, cudaMemcpyDeviceToHost));
99     printf("-----_B=");
100    for (int i = 0; i < 8; ++i)
101    {
102        printf("\n");
103        for (int j = 0; j < 8; ++j)
104            printf("res_%d=%d_\t", i * 8 + j, res[i * 8 + j]);
105    }
106    printf("\n");
107    delete pA;
108    delete pB;
109    return 0;
110 }

```

Note: this is a copy of the original instance! It is ONLY OK to change the content of *pDevicePtr* — *> m_pOtherPtr*, NOT *pDevicePtr* — *> somevalue*

6.1.3 device virtual member function

According to <https://stackoverflow.com/questions/26812913/how-to-implement-device-side-cuda-virtual-functions>

To call a device virtual member function, unlike Sec. 6.1.2, the pointer to the virtual function table should also be on device,

- First, cudaMalloc a sizeof(void*), for the device pointer.
- Then, use a kernel function to new the instance on device, and assign it to the device pointer created by cudaMalloc.
- One can copy the pointer, by using cudaMemcpy(void**, void**, sizeof(void*), devicetodevice).
- When copy it to elsewhere, one need to copy it back to host, then copy it again to device. The example shows how to copy it to constant.

in other words, it will work as

```

1
2 class CA
3 {
4 public:
5     __device__ CA() { ; }
6     __device__ ~CA() { ; }
7     __device__ virtual void CallMe() { printf("This_is_A\n"); }
8 };
9
10 class CB : public CA
11 {
12 public:
13     __device__ CB() : CA() { ; }
14     __device__ ~CB() { ; }
15     __device__ virtual void CallMe() { printf("This_is_B\n"); }
16 };
17
18 __global__ void _kernelCreateInstance(CA** pptr)
19 {
20     (*pptr) = new CB();
21 }
22
23 __global__ void _kernelDeleteInstance(CA** pptr)
24 {
25     delete (*pptr);
26 }
27
28 extern "C" {
29     void _kCreateInstance(CA** pptr)
30     {
31         _kernelCreateInstance << <1, 1 >> >(pptr);
32     }
33
34     void _kDeleteInstance(CA** pptr)
35     {
36         _kernelDeleteInstance << <1, 1 >> >(pptr);
37     }
38 }
39
40 __constant__ CA* m_pA;
41
42 __global__ void _kernelCallConstantFunction()
43 {
44     m_pA->CallMe();
45 }
46
47
48 extern "C" {
49     void _cKernelCallConstantFunction()
50     {
51         _kernelCallConstantFunction << <1, 1 >> > ();
52     }
53 }
54
55 int main()
56 {
57     CA** pptr;

```

```
58     cudaMalloc((void**)&pptr, sizeof(CA*));
59     _kCreateInstance(pptr);
60
61     //I can NOT use a kernel to set m_pA = (*pptr), because it is constant.
62     //I can NOT use cudaMemcpyToSymbol(m_pA, (*pptr)), because * operator on host is incorrect when pptr is a device ptr.
63     //I can NOT use cudaMemcpyToSymbol(m_pA, (*pptr)) in kernel, because cudaMemcpyToSymbol is a __host__ function
64     //I have to at first copy it back to host, then copy it back back again to constant
65     CA* pptrHost[1];
66     cudaMemcpy(pptrHost, pptr, sizeof(CA**), cudaMemcpyDeviceToHost);
67     cudaMemcpyToSymbol(m_pA, pptrHost, sizeof(CA*));
68     _cKernelCallConstantFunction();
69
70     _kDeleteInstance(pptr);
71     cudaFree(pptr);
72     return 0;
73 }
```

7 Testing

7.1 random number

8 Applications

8.1 Rotating Frame

We follow Ref. [31].

The matrix element can be written as

$$\mathcal{M} = \int \mathcal{D}(A_\mu \psi) \exp \left(i \int d^4x \mathcal{L} \right) \quad (485)$$

with

$$\mathcal{L} = \bar{\psi} (i \not{D} - m) \psi - \frac{1}{4} (F_{\mu\nu}^a)^2, \quad D_\mu \equiv \partial_\mu + i g_{YM} \sum_a T_a A_\mu^a \quad (486)$$

The first few steps are as usual, defining

$$A_\mu = g_{YM} \sum_a T_a A_\mu^a, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \quad (487)$$

using $\text{tr}[T_i T_j] = \frac{1}{2} \delta_{ij}$

$$F_{\mu\nu} = g_{YM} \sum_a T_a F_{\mu\nu}^a$$

$$\frac{1}{4} (F_{\mu\nu}^a)^2 = \frac{1}{2g_{YM}^2} \text{tr} [F_{\mu\nu}^2] \quad (488)$$

and

$$\mathcal{L} = \bar{\psi} (i \not{D} - m) \psi - \frac{1}{2g_{YM}^2} \text{tr} [F_{\mu\nu}^2]$$

$$D_\mu = \partial_\mu + i A_\mu \quad (489)$$

For rotational frame, the metric and frame can be defined as

$$h_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$g_{\mu\nu} = \begin{pmatrix} 1 - r^2 \Omega^2 & y\Omega & -x\Omega & 0 \\ y\Omega & -1 & 0 & 0 \\ -x\Omega & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-g_{\mu\nu}} = 1 \quad (490)$$

$$e_0 = (1, y\Omega, -x\Omega, 0)$$

$$e_1 = (0, 1, 0, 0)$$

$$e_2 = (0, 0, 1, 0)$$

$$e_3 = (0, 0, 0, 1)$$

8.1.1 The rotating gauge action

Note that $g^{\mu\nu} \neq g_{\mu\nu}$. Using

$$g^{\mu\nu} = \begin{pmatrix} 1 & y\Omega & -x\Omega & 0 \\ y\Omega & y^2 \Omega^2 - 1 & -xy\Omega^2 & 0 \\ -x\Omega & -xy\Omega^2 & x^2 \Omega^2 - 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-\det(g_{\mu\nu})} = 1 \quad (491)$$

Considering the case of pure gauge, the action can be written as

$$\mathcal{L}_G = -\sqrt{\det(-g_{\alpha\beta})} \frac{1}{2g_{YM}^2} g^{\mu\nu} g^{\rho\sigma} \text{tr}[F_{\mu\rho} F_{\nu\sigma}]$$

$$= -\frac{1}{2g_{YM}^2} \left(\sum_{ijkl=0}^3 \eta_{ij} \eta_{kl} \text{tr}[F_{ik} F_{jl}] - 2\Omega^2 \text{tr}[(xF_{23} - yF_{13})^2 + r^2 F_{12}^2] \right. \quad (492)$$

$$\left. - 4\Omega(x\text{tr}[F_{01} F_{12}] + y\text{tr}[F_{02} F_{12}] + y\text{tr}[F_{03} F_{13}] - x\text{tr}[F_{03} F_{23}]) \right)$$

- Wick rotation of gauge action

The Wick rotation ($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]$)

$$t \rightarrow -i\tau, \quad \partial_t \rightarrow i\partial_\tau, \quad \Omega \rightarrow i\Omega, \quad A_\mu \rightarrow (iA_0, A_1, A_2, A_3), \quad F_{0i} \rightarrow +iF_{0i} \quad (493)$$

and substitute $x = (t, x, y, z) \rightarrow x_E = (x, y, z, \tau)$. After Wick rotation, we are expecting

$$\exp(-S_G) = \exp(i \int d^4x \mathcal{L}_G) \quad (494)$$

the result is

$$\begin{aligned} -S_G &= i \int d^4x \mathcal{L}_G \\ S_G &= \int d^4x_E \frac{1}{2g_{YM}^2} \left(\sum_{ij=1}^4 \text{tr}[F_{ij}F_{ij}] + 2\Omega^2 \text{tr}[(xF_{23} - yF_{13})^2 + r^2 F_{12}^2] \right. \\ &\quad \left. + 4\Omega (x\text{tr}[F_{41}F_{12}] + y\text{tr}[F_{42}F_{12}] + y\text{tr}[F_{43}F_{13}] - x\text{tr}[F_{43}F_{23}]) \right) \end{aligned} \quad (495)$$

Therefore S_G is real. The $\sum_{ij=1}^4 \text{tr}[F_{ij}F_{ij}]$ is the gauge action in rest frame. **The sign of the terms propotional to Ω is different from Refs.**

- Discretization of gauge action

The discretized version can be derived using compact gauge group

$$U_\mu(x) = \exp(iaA_\mu(x)), \quad U_{-\mu}(x) = U_\mu^{-1}(x - \mu). \quad (496)$$

As usual,

$$\begin{aligned} U_{\mu,\nu}(n) &\equiv U_\mu(n)U_\nu(n+a\mu)U_\mu^{-1}(n+a\nu)U_\nu^{-1}(n) = \exp(ia^2F_{\mu\nu} + \mathcal{O}(a^3)). \\ \text{Re}[U_{\mu\nu}(n)] &= \mathbb{I}_{N_c \times N_c} - \frac{a^4}{2}F_{\mu\nu}^2 + \mathcal{O}(a^6) \\ \frac{1}{2g_{YM}^2} \sum_{\mu \neq \nu} \text{tr}[F_{\mu\nu}^2] &= \frac{1}{a^4 g_{YM}^2} \sum_{\mu \neq \nu} \text{Retr}[1 - U_{\mu\nu}(n)] = \frac{2}{a^4 g_{YM}^2} \sum_{\mu > \nu} \text{Retr}[1 - U_{\mu\nu}(n)] \end{aligned} \quad (497)$$

For those plaquette with coordinate as coefficients, we use the average of plaquette

$$\begin{aligned} \bar{U}_{\mu,\nu}(n) &\equiv \frac{1}{4} (U_{\mu,\nu}(n) + U_{-\mu,\nu}(n) + U_{\mu,-\nu}(n) + U_{-\mu,-\nu}(n)) \\ \text{Retr}[\bar{U}_{\mu,\nu}(n)] &= N_c - \frac{a^4}{2} \text{tr}[F_{\mu\nu}^2] + \mathcal{O}(a^6), \quad \frac{1}{2g_{YM}^2} \text{tr}[F_{\mu\nu}^2] = \frac{2}{a^4 g_{YM}^2} \frac{1}{2} \text{Retr}[1 - \bar{U}_{\mu,\nu}(n)] \end{aligned} \quad (498)$$

The remaining are those has the form $\text{tr}[F_{ab}F_{bc}]$, using

$$\begin{aligned} U_{\mu\nu}^{-1}(n) &= U_\nu(n)U_\mu(n+a\nu)U_\nu^{-1}(n+a\mu)U_\mu^{-1}(n) = U_{\nu\mu}(n) = \exp(-ia^2F_{\mu\nu}) + \mathcal{O}(a^6) \\ U_{a,b}(n)(U_{b,c}(n) - U_{c,b}(n)) &= \exp(-ia^2(F_{ab} + F_{bc})) - \exp(-ia^2(F_{ab} - F_{bc})) \\ \frac{1}{2} \text{Re}[U_{a,b}(n)(U_{c,b}(n) - U_{b,c}(n))] &= a^4 F_{ab}F_{bc} + \mathcal{O}(a^6) \end{aligned} \quad (499)$$

Note that b is not summed.

We can have a more symmetric form to use

$$U_{c,b}(n) = U_{b,-c}(n) + \mathcal{O}(a^4) \quad (500)$$

then it is a chair-type.

$$\frac{1}{2} \text{Re}[U_{a,b}(n)(U_{b,-c}(n) - U_{b,c}(n))] = a^4 F_{ab}F_{bc} + \mathcal{O}(a^6) \quad (501)$$

Similarly, we can use the average of chairs, and define

$$\begin{aligned} V_{\mu\nu\sigma} &= \frac{1}{8} ((U_{\mu,\nu} - U_{-\mu,\nu})(U_{\nu,\sigma} - U_{-\nu,\sigma}) + (U_{\mu,-\nu} - U_{-\mu,-\nu})(U_{-\nu,\sigma} - U_{-\nu,-\sigma})) \\ \text{Retr}[V_{\mu\nu\rho}] &= -a^4 \text{tr}[F_{\mu\nu}F_{\nu\rho}] + \mathcal{O}(a^6), \quad \frac{1}{2g_{YM}^2} \text{tr}[F_{\mu\nu}F_{\nu\rho}] = -\frac{2}{a^4 g_{YM}^2} \frac{1}{4} \text{Retr}[V_{\mu\nu\rho}] \end{aligned} \quad (502)$$

- Final result of gauge action

The discretized and Wick rotated gauge action is

$$\begin{aligned}
 S_G &= \frac{2}{a^4 g_{YM}^2} \sum_n \left(\sum_{\mu > \nu} \text{Retr}[1 - U_{\mu\nu}(n)] - \Omega (x \text{Retr}[V_{412} + V_{432}] - y \text{Retr}[V_{421} + V_{431}]) \right. \\
 &\quad \left. + \Omega^2 (x^2 \text{Retr}[1 - \bar{U}_{23}(n)] + y^2 \text{Retr}[1 - \bar{U}_{13}(n)] + r^2 \text{Retr}[1 - \bar{U}_{34}(n)] - xy \text{Retr}[V_{132}]) \right) \\
 &= \frac{\beta}{N_c} \sum_n \left(\sum_{\mu > \nu} \text{Retr}[1 - U_{\mu\nu}(n)] - \Omega (x \text{Retr}[V_{412} + V_{432}] - y \text{Retr}[V_{421} + V_{431}]) \right. \\
 &\quad \left. + \Omega^2 (x^2 \text{Retr}[1 - \bar{U}_{23}(n)] + y^2 \text{Retr}[1 - \bar{U}_{13}(n)] + r^2 \text{Retr}[1 - \bar{U}_{34}(n)] - xy \text{Retr}[V_{132}]) \right)
 \end{aligned} \tag{503}$$

with $\frac{\beta}{N_c} \equiv \frac{2}{a^4 g_{YM}^2}$.

8.1.2 Rotating Fermion action

$$D_R = \left[i\gamma^\mu \left((\partial_\mu + ieA_\mu) - \frac{i}{4} \sigma^{ij} w_{\mu ij} \right) - m \right] \tag{504}$$

with

$$\begin{aligned}
 w_{\mu ij} &= g_{\alpha\beta} e_i^\alpha (\partial_\mu e_j^\beta + \Gamma_{\mu\nu}^\beta e_j^\nu) \\
 \Gamma_{\mu\nu}^\beta &= \frac{1}{2} g^{\beta\alpha} \left(\frac{\partial g_{\alpha\mu}}{\partial x^\nu} + \frac{\partial g_{\alpha\nu}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\alpha} \right) \\
 \sigma^{ij} &= \frac{i}{2} [\gamma^i, \gamma^j] \\
 e_0 &= (1, y\Omega, -x\Omega, 0) \\
 e_1 &= (0, 1, 0, 0) \\
 e_2 &= (0, 0, 1, 0) \\
 e_3 &= (0, 0, 0, 1)
 \end{aligned} \tag{505}$$

so

$$\frac{i}{4} \sigma^{ij} w_{\mu ij} = \left(\frac{i}{2} \Omega \sigma^{12}, 0, 0, 0 \right) \tag{506}$$

and

$$D_R = \left[i\gamma^x (\partial_x + ieA_x) + i\gamma^y (\partial_y + ieA_y) + i\gamma^z (\partial_z + ieA_z) + i\gamma^t (\partial_t + ieA_t - \frac{i}{2} \Omega \sigma^{12}) - m \right] \tag{507}$$

using $\gamma^\mu = \gamma^i e_i^\mu$, it is

$$\begin{aligned}
 D_R &= [i(\gamma^1 + y\Omega\gamma^0)(\partial_x + ieA_x) + i(\gamma^2 - x\Omega\gamma^0)(\partial_y + ieA_y) + i\gamma^3(\partial_z + ieA_z) \\
 &\quad + i\gamma^0(\partial_t + ieA_t - \frac{i}{2} \Omega \sigma^{12}) - m]
 \end{aligned} \tag{508}$$

- The Wick rotation of Fermion action

The Wick rotation

$$t \rightarrow -i\tau, \quad \gamma_i^M \rightarrow i\gamma_i^E, \quad \gamma_4 = \gamma_0, \quad \gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4, \quad A_t \rightarrow iA_\tau, \quad \partial_t \rightarrow i\partial_\tau, \quad \Omega \rightarrow i\Omega, \quad \sigma^{12} \rightarrow -\sigma^{12} \tag{509}$$

where the superscript of gamma matrix stands for Minkowski or Euclidian. So

$$\begin{aligned}
 D_R &= -[(\gamma_1 + y\Omega\gamma_4)(\partial_x + ieA_x) + (\gamma_2 - x\Omega\gamma_4)(\partial_y + ieA_y) + \gamma_3(\partial_z + ieA_z) \\
 &\quad + \gamma_4(\partial_\tau + ieA_\tau + \frac{i}{2} \Omega \sigma^{12}) + m] \\
 &= -[\gamma_1(\partial_x + ieA_x) + \gamma_2(\partial_y + ieA_y) + \gamma_3(\partial_z + ieA_z) + \gamma_4(\partial_\tau + ieA_\tau) + m \\
 &\quad + \gamma_4(y\Omega(\partial_x + ieA_x) - x\Omega(\partial_y + ieA_y)) + \frac{i}{2} \gamma_4 \Omega \sigma^{12}]
 \end{aligned} \tag{510}$$

And

$$\begin{aligned}
 -S_F &= i \int d^4x \sqrt{-g_{\alpha\beta}} \bar{\psi} D_R \psi \\
 S_F &= \int d^4x_E \bar{\psi} [\gamma_1(\partial_x + ieA_x) + \gamma_2(\partial_y + ieA_y) + \gamma_3(\partial_z + ieA_z) + \gamma_4(\partial_\tau + ieA_\tau) + m \\
 &\quad + \gamma_4(y\Omega(\partial_x + ieA_x) - x\Omega(\partial_y + ieA_y)) + \frac{i}{2}\gamma_4\Omega\sigma^{12}] \psi
 \end{aligned} \tag{511}$$

- The discretization of Fermion action

The naive discretization yields

$$\partial_\mu \psi(n) = \frac{\psi(n+a\mu) - \psi(n-a\mu)}{2a} \tag{512}$$

and

$$\begin{aligned}
 U_\mu(n) &= \exp(iaA_\mu) \approx 1 + iaA_\mu(n), \quad U_\mu^{-1}(n) \approx 1 - iaA_\mu(n) \\
 iA_\mu(n) &= \frac{2iaA_\mu(n)}{2a} \approx \frac{(U_\mu(n) - 1) - (U_\mu^{-1}(n) - 1)}{2a} \approx \frac{(U_\mu(n) - 1) - (U_{-\mu}(n) - 1)}{2a} \\
 iA_\mu(n)\psi(n) &= \frac{(U_\mu(n) - 1)\psi(n+a\mu) - (U_{-\mu}(n) - 1)\psi(n-a\mu)}{2a} + \mathcal{O}(a)
 \end{aligned} \tag{513}$$

Therefor

$$(\partial_\mu + iA_\mu)\psi(n) = \frac{U_\mu(n)\psi(n+a\mu) - U_{-\mu}(n)\psi(n-a\mu)}{2a} + \mathcal{O}(a) \tag{514}$$

The Wilson term is

$$W\psi(n) = - \sum_\mu \frac{U_\mu(n)\psi(n+a\mu) + U_{-\mu}(n)\psi(n-a\mu) - 2\psi(n)}{2a} \rightarrow 0 \tag{515}$$

considering the rotation, we add a modified Wilson term as

$$\begin{aligned}
 W_R\psi(n) &= - \sum_\mu \frac{U_\mu(n)\psi(n+a\mu) + U_{-\mu}(n)\psi(n-a\mu) - 2\psi(n)}{2a} \\
 &\quad - y\Omega \frac{U_x(n)\psi(n+ax) + U_{-x}(n)\psi(n-ax) - 2\psi(n)}{2a} \\
 &\quad + x\Omega \frac{U_y(n)\psi(n+ay) + U_{-y}(n)\psi(n-ay) - 2\psi(n)}{2a}
 \end{aligned} \tag{516}$$

Similar as the Wilson term, the last two terms also decouples when approaching the continuum limit. And the Wilson-Dirac operator becomes

$$\begin{aligned}
 S_F &= \sum_{n,m} \bar{\psi}(n) D_W(n|m) \psi(m) \\
 D_W(n|m) &= \left(m + \frac{4}{a} + \frac{y\Omega}{a} - \frac{x\Omega}{a}\right) \delta_{n,m} - \sum_\mu \frac{(1-\gamma_\mu)U_\mu(n)\delta_{n+a\mu,m} + (1+\gamma_\mu)U_{-\mu}(n)\delta_{n-a\mu,m}}{2a} \\
 &\quad - y\Omega \frac{(1-\gamma_4)U_x(n)\delta_{n+ax,m} + (1+\gamma_4)U_{-x}(n)\delta_{n-ax,m}}{2a} \\
 &\quad + x\Omega \frac{(1-\gamma_4)U_y(n)\delta_{n+ay,m} + (1+\gamma_4)U_{-y}(n)\delta_{n-ay,m}}{2a} + \frac{i}{2}\gamma_4\Omega\sigma^{12}\delta_{n,m}
 \end{aligned} \tag{517}$$

Note that,

$$\begin{aligned}
 (U(1)\delta_{n+1,m,n=1,m=2})^\dagger &= U^\dagger(1)\delta_{n,m+1,n=2,m=1} = U^\dagger(m)\delta_{n,m+1} \text{ or } U^\dagger(n-1)\delta_{n-1,m} \\
 (U_\mu(n)\delta_{n+a\mu,m})^\dagger &= U_\mu^{-1}(n-a\mu)\delta_{n,m+a\mu} = U_{-\mu}(n)\delta_{n-a\mu,m}
 \end{aligned} \tag{518}$$

Let's check whether the periodic condition for gauge field or infinite lattice volume is necessary

$$\begin{aligned}
 \sum_{n=1,2,3,m=1,2,3} (U_\mu(n)\delta_{n+1,m} + U_{-\mu}(n)\delta_{n-1,m})^\dagger &= (U_\mu(1))^\dagger + (U_\mu(2))^\dagger + (U_{-\mu}(2))^\dagger + (U_{-\mu}(3))^\dagger \\
 &= U_\mu^{-1}(2-1) + U_\mu^{-1}(3-1) + (U_\mu^{-1}(1))^\dagger + (U_\mu^{-1}(2))^\dagger \\
 &= U_{-\mu}(2) + U_{-\mu}(3) + U_\mu(1) + U_\mu(2) \\
 &= \sum_{n=1,2,3,m=1,2,3} (U_\mu(n)\delta_{n+1,m} + U_{-\mu}(n)\delta_{n-1,m})
 \end{aligned} \tag{519}$$

So we can conclude The γ_5 -hermiticity is kept with open(Dirichlet) boundary condition and finite volume.

Both the naive discretization and the Wilson term satisfy the γ_5 -hermiticity (separately).

$$\begin{aligned} \gamma_5 \gamma_\nu \gamma_5 &= -\gamma_\nu, \quad \gamma_\mu^\dagger = \gamma_\mu, \quad \gamma_5^2 = 1 \\ \sum_{n,m} (\gamma_\nu U_\mu(n) \delta_{n+a\mu,m} - \gamma_\nu U_{-\mu}(n) \delta_{n,m+a\mu})^\dagger &= \sum_{n,m} (\gamma_5 \gamma_\nu \gamma_5 U_\mu(n) \delta_{n+a\mu,m} - \gamma_5 \gamma_\nu \gamma_5 U_{-\mu}(n) \delta_{n-a\mu,m}) \\ \sum_{n,m} (U_\mu(n) \delta_{n+a\mu,m} + U_{-\mu}(n) \delta_{n,m+a\mu})^\dagger &= \sum_{n,m} (\gamma_5^2 U_\mu(n) \delta_{n+a\mu,m} + \gamma_5^2 U_{-\mu}(n) \delta_{n-a\mu,m}) \end{aligned} \quad (520)$$

Apart from that

$$\left(\frac{i}{2} \gamma_4 \Omega \sigma^{12} \delta_{n,m} \right)^\dagger = \gamma_5 \frac{i}{2} \gamma_4 \Omega \sigma^{12} \delta_{n,m} \gamma_5 \quad (521)$$

Therefor, the new Wilson-Dirac operator is also γ_5 -hermiticity.

- The doubler problem

Note that the naive action and Wilson term both satisfy the γ_5 -hermiticity, the traditional Wilson term will also lead to a γ_5 -hermiticity fermion action, with

$$\begin{aligned} D_W(n|m) &= \left(m + \frac{4}{a} \right) \delta_{n,m} - \sum_\mu \frac{(1 - \gamma_\mu) U_\mu(n) \delta_{n+a\mu,m} + (1 + \gamma_\mu) U_{-\mu}(n) \delta_{n-a\mu,m}}{2a} \\ &+ y \Omega \frac{\gamma_4 U_x(n) \delta_{n+ax,m} - \gamma_4 U_{-x}(n) \delta_{n-ax,m}}{2a} - x \Omega \frac{\gamma_4 U_y(n) \delta_{n+ay,m} - \gamma_4 U_{-y}(n) \delta_{n-ay,m}}{2a} + \frac{i}{2} \gamma_4 \Omega \sigma^{12} \delta_{n,m} \end{aligned} \quad (522)$$

This action also does not suffer from the doubler problem.

- The final action of fermions

As usual, we define the hopping parameter as $\kappa = \frac{1}{2am+8}$, then rescale the fermion field, the action is

$$\begin{aligned} S_F &= \sum_{n,m} \bar{\psi}(n) D_W(n|m) \psi(m) \\ D_W(n|m) &= (1 + 2\kappa(y-x)\Omega) \delta_{n,m} - \kappa \sum_\mu ((1 - \gamma_\mu) U_\mu(n) \delta_{n+a\mu,m} + (1 + \gamma_\mu) U_{-\mu}(n) \delta_{n-a\mu,m}) \\ &- \kappa y \Omega ((1 - \gamma_4) U_x(n) \delta_{n+ax,m} + (1 + \gamma_4) U_{-x}(n) \delta_{n-ax,m}) \\ &+ \kappa x \Omega ((1 - \gamma_4) U_y(n) \delta_{n+ay,m} + (1 + \gamma_4) U_{-y}(n) \delta_{n-ay,m}) + \kappa i \gamma_4 a \Omega \sigma^{12} \delta_{n,m} \end{aligned} \quad (523)$$

8.1.3 The exponential chemical potential

On the other hand, the $i\kappa\gamma_4\hat{\Omega}\sigma^{12}$ term can also be modified. The σ^{12} term can be considered as a chemical potential ($\bar{\psi}\gamma_0\psi$ and then do the Wick rotation $\gamma_0 \rightarrow \gamma_4$. The sign is after Wick rotation and relative to the mass term)

$$\mu \bar{\psi} \gamma_4 \psi, \quad \mu = \frac{i\Omega}{2} \sigma^{12} \quad (524)$$

and discretized as

$$\begin{aligned} D_\tau + \mu \bar{\psi} \gamma_4 \psi &\rightarrow -\kappa (e^{\mu a} (1 - \gamma_4) U_\tau(n) \delta_{n,n+t} + e^{-\mu a} (1 + \gamma_4) U_{-\tau}(n) \delta_{n-t,n}) \\ &= -\kappa \left(e^{+\frac{ia\Omega\sigma^{12}}{2}} (1 - \gamma_4) U_\tau(n) \delta_{n,n+t} + e^{-\frac{ia\Omega\sigma^{12}}{2}} (1 + \gamma_4) U_{-\tau}(n) \delta_{n-t,n} \right) \end{aligned} \quad (525)$$

It looks not satisfy the γ_5 -hermiticity. However, using $(\sigma^{12})^2 = 1$, it is in fact

$$\begin{aligned} D_\tau + \mu \bar{\psi} \gamma_4 \psi &\rightarrow -\kappa \left[\left(\cos\left(\frac{a\Omega}{2}\right) + i \sin\left(\frac{a\Omega}{2}\right) \sigma^{12} \right) (1 - \gamma_4) U_\tau(n) \delta_{n,n+t} \right. \\ &\left. + \left(\cos\left(\frac{a\Omega}{2}\right) - i \sin\left(\frac{a\Omega}{2}\right) \sigma^{12} \right) (1 + \gamma_4) U_{-\tau}(n) \delta_{n-t,n} \right] \end{aligned} \quad (526)$$

The 1 in $1 \pm \frac{ia\Omega\sigma^{12}}{2}$ is the usual D_τ . So the additional term is in fact

$$\begin{aligned} &- \kappa \left[\left(\cos\left(\frac{a\Omega}{2}\right) - 1 + i \sin\left(\frac{a\Omega}{2}\right) \sigma^{12} \right) (1 - \gamma_4) U_\tau(n) \delta_{n,n+t} \right. \\ &\left. + \left(\cos\left(\frac{a\Omega}{2}\right) - 1 - i \sin\left(\frac{a\Omega}{2}\right) \sigma^{12} \right) (1 + \gamma_4) U_{-\tau}(n) \delta_{n-t,n} \right] \end{aligned} \quad (527)$$

Another way to do so

$$-\frac{U_\mu(n)\psi(n+a\mu)+U_{-\mu}(n)\psi(n-a\mu)-2\psi(n)}{2a}\rightarrow 0$$

$$\psi(n)\rightarrow (U_\mu(n)\psi(n+a\mu)+U_{-\mu}(n)\psi(n-a\mu))$$
(528)

so

$$\begin{aligned} \kappa\gamma_4 ia\Omega\sigma^{12}\psi(n) &= -\kappa(-ia\Omega\sigma^{12})\gamma_4\psi(n) \\ &\approx -\kappa\left(\frac{-ia\Omega\sigma^{12}}{2}\right)(\gamma_4 U_\mu(n)\psi(n+\tau)+\gamma_4 U_{-\mu}(n)\psi(n-\tau)) \\ &\approx -\kappa\left(\frac{-ia\Omega\sigma^{12}}{2}\right)(\gamma_4 U_\mu(n)\psi(n+\tau)+\gamma_4 U_{-\mu}(n)\psi(n-\tau)-U_\mu(n)\psi(n+\tau)+U_{-\mu}(n)\psi(n-\tau)) \\ &= -\kappa\left(\frac{-ia\Omega\sigma^{12}}{2}\right)((\gamma_4-1)U_\mu(n)\psi(n+\tau)+(\gamma_4+1)U_{-\mu}(n)\psi(n-\tau)) \\ &= -\kappa\left(\frac{-ia\Omega\sigma^{12}}{2}(\gamma_4-1)U_\mu(n)\psi(n+\tau)+\frac{-ia\Omega\sigma^{12}}{2}(\gamma_4+1)U_{-\mu}(n)\psi(n-\tau)\right) \\ &= -\kappa\left(\frac{ia\Omega\sigma^{12}}{2}(1-\gamma_4)U_\mu(n)\psi(n+\tau)+\frac{-ia\Omega\sigma^{12}}{2}(1+\gamma_4)U_{-\mu}(n)\psi(n-\tau)\right) \end{aligned}$$
(529)

so

$$\begin{aligned} &-\kappa\gamma_4 X_4 + \kappa\gamma_4 ia\Omega\sigma^{12}\psi(n) \\ &= -\kappa((1-\gamma_4)U_\mu(n)\psi(n+\tau)+(1+\gamma_4)U_{-\mu}(n)\psi(n-\tau)) \\ &-\kappa\left(\frac{ia\Omega\sigma^{12}}{2}(1-\gamma_4)U_\mu(n)\psi(n+\tau)+\frac{-ia\Omega\sigma^{12}}{2}(1+\gamma_4)U_{-\mu}(n)\psi(n-\tau)\right) \\ &= -\kappa\left(\left(1+\frac{ia\Omega\sigma^{12}}{2}\right)(1-\gamma_4)U_\mu(n)\psi(n+\tau)+\left(1-\frac{ia\Omega\sigma^{12}}{2}\right)(1+\gamma_4)U_{-\mu}(n)\psi(n-\tau)\right) \\ &\approx -\kappa\left(e^{\frac{ia\Omega\sigma^{12}}{2}}(1-\gamma_4)U_\mu(n)\psi(n+\tau)+e^{\frac{-ia\Omega\sigma^{12}}{2}}(1+\gamma_4)U_{-\mu}(n)\psi(n-\tau)\right) \end{aligned}$$
(530)

which go back to the σ^{12} term.

We still check the γ_5 -hermiticity, using

$$\begin{aligned} \sum_{n=1,2,3,m=1,2,3} (U_\mu(n)\delta_{n+1,m}+U_{-\mu}(n)\delta_{n-1,m})^\dagger &= \sum_{n=1,2,3,m=1,2,3} (U_\mu(n)\delta_{n+1,m}+U_{-\mu}(n)\delta_{n-1,m}) \\ \gamma_5\gamma_4\gamma_5 &= -\gamma_4^\dagger \\ \gamma_5\frac{ia\Omega\sigma^{12}}{2}\gamma_5 &= -\left(\frac{ia\Omega\sigma^{12}}{2}\right)^\dagger \end{aligned}$$
(531)

It is γ_5 -hermite.

8.1.4 The final action of rotation

Defining $\frac{\beta}{N_c} \equiv \frac{2}{a^4 g_{YM}^2}$, $\kappa \equiv \frac{1}{2am+8}$, $\hat{\mu} \equiv \frac{\mu}{a}$, $\hat{\Omega} \equiv a\Omega$, (here $\mu = x, y, z, t$ is the coordinate) we have

$$Z = \exp(-S_G - S_F)$$
(532)

with

$$\begin{aligned} S_G &= \frac{\beta}{N_c} \sum_n \left(\sum_{\mu>\nu} \text{Retr}[1-U_{\mu\nu}(n)] - \hat{\Omega}(\hat{x}\text{Retr}[V_{412}+V_{432}] - \hat{y}\text{Retr}[V_{421}+V_{431}]) \right. \\ &\quad \left. + \hat{\Omega}^2(\hat{x}^2\text{Retr}[1-\bar{U}_{23}(n)] + \hat{y}^2\text{Retr}[1-\bar{U}_{13}(n)] + (\hat{x}^2 + \hat{y}^2)\text{Retr}[1-\bar{U}_{12}(n)] - \hat{x}\hat{y}\text{Retr}[V_{132}]) \right) \\ U_{\mu,\nu}(n) &\equiv U_\mu(n)U_\nu(n+a\hat{\mu})U_\nu^{-1}(n+a\hat{\nu})U_\mu^{-1}(n) \\ \bar{U}_{\mu,\nu}(n) &\equiv \frac{1}{4}(U_{\mu,\nu}(n)+U_{-\mu,\nu}(n)+U_{\mu,-\nu}(n)+U_{-\mu,-\nu}(n)) \\ V_{\mu\nu\sigma}(n) &= \frac{1}{8}((U_{\mu,\nu}(n)-U_{-\mu,\nu}(n))(U_{\nu,\sigma}(n)-U_{-\nu,\sigma}(n)) \\ &\quad + (U_{\mu,-\nu}(n)-U_{-\mu,-\nu}(n))(U_{-\nu,\sigma}(n)-U_{-\nu,-\sigma}(n))) \end{aligned}$$
(533)

and

$$\begin{aligned}
S_F &= \sum_{n,m} \bar{\psi}(n) D(n|m) \psi(m) \\
D(n|m) &= \left(1 + 2\kappa(\hat{y} - \hat{x})\hat{\Omega} + i\kappa\gamma_4\hat{\Omega}\sigma^{12}\right) \delta_{n,m} \\
&\quad - \kappa \sum_{\mu} [(1 - \gamma_{\mu})U_{\mu}(n)\delta_{n+a\hat{\mu},m} + (1 + \gamma_{\mu})U_{-\mu}(n)\delta_{n-a\hat{\mu},m}] \\
&\quad - \kappa\hat{y}\hat{\Omega}((1 - \gamma_4)U_x(n)\delta_{n+a\hat{x},m} + (1 + \gamma_4)U_{-x}(n)\delta_{n-a\hat{x},m}) \\
&\quad + \kappa\hat{x}\hat{\Omega}((1 - \gamma_4)U_y(n)\delta_{n+a\hat{y},m} + (1 + \gamma_4)U_{-y}(n)\delta_{n-a\hat{y},m})
\end{aligned} \tag{534}$$

such that $\gamma_5 D \gamma_5 = D^\dagger$.

or (As in Ref. [31], the naive discretization is used.)

$$\begin{aligned}
S_F &= \sum_{n,m} \bar{\psi}(n) D(n|m) \psi(m) \\
D(n|m) &= \left(1 + i\kappa\gamma_4\hat{\Omega}\sigma^{12}\right) \delta_{n,m} \\
&\quad - \kappa \sum_{\mu} ((1 - \gamma_{\mu})U_{\mu}(n)\delta_{n+a\hat{\mu},m} + (1 + \gamma_{\mu})U_{-\mu}(n)\delta_{n-a\hat{\mu},m}) \\
&\quad - \kappa\hat{y}\hat{\Omega}((- \gamma_4)U_x(n)\delta_{n+a\hat{x},m} + (+\gamma_4)U_{-x}(n)\delta_{n-a\hat{x},m}) \\
&\quad + \kappa\hat{x}\hat{\Omega}((- \gamma_4)U_y(n)\delta_{n+a\hat{y},m} + (+\gamma_4)U_{-y}(n)\delta_{n-a\hat{y},m})
\end{aligned} \tag{535}$$

If using the exponential spin coupling term it is

$$\begin{aligned}
S_F &= \sum_{n,m} \bar{\psi}(n) D(n|m) \psi(m) \\
D(n|m) &= \delta_{n,m} - \kappa \sum_{\mu=1,2,3} ((1 - \gamma_{\mu})U_{\mu}(n)\delta_{n+a\hat{\mu},m} + (1 + \gamma_{\mu})U_{-\mu}(n)\delta_{n-a\hat{\mu},m}) \\
&\quad - \kappa\hat{y}\hat{\Omega}((- \gamma_4)U_x(n)\delta_{n+a\hat{x},m} + (+\gamma_4)U_{-x}(n)\delta_{n-a\hat{x},m}) \\
&\quad + \kappa\hat{x}\hat{\Omega}((- \gamma_4)U_y(n)\delta_{n+a\hat{y},m} + (+\gamma_4)U_{-y}(n)\delta_{n-a\hat{y},m}) \\
&\quad - \kappa \left(e^{\frac{ia\Omega\sigma^{12}}{2}} (1 - \gamma_4)U_{\tau}(n)\delta_{n,n+t} + e^{\frac{-ia\Omega\sigma^{12}}{2}} (1 + \gamma_4)U_{-\tau}(n)\delta_{n-t,n} \right)
\end{aligned} \tag{536}$$

For convenient, it is

$$\begin{aligned}
D(n|m) &= \delta_{n,m} - \kappa M \\
M &= \sum_{\mu=1,2,3} ((1 - \gamma_{\mu})U_{\mu}(n)\delta_{n+a\hat{\mu},m} + (1 + \gamma_{\mu})U_{-\mu}(n)\delta_{n-a\hat{\mu},m}) \\
&\quad - \hat{y}\hat{\Omega}\gamma_4 U_x(n)\delta_{n+a\hat{x},m} + \hat{y}\hat{\Omega}\gamma_4 U_{-x}(n)\delta_{n-a\hat{x},m} \\
&\quad + \hat{x}\hat{\Omega}\gamma_4 U_y(n)\delta_{n+a\hat{y},m} - \hat{x}\gamma_4 U_{-y}(n)\delta_{n-a\hat{y},m} \\
&\quad + \left(\cos\left(\frac{ia\Omega}{2}\right) + i \sin\left(\frac{ia\Omega}{2}\right) \sigma^{12} \right) (1 - \gamma_4)U_{\tau}(n)\delta_{n,n+t} \\
&\quad + \left(\cos\left(\frac{ia\Omega}{2}\right) - i \sin\left(\frac{ia\Omega}{2}\right) \sigma^{12} \right) (1 + \gamma_4)U_{-\tau}(n)\delta_{n-t,n}
\end{aligned} \tag{537}$$

8.1.5 The force from gauge action

$$\begin{aligned}
U_{\mu,\nu}(n) &= U_{\mu}(n)U_{\nu}(n+a\mu)U_{\mu}^{-1}(n+a\nu)U_{\nu}^{-1}(n). \quad U_{\mu,\nu}^{\dagger}(n) = U_{\mu,\nu}^{-1}(n) \\
U_{\mu,\nu}^{-1}(n) &= U_{\nu}(n)U_{\mu}(n+a\nu)U_{\nu}^{-1}(n+a\mu)U_{\mu}^{-1}(n) = U_{\nu,\mu}(n). \\
\text{tr}[U_{\nu,\mu}(n)] &= \text{tr}[U_{\nu}(n)U_{\mu}(n+a\nu)U_{\nu}^{-1}(n+a\mu)U_{\mu}^{-1}(n)]. \\
\text{tr}[U_{-\mu,\nu}(n)] &= \text{tr}[U_{\nu}(n-a\mu)U_{-\mu}^{-1}(n+a\nu+a\mu-a\mu)U_{\nu}^{-1}(n)U_{-\mu}(n)] \\
&= \text{tr}[U_{\nu}(n-a\mu)U_{\mu}(n+a\nu-a\mu)U_{\nu}^{-1}(n)U_{\mu}^{-1}(n-a\mu)] = \text{tr}[U_{\nu,\mu}(n-a\mu)] = \text{tr}[U_{\mu,\nu}^{\dagger}(n-a\mu)] \\
\text{tr}[U_{\mu,-\nu}(n)] &= \text{tr}[U_{-\nu,\mu}^{\dagger}(n)] = (\text{tr}[U_{-\nu,\mu}(n)])^* = (\text{tr}[U_{\mu,\nu}(n-a\nu)])^* = \text{tr}[U_{\mu,\nu}^{\dagger}(n-a\nu)] \\
\text{tr}[U_{-\mu,-\nu}(n)] &= \text{tr}[U_{\mu,\nu}(n-a\mu-a\nu)]
\end{aligned} \tag{538}$$

so

$$\text{Retr}[\bar{U}_{\mu,\nu}(n)] = \frac{1}{4} \text{Retr}[U_{\mu\nu}(n) + U_{\mu\nu}(n - a\mu) + U_{\mu\nu}(n - a\nu) + U_{\mu\nu}(n - a\mu - a\nu)] \quad (539)$$

and

$$\sum_n f(n) \text{Retr}[1 - \bar{U}_{\mu,\nu}(n)] = \sum_n \frac{f(n) + f(n + a\mu) + f(n + a\nu) + f(n + a\mu + a\nu)}{4} \text{Retr}[1 - U_{\mu\nu}(n)] \quad (540)$$

so (**Note, this is for finite lattice size, the boundary condition should be considered**)

$$\begin{aligned} \sum_n \hat{\Omega}^2 \hat{x}^2 \text{Retr}[1 - \bar{U}_{1,4}(n)] &= \sum_n \hat{\Omega}^2 \frac{2\hat{x}^2 + 2\hat{x} + 1}{2} \text{Retr}[1 - U_{1,4}(n)] \\ \sum_n \hat{\Omega}^2 \hat{y}^2 \text{Retr}[1 - \bar{U}_{2,4}(n)] &= \sum_n \hat{\Omega}^2 \frac{2\hat{y}^2 + 2\hat{y} + 1}{2} \text{Retr}[1 - U_{2,4}(n)] \\ \sum_n \hat{\Omega}^2 (\hat{x}^2 + \hat{y}^2) \text{Retr}[1 - \bar{U}_{3,4}(n)] &= \sum_n \hat{\Omega}^2 (\hat{x}^2 + \hat{y}^2) \text{Retr}[1 - U_{3,4}(n)] \end{aligned} \quad (541)$$

Using

$$\begin{aligned} \text{Retr}[U_{\mu,\nu}(n - a\nu)] &= \text{Retr}[U_\mu(n - a\nu)U_\nu(n + a\mu - a\nu)U_\mu^{-1}(n)U_\nu^{-1}(n - a\nu)] \\ &= \text{Retr}[U_\nu(n - a\nu)U_\mu(n)U_\nu^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)] \\ &= \text{Retr}[U_\mu(n)U_\nu^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)U_\nu(n - a\nu)] \end{aligned} \quad (542)$$

$$\begin{aligned} \sum_n g(n) \text{Retr}[1 - U_{\mu,\nu}(n)] &= N \times N_c - \sum_n \text{Retr}[U_\mu(n)\Sigma_\mu^\dagger(n)] \\ \Sigma_{\mu,i}(n, \nu) &= g_i(n)U_\nu(n)U_\mu(n + a\nu)U_\nu^{-1}(n + a\mu) + g_i(n - a\nu)U_\nu^{-1}(n - a\nu)U_\mu(n - a\nu)U_\nu(n + a\mu - a\nu) \end{aligned} \quad (543)$$

The product is just same as the definition of staples. However, there are two differences, (i) there is a coefficient function for each term of the sum. (ii) there is no sum over ν .

The following is usual, with the new definition of the staple, one have

$$F_\mu(n) = -\frac{\beta}{2N_c} \left\{ U_\mu(n) \Sigma_{\mu,i}^\dagger(n, \nu) \right\}_{TA} \quad (544)$$

with $i = 1, 2, 3$ and (**Note, this is for infinite lattice size, the boundary condition should be considered**)

$$\begin{aligned} g_1(n) &= \frac{\Omega^2(2x^2 + 2x + 1)}{2}, \quad g_2(n) = \frac{\Omega^2(2y^2 + 2y + 1)}{2}, \quad g_3(n) = \Omega^2(x^2 + y^2), \\ F_{\mu=1,2,3}(n) &= -\frac{\beta}{2N_c} \left\{ U_\mu(n) \Sigma_{\mu,\mu}^\dagger(n, 4) \right\}_{TA} \\ F_4(n) &= -\frac{\beta}{2N_c} \left\{ U_4(n) \sum_{i=1,2,3} \Sigma_{4,i}^\dagger(n, i) \right\}_{TA} \end{aligned} \quad (545)$$

Now we consider the force of V

$$\begin{aligned} V_{\mu\nu\sigma} &= \frac{1}{8} (U_{\mu,\nu}U_{\nu,\sigma} + U_{-\mu,\nu}U_{\nu,-\sigma} + U_{\mu,-\nu}U_{-\nu,\sigma} + U_{-\mu,-\nu}U_{-\nu,-\sigma} \\ &\quad - U_{\mu,\nu}U_{\nu,-\sigma} - U_{-\mu,\nu}U_{\nu,\sigma} - U_{\mu,-\nu}U_{-\nu,-\sigma} - U_{-\mu,-\nu}U_{-\nu,\sigma}) \end{aligned} \quad (546)$$

Using

$$\text{Retr}[U_{\mu,\nu}U_{\nu,\sigma}] = \text{Retr}[U_{\sigma,\nu}U_{\nu,\mu}], \quad \text{Retr}[U_{\mu,\nu}U_{\nu,-\sigma}] = \text{Retr}[U_{-\sigma,\nu}U_{\nu,\mu}] \quad (547)$$

one have

$$\text{Retr}[V_{\mu\nu\rho}] = \text{Retr}[V_{\rho\nu\mu}] \quad (548)$$

So we only need to calculate $\frac{\partial}{\partial\omega_\mu} V_{\mu\nu\rho}$ and $\frac{\partial}{\partial\omega_\nu} V_{\mu\nu\rho}$.

One can find

$$\sum_n \text{Retr}[g(n)V_{\mu\nu\rho}(n)] \rightarrow S[U_\mu(n)] = \text{Retr}[U_\mu(n)M(n)] \quad (549)$$

with

$$\begin{aligned}
M(n) = & \frac{1}{8} \left((g(n) + g(n + a\nu))U_\nu(n + a\mu)U_\mu^{-1}(n + a\nu)U_\rho(n + a\nu)U_\nu^{-1}(n + a\rho)U_\rho^{-1}(n) \right. \\
& + (g(n) + g(n - a\nu))U_\nu^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)U_\rho(n - a\nu)U_\nu(n - a\nu + a\rho)U_\rho^{-1}(n) \\
& + (g(n + a\mu - a\nu) + g(n + a\mu))U_\rho^{-1}(n + a\mu - a\rho)U_\nu^{-1}(n + a\mu - a\rho - a\nu) \\
& \times U_\rho(n + a\mu - a\rho - a\nu)U_\mu^{-1}(n - a\nu)U_\nu(n - a\nu) \\
& + (g(n + a\mu + a\nu) + g(n + a\mu))U_\rho^{-1}(n + a\mu - a\rho)U_\nu(n + a\mu - a\rho) \\
& \times U_\rho(n + a\mu - a\rho + a\nu)U_\mu^{-1}(n + a\nu)U_\nu^{-1}(n) \\
& - (g(n + a\mu) + g(n + a\mu + a\nu))U_\rho(n + a\mu)U_\nu(n + a\mu + a\rho)U_\rho^{-1}(n + a\mu + a\nu)U_\mu^{-1}(n + a\nu)U_\nu^{-1}(n) \\
& - (g(n + a\mu) + g(n + a\mu - a\nu))U_\rho(n + a\mu)U_\nu^{-1}(n + a\mu + a\rho - a\nu) \\
& \times U_\rho^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)U_\nu(n - a\nu) \\
& - (g(n) + g(n + a\nu))U_\nu(n + a\mu)U_\mu^{-1}(n + a\nu)U_\rho^{-1}(n + a\nu - a\rho)U_\nu^{-1}(n - a\rho)U_\rho(n - a\rho) \\
& \left. - (g(n) + g(n - a\nu))U_\nu^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)U_\rho^{-1}(n - a\nu - a\rho)U_\nu(n - a\nu - a\rho)U_\rho(n - a\rho) \right)
\end{aligned} \tag{550}$$

It can be simplified as

$$\begin{aligned}
M(n) = & \frac{1}{8} \left((g(n) + g(n + a\nu))U_\nu(n + a\mu)U_\mu^{-1}(n + a\nu)S_1 \right. \\
& + (g(n) + g(n - a\nu))U_\nu^{-1}(n + a\mu - a\nu)U_\mu^{-1}(n - a\nu)S_2 \\
& + (g(n + a\mu) + g(n + a\mu + a\nu))S_3U_\mu^{-1}(n + a\nu)U_\nu^{-1}(n) \\
& \left. + (g(n + a\mu) + g(n + a\mu - a\nu))S_4U_\mu^{-1}(n - a\nu)U_\nu(n - a\nu) \right) \\
S_1 = & U_\rho(n + a\nu)U_\nu^{-1}(n + a\rho)U_\rho^{-1}(n) - U_\rho^{-1}(n + a\nu - a\rho)U_\nu^{-1}(n - a\rho)U_\rho(n - a\rho) \\
S_2 = & U_\rho(n - a\nu)U_\nu(n - a\nu + a\rho)U_\rho^{-1}(n) - U_\rho^{-1}(n - a\nu - a\rho)U_\nu(n - a\nu - a\rho)U_\rho(n - a\rho) \\
S_3 = & U_\rho^{-1}(n + a\mu - a\rho)U_\nu(n + a\mu - a\rho)U_\rho(n + a\mu - a\rho + a\nu) \\
& - U_\rho(n + a\mu)U_\nu(n + a\mu + a\rho)U_\rho^{-1}(n + a\mu + a\nu) \\
S_4 = & U_\rho^{-1}(n + a\mu - a\rho)U_\nu^{-1}(n + a\mu - a\rho - a\nu)U_\rho(n + a\mu - a\rho - a\nu) \\
& - U_\rho(n + a\mu)U_\nu^{-1}(n + a\mu + a\rho - a\nu)U_\rho^{-1}(n + a\mu - a\nu)
\end{aligned} \tag{551}$$

Similarly, one also have

$$\sum_n \text{Retr}[g(n)V_{\mu\nu\rho}(n)] \rightarrow S[U_\nu(n)] = \text{Retr}[U_\nu(n)N(n)] \tag{552}$$

where

$$\begin{aligned}
N(n) = & \frac{1}{8} \left\{ (g(n + a\mu) + g(n + a\nu + a\mu))U_\mu(n + a\nu)T_1U_\mu^{-1}(n) \right. \\
& + (g(n - a\mu) + g(n + a\nu - a\mu))U_\mu^{-1}(n + a\nu - a\mu)T_2U_\mu(n - a\mu) \\
& + (g(n + a\rho) + g(n + a\nu + a\rho))U_\rho(n + a\nu)T_3U_\rho^{-1}(n) \\
& \left. + (g(n - a\rho) + g(n + a\nu - a\rho))U_\rho^{-1}(n + a\nu - a\rho)T_4U_\rho(n - a\rho) \right\}, \\
T_1 = & U_\rho^{-1}(n + a\nu + a\mu - a\rho)U_\nu^{-1}(n + a\mu - a\rho)U_\rho(n + a\mu - a\rho) \\
& - U_\rho(n + a\nu + a\mu)U_\nu^{-1}(n + a\mu + a\rho)U_\rho^{-1}(n + a\mu), \\
T_2 = & U_\rho(n + a\nu - a\mu)U_\nu^{-1}(n - a\mu + a\rho)U_\rho^{-1}(n - a\mu) \\
& - U_\rho^{-1}(n + a\nu - a\mu - a\rho)U_\nu^{-1}(n - a\mu - a\rho)U_\rho(n - a\mu - a\rho), \\
T_3 = & U_\mu^{-1}(n + a\nu + a\rho - a\mu)U_\nu^{-1}(n + a\rho - a\mu)U_\mu(n + a\rho - a\mu) \\
& - U_\mu(n + a\nu + a\rho)U_\nu^{-1}(n + a\mu + a\rho)U_\mu^{-1}(n + a\rho), \\
T_4 = & U_\mu(n + a\nu - a\rho)U_\nu^{-1}(n - a\rho + a\mu)U_\mu^{-1}(n - a\rho) \\
& - U_\mu^{-1}(n + a\nu - a\mu - a\rho)U_\nu^{-1}(n - a\mu - a\rho)U_\mu(n - a\mu - a\rho),
\end{aligned} \tag{553}$$

Note that the T_3, T_4 terms are T_1, T_2 terms with $\mu \leftrightarrow \rho$.

One can further reduce the dagger operation by defining

$$S[U_\mu(n)] = \text{Retr}[U_\mu(n)M^\dagger(n)], \quad S[U_\nu(n)] = \text{Retr}[U_\nu(n)N^\dagger(n)] \tag{554}$$

with

$$\begin{aligned}
M(n) &= \frac{1}{8} \left((g(n) + g(n + a\nu)) S_1 U_\mu(n + a\nu) U_\nu^{-1}(n + a\mu) \right. \\
&\quad + (g(n) + g(n - a\nu)) S_2 U_\mu(n - a\nu) U_\nu(n + a\mu - a\nu) \\
&\quad + (g(n + a\mu) + g(n + a\mu + a\nu)) U_\nu(n) U_\mu(n + a\nu) S_3 \\
&\quad \left. + (g(n + a\mu) + g(n + a\mu - a\nu)) U_\nu^{-1}(n - a\nu) U_\mu(n - a\nu) S_4 \right) \\
S_1 &= U_\rho(n) U_\nu(n + a\rho) U_\rho^{-1}(n + a\nu) - U_\rho^{-1}(n - a\rho) U_\nu(n - a\rho) U_\rho(n + a\nu - a\rho) \\
S_2 &= U_\rho(n) U_\nu^{-1}(n - a\nu + a\rho) U_\rho^{-1}(n - a\nu) - U_\rho^{-1}(n - a\rho) U_\nu^{-1}(n - a\nu - a\rho) U_\rho(n - a\nu - a\rho) \\
S_3 &= U_\rho^{-1}(n + a\mu - a\rho + a\nu) U_\nu^{-1}(n + a\mu - a\rho) U_\rho(n + a\mu - a\rho) \\
&\quad - U_\rho(n + a\mu + a\nu) U_\nu^{-1}(n + a\mu + a\rho) U_\rho^{-1}(n + a\mu) \\
S_4 &= U_\rho^{-1}(n + a\mu - a\rho - a\nu) U_\nu(n + a\mu - a\rho - a\nu) U_\rho(n + a\mu - a\rho) \\
&\quad - U_\rho(n + a\mu - a\nu) U_\nu(n + a\mu + a\rho - a\nu) U_\rho^{-1}(n + a\mu)
\end{aligned} \tag{555}$$

and

$$\begin{aligned}
N(n) &= \frac{1}{8} (N(\mu, \rho)(n) + N(\rho, \mu)(n)) \\
N(\mu, \rho)(n) &= \left\{ (g(n + a\mu) + g(n + a\nu + a\mu)) U_\mu(n) T_1 U_\mu^{-1}(n + a\nu) \right. \\
&\quad \left. + (g(n - a\mu) + g(n + a\nu - a\mu)) U_\mu^{-1}(n - a\mu) T_2 U_\mu(n + a\nu - a\mu) \right\}, \\
T_1 &= U_\rho^{-1}(n + a\mu - a\rho) U_\nu(n + a\mu - a\rho) U_\rho(n + a\nu + a\mu - a\rho) \\
&\quad - U_\rho(n + a\mu) U_\nu(n + a\mu + a\rho) U_\rho^{-1}(n + a\nu + a\mu), \\
T_2 &= U_\rho(n - a\mu) U_\nu(n - a\mu + a\rho) U_\rho^{-1}(n + a\nu - a\mu) \\
&\quad - U_\rho^{-1}(n - a\mu - a\rho) U_\nu(n - a\mu - a\rho) U_\rho(n + a\nu - a\mu - a\rho),
\end{aligned} \tag{556}$$

Note, instead of $S_G[U_\mu] = -U_\mu \Sigma_\mu^\dagger$, here it is $S_G[U_\mu] = +U_\mu M_\mu^\dagger$ and $S_G[U_\mu] = +U_\mu N_\mu^\dagger$.

8.1.6 The force from fermion action

The first step is to shift the second term to factorize $U_\mu(n)$ out

$$\begin{aligned}
\sum_{m,n} g(n) (1 + \gamma_\mu) U_{-\mu}(n) \delta_{n-a\mu,m} &= \sum_{m,n} g(n) (1 + \gamma_\mu) U_\mu^{-1}(n - a\mu) \delta_{n-a\mu,m} \\
&= \sum_{m,n} g(n + a\mu) (1 + \gamma_\mu) U_\mu^{-1}(n) \delta_{n,m}
\end{aligned} \tag{557}$$

It is more convenient to split the D operator as (note that for yU_x , $g(n) = y$, and $g(n) = g(n + x)$, xU_y is similar. Also, note that, it is also true for open boundary)

$$\begin{aligned}
M^a &= \left\{ (1 - \gamma_\mu) T^a U_\mu \delta_{x_L, (x+\mu)_R} - (1 + \gamma_\mu) U_\mu^{-1} T^a \delta_{(x+\mu)_L, x_R} \right\} \\
&\quad - y \Omega \delta_{\mu,x} \left\{ (1 - \gamma_4) T^a U_\mu \delta_{x_L, (x+\mu)_R} - (1 + \gamma_4) U_\mu^{-1} T^a \delta_{(x+\mu)_L, x_R} \right\} \\
&\quad + x \Omega \delta_{\mu,y} \left\{ (1 - \gamma_4) T^a U_\mu \delta_{x_L, (x+\mu)_R} - (1 + \gamma_4) U_\mu^{-1} T^a \delta_{(x+\mu)_L, x_R} \right\} \\
F_{pf} &= 2i\kappa \sum_a \text{Im} \left[\left(\phi_1^\dagger M_a \phi_2 \right) \right] T_a
\end{aligned} \tag{558}$$

with

$$\phi_1 = \left(\left(\hat{D} \hat{D}^\dagger \right)^{-1} \phi \right), \quad \phi_2 = \hat{D}^\dagger \phi_1, \tag{559}$$

similarly, with

$$\begin{aligned}
\phi_{L1}(n) &= \phi_1(n), \quad \phi_{R1}(n) = \{ (1 - \gamma_\mu) + (x \Omega \delta_{\mu,y} - y \Omega \delta_{\mu,x}) (1 - \gamma_4) \} \phi_2(n + \mu), \\
\phi_{L2}(n) &= \phi_1(n + \mu), \quad \phi_{R1}(n) = \{ (1 + \gamma_\mu) + (x \Omega \delta_{\mu,y} - y \Omega \delta_{\mu,x}) (1 + \gamma_4) \} \phi_2(n),
\end{aligned} \tag{560}$$

$$F_\mu^{pf}(n) = \kappa \left\{ U_\mu(n) \left(\phi_{R1} \phi_{L1}^\dagger + \phi_{R2} \phi_{L2}^\dagger \right) \right\} \Big|_{TA} \tag{561}$$

Note that **Both the force from gauge and fermion actions are kept anti-hermitian traceless.**

8.1.7 The angular momentum

The angular momentum operator is defined as

$$\begin{aligned}
 J &\equiv \left. \frac{\delta \mathcal{L}}{\delta \Omega} \right|_{\Omega=0} \\
 &= J_G + J_{FL} + J_{FS} \\
 J_G &= \frac{\beta}{N_c} \sum_n (\hat{x} \text{Retr}[V_{412}(n) + V_{432}(n)] - \hat{y} \text{Retr}[V_{421}(n) + V_{431}(n)]) \\
 J_{FL} &= \bar{\psi} \{ -\kappa \hat{y} ((-\gamma_4) U_x(n) \delta_{n+a\hat{x},m} + (+\gamma_4) U_{-x}(n) \delta_{n-a\hat{x},m}) \\
 &\quad + \hat{x} ((-\gamma_4) U_y(n) \delta_{n+a\hat{y},m} + (+\gamma_4) U_{-y}(n) \delta_{n-a\hat{y},m}) \} \\
 &= -\kappa \bar{\psi} \gamma_4 (\hat{y} D_x - x D_y) \psi, \\
 J_{FS} &= -i \kappa \bar{\psi} \gamma_4 \sigma^{12} \psi.
 \end{aligned} \tag{562}$$

The result is derived as $\delta \mathcal{L} / \delta \hat{\Omega}$, therefor, the result has unit as a^{-3} .

The measurement of $\langle J_G \rangle$ is straightforward. The measurement of J_{FL} and J_{FS} are inertia mass densities of quark-antiquark pairs. So, for the $u - \bar{u}$ pair. On the other hand, J is **NOT** a local operator. $\langle J_F(n|m) \rangle$ can be written as (in the spinor space, where a, b are spinor indices)

$$\begin{aligned}
 \langle J_F(n|m) \rangle &= \langle \bar{u}(n) O(n|m) u(m) \rangle \\
 &= \sum_{a,b} O_{a,b}(n|m) \langle \bar{u}_a(n) u_b(m) \rangle = - \sum_{a,b} O_{a,b}(n|m) \langle u_b(m) \bar{u}_a(n) \rangle \\
 &= - \sum_{a,b} O_{a,b}(n|m) D^{-1}(m|n)_{b,a} = -\text{tr}_{c,s} [O(n|m) D^{-1}(m|n)]
 \end{aligned} \tag{563}$$

So the definition of local angular momentum density should be

$$\langle J_F(n) \rangle = - \sum_{a,b,m} O_{a,b}(n|m) D^{-1}(m|n)_{b,a} = -\text{tr}_{c,s,m} [O(n|m) D^{-1}(m|n)] \tag{564}$$

So, we need to calculate $\sum_n O(m_0|n) D^{-1}(n|m_0)$ as a matrix in color and spinor space. It can be done by introduce the source

$$\phi_{m_0, c_2, s_2}^S(m)_{c,s} = \delta(m - m_0) \delta(c - c_2) \delta(s - s_2) \tag{565}$$

and $D^{-1}(n|m_0)_{c,s}$ as a vector $\vec{v}(n)$ can be written as

$$\begin{pmatrix} D_{1,cs}^{-1}(n) \\ D_{2,cs}^{-1}(n) \\ D_{3,cs}^{-1}(n) \\ \dots \\ D_{10,cs}^{-1}(n) \\ D_{11,cs}^{-1}(n) \\ D_{12,cs_2}^{-1}(n) \end{pmatrix} = \begin{pmatrix} D_{1,1}^{-1} & D_{1,2}^{-1} & \dots & D_{1,cs}^{-1}(n|m_0) & \dots & D_{1,11}^{-1} & D_{1,12}^{-1} \\ D_{2,1}^{-1} & D_{2,2}^{-1} & \dots & D_{2,cs}^{-1}(n|m_0) & \dots & D_{2,11}^{-1} & D_{2,12}^{-1} \\ D_{3,1}^{-1} & D_{3,2}^{-1} & \dots & D_{3,cs}^{-1}(n|m_0) & \dots & D_{3,11}^{-1} & D_{3,12}^{-1} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ D_{10,1}^{-1} & D_{10,2}^{-1} & \dots & D_{10,cs}^{-1}(n|m_0) & \dots & D_{10,11}^{-1} & D_{10,12}^{-1} \\ D_{11,1}^{-1} & D_{11,2}^{-1} & \dots & D_{11,cs}^{-1}(n|m_0) & \dots & D_{11,11}^{-1} & D_{11,12}^{-1} \\ D_{12,1}^{-1} & D_{12,2}^{-1} & \dots & D_{12,cs}^{-1}(n|m_0) & \dots & D_{12,11}^{-1} & D_{12,12}^{-1} \end{pmatrix} \begin{pmatrix} 0 \\ \dots \\ 0 \\ 1_{idx=cs, x=m_0} \\ 0 \\ \dots \\ 0 \end{pmatrix} \tag{566}$$

and $\sum_n O(m|n) D^{-1}(n|m_0)_{c,s}$ as a vector $\vec{v}(m)$ can be written as

$$\begin{pmatrix} OD_{1,cs}^{-1}(m) \\ OD_{2,cs}^{-1}(m) \\ \dots \\ OD_{11,cs}^{-1}(m) \\ OD_{12,cs_2}^{-1}(m) \end{pmatrix} = \begin{pmatrix} O_{1,1} & O_{1,2} & \dots & O_{1,11} & O_{1,12} \\ O_{2,1} & O_{2,2} & \dots & O_{2,11} & O_{2,12} \\ O_{3,1} & O_{3,2} & \dots & O_{3,11} & O_{3,12} \\ \dots & \dots & \dots & \dots & \dots \\ O_{10,1} & O_{10,2} & \dots & O_{10,11} & O_{10,12} \\ O_{11,1} & O_{11,2} & \dots & O_{11,11} & O_{11,12} \\ O_{12,1} & O_{12,2} & \dots & O_{12,11} & O_{12,12} \end{pmatrix} \begin{pmatrix} D_{1,cs}^{-1} \\ D_{2,cs}^{-1} \\ \dots \\ D_{11,cs}^{-1} \\ D_{12,cs_2}^{-1} \end{pmatrix} \tag{567}$$

And the trace is just

$$\sum_{i=1}^{12} (OD^{-1})_{i,i}(n) \tag{568}$$

Also, note that, the D and ψ are scaled.

Now, we can consider the physical meaning of J_G . Using

$$F_{0i} = E^i, \quad F_{ij} = 2c\epsilon_{ijk}B_i \quad (569)$$

therefor

$$\begin{aligned} \mathbf{j} &= (x, y, 0) \times (\mathbf{E} \times \mathbf{B}) \\ \mathbf{j} &= (x, y, 0) \times ((F_{01}, F_{02}, F_{03}), F_{23}, F_{31}, F_{12}) \\ j_z &= -2c(xF_{01}F_{12} + yF_{02}F_{12} - xF_{03}F_{23} + yF_{03}F_{13}) \end{aligned} \quad (570)$$

Details about Angular momentum

In the lattice code, we use

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] = g_{YM} \sum_a T^a F_{\mu\nu}^a \quad (571)$$

and

$$F_{0i}^a = E_i^a, \quad F_{ij}^a = \epsilon_{ijk}B_i^a \quad (572)$$

and

$$\begin{aligned} \mathbf{J}_G &= \sum_a \mathbf{r} \times (\mathbf{E}^a \times \mathbf{B}^a) \\ &= - \begin{pmatrix} y(F_{02}^a F_{23}^a - F_{01}^a F_{31}^a) + z(F_{03}^a F_{23}^a - F_{01}^a F_{12}^a) \\ x(F_{01}^a F_{31}^a - F_{02}^a F_{23}^a) + z(F_{03}^a F_{31}^a - F_{02}^a F_{12}^a) \\ xF_{01}^a F_{12}^a + yF_{02}^a F_{12}^a - xF_{03}^a F_{23}^a + yF_{03}^a F_{13}^a \end{pmatrix} \end{aligned} \quad (573)$$

Using $\text{tr}[T_i T_j] = \frac{1}{2}\delta_{ij}$, one have

$$\frac{2}{g_{YM}^2} \text{tr}[F_{\mu\nu} F_{\rho\sigma}] = \sum_a F_{\mu\nu}^a F_{\rho\sigma}^a \quad (574)$$

with $\frac{\beta}{N_c} \equiv \frac{2}{g_{YM}^2}$, it is

$$\mathbf{J}_G = -\frac{\beta}{N_c} \text{tr}_c \left[\begin{pmatrix} y(F_{02}F_{23} - F_{01}F_{31}) + z(F_{03}F_{23} - F_{01}F_{12}) \\ x(F_{01}F_{31} - F_{02}F_{23}) + z(F_{03}F_{31} - F_{02}F_{12}) \\ xF_{01}F_{12} + yF_{02}F_{12} - xF_{03}F_{23} + yF_{03}F_{13} \end{pmatrix} \right], \quad (575)$$

After Wick rotation, which is to replace $F_{0i} \rightarrow iF_{0i}$, one have

$$\mathbf{J}_G^E = -i\frac{\beta}{N_c} \text{tr}_c \left[\begin{pmatrix} y(F_{02}F_{23} - F_{01}F_{31}) + z(F_{03}F_{23} - F_{01}F_{12}) \\ x(F_{01}F_{31} - F_{02}F_{23}) + z(F_{03}F_{31} - F_{02}F_{12}) \\ xF_{01}F_{12} + yF_{02}F_{12} - xF_{03}F_{23} + yF_{03}F_{13} \end{pmatrix} \right], \quad (576)$$

where tr_c is trace in color space. Then J_{Gz} is

$$J_{Gz}^E = -i\frac{\beta}{N_c} (x\text{tr}[F_{01}F_{12}] + y\text{tr}[F_{02}F_{12}] - x\text{tr}[F_{03}F_{23}] + y\text{tr}[F_{03}F_{13}]) \quad (577)$$

Then discretize using $\text{Retr}[V_{\mu\nu\rho}] = -a^4 \text{tr}[F_{\mu\nu}F_{\nu\rho}] + \mathcal{O}(a^6)$, and make the index of time $0 \rightarrow 4$, and use dimensionless $\hat{x} = a^{-1}x$, it is

$$J_{Gz}^E = ia^{-3} \frac{\beta}{N_c} (\hat{x}\text{Retr}[V_{412} + V_{432}] - \hat{y}\text{Retr}[V_{421} + V_{431}]) \quad (578)$$

If we use $\partial\mathcal{L}/\partial\Omega$, it is

$$\begin{aligned} J'_G &= \frac{\partial\mathcal{L}_G}{\partial\Omega} \Big|_{\Omega=0} = a \left(\frac{\partial\mathcal{L}_G}{\partial(a\Omega)} \Big|_{\Omega=0} \right) \\ &= a^{-3} \left\{ \frac{\beta}{N_c} (\hat{x}\text{Retr}[V_{412} + V_{432}] - \hat{y}\text{Retr}[V_{421} + V_{431}]) \right\} \end{aligned} \quad (579)$$

So, the iJ'_G is just the z component of J_G^E in Ji decomposition [32]

Now, considering

$$\mathbf{L}_F = \frac{1}{i}\psi^\dagger \mathbf{r} \times \mathbf{D}\psi = \frac{1}{i}\bar{\psi}\gamma_0 \mathbf{r} \times \mathbf{D}\psi \quad (580)$$

using

$$\langle \bar{\psi}_{c_1, s_1}(m)\psi_{c_2, s_2}(n) \rangle_F \equiv \frac{1}{Z} \int \mathcal{D}[\bar{\psi}\psi] \bar{\psi}_i \psi_j \exp(-\sum_n \bar{\psi}(n)A\psi(n)) = -(A^{-1})_{c_2, s_2; c_1, s_1}(n|m) \quad (581)$$

with $\hat{D} = m - \not{D}$ and Wick rotated $S_F = a^4 \sum_n \bar{\psi}(n)\hat{D}\psi(n)$ (here S is not spin but action, once change the integral over a Minkovski space to a summation, the Wick rotation is already assumed. However, Wick rotation will not change $\mathbf{D}$.) Using that

$$\langle \bar{\psi}_{c_1, s_1}(m)\psi_{c_2, s_2}(n) \rangle_F \equiv \frac{1}{Z} \int \mathcal{D}[\bar{\psi}\psi] \bar{\psi}_i \psi_j \exp(-S_F) = -a^{-4}(\hat{D}^{-1})_{c_2, s_2; c_1, s_1}(n|m) \quad (582)$$

therefor

$$\begin{aligned} & \sum_{c_1, c_2, s_1, s_2} \delta_{c_1, c_2} \delta_{s_1, s_2} \langle \bar{\psi}_{c_1, s_1}(m) A_{c_1, s_1; c_2, s_2}(m|n) \psi_{c_2, s_2}(n) \rangle_F \\ &= -a^{-4}(\hat{D}^{-1})_{c_2, s_2; c_1, s_1}(n|m) A_{c_1, s_1; c_2, s_2}(m|n) \end{aligned} \quad (583)$$

consider a local operator it is

$$\begin{aligned} & \sum_{c_1, c_2, s_1, s_2} \delta_{c_1, c_2} \delta_{s_1, s_2} \langle \bar{\psi}_{c_1, s_1}(n) A_{c_1, s_1; c_2, s_2}(n|n) \psi_{c_2, s_2}(n) \rangle_F \\ &= -a^{-4}(\hat{D}^{-1})_{c_2, s_2; c_1, s_1}(n|n) A_{c_1, s_1; c_2, s_2}(n|n) = -a^{-4} \text{tr}_{c, s} [A(n) \hat{D}^{-1}(n)] \end{aligned} \quad (584)$$

Specifically, we are considering

$$\langle L_{F_z}^E \rangle = ia^{-4} \text{tr}_{c, s} [\gamma^0 (x(\partial_y + iA_y(n)) - y(\partial_x + iA_x(n))) \hat{D}^{-1}(n)] \quad (585)$$

The D operator and ψ fields are scaled ones. The $\tilde{D}$ and $\tilde{\psi}$ are the original ones.

The discretized Wilson operator is

$$\begin{aligned} D_W &\approx \frac{1}{m + \frac{4}{a}} \hat{D} \\ D_W(n|m) &= \delta_{n, m} - \kappa \sum_{\mu} ((1 - \gamma_{\mu}) U_{\mu}(n) \delta_{n+a\mu, m} + (1 + \gamma_{\mu}) U_{-\mu}(n) \delta_{n-a\mu, m}) \\ &+ y\Omega\gamma_4\kappa (U_x(n) \delta_{n+ax, m} - U_{-x}(n) \delta_{n-ax, m}) - x\Omega\gamma_4\kappa (U_y(n) \delta_{n+ay, m} - U_{-y}(n) \delta_{n-ay, m}) - i\kappa\gamma_4 a\Omega\sigma^{12} \delta_{n, m} \\ \kappa &= \frac{1}{2am + 8} \end{aligned} \quad (586)$$

so

$$\langle L_{F_z}^E \rangle = ia^{-4} \frac{1}{m + \frac{4}{a}} \text{tr}_{c, s} [\gamma^0 (x(\partial_y + iA_y(n)) - y(\partial_x + iA_x(n))) \hat{D}_W^{-1}(n)] \quad (587)$$

and use

$$(\partial_{\mu} + iA_{\mu})\psi(n) = \frac{U_{\mu}(n)\psi(n+a\mu) - U_{-\mu}(n)\psi(n-a\mu)}{2a} + \mathcal{O}(a) \quad (588)$$

so

$$\begin{aligned} \langle L_{F_z}^E \rangle &= ia^{-3} \kappa \text{tr}_{c, s} [\gamma^4 (\hat{x} (U_y(n) \delta_{n+ay, m} - U_{-y}(n) \delta_{n-ay, m}) \\ &- \hat{y} (U_x(n) \delta_{n+ax, m} - U_{-x}(n) \delta_{n-ax, m})) \hat{D}_W^{-1}(n)] \end{aligned} \quad (589)$$

Similarly, using

$$\begin{aligned} \left. \frac{\partial \mathcal{L}}{\partial \Omega} \right|_{\Omega=0} &= \bar{\psi} \{ -\kappa y ((-\gamma_4) U_x(n) \delta_{n+a\hat{x}, m} + (+\gamma_4) U_{-x}(n) \delta_{n-a\hat{x}, m}) \\ &+ \kappa x ((-\gamma_4) U_y(n) \delta_{n+a\hat{y}, m} + (+\gamma_4) U_{-y}(n) \delta_{n-a\hat{y}, m}) \} \psi \end{aligned} \quad (590)$$

and

$$\begin{aligned}
\langle L'_F \rangle &= \left\langle \frac{\partial \mathcal{L}}{\partial \Omega} \right|_{\Omega=0} \rangle = -a^{-4} \text{tr} \left[-\kappa y \left((-\gamma_4) U_x(n) \delta_{n+a\hat{x},m} + (+\gamma_4) U_{-x}(n) \delta_{n-a\hat{x},m} \right) \right. \\
&\quad \left. + \kappa x \left((-\gamma_4) U_y(n) \delta_{n+a\hat{y},m} + (+\gamma_4) U_{-y}(n) \delta_{n-a\hat{y},m} \right) \hat{D}_W \right] \\
&= \kappa a^{-3} \text{tr}_{c,s} \left[\gamma^4 \left(\hat{x} (U_y(n) \delta_{n+ay,m} - U_{-y}(n) \delta_{n-ay,m}) \right. \right. \\
&\quad \left. \left. - \hat{y} (U_x(n) \delta_{n+ax,m} - U_{-x}(n) \delta_{n-ax,m}) \right) \hat{D}_W^{-1}(n) \right]
\end{aligned} \tag{591}$$

Again, $i\langle L'_F \rangle$ is the z component of L_F in Ji decomposition [32].

Note that κ entered because of $\hat{D}_W^{-1}$. Later we will calculate D_{pure} acting on A_{phys} , where κ shall not show up.

Similarly, considering (here $\gamma_{1,2}$ have already been Wick rotated)

$$\begin{aligned}
\mathbf{S}_F &= \tilde{\psi}^\dagger \frac{1}{2} \boldsymbol{\Sigma} \tilde{\psi} \\
\Sigma_z &\equiv \begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix} = \sigma_{12}^E \equiv \frac{i}{2} (\gamma_2^E \gamma_1^E - \gamma_1^E \gamma_2^E) \\
S_{F_z} &= \psi^\dagger \frac{1}{2} \sigma_{12}^E \psi = \bar{\psi} \gamma_4 \frac{1}{2} \sigma_{12}^E \psi
\end{aligned} \tag{592}$$

therefor

$$\begin{aligned}
\langle S_{F_z} \rangle &= -a^{-4} \text{tr}_{c,s} \left[\gamma_4 \frac{1}{2} \sigma_{12}^E \hat{D}_W^{-1}(n) \right] = -a^{-4} \frac{1}{m + \frac{4}{a}} \text{tr}_{c,s} \left[\gamma_0 \frac{1}{2} \sigma^{12} \hat{D}_W^{-1}(n) \right] = -a^{-3} \kappa \text{tr}_{c,s} \left[\gamma_4 \sigma^{12} \hat{D}_W^{-1}(n) \right] \\
\langle S_{F_z} \rangle &= ia^{-3} \kappa \text{tr}_{c,s} \left[\gamma_4 i \sigma^{12} \hat{D}_W^{-1}(n) \right]
\end{aligned} \tag{593}$$

On the other hand

$$\begin{aligned}
\left\langle \frac{\partial \mathcal{L}}{\partial \Omega} \right|_{\Omega=0} \rangle &= \langle \bar{\psi} (-i\kappa \gamma_4 a \sigma^{12}) \psi \rangle \\
&= -a^{-4} \text{tr} \left[-i\kappa \gamma_4 a \sigma^{12} \hat{D}_W^{-1} \right] = a^{-3} \text{tr} \left[i\kappa \gamma_4 \sigma^{12} \hat{D}_W^{-1} \right]
\end{aligned} \tag{594}$$

Note that, finally, this term is discretized as an imaginary chiral potential such that

$$\begin{aligned}
\left\langle \frac{\partial \mathcal{L}}{\partial \Omega} \right|_{\Omega=0} \rangle &= \langle \bar{\psi} \left(-\kappa \frac{ia\sigma^{12}}{2} ((\gamma_4 - 1) U_\tau(n) \delta_{n,n+t} + (\gamma_4 + 1) U_{-\tau}(n) \delta_{n-t,n}) \right) \psi \rangle \\
&= a^{-3} \text{tr} \left[\frac{i\kappa \sigma^{12}}{2} ((\gamma_4 - 1) U_\tau(n) \delta_{n,n+t} + (\gamma_4 + 1) U_{-\tau}(n) \delta_{n-t,n}) \hat{D}_W^{-1} \right]
\end{aligned} \tag{595}$$

We can do it in Coulomb gauge, since in Coulomb gauge, $\mathbf{A} = \mathbf{A}_{phys}$, and $\mathbf{A}_{phys}$ is unchanged under gauge transformation.

Using the fact that

$$\begin{aligned}
E_{0i} &= F_{0i} = g_{YM} \sum_a T^a F_{0i}^a, \quad \mathbf{A} = g_{YM} \sum_a T^a \mathbf{A}^a, \quad \text{tr}[T^a T^b] = \frac{1}{2} \delta_{ab} \\
\sum_a \mathbf{E}^a \times \mathbf{A}^a &= \frac{2}{g_{YM}^2} \text{tr}[\mathbf{F}_{0i} \times \mathbf{A}] \\
\left(\sum_a \mathbf{E}^a \times \mathbf{A}^a \right)_z &= \frac{2}{g_{YM}^2} \text{tr}[F_{0x} A_y - F_{0y} A_x]
\end{aligned} \tag{596}$$

with

$$\begin{aligned}
\{M\}_{TA} &= \frac{1}{2} \left(M - M^\dagger - \frac{1}{3} \text{tr}(M - M^\dagger) \right). \\
U_\mu(n) &= \exp(iaA_\mu(n)) \\
A_\mu(n) &\approx a^{-1} \frac{1}{i} \{U_\mu(n)\}_{TA} \\
U_{\mu,\nu}(n) &\equiv U_\mu(n) U_\nu(n + a\mu) U_\mu^{-1}(n + a\nu) U_\nu^{-1}(n) = \exp(ia^2 F_{\mu\nu}(n) + \mathcal{O}(a^3)) \\
F_{\mu\nu}(n) &\approx a^{-2} \frac{1}{i} (U_{\mu\nu}(n))_{TA}
\end{aligned} \tag{597}$$

or we can use the Clover gauge field

$$F_{\mu\nu}^{clover}(n) = a^{-2} \frac{1}{4i} [U_{\mu,\nu}(n) + U_{\nu,-\mu}(n) + U_{-\mu,-\nu}(n) + U_{-\nu,\mu}(n)]_{TA} \tag{598}$$

so after Wick rotation $F_{0i} \rightarrow iF_{0i}$ it is

$$\left\langle \left(\sum_a \mathbf{E}^a \times \mathbf{A}^a \right)_z \right\rangle = ia^{-3} \frac{\beta}{N_c} \text{tr}[F_{0x}^{\text{clover}} \hat{A}_y - F_{0y}^{\text{clover}} \hat{A}_x] \quad (599)$$

where $\hat{A} = aA$

Note that, for all angular momentums, the lattice version is $i\langle \dots \rangle_{\text{lat}} = \langle \dots \rangle$, therefor, our final form of angular momentum is

$$\begin{aligned} \left\langle \left(\sum_a \mathbf{E}^a \times \mathbf{A}^a \right)_z \right\rangle_{\text{lat}} &= a^{-3} \frac{\beta}{N_c} \text{tr}[F_{4x}^{\text{clover}} \hat{A}_y - F_{4y}^{\text{clover}} \hat{A}_x] \\ &= -a^{-3} \frac{\beta}{N_c} \text{tr}[\{U_{4x}^{\text{clover}}\}_{TA} \{U_{phys,y}\}_{TA} - \{U_{4y}^{\text{clover}}\}_{TA} \{U_{phys,x}\}_{TA}] \end{aligned} \quad (600)$$

The gauge invariant D_{pure} for fermion can be obtained by

$$(\partial_\mu + iA_\mu)\psi(n) = \frac{U_\mu(n)\psi(n+a\mu) - U_{-\mu}(n)\psi(n-a\mu)}{2a} + \mathcal{O}(a) \quad (601)$$

so

$$(\partial_\mu + i(A_\mu - A_{phys,\mu})\psi(n) = \frac{U_\mu(n)\psi(n+a\mu) - U_{-\mu}(n)\psi(n-a\mu)}{2a} - \frac{2iaA_{phys,\mu}(n)}{2a}\psi(n) + \mathcal{O}(a) \quad (602)$$

Note that $2iaA_{phys}$ is just $2\{U_{phys}\}_{TA}$.

Another momentum angular is

$$\sum_{a,i} E^{ai}(\mathbf{x} \times \mathbf{D}_{\text{pure}})A^{ai} \quad (603)$$

with $D_{\text{pure}} = \partial_\mu + ig[\tilde{A}_{\text{pure},\mu}, \cdot]$ (in our definition, $A = g\tilde{A}$)

After Wick rotation, it is

$$i \sum_{a,i} F_{0i}^a(\mathbf{x} \times \mathbf{D}_{\text{pure}})A^{ai} = i \frac{2}{g_{YM}^2} \sum_i \text{tr}[F_{0i}(\mathbf{x} \times \mathbf{D}_{\text{pure}})A^i] \quad (604)$$

Note that

$$\begin{aligned} &U_\mu(n)A_{phys,\nu}(n+\mu)U_\mu^\dagger(n) - U_\mu^\dagger(n-\mu)A_{phys,\nu}(n-\mu)U_\mu(n-\mu) \\ &= (1 + iaA_\mu(n) + \mathcal{O}(a^2))A_{phys,\nu}(n+\mu)(1 - iaA_\mu(n) + \mathcal{O}(a^2)) \\ &\quad - (1 - iaA_\mu(n-\mu) + \mathcal{O}(a^2))A_{phys,\nu}(n-\mu)(1 - iaA_\mu(n-\mu) + \mathcal{O}(a^2)) \\ &= A_{phys,\nu}(n+\mu) - A_{phys,\nu}(n-\mu) \\ &\quad + ia[A_\mu(n), A_{phys,\nu}(n+\mu)] + ia[A_\mu(n-\mu), A_{phys,\nu}(n-\mu)] + \mathcal{O}(a^2) \\ &= A_{phys,\nu}(n+\mu) - A_{phys,\nu}(n-\mu) + 2ia[A_\mu(n), A_{phys,\nu}(n)] + \mathcal{O}(a^2) \end{aligned} \quad (605)$$

in the last step, $A(n+\mu) \approx A(n) + \mathcal{O}(a)$ is used. Then

$$\begin{aligned} &U_\mu(n)A_{phys,\nu}(n+\mu)U_\mu^\dagger(n) - U_\mu^\dagger(n-\mu)A_{phys,\nu}(n-\mu)U_\mu(n-\mu) \\ &= 2a \left(\frac{A_{phys,\nu}(n+\mu) - A_{phys,\nu}(n-\mu)}{2a} + i[A_\mu(n), A_{phys,\nu}(n)] \right) + \mathcal{O}(a^2) \\ &\approx 2a(\partial_\mu A_{phys,\nu} + i[A_{\text{pure},\mu}, A_{phys,\nu}]) = 2aD_{\text{pure},\mu}A_{phys,\nu} \end{aligned} \quad (606)$$

So

$$\begin{aligned} D_{\text{pure},\mu}A_{phys,\nu} &= \frac{1}{2a} (U_\mu(n)A_{phys,\nu}(n+\mu)U_\mu^\dagger(n) - U_\mu^\dagger(n-\mu)A_{phys,\nu}(n-\mu)U_\mu(n-\mu)) \\ &= \frac{1}{2ia^2} (U_\mu(n)\{U_{phys,\nu}\}_{TA}(n+\mu)U_\mu^\dagger(n) - U_\mu^\dagger(n-\mu)\{U_{phys,\nu}\}_{TA}(n-\mu)U_\mu(n-\mu)) \\ &\equiv \frac{1}{2ia^2}(DA)_{\mu\nu} \end{aligned} \quad (607)$$

and

$$\begin{aligned} i \sum_{a,j} F_{0j}^a(\mathbf{x} \times \mathbf{D}_{\text{pure}})A^{aj} &= i \sum_j \frac{1}{i} a^{-2} \frac{\beta}{N_C} \text{tr}[\{U_{0j}\}_{TA}(\mathbf{x} \times \mathbf{D}_{\text{pure}})A^{aj}] \\ &= -i \sum_j \frac{1}{2} a^{-4} \frac{\beta}{N_C} \text{tr}[\{U_{0j}\}_{TA}(x(DA)_{yj} - y(DA)_{xj})] \\ &= -i \frac{1}{2} \frac{\beta}{N_C} a^{-3} \sum_j \text{tr}[\{U_{4j}\}_{TA}(\hat{x}(DA)_{1j} - \hat{y}(DA)_{0j})] \end{aligned} \quad (608)$$

8.1.8 The Current density and Charge density

In the case of exponential σ^{12} term, it is interesting to also measure

$$J_{12} = -i\kappa \langle \bar{\psi} \gamma_4 \sigma^{12} \psi \rangle \quad (609)$$

Also the currents defined as

$$J_\mu = \langle \bar{\psi} \gamma_\mu \psi \rangle \quad (610)$$

is measured, such that

$$\begin{aligned} J_x &= \langle \bar{\psi} (\gamma_1 + y\Omega\gamma_4) \psi \rangle \\ J_y &= \langle \bar{\psi} (\gamma_2 - x\Omega\gamma_4) \psi \rangle \\ J_z &= \langle \bar{\psi} \gamma_3 \psi \rangle \\ J_\tau &= \langle \bar{\psi} \gamma_4 \psi \rangle \end{aligned} \quad (611)$$

we also measure the

$$\begin{aligned} J_1 &= \langle \bar{\psi} \gamma_1 \psi \rangle \\ J_2 &= \langle \bar{\psi} \gamma_2 \psi \rangle \end{aligned} \quad (612)$$

and the chiral charge density

$$n_5 = a^3 \langle \bar{\psi} \gamma_4 \gamma_5 \psi \rangle \quad (613)$$

8.1.9 The Topological Density

The topological charge is defined as (**This might has to be modified in the rotating frame!**)

$$\begin{aligned} Q &= \frac{1}{32\pi^2} a^4 \sum_n \epsilon_{\mu\nu\rho\sigma} \text{tr} [C_{\mu\nu}(n) C_{\rho\sigma}(n)] \\ C_{\mu\nu}(n) &= \text{Im} [U_{\mu\nu}(n)] \end{aligned} \quad (614)$$

Another definition is

$$\begin{aligned} Q &= \frac{1}{32\pi^2} a^4 \sum_n \epsilon_{\mu\nu\rho\sigma} \text{tr} [C_{\mu\nu}^{clover}(n) C_{\rho\sigma}^{clover}(n)] \\ C_{\mu\nu}^{clover}(n) &= \frac{1}{4} \text{Im} [U_{\mu,\nu}(n) + U_{\nu,-\mu}(n) + U_{-\mu,-\nu}(n) + U_{-\nu,\mu}(n)] \end{aligned} \quad (615)$$

Note for both $C_{\mu\nu}$ and $C_{\mu\nu}^{clover}$, one have $C_{\mu\nu} = -C_{\nu\mu}$, alone with $\epsilon_{\mu\nu\rho\sigma}$, it doubles the term. Therefor

$$\sum \epsilon_{\mu\nu\rho\sigma} \text{tr} [C_{\mu\nu}(n) C_{\rho\sigma}(n)] = 8 (\text{tr} [C_{12}(n) C_{34}(n)] - \text{tr} [C_{13}(n) C_{24}(n)] + \text{tr} [C_{14}(n) C_{23}(n)]) \quad (616)$$

8.1.10 The Polyakov loop

Polyakov loop is measured straight forwardly.

8.1.11 The Chiral Condensate

The Chiral condensate can be calculate by Grassman number integral

$$\langle \bar{u}u \rangle = \text{tr} [D_u^{-1}] \quad (617)$$

We are using two degenerate fermions, so

$$\langle \bar{\psi} \psi \rangle = \text{tr} [D^{-1}] = a^{-4} \frac{1}{m + \frac{4}{a}} \text{tr} [\hat{D}^{-1}] = a^{-4} \times 2a\kappa \text{tr} [\hat{D}^{-1}] = 2a^{-3} \kappa \text{tr} [\hat{D}^{-1}] \quad (618)$$

8.2 Sample Producer

In HMC, the most time-consuming operation is $(DD^\dagger)^{-1}\phi$, which need to solve the Wilson-Dirac equation, a matrix equation $\mathbf{b} = A\mathbf{x}$, where $A = DD^\dagger$ is a matrix depending on the gauge field and acting on the pseudo-fermion field.

At the same time, applying machine learning algorithms to physics problems has gained more and more attentions. The machine learning algorithms has been applied to solve partial differential equations [33]. In Ref. [34], deep learning is applied to map between potential and energy bypassing the need to solve the Schrödinger equation, in other words, the Schrödinger equation is implicitly solved by the network. So, it is reasonable to ask whether the machine learning can also help to solve the Wilson-Dirac equation? For example, is it possible to train the network to output eigenvectors by inputting a gauge field, or even better output $\mathbf{x}$ by inputting a gauge field and a pseudo-fermion field $\mathbf{b}$?

8.3 Data Analyse

We write the data analyse code based on Ref. [35] in Mathematica.

8.3.1 What is autocorrelation

The problem to address is that, the configurations generated are not statistically independent. So one need to take the relations between configurations into account.

To consider the relation between two sets, correlation functions are used, assuming two sets $a_{\alpha,\beta}$, assume $a_{\alpha,\beta} - \bar{a}_{\alpha,\beta}$ is a normal distribution.

$$\langle (a_\alpha - \bar{a}_\alpha)(a_\beta - \bar{a}_\beta) \rangle = \frac{1}{N^2} \sum_{i,j} \Gamma_{\alpha\beta}(j-i) \quad (619)$$

and

$$C_{\alpha\beta} = \sum_{t=-\infty}^{\infty} \Gamma_{\alpha\beta}(t) \quad (620)$$

Note that, $C_{\alpha\alpha}(0) = N\langle\delta_\alpha^2\rangle$ is the standard error.

For one single observable, one can define a correlation of a set of itself with delayed Markov time as

$$\tau_\alpha = \frac{1}{2\Gamma_{\alpha\alpha}(0)} \sum_{t=-\infty}^{\infty} \Gamma_{\alpha\alpha}(t) \quad (621)$$

For a purely exponential behaviour, $\Gamma_{\alpha\beta}(t) \sim \exp(-|t|/\tau)$. Generally, we can estimate $2\tau_\alpha$ as an interval such that two configurations are effectively independent [35].

Here, $\Gamma_{\alpha\beta}(t)$ is **autocorrelation**.

8.3.2 How to calculate autocorrelation, and how to use it to obtain the interval

Considering, we have already obtained a set of measurements by using configurations generated with Markov chain $\{a_\alpha^{i,r}\}$, where α indicating different observables, $r = 1 \rightarrow R$ indicating different replicas (usually, different replicas are obtained by running multi-times starting from same parameters, or running parallelly starting from same parameters), and $i = 1 \rightarrow N_r$ is index of each value in the replica.

Assume we measured $a_\alpha^{i,r}$, and want to obtain $F = f(a_\alpha)$.

In Ref. [35], a biased estimator is used such that

$$\begin{aligned} N &= \sum_r^R N_r, \\ \bar{a}_\alpha^r &= \frac{1}{N_r} \sum_i a_\alpha^{i,r} \\ \bar{\bar{a}}_\alpha &= \frac{1}{N} \sum_r^R N_r \bar{a}_\alpha^r \\ \bar{F} &= \frac{1}{N} \sum_r^R N_r f(\bar{a}_\alpha^r), \\ \bar{\bar{F}} &= f(\bar{\bar{a}}_\alpha) \end{aligned} \quad (622)$$

and

$$F_{mean} = \begin{cases} \bar{\bar{F}}, & R = 1 \\ \frac{R\bar{\bar{F}} - \bar{F}}{R-1} & R \geq 2 \end{cases} \quad (623)$$

The error is related to a correlation

$$\bar{\bar{\Gamma}}_{\alpha\beta}(t) = \frac{1}{N - Rt} \sum_{r=1}^R \sum_{i=1}^{N_r-t} (a_{\alpha}^{i,r} - \bar{a}_{\alpha}) (a_{\beta}^{i+t,r} - \bar{a}_{\beta}) \quad (624)$$

at first we need to project it onto single variable, for this purpose, we need to calculate gradient as

$$h_{\alpha} = \sqrt{\frac{\Gamma_{\alpha\alpha}(0)}{N}}, \quad (625)$$

$$\bar{\bar{f}}_{\alpha} \approx \frac{1}{2h_{\alpha}} (f(\bar{a}_1, \bar{a}_2, \dots, \bar{a}_{\alpha} + h_{\alpha}, \dots) - f(\bar{a}_1, \bar{a}_2, \dots, \bar{a}_{\alpha} - h_{\alpha}, \dots))$$

and

$$\bar{\bar{\Gamma}}_F(t) = \sum_{\alpha\beta} \bar{\bar{f}}_{\alpha} \bar{\bar{f}}_{\beta} \bar{\bar{\Gamma}}_{\alpha\beta}(t) \quad (626)$$

then the sum from $t = -\infty \rightarrow \infty$ is approximated as

$$\bar{\bar{C}}_F(W) = \bar{\bar{\Gamma}}_F(0) + 2 \sum_{t=1}^W \bar{\bar{\Gamma}}_F(t) \quad (627)$$

By definition, if window W is known, then

$$\bar{\bar{\tau}}_{int}(W) = \frac{\bar{\bar{C}}_F(W)}{2\bar{\bar{C}}_F(0)} \quad (628)$$

To get τ one need to use a factor S to fit the exponential, assuming

$$2\bar{\bar{\tau}}_{int}(W) = \sum_{t=-\infty}^{\infty} \exp\left(-\frac{S|t|}{\bar{\bar{\tau}}(W)}\right) \quad (629)$$

with S as a constant usually $S = 1 \rightarrow 2$.

Then one can let $\bar{\bar{\tau}}(W) = S\bar{\bar{\tau}}_{int}(W)$, and calculate

$$g(W) = \exp\left(-\frac{W}{\bar{\bar{\tau}}(W)}\right) - \frac{\bar{\bar{\tau}}(W)}{\sqrt{WN}} \quad (630)$$

and fix W as the first index such that $g(W) < 0$ change sign.

Once W is obtained, one can calculate $2\bar{\bar{\tau}}_{int}(W)$ which is the Markov time separation such that two configurations can be considered as independent. Also, the error estimate is

$$\delta_F^2 = \frac{\bar{\bar{C}}(W)}{N} \quad (631)$$

8.4 Constant Acceleration

8.4.1 Discretization

The metric of constant acceleration g at $\mathbf{z}$ direction is

$$g_{\mu\nu} = \begin{pmatrix} 1 - g^2 t^2 & 0 & 0 & gt \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ gt & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-\det(g_{\mu\nu})} = i \quad (632)$$

The analytical extension is simply add -2 to g_{00} ,

$$g_{\mu\nu} = \begin{pmatrix} -1 - g^2 t^2 & 0 & 0 & gt \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ gt & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-\det(g_{\mu\nu})} = 1 \quad (633)$$

Then

$$\begin{aligned} \mathcal{L} = & \frac{1}{2g_{YM}^2} \left(\sum_{ijkl=0}^3 \delta_{ij} \delta_{kl} \text{tr}[F_{ik} F_{jl}] + 2g^2 t^2 \text{tr}[F_{01}^2 + F_{02}^2] \right. \\ & \left. + 4gt (\text{tr}[F_{01} F_{13}] + \text{tr}[F_{02} F_{23}]) \right). \end{aligned} \quad (634)$$

And

$$\begin{aligned} i \int d^4x \sqrt{-\det(g_{\mu\nu})} \mathcal{L} = & -\frac{1}{2g_{YM}^2} \int d^4x \left(\sum_{ijkl=0}^3 \delta_{ij} \delta_{kl} \text{tr}[F_{ik} F_{jl}] + 2g^2 t^2 \text{tr}[F_{01}^2 + F_{02}^2] \right. \\ & \left. + 4gt (\text{tr}[F_{01} F_{13}] + \text{tr}[F_{02} F_{23}]) \right). \end{aligned} \quad (635)$$

Let $\exp(i \int d^4x \mathcal{L}) \rightarrow \exp(-S_F)$, it is

$$\begin{aligned} S_G = & \frac{1}{2g_{YM}^2} \int d^4x \left(\sum_{ijkl=0}^3 \delta_{ij} \delta_{kl} \text{tr}[F_{ik} F_{jl}] + 2g^2 t^2 \text{tr}[F_{01}^2 + F_{02}^2] \right. \\ & \left. + 4gt (\text{tr}[F_{01} F_{13}] + \text{tr}[F_{02} F_{23}]) \right). \end{aligned} \quad (636)$$

The discretized version is then

$$\begin{aligned} \sum_{ijkl=0}^3 \delta_{ij} \delta_{kl} \text{tr}[F_{ik} F_{jl}] &= 2 \sum_{\mu>\nu} \text{tr}[F_{\mu\nu}^2] \\ \text{Retr}[1 - U_{\mu\nu}] &= \text{Retr}[1 - \bar{U}_{\mu\nu}] = 2\text{tr}[F_{\mu\nu}^2] \\ \text{Retr}V_{\mu\nu\rho} &= 4\text{tr}[F_{\mu\nu} F_{\nu\rho}] \\ S_G &= a^4 \frac{\beta}{N_c} \sum_n \left(\sum_{\mu>\nu} \text{Retr}[1 - U_{\mu\nu}] + g^2 t^2 (\text{Retr}[1 - \bar{U}_{14}] + \text{Retr}[1 - \bar{U}_{24}]) \right. \\ & \left. + gt (\text{Retr}[V_{413}] + \text{Retr}[V_{423}]) \right). \end{aligned} \quad (637)$$

Note that $\text{Retr}[1 - U_{\mu\nu}] = \text{Retr}[1 - U_{\nu\mu}]$, $\text{Retr}[V_{\mu\nu\rho}] = \text{Retr}[V_{\rho\nu\mu}]$

For the fermions, we choose

$$\begin{aligned} e_0 &= i(1, 0, 0, gt), \quad e_1 = i(0, 1, 0, 0), \\ e_2 &= i(0, 0, 1, 0), \quad e_3 = i(0, 0, 0, 1). \end{aligned} \quad (638)$$

one can verify $\delta_{ab} = e_a^\mu e_b^\nu g_{\mu\nu}^E$.

The only non-zero derivative of metric is $\partial g_{11}/\partial x_1 = -2g^2 t$, $\partial g_{14}/\partial x_1 = g$, so the non-zero Christoffel connection are

$$\begin{aligned} \Gamma_{1,14} &= \frac{1}{2} \left(\frac{\partial g^{11}}{\partial x_4} + \frac{\partial g^{14}}{\partial x_1} - \frac{\partial g^{14}}{\partial x_1} \right) \\ \Gamma_{4,11} &= \frac{1}{2} \left(\frac{\partial g^{41}}{\partial x_1} + \frac{\partial g^{41}}{\partial x_1} - \frac{\partial g^{11}}{\partial x_4} \right) \\ \Gamma_{1,41} &= \frac{1}{2} \left(\frac{\partial g^{14}}{\partial x_1} + \frac{\partial g^{11}}{\partial x_4} - \frac{\partial g^{41}}{\partial x_1} \right) \\ \Gamma_{1,11} &= \frac{1}{2} \left(\frac{\partial g^{11}}{\partial x_1} \right) \end{aligned} \quad (639)$$

$$\begin{aligned} \Gamma_{1,14} &= \frac{1}{2} \left(\frac{\partial g^{14}}{\partial x_1} - \frac{\partial g^{14}}{\partial x_1} \right) = 0 \\ \Gamma_{4,11} &= \frac{1}{2} \left(\frac{\partial g^{41}}{\partial x_1} + \frac{\partial g^{41}}{\partial x_1} \right) = g \\ \Gamma_{1,41} &= \frac{1}{2} \left(\frac{\partial g^{14}}{\partial x_1} - \frac{\partial g^{41}}{\partial x_1} \right) = 0 \\ \Gamma_{1,11} &= \frac{1}{2} \left(\frac{\partial g^{11}}{\partial x_1} \right) = -g^2 t \end{aligned}$$

Then $\Gamma_{jk}^i = g_{il} \Gamma_{l,jk}$, jk can only be 11.

$$\Gamma_{11}^1 = \sum_n g_{1n} \Gamma_{n,11} = g_{11} \Gamma_{1,11} + g_{14} \Gamma_{4,11} = -1 \times (-g^2 t) + (-gt) \times g = 0 \quad (640)$$

The only non-zero is Γ_{11}^4 .

$$w_{\mu ij} = g_{\alpha\beta} e_i^\alpha (\partial_\mu e_j^\beta + \Gamma_{\mu\nu}^\beta e_j^\nu) \quad (641)$$

The μ of w can only be 1.

$$\begin{aligned} w_{1ij} &= g_{\alpha\beta} e_i^\alpha (\partial_t e_j^\beta + \Gamma_{1\nu}^\beta e_j^\nu) \\ &= g_{\alpha 4} e_i^\alpha (\partial_t e_j^\alpha + \Gamma_{11}^4 e_j^t) \\ &= g_{14} e_i^t (\partial_t e_j^z + \Gamma_{11}^4 e_j^t) + g_{44} e_i^z (\partial_t e_j^z + \Gamma_{11}^4 e_j^t) \\ &= g_{14} e_i^t (\partial_t e_j^z - g e_j^t) + g_{44} e_i^z (\partial_t e_j^z - g e_j^t) \end{aligned} \quad (642)$$

Just look at $\partial_t e_j^z - g e_j^t$, for any j , it is zero. So $w = 0$.

So the result is just

$$D_g = \left[i\gamma^\mu \left((\partial_\mu + ieA_\mu) - \frac{i}{4} \sigma^{ij} w_{\mu ij} \right) - m \right] \quad (643)$$

with $\gamma^\mu = \gamma^i e_i^\mu$ and $w = 0$. It is

$$D_g = \left[- \sum_i \gamma^i (\partial_i + ieA_i) - gt\gamma_0 (\partial_z + ieA_z) - m \right] \quad (644)$$

As a summary the Wick rotation is

$$\begin{aligned} -S_F &= i \int d^4x \sqrt{-g_{\alpha\beta}} \bar{\psi} D_a \psi \\ S_F &= \int d^4x_E \bar{\psi} [\gamma_1 (\partial_x + ieA_x) + \gamma_2 (\partial_y + ieA_y) + \gamma_3 (\partial_z + ieA_z) + \gamma_4 (\partial_\tau + ieA_\tau) + m \\ &\quad + gt\gamma_4 (\partial_z + ieA_z)] \psi \end{aligned} \quad (645)$$

The first line is Wilson-Dirac operator, satisfying γ_5 - hermiticity, the last line is discretized as

$$\begin{aligned} &\left(\gamma_4 \frac{U_\mu(n) \delta_{n,n+\mu} - U_{-\mu}(n) \delta_{n,n-\mu}}{2a} \right)^\dagger \\ &= \left(\frac{\delta_{n,n+\mu}^\dagger U_\mu^\dagger(n) - \delta_{n,n-\mu}^\dagger U_{-\mu}^\dagger(n)}{2a} \right) \gamma_4^\dagger \\ &= \left(\frac{\delta_{n+\mu,n} U_{-\mu}(n+\mu) - \delta_{n-\mu,n} U_\mu(n-\mu)}{2a} \right) \gamma_4 \end{aligned} \quad (646)$$

Do the shift then resemble, it is

$$\begin{aligned} &= \left(\frac{\delta_{n,n-\mu} U_{-\mu}(n) - \delta_{n,n+\mu} U_\mu(n)}{2a} \right) \gamma_4 \\ &= \gamma_4 \left(\frac{\delta_{n,n-\mu} U_{-\mu}(n) - \delta_{n,n+\mu} U_\mu(n)}{2a} \right) = - \left(\gamma_4 \frac{U_\mu(n) \delta_{n,n+\mu} - U_{-\mu}(n) \delta_{n,n-\mu}}{2a} \right) \end{aligned} \quad (647)$$

So it is anti-hermiticity, note that $\gamma_5 \gamma_4 \gamma_5 = -\gamma_4$, it is then γ_5 - hermiticity.

8.4.2 Force

- Gauge force

The $\bar{U}_{\mu\nu}$ term:

$$\sum_n f(n) \text{Retr}[1 - \bar{U}_{\mu,\nu}(n)] = \sum_n \frac{f(n) + f(n+a\mu) + f(n+a\nu) + f(n+a\mu+a\nu)}{4} \text{Retr}[1 - U_{\mu\nu}(n)] \quad (648)$$

So

$$\begin{aligned} g^2 t^2 \text{Retr}[1 - \bar{U}_{4,1}(n)] &= g^2 \frac{t^2 + (t+1)^2 + t^2 + (t+1)^2}{4} \text{Retr}[1 - U_{4,1}(n)] = \frac{g^2(2t^2 + 2t + 1)}{2} \text{Retr}[1 - U_{4,1}(n)] \\ g^2 t^2 \text{Retr}[1 - \bar{U}_{4,2}(n)] &= \frac{g^2(2t^2 + 2t + 1)}{2} \text{Retr}[1 - U_{4,2}(n)] \end{aligned} \quad (649)$$

Then

$$\begin{aligned}
 g(n) &= \frac{g^2(2t^2 + 2t + 1)}{2} \\
 \sum_n g(n) \text{Retr}[1 - U_{\mu,\nu}(n)] &= N \times N_c - \sum_n \text{Retr}[U_\mu(n) \Sigma_\mu^\dagger(n)] \\
 \Sigma_\mu(n, \nu) &= g(n) U_\nu(n) U_\mu(n + a\nu) U_\nu^{-1}(n + a\mu) + g(n - a\nu) U_\nu^{-1}(n - a\nu) U_\mu(n - a\nu) U_\nu(n + a\mu - a\nu)
 \end{aligned} \tag{650}$$

Note that there is no sum over ν , and $\nu = 1$ or $\nu = 2$.

Note that, if we put rotation and constant acceleration together, $f(n)$ function will not touch the Dirichlet boundary condition, while the $g(n)$ function will. So although $g(n)$ and $g(n - a\nu)$ have the same form $\frac{g^2(2t^2+2t+1)}{2}$, we will not combine them, because it is possible $g(n) = 0$ for Dirichlet boundary.

Finally

$$\begin{aligned}
 F_{\mu=1,2}(n) &= -\frac{\beta}{2N_c} \{U_\mu(n) \Sigma_\mu^\dagger(n, 4)\}_{TA}, \quad F_3 = 0 \\
 F_4(n) &= -\frac{\beta}{2N_c} \left\{ U_4(n) \sum_{i=1,2} \Sigma_4^\dagger(n, i) \right\}_{TA}
 \end{aligned} \tag{651}$$

The $V_{\mu\nu\rho}$ force is derived in the ‘Rotation’ chapter.

We have two kinds of V to calculate, one for μ, ρ and the other is for ν .

The result is

$$S[U_\mu(n)] = \text{Retr}[U_\mu(n) M^\dagger(n)], \quad S[U_\nu(n)] = \text{Retr}[U_\nu(n) N^\dagger(n)] \tag{652}$$

with

$$\begin{aligned}
 M(n) &= \frac{1}{8} \left((g(n) + g(n + a\nu)) S_1 U_\mu(n + a\nu) U_\nu^{-1}(n + a\mu) \right. \\
 &\quad + (g(n) + g(n - a\nu)) S_2 U_\mu(n - a\nu) U_\nu(n + a\mu - a\nu) \\
 &\quad + (g(n + a\mu) + g(n + a\mu + a\nu)) U_\nu(n) U_\mu(n + a\nu) S_3 \\
 &\quad \left. + (g(n + a\mu) + g(n + a\mu - a\nu)) U_\nu^{-1}(n - a\nu) U_\mu(n - a\nu) S_4 \right) \\
 S_1 &= U_\rho(n) U_\nu(n + a\rho) U_\rho^{-1}(n + a\nu) - U_\rho^{-1}(n - a\rho) U_\nu(n - a\rho) U_\rho(n + a\nu - a\rho) \\
 S_2 &= U_\rho(n) U_\nu^{-1}(n - a\nu + a\rho) U_\rho^{-1}(n - a\nu) - U_\rho^{-1}(n - a\rho) U_\nu^{-1}(n - a\nu - a\rho) U_\rho(n - a\nu - a\rho) \\
 S_3 &= U_\rho^{-1}(n + a\mu - a\rho + a\nu) U_\nu^{-1}(n + a\mu - a\rho) U_\rho(n + a\mu - a\rho) \\
 &\quad - U_\rho(n + a\mu + a\nu) U_\nu^{-1}(n + a\mu + a\rho) U_\rho^{-1}(n + a\mu) \\
 S_4 &= U_\rho^{-1}(n + a\mu - a\rho - a\nu) U_\nu(n + a\mu - a\rho - a\nu) U_\rho(n + a\mu - a\rho) \\
 &\quad - U_\rho(n + a\mu - a\nu) U_\nu(n + a\mu + a\rho - a\nu) U_\rho^{-1}(n + a\mu)
 \end{aligned} \tag{653}$$

and

$$\begin{aligned}
 N(n) &= \frac{1}{8} (N(\mu, \rho)(n) + N(\rho, \mu)(n)) \\
 N(\mu, \rho)(n) &= \{ (g(n + a\mu) + g(n + a\nu + a\mu)) U_\mu(n) T_1 U_\mu^{-1}(n + a\nu) \\
 &\quad + (g(n - a\mu) + g(n + a\nu - a\mu)) U_\mu^{-1}(n - a\mu) T_2 U_\mu(n + a\nu - a\mu) \}, \\
 T_1 &= U_\rho^{-1}(n + a\mu - a\rho) U_\nu(n + a\mu - a\rho) U_\rho(n + a\nu + a\mu - a\rho) \\
 &\quad - U_\rho(n + a\mu) U_\nu(n + a\mu + a\rho) U_\rho^{-1}(n + a\nu + a\mu), \\
 T_2 &= U_\rho(n - a\mu) U_\nu(n - a\mu + a\rho) U_\rho^{-1}(n + a\nu - a\mu) \\
 &\quad - U_\rho^{-1}(n - a\mu - a\rho) U_\nu(n - a\mu - a\rho) U_\rho(n + a\nu - a\mu - a\rho),
 \end{aligned} \tag{654}$$

Note, instead of $S_G[U_\mu] = -U_\mu \Sigma_\mu^\dagger$, here it is $S_G[U_\mu] = +U_\mu M_\mu^\dagger$ and $S_G[U_\mu] = +U_\mu N_\mu^\dagger$.

8.4.3 Rigid Acceleration

Another metric of constant acceleration g at $\mathbf{z}$ direction is

$$g_{\mu\nu} = \begin{pmatrix} (1 + gz)^2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-\det(g_{\mu\nu})} = (1 + gz) \tag{655}$$

The tangent space Wick rotation yields,

$$g_{\mu\nu} = \begin{pmatrix} -(1+gz)^2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \sqrt{-\det(g_{\mu\nu})} = i(1+gz) \quad (656)$$

Then

$$\begin{aligned} S_G &= \frac{1}{2g_{YM}^2} (1+gz) 2 (F_{12}^2 + F_{13}^2 + F_{23}^2 + (1+gz)^2 F_{01}^2 + (1+gz)^2 F_{02}^2 + (1+gz)^2 F_{03}^2) \\ &= \frac{\beta}{N_c} (1+gz) \text{Retr} [3 - \bar{U}_{12} - \bar{U}_{13} - \bar{U}_{23} + (1+gz)^2 (3 - \bar{U}_{41} - \bar{U}_{42} - \bar{U}_{43})] \end{aligned} \quad (657)$$

Since every plaquette is related to coordinate, we use $\bar{U}$ for every coordinate.

The Dirac equation becomes

$$\begin{aligned} D &= (1+gz) \left[\frac{1}{1+gz} \gamma^4 D_4 + \gamma^i D_i + i \frac{g}{2(1+gz)} \gamma_4 \sigma^{43} + m \right] \\ &= \left[\gamma^4 D_4 + (1+gz) \gamma^i D_i + i \frac{g}{2} \gamma_4 \sigma^{43} + m(1+gz) \right] \end{aligned} \quad (658)$$

As usual, the γ_4 term is realized as chemical potential $\mu = \frac{ig}{2} \sigma^{43}$,

$$\begin{aligned} D(n|m) &= (1+gz) - \kappa(1+gz) \sum_{i=1,2,3} T_i \\ &\quad - \kappa \left(e^{\frac{i}{2} ag \sigma^{12}} (1 - \gamma_4^E) U_\tau(n) \delta_{n+\tau, m} + e^{-\frac{i}{2} ag \sigma^{12}} (1 + \gamma_4^E) U_{-\tau}(n) \delta_{n-\tau, m} \right), \\ T_i &\equiv (1 - \gamma_i^E) U_i(n) \delta_{n+i, m} + (1 + \gamma_i^E) U_{-i}(n) \delta_{n-i, m}, \end{aligned} \quad (659)$$

The detail of the last term, we start from

$$\gamma^4 D_4 + i \frac{g}{2} \gamma_4 \sigma^{43} \quad (660)$$

The D_μ in Wilson-Dirac operator is

$$\begin{aligned} (\partial_\mu + iA_\mu(n)) \psi(n) &\approx \frac{U_\mu(n) \psi(n+a\mu) - U_{-\mu}(n) \psi(n-a\mu)}{2a} \\ &\quad - \frac{U_\mu(n) \psi(n+a\mu) + U_{-\mu}(n) \psi(n-a\mu) - 2\psi(n)}{2a} \rightarrow 0 \\ \gamma_\mu (\partial_\mu + iA_\mu(n)) \psi(n) & \\ \approx \gamma_\mu \frac{U_\mu(n) \psi(n+a\mu) - U_{-\mu}(n) \psi(n-a\mu)}{2a} &- \frac{U_\mu(n) \psi(n+a\mu) + U_{-\mu}(n) \psi(n-a\mu) - 2\psi(n)}{2a} \\ = \frac{\psi}{a} - \frac{(1+\gamma_\mu) U_\mu(n) \psi(n+a\mu) + (1-\gamma_\mu) U_{-\mu}(n) \psi(n-a\mu)}{2a} \end{aligned} \quad (661)$$

The $4 \frac{\psi}{a}$ combined with m is $\frac{am+4}{a}$, this is why the κ has this 4.

Now, we have

$$\begin{aligned} D &= m(1+gz) + 3(1+gz) \frac{1}{a} + \frac{1}{a} - (1+gz) \sum_{\mu=x,y,z} \frac{(1+\gamma_\mu) U_\mu(n) \psi(n+a\mu) + (1-\gamma_\mu) U_{-\mu}(n) \psi(n-a\mu)}{2a} \\ &\quad - \frac{(1+\gamma_t) U_t(n) \psi(n+at) + (1-\gamma_t) U_{-t}(n) \psi(n-at)}{2a} - i \frac{g}{2} \sigma^{43} \gamma_4 \end{aligned} \quad (662)$$

The ‘-’ of last term is from anti-commutation. Then

$$\begin{aligned} D &= (1+gz) \left(m + 4 \frac{1}{a} \right) - \frac{gz}{a} - (1+gz) \sum_{\mu=x,y,z} \frac{(1+\gamma_\mu) U_\mu(n) \psi(n+a\mu) + (1-\gamma_\mu) U_{-\mu}(n) \psi(n-a\mu)}{2a} \\ &\quad - \frac{(1+\gamma_t) U_t(n) \psi(n+at) + (1-\gamma_t) U_{-t}(n) \psi(n-at)}{2a} + i \frac{g}{2} \sigma^{34} \gamma_4 \end{aligned} \quad (663)$$

Still let $\kappa = \frac{1}{2am+8}$

$$\begin{aligned} 2a\kappa D &= (1+gz) - 2gz\kappa - \kappa(1+gz) \sum_{\mu=x,y,z} ((1+\gamma_\mu) U_\mu(n) \psi(n+a\mu) + (1-\gamma_\mu) U_{-\mu}(n) \psi(n-a\mu)) \\ &\quad - \kappa ((1+\gamma_t) U_t(n) \psi(n+at) + (1-\gamma_t) U_{-t}(n) \psi(n-at)) + ia\kappa g \sigma^{34} \gamma_4 \end{aligned} \quad (664)$$

Still use

$$-\frac{U_\mu(n)\psi(n+a\mu)+U_{-\mu}(n)\psi(n-a\mu)-2\psi(n)}{2a}\rightarrow 0$$

$$\psi(n)\rightarrow \frac{U_\mu(n)\psi(n+a\mu)+U_{-\mu}(n)\psi(n-a\mu)}{2}$$
(665)

The last term is

$$\begin{aligned} & \kappa\gamma_4 iag\sigma^{43}\psi(n) \\ & - \approx \kappa \left(\frac{-iag\sigma^{34}}{2} \right) (\gamma_4 U_\mu(n)\psi(n+\tau) + \gamma_4 U_{-\mu}(n)\psi(n-\tau)) \\ & \approx -\kappa \left(\frac{-iag\sigma^{34}}{2} \right) (\gamma_4 U_\mu(n)\psi(n+\tau) + \gamma_4 U_{-\mu}(n)\psi(n-\tau) - U_\mu(n)\psi(n+\tau) + U_{-\mu}(n)\psi(n-\tau)) \\ & = -\kappa \left(\frac{-iag\sigma^{34}}{2} \right) ((\gamma_4 - 1)U_\mu(n)\psi(n+\tau) + (\gamma_4 + 1)U_{-\mu}(n)\psi(n-\tau)) \\ & = -\kappa \left(\frac{-iag\sigma^{34}}{2} (\gamma_4 - 1)U_\mu(n)\psi(n+\tau) + \frac{-iag\sigma^{34}}{2} (\gamma_4 + 1)U_{-\mu}(n)\psi(n-\tau) \right) \\ & = -\kappa \left(\frac{iag\sigma^{34}}{2} (1 - \gamma_4)U_\mu(n)\psi(n+\tau) + \frac{-iag\sigma^{34}}{2} (1 + \gamma_4)U_{-\mu}(n)\psi(n-\tau) \right) \end{aligned}$$
(666)

Then $D_t + \gamma_4 i \frac{g}{2} \sigma^{43} \psi(n)$ is

$$\begin{aligned} & = -\kappa \left(\left(1 + \frac{iag\sigma^{34}}{2} \right) (1 - \gamma_4)U_\mu(n)\psi(n+\tau) + \left(1 - \frac{iag\sigma^{34}}{2} \right) (1 + \gamma_4)U_{-\mu}(n)\psi(n-\tau) \right) \\ & \approx -\kappa \left(e^{\frac{iag\sigma^{34}}{2}} (1 - \gamma_4)U_\mu(n)\psi(n+\tau) + e^{-\frac{iag\sigma^{34}}{2}} (1 + \gamma_4)U_{-\mu}(n)\psi(n-\tau) \right) \end{aligned}$$
(667)

Finally

$$\begin{aligned} D & = 1 + gz - 2gz\kappa - \kappa(1 + gz) \sum_{\mu=x,y,z} ((1 + \gamma_\mu)U_\mu(n)\psi(n+a\mu) + (1 - \gamma_\mu)U_{-\mu}(n)\psi(n-a\mu)) \\ & - \kappa \left(e^{\frac{iag\sigma^{34}}{2}} (1 - \gamma_4)U_\mu(n)\psi(n+\tau) + e^{-\frac{iag\sigma^{34}}{2}} (1 + \gamma_4)U_{-\mu}(n)\psi(n-\tau) \right) \end{aligned}$$
(668)

To implement it, note $(\sigma^{34})^2 = 1$

$$e^{\pm ix\sigma^{34}} = \cos(x\sigma^{34}) \pm i \sin(x\sigma^{34}) = \cos(x) \pm i\sigma^{34} \sin(x)$$
(669)

We have sign problem here. There is an $\mathcal{O}(a)$ sign problem: $f(n)(1 - \gamma_\mu)U_\mu(n)\delta_{n,n+\mu} + f(n+\mu)(1 + \gamma_\mu)U_\mu^{-1}(n)\delta_{n+\mu,n}$ such that $f(n) \neq f(n+\mu)$. There is another chemical potential like sign problem, $i\gamma_4\sigma^{34}$ is NOT γ_5 -hermitian.

8.4.4 Force Terms

The force term of gauge field is obtained by

$$\sum_n f(n) \text{Retr}[1 - \bar{U}_{\mu,\nu}(n)] = \sum_n \frac{f(n) + f(n+a\mu) + f(n+a\nu) + f(n+a\mu+a\nu)}{4} \text{Retr}[1 - U_{\mu\nu}(n)]$$
(670)

and

$$\Sigma_\mu(n, \nu) = g(n)U_\nu(n)U_\mu(n+a\nu)U_\nu^{-1}(n+a\mu) + g(n-a\nu)U_\nu^{-1}(n-a\nu)U_\mu(n-a\nu)U_\nu(n+a\mu-a\nu)$$
(671)

The force of Fermion field is

$$\begin{aligned} \frac{\partial D}{\partial \omega_\mu^a} & = -i\kappa M_a, \\ \frac{\partial}{\partial \omega_a} \left(\phi^\dagger \left(\hat{D} \hat{D}^\dagger \right)^{-1} \phi \right) & = -2\kappa \text{Im} \left[\left(\phi_1^\dagger M \phi_2 \right) \right] \\ \phi_1 & = \left(\left(\hat{D} \hat{D}^\dagger \right)^{-1} \phi \right), \quad \phi_2 = D^{-1} \phi, \end{aligned}$$
(672)

And use

$$\begin{aligned} D(U_\mu(n)) & = f(n)(1 - \gamma_\mu)U_\mu(n)\delta_{n,n+\mu} + f(n)(1 + \gamma_\mu)U_{-\mu}(n)\delta_{n,n-\mu} \\ & = f(n)(1 - \gamma_\mu)U_\mu(n)\delta_{n,n+\mu} + f(n)(1 + \gamma_\mu)U_\mu^{-1}(n-\mu)\delta_{n,n-\mu} \\ & = f(n)(1 - \gamma_\mu)U_\mu(n)\delta_{n,n+\mu} + f(n+\mu)(1 + \gamma_\mu)U_\mu^{-1}(n)\delta_{n+\mu,n} \end{aligned}$$
(673)

8.5 PSU3

8.5.1 Action

The link variables are represented by matrices $U_\mu(n) \in SU(3)$, but the action is chosen to be insensitive to the center of $SU(3)$. For $z \in Z_3$ the adjoint representation satisfies

$$D_{\text{adj}}(zU) = D_{\text{adj}}(U). \quad (674)$$

Equivalently, the character identity following from $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$ is

$$\text{Tr}_{\text{adj}} U = |\text{Tr}_F U|^2 - 1. \quad (675)$$

The adjoint Wilson plaquette action can therefore be evaluated using only fundamental 3×3 matrices:

$$S_A = -\beta_A \sum_n \sum_{\mu < \nu} \frac{1}{8} (|\text{Tr} U_{\mu\nu}(n)|^2 - 1), \quad (676)$$

where $1/8$ is the normalization by the dimension of the adjoint representation and

$$U_{\mu\nu}(n) = U_\mu(n)U_\nu(n + \hat{\mu})U_\mu^\dagger(n + \hat{\nu})U_\nu^\dagger(n). \quad (677)$$

This remains a one-plaquette action. The squared quantity is the modulus of the fundamental trace; it is neither $U_{\mu\nu}^2$ nor a geometrically doubled plaquette. It appears only because the adjoint character is being written in terms of fundamental matrices.

Indeed, the links may be transformed independently as

$$U_\mu(n) \longrightarrow z_\mu(n)U_\mu(n), \quad z_\mu(n) \in Z_3. \quad (678)$$

For a plaquette this gives

$$U_{\mu\nu}(n) \longrightarrow z_{\mu\nu}(n)U_{\mu\nu}(n), \quad z_{\mu\nu}(n) \in Z_3, \quad (679)$$

and hence

$$|\text{Tr} U_{\mu\nu}(n)|^2 \longrightarrow |z_{\mu\nu}(n)|^2 |\text{Tr} U_{\mu\nu}(n)|^2 = |\text{Tr} U_{\mu\nu}(n)|^2. \quad (680)$$

Thus center-related $SU(3)$ link matrices represent the same physical $PSU(3) = SU(3)/Z_3$ link variable. The continuous Lie algebra is nevertheless the usual $\mathfrak{su}(3)$ algebra, so the HMC momenta and tangent-space projection are the same as for an $SU(3)$ link model.

The additive constant in S_A does not affect molecular dynamics. Defining

$$\tilde{\beta} = \frac{\beta_A}{8}, \quad (681)$$

the dynamic part used below is

$$S_{\text{dyn}} = -\tilde{\beta} \sum_n \sum_{\mu < \nu} |\text{Tr} U_{\mu\nu}(n)|^2. \quad (682)$$

8.5.2 HMC Force

Fix one link and abbreviate

$$U = U_\mu(n). \quad (683)$$

For every $\nu \neq \mu$, two plaquettes in the $\mu\nu$ plane contain this link. After cyclically rotating the trace, both can be written with U as their first factor:

$$W_+(\mu, \nu, n) = UB_+(\mu, \nu, n), \quad (684)$$

$$W_-(\mu, \nu, n) = UB_-(\mu, \nu, n), \quad (685)$$

where

$$B_+(\mu, \nu, n) = U_\nu(n + \hat{\mu})U_\mu^\dagger(n + \hat{\nu})U_\nu^\dagger(n), \quad (686)$$

$$B_-(\mu, \nu, n) = U_\nu^\dagger(n + \hat{\mu} - \hat{\nu})U_\mu^\dagger(n - \hat{\nu})U_\nu(n - \hat{\nu}). \quad (687)$$

Their traces satisfy

$$t_+ \equiv \text{Tr } W_+ = \text{Tr } U_{\mu\nu}(n), \quad (688)$$

$$t_- \equiv \text{Tr } W_- = \text{Tr } U_{\nu\mu}(n - \hat{\nu}). \quad (689)$$

We first calculate the raw component derivative by regarding U and $U^\dagger$ as independent variables. In particular, $U^\dagger$ is held fixed in $\partial S/\partial U$. For a generic path $W = UB$,

$$\text{Tr } W = U^l{}_k B^k{}_l, \quad (690)$$

and therefore

$$\frac{\partial \text{Tr } W}{\partial U^j{}_i} = \delta^l{}_j \delta^i{}_k B^k{}_l = B^i{}_j. \quad (691)$$

Since the conjugate trace is held fixed in this raw derivative,

$$\left. \frac{\partial |\text{Tr } W|^2}{\partial U^j{}_i} \right|_{U^\dagger} = (\text{Tr } W)^* B^i{}_j. \quad (692)$$

It follows that the raw derivative of the action with respect to the chosen link is

$$\left. \left(\frac{\partial S}{\partial U_\mu(n)} \right)^i{}_j \right|_{U^\dagger} = -\tilde{\beta} \sum_{\nu \neq \mu} [t_+^* B_+^i{}_j + t_-^* B_-^i{}_j]. \quad (693)$$

Equivalently, in terms of the original plaquette notation,

$$\left. \left(\frac{\partial S}{\partial U_\mu(n)} \right)^i{}_j \right|_{U^\dagger} = -\tilde{\beta} \sum_{\nu \neq \mu} [\text{Tr } U_{\nu\mu}(n) B_+^i{}_j + \text{Tr } U_{\mu\nu}(n - \hat{\nu}) B_-^i{}_j]. \quad (694)$$

The coefficient is $-\tilde{\beta}$, rather than $-\tilde{\beta}/2$. One may write the action as

$$S_{\text{dyn}} = -\frac{\tilde{\beta}}{2} \sum_n \sum_{\mu \neq \nu} \text{Tr } U_{\mu\nu}(n) \text{Tr } U_{\nu\mu}(n), \quad (695)$$

but the ordered sum contains both (μ, ν) and (ν, μ) . Their identical derivatives cancel the explicit factor $1/2$.

To compare the component derivative with the force field stored by the action, define the code staple by

$$\Sigma_P = B_P^\dagger, \quad U_P = U \Sigma_P^\dagger = U B_P. \quad (696)$$

The raw field is

$$Y_\mu(n) = -\tilde{\beta} \sum_{P \ni (n, \mu)} \text{Tr}(U_P) \Sigma_P(n). \quad (697)$$

Consequently, the tensor-element result is precisely

$$\left. \frac{\partial S}{\partial U_\mu(n)} \right|_{U^\dagger} = Y_\mu^\dagger(n), \quad (698)$$

with the index placement understood as in the component equations above.

For completeness, the same result may be expressed directly as an $\mathfrak{su}(3)$ variation. Let A be traceless and anti-Hermitian and vary the link from the left,

$$U \longrightarrow e^A U, \quad \delta U = AU. \quad (699)$$

For one oriented plaquette $U_P = U B_P$ this gives

$$\delta S_P = -\tilde{\beta} \text{Tr} \left\{ A \left[\text{Tr}(U_P^\dagger) U_P - \text{Tr}(U_P) U_P^\dagger \right] \right\}. \quad (700)$$

Defining the mathematical force by $\delta S = -\text{Tr}(A F_\mu(n))$, one obtains

$$F_\mu(n) = \tilde{\beta} \sum_{P \ni (n, \mu)} X_P, \quad X_P = \text{Tr}(U_P^\dagger) U_P - \text{Tr}(U_P) U_P^\dagger. \quad (701)$$

The matrix X_P is automatically traceless and anti-Hermitian. This last form is only the tangent-space version of the tensor derivative above; the two expressions are not independent force formulas.

8.5.3 CLGLib force convention

The matrix $F_\mu(n)$ defined above is the mathematical force in

$$\delta S = -\text{Tr}(AF_\mu(n)). \quad (702)$$

CLGLib stores a different, but directly related, matrix in its force field. The gauge momentum is anti-Hermitian and has kinetic energy

$$K = -\text{Tr}(P^2). \quad (703)$$

With the molecular-dynamics equation $\dot{U} = PU$, one has

$$\dot{S} = -\text{Tr}(PF), \quad \dot{K} = -2\text{Tr}(P\dot{P}). \quad (704)$$

Energy conservation therefore requires

$$\boxed{\dot{P} = -\frac{1}{2}F.} \quad (705)$$

Thus the matrix added to the momentum by the integrator is

$$F_\mu^{\text{code}}(n) = -\frac{1}{2}F_\mu(n). \quad (706)$$

For any matrix M , CLGLib's operation $\text{TA}()$ is

$$\text{TA}(M) = \frac{1}{2}(M - M^\dagger) - \frac{1}{3}\text{Tr}\left[\frac{1}{2}(M - M^\dagger)\right]\mathbf{1}. \quad (707)$$

The action kernel stores the raw matrix $Y_\mu(n)$, after which the common integrator performs

$$\boxed{F_\mu^{\text{code}}(n) = \text{TA}\left(U_\mu(n)Y_\mu^\dagger(n)\right).} \quad (708)$$

There is no additional factor $-1/2$ outside $\text{TA}()$; the factor $1/2$ is already part of the anti-Hermitian projection.

Indeed, using

$$Y_\mu(n) = -\tilde{\beta} \sum_{P \ni (n, \mu)} \text{Tr}(U_P)\Sigma_P(n), \quad U_P = U_\mu(n)\Sigma_P^\dagger(n), \quad (709)$$

one obtains

$$\begin{aligned} U_\mu(n)Y_\mu^\dagger(n) &= -\tilde{\beta} \sum_P \text{Tr}(U_P^\dagger)U_\mu(n)\Sigma_P^\dagger(n) \\ &= -\tilde{\beta} \sum_P \text{Tr}(U_P^\dagger)U_P. \end{aligned} \quad (710)$$

Taking the traceless anti-Hermitian part gives

$$\boxed{F_\mu^{\text{code}}(n) = -\frac{\tilde{\beta}}{2} \sum_{P \ni (n, \mu)} X_P = -\frac{1}{2}F_\mu(n).} \quad (711)$$

8.5.4 CUDA implementation

The implementation in `CActionGaugePlaquettePSU3.cu` follows the preceding equations in two stages.

For each link, `_kernelForce_PSU3` loops over the surrounding three-link paths supplied by the staple cache. For each path it performs

$$\Sigma_P \leftarrow \text{cached three-link path}, \quad (712)$$

$$U_P \leftarrow U_\mu(n)\Sigma_P^\dagger, \quad (713)$$

$$t_P \leftarrow \text{Tr}(U_P), \quad (714)$$

$$Y_\mu(n) \leftarrow Y_\mu(n) - \tilde{\beta} t_P \Sigma_P. \quad (715)$$

After all actions have contributed their raw fields, the common integrator performs `LeftMul(gauge, FALSE, TRUE)`, which replaces the raw field by $U_\mu Y_\mu^\dagger$, and then calls `TA()`.

The trace t_P must be computed separately for every plaquette. Although the trace itself is linear, the force contains the weighted sum

$$\sum_P \text{Tr}(U_P) \Sigma_P, \quad (716)$$

whose weight depends on P . It cannot be reconstructed from one trace of a pre-summed staple.

For the $SU(N)$ fields supported by this class, the normalization uses the adjoint dimension:

$$\text{Beta} = \beta_A, \quad \tilde{\beta} = \frac{\beta_A}{N^2 - 1} = \begin{cases} \beta_A/8, & SU(3), \\ \beta_A/3, & SU(2). \end{cases} \quad (717)$$

There are two implementation qualifications:

- Raw gauge forces in CLGLib are additive because several actions may contribute to the same link. The kernel therefore *adds* its raw contribution to the shared force field via `_add(pForceData[uiLinkIndex], res)`, exactly as the standard plaquette kernel does; the field is zeroed once by the integrator before any action is evaluated.
- The coupling is normalized by the adjoint dimension $N^2 - 1$, so the same class works for $SU(3)$ ($\tilde{\beta} = \beta_A/8$) and $SU(2)$ ($\tilde{\beta} = \beta_A/3$). Other matrix sizes are reported as unsupported at initialization and are rejected by the force and energy routines.

8.5.5 Coupling to fermions

At the level of the HMC program, this gauge action can be combined with a fermion action using the same $SU(3)$ link matrices: each action contributes to the total energy and adds its force to the shared force field. However, the resulting theory is a genuine $PSU(3)$ gauge theory only if the fermions are neutral under the center Z_3 .

For example, a fundamental-fermion hopping term contains

$$\bar{\psi}(n) U_\mu(n) \psi(n + \hat{\mu}). \quad (718)$$

Under the independent link-center transformation $U_\mu(n) \rightarrow z_\mu(n) U_\mu(n)$, this term acquires the factor $z_\mu(n)$ and is not invariant. A fundamental **3** is therefore a representation of $SU(3)$, but not a representation of $PSU(3) = SU(3)/Z_3$.

Consequently, combining the adjoint plaquette action with ordinary fundamental $SU(3)$ fermions is a valid lattice model, but it should be interpreted as an $SU(3)$ gauge theory with an adjoint gauge action. The fermion action removes the independent local Z_3 identification that made the pure-gauge sector center blind. To preserve the $PSU(3)$ interpretation, one should use center-neutral, zero-triality matter, such as adjoint fermions. Introducing fundamental matter while retaining a quotient-gauge interpretation requires additional structure and is not achieved by simply adding a standard fundamental-fermion action.

8.6 Cylindrical coordinates

8.6.1 The gauge action in cylindrical coordinates

We follow the same convention as the rotating frame. In Euclidean space time, the gauge action is

$$S_G = \frac{1}{2g_{YM}^2} \int d^4x \text{tr} [F_{\mu\nu} F_{\mu\nu}] \quad (719)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]$$

The cylindrical coordinates (r, ϕ, z, t) are defined by $x = r \cos \phi$, $y = r \sin \phi$, and the metric is

$$g_{\mu\nu} = \text{diag}(1, r^2, 1, 1), \quad g^{\mu\nu} = \text{diag}\left(1, \frac{1}{r^2}, 1, 1\right), \quad \sqrt{\det(g_{\mu\nu})} = r \quad (720)$$

The gauge action in cylindrical coordinates is

$$S_G = \frac{1}{2g_{YM}^2} \int dr d\phi dz dt \sqrt{\det(g_{\alpha\beta})} g^{\mu\rho} g^{\nu\sigma} \text{tr} [F_{\mu\nu} F_{\rho\sigma}]$$

$$= \frac{1}{g_{YM}^2} \int dr d\phi dz dt \sum_{\mu < \nu} w_{\mu\nu}(r) \text{tr} [F_{\mu\nu}^2] \quad (721)$$

where the weight of a plaquette plane depends on whether the plane contains the angular direction ϕ ,

$$w_{r\phi} = w_{\phi z} = w_{\phi t} = \frac{1}{r} \quad (722)$$

$$w_{rz} = w_{rt} = w_{zt} = r$$

8.6.2 The lattice action

The lattice directions $0, 1, 2, 3$ (also known as x, y, z, w or x, y, z, t) are interpreted as (r, ϕ, z, t) . The lattice spacing is set to be $a = 1$. Since $r = 0$ is a coordinate singularity, the radial direction can not start from $r = 0$. The radial coordinate of a site $n = (n_0, n_1, n_2, n_3)$ is

$$r(n) = r_{start} + n_0, \quad r_{start} > 0, \quad r_{end} = r_{start} + L_0 \quad (723)$$

and the r direction is Dirichlet by default.

Using $\text{ReTr}[U_{\mu\nu}(n)] \approx N - \frac{1}{2}\text{tr}[F_{\mu\nu}^2]$, the lattice action with position dependent coupling is

$$\begin{aligned} S_G &= \sum_n \sum_{\mu < \nu} \bar{\beta}_{\mu\nu}(n) \left(1 - \frac{1}{N} \text{ReTr}[U_{\mu\nu}(n)] \right) \\ K_{\mu\nu}(n) &= \beta(r(n)) w_{\mu\nu}(r(n)), \quad \beta(r) = \frac{2N}{g_{YM}^2} \\ \bar{\beta}_{\mu\nu}(n) &= \frac{1}{4} \sum_{c \in \text{corners}(P)} K_{\mu\nu}(r(c)) \end{aligned} \quad (724)$$

- Since the physical size of a cell depends on r , β is not a constant but a numeric function $\beta(r)$, configured as an array, one value for each r slice.
- The coupling of one plaquette is the average of the coupling of its 4 corners. For a plane not containing r , the 4 corners are on the same r slice, so $\bar{\beta}_{\mu\nu}(n) = K_{\mu\nu}(r(n))$. For a plane containing r , two corners are at r and the other two are at $r + 1$, so $\bar{\beta}_{\mu\nu}(n) = \frac{1}{2} (K_{\mu\nu}(r) + K_{\mu\nu}(r + 1))$.

8.6.3 The force

The force is calculated using staples. A staple of link $U_\mu(n)$ in direction ν is

$$C_{\mu,\nu}(n) = U_\nu(n + \mu) U_\mu^\dagger(n + \nu) U_\nu^\dagger(n) + U_\nu^\dagger(n + \mu - \nu) U_\mu^\dagger(n - \nu) U_\nu(n - \nu) \quad (725)$$

With the position dependent coupling, every staple is weighted by $-\frac{1}{2}$ of the corner-averaged coupling of the plaquette,

$$F_\mu(n) = \sum_{\nu \neq \mu} \left(-\frac{1}{2} \right) (\bar{\beta}_{\mu\nu}(n) U_\mu(n) C_{\mu,\nu}(n))_{\text{TA}} \quad (726)$$

where $\bar{\beta}_{\mu\nu}$ is defined as above, and TA is the traceless anti-Hermitian projection. For a plaquette not containing r , $\bar{\beta}_{\mu\nu} = K_{\mu\nu}(r(n))$. For a plaquette containing r with corners at r_1 and $r_2 = r_1 + 1$, $\bar{\beta}_{\mu\nu} = \frac{1}{2} (K_{\mu\nu}(r_1) + K_{\mu\nu}(r_2))$. This is the same convention as the flat action, where every staple is weighted by $-\frac{\beta}{2N}$.

8.7 Anisotropic Symanzik gauge action

8.7.1 The tree-level action

For anisotropic lattices with bare gauge anisotropy ξ_0 , the tree-level Symanzik gauge action is

$$S_G = \beta \sum_n \left[\sum_{i < j} \frac{1}{\xi_0} (c_P P_{ij}(n) + c_R R_{ij}(n)) + \sum_i \xi_0 (c_P P_{i4}(n) + c_R R_{i4}(n)) \right] \quad (727)$$

$$c_P = 1, \quad c_R = -\frac{1}{20}, \quad \beta = \frac{10N}{3g^2} \quad \left(\beta = \frac{10}{g^2} \text{ for SU}(3) \right)$$

where $P_{\mu\nu}$ and $R_{\mu\nu}$ are the plaquette and the 2×1 rectangle in the μ - ν plane. Planes made of only spatial directions carry the weight $1/\xi_0$, and planes containing the temporal direction carry ξ_0 . The field level kernels use the inverse convention, so $q = 1/\xi_0$ is passed to the kernels.

8.7.2 The twisted (parallelogram) loop term

The one-loop improved action also contains the twisted loop with six links $[\mu, \nu, \rho, -\mu, -\nu, -\rho]$ (also called a parallelogram or "rectangle without right angles"). There are four direction triples: the purely spatial triple $(1, 2, 3)$, and the three triples $(1, 2, 4)$, $(1, 3, 4)$, $(2, 3, 4)$ containing the temporal direction.

A common mistake is to weight the whole loop by "whether the triple contains the temporal direction". In the weak field expansion of one loop,

$$1 - \frac{1}{N} \text{Re tr} [U_{\mu\nu\rho}(n)] \sim F_{\mu\nu}^2 + F_{\mu\rho}^2 + F_{\nu\rho}^2 + \text{cross terms} \quad (728)$$

and the cross terms cancel after summing the four sign variants, so every twisted loop contributes to all the three planes of its triple. Defining Q_{123} as the sum of the purely spatial triple and $Q_{\sigma\sigma\tau} = Q_{124} + Q_{134} + Q_{234}$, their expansions are

$$Q_{123} \sim F_{12}^2 + F_{13}^2 + F_{23}^2 \quad (729)$$

$$Q_{\sigma\sigma\tau} \sim (F_{12}^2 + F_{13}^2 + F_{23}^2) + 2(F_{14}^2 + F_{24}^2 + F_{34}^2)$$

Therefore a time-containing twisted loop is not a purely temporal operator, it still contains one spatial F_{ij}^2 . Requiring the spatial planes to carry $1/\xi_0$ and the temporal planes to carry ξ_0 , the minimal generalization of the isotropic term $S_T^{iso} = \beta c_T (Q_{123} + Q_{\sigma\sigma\tau})$ with the correct continuum limit is

$$S_T^{ani} = \beta c_T \left[\left(\frac{2}{\xi_0} - \xi_0 \right) Q_{123} + \xi_0 Q_{\sigma\sigma\tau} \right] \quad (730)$$

which reduces to the isotropic one at $\xi_0 = 1$. Equivalently, with the separated operators

$$Q_s = Q_{123}, \quad Q_t = \frac{1}{2} (Q_{\sigma\sigma\tau} - Q_{123}), \quad S_T^{ani} = 2\beta c_T \left(\frac{1}{\xi_0} Q_s + \xi_0 Q_t \right) \quad (731)$$

which shows why the subtraction of Q_{123} is unavoidable. In the kernel convention $q = 1/\xi_0$, the weights are

$$w_{123} = 2q - \frac{1}{q}, \quad w_{\sigma\sigma\tau} = \frac{1}{q} \quad (732)$$

for the purely spatial and the time-containing triples, and the energy and the force kernels use exactly the same weights.

8.7.3 Limitations

The twisted term above is a continuum-matched (heuristic) extension, not a complete one-loop anisotropic Symanzik improvement. A true one-loop action must split all operator classes,

$$S_g = \beta \sum_n \left[c_{P,s} P_{ij} + c_{P,t} P_{i4} + c_{R,s} R_{ij} + c_{R,st}^{(s)} R_{ij4} + c_{R,st}^{(t)} R_{i44} + c_{T,s} Q_s + c_{T,t} Q_t + \dots \right] \quad (733)$$

with coefficients $c_a = c_a^{(0)} + \alpha_s c_a^{(1)}(\xi_0, N_f)$ rematched by anisotropic lattice perturbation theory, and in general two tadpole factors u_s and u_t . The isotropic MILC coefficients

$$c_R = -\frac{1}{20u_0^2} [1 - (0.6264 - 1.1746N_f) \ln u_0], \quad c_T = \frac{(0.0433 - 0.0156N_f) \ln u_0}{u_0^2} \quad (734)$$

are strictly valid only at $\xi_0 = 1$. For the strict implementation, use the tree-level action with $c_T = 0$ and $c_R = -1/20$.

β	r_0/a	r_1/a	a^{-1} (MeV)	$c(\beta)r_0$	$c(\beta)a$
5.45	1.727(5)	1.193(22)	681(2)	-1.070(62)	-0.619(36)
5.5	1.884(4)	1.325(8)	742(2)	-1.160(31)	-0.616(16)
5.55	2.065(3)	1.458(3)	814(1)	-1.307(15)	-0.633(7)
5.6	2.312(3)	1.637(1)	911(1)	-1.507(8)	-0.652(3)
5.65	2.595(2)	1.845(1)	1022(1)	-1.716(5)	-0.661(2)
5.7	2.932(3)	2.091(1)	1155(1)	-1.965(4)	-0.670(1)
5.75	3.317(3)	2.368(1)	1307(1)	-2.251(3)	-0.679(1)
5.8	3.693(3)	2.644(2)	1455(1)	-2.501(3)	-0.677(1)
5.85	4.119(4)	2.953(3)	1623(2)	-2.791(3)	-0.678(1)
5.9	4.532(5)	3.255(3)	1785(2)	-3.053(4)	-0.674(1)
5.95	4.958(5)	3.568(3)	1953(2)	-3.311(4)	-0.668(1)
6.0	5.435(6)	3.919(4)	2141(2)	-3.596(4)	-0.662(1)
6.05	5.867(8)	4.238(5)	2312(3)	-3.833(5)	-0.653(1)

表 11: Result of matching for quenched approximation I.

β	r_0/a	r_1/a	a^{-1} (MeV)	$c(\beta)r_0$	$c(\beta)a$
5.6	2.3086(13)	1.6376(60)	909.6 ± 0.5	-1.496(17)	-0.6478(76)
5.7	2.9273(13)	2.0982(5)	1153.4 ± 0.8	-1.930(3)	-0.6591(10)
5.8	3.6758(13)	2.6413(7)	1448.3 ± 0.5	-2.457(2)	-0.6685(4)
5.9	4.4930(17)	3.2377(10)	1770.3 ± 0.7	-2.989(1)	-0.6652(2)
6.0	5.3340(22)	3.8568(14)	2101.6 ± 0.9	-3.486(2)	-0.6535(2)
6.1	6.2577(30)	4.5382(20)	2465.5 ± 1.2	-4.003(2)	-0.6397(1)
6.2	7.2903(38)	5.3014(26)	2872.4 ± 1.5	-4.557(2)	-0.6250(1)
6.3	8.0975(45)	5.9112(32)	3190.4 ± 1.8	-4.899(3)	-0.6050(1)
6.4	9.0717(55)	6.6410(39)	3574.3 ± 2.2	-5.335(3)	-0.5881(1)
6.5	9.8059(63)	7.2024(45)	3863.5 ± 2.5	-5.579(4)	-0.5690(1)

表 12: Result of matching for quenched approximation II.

9 Some Results

9.1 Matching quenched

The first set of result is for $12^3 \times 48$, in the quenched approximation, torus boundary.

1	GaugeSmearing:
2	
3	SmearingName : CGaugeSmearingAPEProj
4	AlphaLeft : 2.3
5	AlphaRight : 1
6	Iterate : 26
7	# Cabibbo-Marinani projection is not correctly implemented
8	Cabibbo : 1
9	ProjIterate : 15
10	HasT : 0

The Wilson loop is calculated using $\Delta t = [2a, 6a]$, and $r^2 \leq 20a^2$, $200 + (100 + 1900) \times 13$ configurations are generated, the statistical error is calculated using only jackknife. Autocorrelation shows that the $\tau_{\text{int}} \approx 0.5$ for all beta. The results are in Table. 17.

Another run uses Wilson loop is calculated using $\Delta t = [2a, 8a]$, and $r^2 \leq 54a^2$, $200 + (100 + 16900) \times 10$ configurations are

β	r_0/a	r_1/a	a^{-1} (MeV)	$c(\beta)r_0$	$c(\beta)a$	T
6.8	8.456(9)	6.344(6)	3337 ± 4	$-4.014(5)$	$-0.4747(3)$	278.1 ± 0.3
6.9	8.792(10)	6.607(7)	3470 ± 4	$-4.058(5)$	$-0.4616(3)$	289.1 ± 0.3
7.0	9.212(10)	6.927(7)	3636 ± 4	$-4.163(5)$	$-0.4520(2)$	303.0 ± 0.3
7.1	9.335(12)	7.029(8)	3684 ± 5	$-4.104(6)$	$-0.4396(3)$	307.0 ± 0.4
7.2	9.453(12)	7.134(8)	3731 ± 5	$-4.022(6)$	$-0.4254(2)$	310.9 ± 0.4
7.3	9.849(13)	7.436(9)	3887 ± 5	$-4.110(6)$	$-0.4173(2)$	323.9 ± 0.4
7.4	10.196(12)	7.701(9)	4024 ± 5	$-4.175(5)$	$-0.4095(2)$	335.3 ± 0.4
7.5	10.316(12)	7.804(9)	4071 ± 5	$-4.108(5)$	$-0.3982(2)$	339.3 ± 0.4
7.6	10.637(13)	8.051(9)	4198 ± 5	$-4.155(5)$	$-0.3906(2)$	349.8 ± 0.4

表 13: Result of matching for quenched approximation III, $24^2 \times 7 \times 48$. a^{-1} is determined by r_0 , T is temperature at $L_t = 12$.

$\beta(10/g^2)$	r_0/a	r_1/a	a^{-1} (MeV)	$c(\beta)r_0$	$c(\beta)a$	T
6.7	2.6731(35)	1.9710(516)	1055.0 ± 1.4	$-1.567(151)$	$-0.5862(564)$	175.83 ± 0.23
6.8	2.9980(46)	2.1517(76)	1183.2 ± 1.8	$-1.944(29)$	$-0.6485(95)$	197.19 ± 0.30
6.9	3.3448(32)	2.4077(13)	1320.0 ± 1.3	$-2.173(10)$	$-0.6496(29)$	220.01 ± 0.21
7.0	3.7112(24)	2.6723(6)	1464.7 ± 1.0	$-2.418(5)$	$-0.6515(12)$	244.11 ± 0.16
7.1	4.1006(21)	2.9558(8)	1618.3 ± 0.8	$-2.664(3)$	$-0.6496(6)$	269.72 ± 0.14
7.2	4.5024(19)	3.2493(9)	1776.9 ± 0.8	$-2.905(2)$	$-0.6453(4)$	296.15 ± 0.13
7.3	4.9166(19)	3.5536(11)	1940.4 ± 0.8	$-3.140(2)$	$-0.6387(3)$	323.39 ± 0.13
7.4	5.3773(20)	3.8917(12)	2122.2 ± 0.8	$-3.400(2)$	$-0.6323(2)$	353.70 ± 0.13
7.5	5.8841(22)	4.2622(14)	2322.2 ± 0.9	$-3.685(2)$	$-0.6262(2)$	387.03 ± 0.15
7.6	6.3138(24)	4.5849(16)	2491.8 ± 0.9	$-3.881(2)$	$-0.6147(1)$	415.30 ± 0.16
7.7	6.9754(27)	5.0680(18)	2752.9 ± 1.1	$-4.250(2)$	$-0.6093(1)$	458.81 ± 0.18
7.8	7.4987(31)	5.4592(21)	2959.4 ± 1.2	$-4.489(2)$	$-0.5986(1)$	493.23 ± 0.20

表 14: Result of matching for quenched approximation with tree-improved, the action is $\beta(plaqute - \frac{1}{20}rect)$, $12^3 \times 48$. $TU = 500 + (100 + 68900) \times 11$. a^{-1} is determined by r_0 , in fitting, $\Delta t \in [3a, 8a]$, $r < \sqrt{108a}$, T is temperature at $L_t = 6$.

generated, the statistical error is calculated using only jackknife. Autocorrelation shows that the $\tau_{\text{int}} \approx 0.5$ for all beta. The results are in Table 12.

$24^2 \times 7 \times 48$, with $\Delta t = [2a, 8a]$, and $r^2 \leq 300a^2$, $300 + (100 + 1500) \times 9$, autocorrelation not included but should not be significant, results are in Table 13.

Note that by using Ref. [36], $1 = 197.3269804 \text{ MeV} \times \text{fm}$, $0.5 \text{ fm} = 0.00253387 \text{ MeV}^{-1}$.

Tree-improved case.

9.2 Matching Wilson Dirac

The matching is almost always using m_ρ .

9.3 Matching for $N_f = 2$ staggered fermion

The first set of result is for $12^3 \times 48$, using $am = 0.1$, torus boundary and $N_f = 2$.

```

1 GaugeSmearing:
2
3 SmearingName : CGaugeSmearingAPEProj
4 AlphaLeft : 2.3
5 AlphaRight : 1
6 Iterate : 26

```

$L_{x,y}$	14	18	22	26
L_z	6	8	10	12
L_τ	24	32	40	48
β	2.5	5.02	5.283	5.35
κ	0.153	0.161	0.161	0.161
TU	200 + 500	200 + 500	300 + 500	300 + 335
am_π	2.0282(8)	1.5245(19)	1.1490(69)	0.9117(57)
am_ρ	2.1063(10)	1.5845(22)	1.2637(40)	1.0575(23)
a^{-1} (MeV)	362.3 ± 0.8	481.5 ± 1.2	603.8 ± 2.3	721.5 ± 2.1
m_π/m_ρ	0.989	0.962	0.909	0.862

表 15: The simulation parameters, the lattice spacing is matched by the mass of vector meson.

β	am_π	am_ρ	m_π/m_ρ	a^{-1} (MeV)
4.63	1.7045(09)	1.7501(12)	0.9739	436.0 ± 0.9
4.66	1.6962(12)	1.7424(13)	0.9734	437.9 ± 0.9
4.69	1.6897(08)	1.7371(10)	0.9727	439.2 ± 0.9
4.72	1.6805(10)	1.7287(12)	0.9721	441.4 ± 0.9
4.75	1.6702(12)	1.7184(15)	0.9720	444.0 ± 1.0
4.78	1.6622(10)	1.7115(12)	0.9712	445.8 ± 1.0
4.81	1.6489(11)	1.6982(13)	0.9710	449.3 ± 1.0
4.84	1.6393(11)	1.6903(14)	0.9698	451.4 ± 1.0
4.87	1.6267(11)	1.6781(13)	0.9694	454.7 ± 1.0
4.90	1.6142(13)	1.6669(15)	0.9684	457.7 ± 1.0
4.93	1.5987(14)	1.6516(17)	0.9680	462.0 ± 1.1
4.96	1.5844(16)	1.6387(19)	0.9669	465.6 ± 1.1
4.99	1.5691(13)	1.6249(15)	0.9657	469.6 ± 1.1
5.02	1.5538(07)	1.6110(09)	0.9645	473.6 ± 1.1
5.05	1.4600(16)	1.5141(18)	0.9643	503.9 ± 1.2

表 16: Simulation parameters and the matching of lattice spacing on a $12^3 \times 24$ lattice. $N_{\text{th}} + N_{\text{mes}} = 200 + 1000$ trajectories are simulated. The error of masses are calculated by using autocorrelation method. $\kappa = 0.16$ for all.

7	# Cabibbo—Marinani projection is not correctly implemented
8	Cabibbo : 1
9	ProjIterate : 15
10	HasT : 0

The Wilson loop is calculated using $\Delta t = [2a, \frac{N_t}{6}a] = [2a, 8a]$, and $r^2 \leq 20a^2$, $200 + (100 + 1900) \times 13$ configurations are generated, the statistical error is calculated using only jackknife. Autocorrelation is calculated with $S = 1.5$ on chiral condensation.

$$\begin{aligned}
 (2a)^4 \sum_h m \bar{q} q &\rightarrow a^4 \sum_n m \bar{\psi} \psi = a^4 \sum_n m \frac{\sqrt{2}\bar{\chi}}{a^{\frac{3}{2}}} \frac{\sqrt{2}\chi}{a^{\frac{3}{2}}} = \sum_n 2am \bar{\chi} \chi \\
 \bar{\psi} \psi &= 2a^{-3} \bar{\chi} \chi \\
 \langle \bar{\psi} \psi \rangle &= \frac{2}{a^3} \langle \bar{\chi} \chi \rangle = -\frac{2}{a^3} \text{tr} [D^{-1}]
 \end{aligned} \tag{735}$$

$$\begin{aligned}
 c &= -\frac{2}{a^3} \times \frac{n_f}{4L_x^3 L_\tau} \langle \text{tr} [D^{-1}] \rangle. \\
 \chi_{disc} &= \frac{4}{a^6} \times a^4 \frac{n_f^2}{16L_x^3 L_\tau} \left(\langle \text{tr} [D^{-1}]^2 \rangle - \langle \text{tr} [D^{-1}] \rangle^2 \right). \\
 \chi_{con} &= -\frac{4}{a^6} \times a^4 \frac{n_f}{4L_x^3 L_\tau} \langle \text{tr} [(D^{-1})^2] \rangle.
 \end{aligned} \tag{736}$$

The a^{-1} can be fitted as,

$$\frac{a^{-1}}{\text{MeV}} = 55173.5 - 23576.4\beta + 2527.39\beta^2. \tag{737}$$

9.4 Phase transition quenched

The first set of results are obtained by using $12^3 \times 6$ with torus periodic boundary. L_{bare} is calculated with autocorrelation, $S = 1.5$, however, when calculate L_{ren} , the $c(\beta)$ is introduced without autocorrelation. $2000 + (100 + 9900) \times 13$ configurations are generated. The results are in Table. 18.

The first set of results are obtained by using $2000 + (100 + 149900) \times 10$. The results are in Table. 19.

New result for $12 \times 12 \times 13 \times 6$, with $2000 + (100 + 29900) \times 8$ in Table 20.

New result for $12 \times 12 \times 12 \times 6$, with $2000 + (100 + 29900) \times 8$ in Table 21.

9.5 Results for $N_f = 2$ staggered fermion

The first two sets of result is for $12^3 \times 6$, using $am = 0.1$, torus boundary and $N_f = 2$. And with $\beta = 5.14, 5.1775, 5.19, 5.215, 5.24, 5.2525, 5.29, 5.3275, 5.34, 5.36$ (marked as origin a). And with $\beta = 5.21, 5.2475, 5.26, 5.2850, 5.31, 5.3225, 5.36, 5.3975, 5.41, 5.4350, 5.46, 5.4725, 5.51$ (marked as origin b). And with $n = 2000 + (100 + 9900)$ configurations. Autocorrelation is calculated with $S = 1.5$ on chiral condensation. Then is the result for $\beta = 5.31, 5.32, 5.33, 5.34, 5.35, 5.36$ (marked as origin c), and with $n = 2000 + (100 + 19900)$ configurations. Then is the result for $\beta = 5.3, 5.32, 5.34, 5.36, 5.38, 5.4$, and with $n = 300 + (100 + 2900)$ configurations, there is only result for chiral condensation.

The temperature is calculated using Eq. (737).

$$\begin{aligned}
 (2a)^4 \sum_h m \bar{q} q &\rightarrow a^4 \sum_n m \bar{\psi} \psi = a^4 \sum_n m \frac{\sqrt{2}\bar{\chi}}{a^{\frac{3}{2}}} \frac{\sqrt{2}\chi}{a^{\frac{3}{2}}} = \sum_n 2am \bar{\chi} \chi \\
 \bar{\psi} \psi &= 2a^{-3} \bar{\chi} \chi \\
 \langle \bar{\psi} \psi \rangle &= \frac{2}{a^3} \langle \bar{\chi} \chi \rangle = -\frac{2}{a^3} \text{tr} [D^{-1}]
 \end{aligned} \tag{738}$$

$$\begin{aligned}
 c &= -\frac{2}{a^3} \times \frac{n_f}{4L_x^3 L_\tau} \langle \text{tr} [D^{-1}] \rangle. \\
 \chi_{disc} &= \frac{4}{a^6} \times a^4 \frac{n_f^2}{16L_x^3 L_\tau} \left(\langle \text{tr} [D^{-1}]^2 \rangle - \langle \text{tr} [D^{-1}] \rangle^2 \right). \\
 \chi_{con} &= -\frac{4}{a^6} \times a^4 \frac{n_f}{4L_x^3 L_\tau} \langle \text{tr} [(D^{-1})^2] \rangle.
 \end{aligned} \tag{739}$$

β	r_{max}^2 (a^2)	$\sqrt{2\tau_{ind}}$	r_0/a	r_1/a	a^{-1} (MeV)	$c(\beta)r_0$	$c(\beta)a$	c/n_f	χ_{disc} (a^4)	χ_{con} (a^4)
5.08	9	3.5855	1.7659(352)	0.8872(883)	697(14)	-1.82(14)	-1.029(76)	0.13431(5)	0.0778(33)	-
5.1	9	6.0047	1.8019(451)	1.0286(713)	711(18)	-1.70(17)	-0.941(93)	0.13131(11)	0.1114(47)	-
5.12	9	5.0911	1.8502(329)	1.1334(343)	730(13)	-1.62(12)	-0.873(62)	0.12792(9)	0.0992(44)	-
5.14	9	3.3401	1.9120(191)	1.2211(141)	755(8)	-1.58(6)	-0.828(32)	0.12438(5)	0.0760(34)	-
5.16	9	6.2901	1.9873(316)	1.3102(153)	784(12)	-1.57(9)	-0.789(45)	0.12042(13)	0.1450(62)	-
5.18	9	4.8493	2.0830(230)	1.3921(85)	822(9)	-1.62(6)	-0.777(27)	0.11618(11)	0.1633(81)	-
5.2	20	5.2630	2.2746(380)	1.4757(238)	898(15)	-1.93(10)	-0.849(44)	0.11143(11)	0.1359(57)	-
5.22	20	6.3062	2.3646(319)	1.6150(146)	933(13)	-1.81(8)	-0.766(32)	0.10621(13)	0.1484(70)	-
5.24	20	7.2217	2.4766(309)	1.7239(134)	977(13)	-1.81(7)	-0.733(26)	0.10138(15)	0.1500(61)	-
5.26	20	6.6872	2.6395(270)	1.8479(123)	1042(11)	-1.93(5)	-0.730(17)	0.09659(15)	0.1636(65)	-
5.28	36	6.7933	2.8607(342)	2.0081(142)	1129(13)	-2.10(6)	-0.735(20)	0.09159(15)	0.1538(67)	-
5.3	36	5.7842	3.0503(278)	2.1499(128)	1204(11)	-2.24(4)	-0.733(12)	0.08773(10)	0.0938(39)	-0.17547(3)
5.32	36	7.2602	3.2193(378)	2.2827(199)	1271(15)	-2.33(5)	-0.723(12)	0.08421(15)	0.1461(66)	-0.16841(4)
5.34	36	8.6395	3.4155(449)	2.4278(239)	1348(18)	-2.46(5)	-0.721(13)	0.08113(15)	0.0942(39)	-0.1623(3)
5.36	36	5.1408	3.5867(246)	2.5542(140)	1415(10)	-2.58(3)	-0.719(6)	0.07857(8)	0.0716(32)	-0.1571(3)
5.38	36	6.8202	3.8335(354)	2.7262(208)	1513(14)	-2.78(4)	-0.726(7)	0.07619(10)	0.0716(31)	-0.1524(3)
5.4	36	5.7238	4.0200(325)	2.8614(195)	1586(13)	-2.92(3)	-0.725(5)	0.07418(7)	0.0528(24)	-0.1484(3)
5.42	36	6.9254	4.2742(518)	3.0428(324)	1687(20)	-3.11(14)	-0.727(6)	0.07224(9)	0.0562(26)	-0.1448(3)
5.44	36	3.6855	4.4474(237)	3.1700(148)	1755(9)	-3.22(2)	-0.724(3)	0.07063(4)	0.0441(20)	-0.1413(2)
5.46	36	6.3526	4.6672(456)	3.3303(290)	1842(18)	-3.37(4)	-0.722(5)	0.06924(8)	0.0501(25)	-0.1385(2)
5.48	36	4.6959	4.7477(352)	3.3954(229)	1874(14)	-3.39(3)	-0.713(3)	0.06799(5)	0.0403(16)	-0.1360(2)
5.5	36	3.6929	5.0160(275)	3.5880(178)	1980(11)	-3.58(2)	-0.714(2)	0.06665(3)	0.0277(11)	-0.1333(2)
5.52	36	5.0005	5.2507(440)	3.7576(292)	2072(17)	-3.74(4)	-0.712(3)	0.06554(5)	0.0321(14)	-0.1311(2)
5.54	36	4.4113	5.3440(351)	3.8326(232)	2109(14)	-3.76(3)	-0.704(2)	0.06455(4)	0.0258(11)	-0.1291(2)
5.56	36	2.2671	5.5738(212)	4.0007(141)	2200(8)	-3.91(2)	-0.702(1)	0.06351(2)	0.0201(9)	-0.1270(2)
5.58	36	3.2041	5.8115(296)	4.1749(197)	2294(12)	-4.06(2)	-0.699(2)	0.06255(3)	0.0220(9)	-0.1251(2)
5.6	36	2.8749	6.0694(302)	4.3606(204)	2395(12)	-4.23(2)	-0.697(1)	0.06169(2)	0.0211(9)	-0.1234(2)
5.62	36	2.3307	6.1995(250)	4.4597(168)	2447(10)	-4.29(2)	-0.692(1)	0.06092(2)	0.0174(8)	-0.1218(1)
5.64	36	2.8987	6.6031(362)	4.7498(247)	2606(14)	-4.57(3)	-0.692(1)	0.06009(2)	0.0155(7)	-0.1202(1)

表 17: Result of matching for $N_f = 2$, note that, the static error of $\chi_{disc,con}$ are jackknife without τ_{ind} . And this table is for $\langle \bar{\chi}\chi \rangle$ instead of $\langle \bar{\psi}\psi \rangle$ therefore dimensionless.

β	T (MeV)	L_{bare}	L_{ren}	χ_P (a^3)
5.45	113.4(3)	0.00770(4)	0.049(5)	0.0277(4)
5.5	123.7(3)	0.00791(4)	0.050(2)	0.0291(5)
5.55	135.6(2)	0.00812(5)	0.054(1)	0.0318(5)
5.6	151.8(2)	0.00869(6)	0.061(1)	0.0361(7)
5.65	170.4(2)	0.00966(9)	0.070(1)	0.0444(12)
5.7	192.5(2)	0.01065(12)	0.080(1)	0.0432(18)
5.75	217.8(2)	0.01309(26)	0.100(2)	0.0804(45)
5.8	242.5(2)	0.01826(82)	0.139(6)	0.1658(220)
5.85	270.4(3)	0.02808(293)	0.215(22)	0.4412(1177)
5.9	297.6(3)	0.05979(314)	0.451(24)	0.8096(1522)
5.95	325.5(3)	0.08117(151)	0.602(11)	0.4689(736)
6.0	356.9(4)	0.09258(149)	0.674(11)	0.4229(612)
6.05	385.2(5)	0.10293(124)	0.731(9)	0.3955(455)

表 18: Result of phase transition, quenched. critical temperature is 297.6 MeV.

β	T (MeV)	τ_{ind}	L_{bare}	L_{ren}	χ_P/T^3
5.6	151.6(1)	0.7835	0.00869(1)	0.0607(14)	7.70 ± 0.04
5.7	192.2(1)	3.5950	0.01100(4)	0.0794(4)	12.55 ± 0.14
5.8	241.4(1)	38.7015	0.01736(22)	0.1290(17)	35.77 ± 1.48
5.9	295.0(1)	233.7995	0.05411(134)	0.3981(98)	213.31 ± 12.73
6.0	350.3(1)	53.4871	0.09317(44)	0.6617(31)	99.26 ± 4.44
6.1	410.9(2)	32.3169	0.11347(31)	0.7732(22)	85.84 ± 2.92
6.2	478.7(2)	29.1558	0.12892(30)	0.8408(20)	85.17 ± 2.56
6.3	531.7(3)	24.4408	0.14375(27)	0.8826(17)	81.71 ± 2.26
6.4	595.7(4)	21.3246	0.15759(25)	0.9198(15)	82.23 ± 2.14
6.5	643.9(4)	30.4303	0.16978(31)	0.9358(17)	89.41 ± 3.54

表 19: Result of phase transition, quenched. critical temperature is 295.0 MeV.

β	L_{bare}	χ_P/T^3	τ_{ind}
5.7	0.01062(9)	12.7 ± 0.3	3.86
5.75	0.01287(19)	19.5 ± 0.9	11.09
5.8	0.01673(48)	36.6 ± 3.2	36.90
5.85	0.02592(169)	102.2 ± 19.7	167.31
5.9	0.05454(320)	211.9 ± 33.6	291.90
5.95	0.07671(131)	127.1 ± 15.5	80.81
6.0	0.09456(100)	130.5 ± 10.9	57.33
6.05	0.10435(61)	80.1 ± 5.0	28.52

表 20: New result for $12 \times 12 \times 13 \times 6$, with $2000 + (100 + 29900) \times 10$, the errors have already taken into account τ_{ind} .

β	L_{bare}	χ_P/T^3	τ_{ind}
5.7	0.01089(8)	12.2 ± 0.3	3.20
5.75	0.01335(17)	18.7 ± 0.7	9.10
5.8	0.01813(52)	38.1 ± 3.5	38.84
5.85	0.03140(202)	123.6 ± 18.5	184.41
5.9	0.06282(199)	165.6 ± 20.7	133.62
5.95	0.08093(151)	121.2 ± 15.1	104.43
6.0	0.09304(80)	94.6 ± 6.9	37.33
6.05	0.10386(87)	94.9 ± 8.3	44.45

表 21: New result for $12 \times 12 \times 13 \times 6$, with $2000 + (100 + 29900) \times 10$, the errors have already taken into account τ_{ind} .

β	τ_{ind}	T (MeV)	P_{bare}	χ_P/T^3	c (a^{-3})	χ_{disc}/T^2
5.14	14.6706	127.2	0.0032(3)	14.1 ± 1.1	0.4980(3)	14.5 ± 1.1
5.1775	18.0624	142.8	0.0038(4)	15.2 ± 1.3	0.4663(4)	17.9 ± 1.6
5.19	14.7510	148.3	0.0044(4)	15.7 ± 1.2	0.4550(4)	17.4 ± 1.4
5.215	17.9125	159.6	0.0053(4)	16.8 ± 1.4	0.4302(4)	19.1 ± 1.7
5.24	17.7415	171.5	0.0066(4)	20.1 ± 1.7	0.4040(5)	21.6 ± 1.9
5.2525	22.1701	177.6	0.0082(5)	22.0 ± 2.1	0.3907(5)	21.5 ± 2.2
5.29	43.2014	196.8	0.0153(9)	35.4 ± 5.4	0.3552(7)	22.2 ± 3.3
5.3275	126.7824	217.2	0.0364(25)	91.2 ± 18.5	0.3210(14)	28.3 ± 5.8
5.34	96.2488	224.2	0.0491(21)	88.5 ± 16.4	0.3086(11)	23.9 ± 4.5
5.3650	29.7971	238.7	0.0700(10)	62.4 ± 6.9	0.2905(5)	14.4 ± 1.4
5.39	23.9353	253.7	0.0813(8)	46.6 ± 4.8	0.2787(3)	8.0 ± 0.8
5.4025	18.1225	261.4	0.0829(6)	42.9 ± 3.8	0.2753(3)	7.7 ± 0.7
5.44	8.3453	285.4	0.0930(4)	41.7 ± 2.5	0.2642(2)	5.2 ± 0.3

表 22: case a.

β	τ_{ind}	T (MeV)	P_{bare}	χ_P/T^3	c (a^{-3})	χ_{disc}/T^2
5.21	29.2206	157.3	0.0051(4)	16.5 ± 1.3	0.4350(6)	23.8 ± 2.6
5.2475	27.7230	175.2	0.0080(5)	22.1 ± 2.0	0.3954(6)	21.2 ± 2.2
5.26	28.1036	181.4	0.0092(4)	24.6 ± 1.9	0.3832(6)	23.8 ± 2.4
5.285	35.4460	194.2	0.0141(6)	32.3 ± 2.8	0.3590(6)	20.2 ± 2.5
5.31	42.7050	207.5	0.0254(7)	54.7 ± 4.5	0.3366(7)	24.1 ± 3.0
5.3225	112.5755	214.4	0.0371(10)	89.0 ± 7.6	0.3224(13)	29.4 ± 5.8
5.36	44.1635	235.8	0.0675(12)	60.9 ± 8.3	0.2931(6)	14.6 ± 1.9
5.3975	32.3623	258.3	0.0812(19)	51.8 ± 12.9	0.2771(4)	9.5 ± 1.2
5.41	19.2787	266.1	0.0861(15)	45.0 ± 8.9	0.2726(3)	7.7 ± 0.7
5.435	9.9262	282.1	0.0920(8)	38.6 ± 4.4	0.2654(2)	5.6 ± 0.3
5.46	6.8235	298.6	0.0986(7)	36.9 ± 3.7	0.2585(1)	4.4 ± 0.2
5.4725	8.3959	307.0	0.0988(7)	44.4 ± 3.9	0.2564(1)	4.5 ± 0.3
5.51	5.5614	333.2	0.1073(4)	38.6 ± 2.2	0.2482(1)	3.3 ± 0.2

表 23: case b.

β	τ_{ind}	T (MeV)	P_{bare}	χ_P/T^3	c (a^{-3})	χ_{disc}/T^2
5.29	48.8121	196.8	0.0163(7)	37.9 ± 3.9	0.3544(6)	23.7 ± 2.5
5.3	43.7558	202.1	0.0197(7)	42.8 ± 4.0	0.3455(5)	23.4 ± 2.2
5.31	54.3458	207.5	0.0253(10)	61.7 ± 6.7	0.3358(6)	26.0 ± 2.7
5.32	89.2433	213.0	0.0360(13)	71.4 ± 9.1	0.3248(8)	26.3 ± 3.4
5.33	117.9276	218.6	0.0473(17)	92.2 ± 14.4	0.3137(10)	28.4 ± 4.1
5.34	64.0406	224.2	0.0546(11)	70.5 ± 7.9	0.3061(6)	21.2 ± 2.3
5.35	50.4272	230.0	0.0636(9)	55.5 ± 5.5	0.2979(5)	15.8 ± 1.6
5.36	31.4626	235.8	0.0686(7)	56.0 ± 4.4	0.2922(3)	13.9 ± 1.1

表 24: case c.

β	τ_{ind}	χ_{disc}/T^2
5.3	41.68	23.0 ± 5.7
5.32	59.27	22.6 ± 5.7
5.34	116.97	31.9 ± 11.7
5.36	19.96	9.2 ± 1.7
5.38	16.64	10.2 ± 1.5
5.4	9.15	7.0 ± 0.8

表 25: case d.

10 Summaries

10.1 Action table

10.1.1 Tree improved gauge

For naive Wilson action, ignore a (let $a = 1$),

$$\begin{aligned} A_{lat} &= g_s A \\ F_{\mu\nu,lat} &= g_s F_{\mu\nu} \\ \sum_{\mu>\nu} F_{\mu\nu,lat}^2 &= \frac{1}{2} \sum_{\mu,\nu} F_{\mu\nu,lat} = \frac{1}{2} g_s^2 F_{\mu\nu} \\ \frac{1}{4} F_{\mu\nu} &= \frac{2}{g_s^2} \sum_{\mu>\nu} F_{\mu\nu,lat} = \frac{\beta}{N_c} \sum_{\mu>\nu} F_{\mu\nu,lat} \end{aligned} \quad (740)$$

so, $\frac{\beta}{N_c} = \frac{2}{g_s^2}$, that is the reason in literatures, $\beta = 6/g^2$.

For tree-level improved case,

$$\begin{aligned} S &= \frac{\beta}{N_c} \left(\frac{5}{3} S_{plaq} - \frac{1}{12} S_{rect} \right) \\ S_{plaq} &= \sum_{plaq} \text{Retr}(1 - U_{plaq}) \\ S_{rect} &= \sum_{rect} \text{Retr}(1 - U_{rect}) \end{aligned} \quad (741)$$

If we redefine $\beta' = \frac{\beta}{N_c} \frac{5}{3}$, then $\beta' = 10/g^2$ and,

$$S = \frac{\beta'}{N_c} \left(S_{plaq} - \frac{1}{20} S_{rect} \right) \quad (742)$$

With tadpole improved, it is

$$S = \frac{\beta'}{N_c u_0^2} \left(S_{plaq} - \frac{1}{20 u_0^2} S_{rect} \right) \quad (743)$$

Usually, one defines $\beta'' = 10/(g^2 u_0^2)$, and

$$\begin{aligned} S &= \frac{\beta''}{N_c} \left(S_{plaq} - \frac{1}{20 u_0^2} S_{rect} \right) \\ S_{plaq} &= \sum_{\mu>\nu} U_{\mu,\nu} \\ S_{rect} &= \sum_{\mu \neq \nu} U_{\mu,\mu,\nu} = \sum_{\mu>\nu} (U_{\mu,\mu,\nu} + U_{\mu,\nu,\nu}) \end{aligned} \quad (744)$$

10.1.2 Stout link smearing Wilson Dirac

Using $\kappa = \frac{1}{2am_f + 8}$, the Wilson-Dirac fermion operator is, (in [Lattice QCD simulations with clover quarks](#), it uses $\bar{\sigma} \bar{F} = \sigma F$, where $\bar{\sigma} = \frac{1}{2}[\gamma, \gamma]$, $\bar{F} = \frac{1}{8}(P - P^\dagger)$, so $\sigma = \frac{i}{2}[\gamma, \gamma]$, $\bar{F} = \frac{1}{8i}(P - P^\dagger)$)

$$\begin{aligned} S &= \bar{\psi} D \psi, \\ D &= 1 - \kappa \left[\sum_{\mu} \left((1 - \gamma_{\mu}) U_{\mu}(x_L) \delta_{x_L, (x+\mu)_R} + (1 + \gamma_{\mu}) U_{\mu}^{-1}(x_L - \mu) \delta_{x_L, (x-\mu)_R} \right) + \frac{1}{2} c_{SW} \sigma_{\mu\nu} F_{\mu\nu} \right] \end{aligned} \quad (745)$$

with

$$F_{\mu\nu} = \frac{1}{8\hat{i}} (P_{\mu\nu} - P_{\mu\nu}^\dagger) \quad (746)$$

There are different definitions of $P_{\mu\nu}$, the original one was,

$$P_{\mu\nu} = U_{\mu,\nu} + U_{\nu,-\mu} - U_{\mu,-\nu} - U_{-\nu,-\mu} \quad (747)$$

Using $U_{a,b} = U_{b,a}^\dagger$, it is equivalent to use,

$$P'_{\mu\nu} = U_{\mu,\nu} - U_{-\mu,\nu} - U_{\mu,-\nu} + U_{-\mu,-\nu} \quad (748)$$

or

$$P''_{\mu\nu} = U_{\mu,\nu} + U_{\nu,-\mu} + U_{-\nu,\mu} + U_{-\mu,-\nu} \quad (749)$$

In the code, we use P' .

Note, σ is in Euclidean space. Because $\gamma_5 \sigma_{41}^E \gamma_5 = (\sigma_{41}^E)^\dagger$, $\gamma_5 \sigma_{41}^M \gamma_5 = -(\sigma_{41}^M)^\dagger$ and $F_{\mu\nu}^\dagger = F_{\mu\nu}$ (note the i in denominator). The Minkowski space $\sigma_{\mu\nu} F_{\mu\nu}$ is not γ_5 Hermitian.

And therefore,

$$\begin{aligned} -(am+4)S &= \frac{1}{2} \sum_{\mu} \bar{\psi}(x) \left((1-\gamma_{\mu})U_{\mu}(x_L)\psi(x+\mu) + (1+\gamma_{\mu})U_{\mu}^{-1}(x_L-\mu)\psi(x-\mu) \right) \\ &+ \bar{\psi} \left[-(am+4) + \frac{1}{4}c_{SW}\sigma_{\mu\nu}F_{\mu\nu} \right] \psi \end{aligned} \quad (750)$$

Using $\sigma_{\nu\mu}F_{\nu\mu} = \sigma_{\mu\nu}F_{\mu\nu}$, it can be simplified as,

$$D = 1 - \kappa \left[\sum_{\mu} \left((1-\gamma_{\mu})U_{\mu}(x_L)\delta_{x_L,(x+\mu)_R} + (1+\gamma_{\mu})U_{\mu}^{-1}(x_L-\mu)\delta_{x_L,(x-\mu)_R} \right) + c_{SW} \sum_{\mu>\nu} \sigma_{\mu\nu}F_{\mu\nu} \right] \quad (751)$$

With tadpole improvement,

$$\begin{aligned} \kappa &= \frac{1}{v_0(2am+8)} \\ c_{SW} &= \frac{1}{v_0^3} \end{aligned} \quad (752)$$

10.1.3 HISQ Action

$$D = 2m\delta_{m,n} + \sum_{\mu} \eta_{\mu}(n) \left[U_{\mu}^{eff,2}(n)\delta_{n,n+\mu} - U_{-\mu}^{eff,2}(n)\delta_{n,n-\mu} + \left(-\frac{1+\epsilon}{24} \right) \left(U_{3\mu}^{eff,1}(n)\delta_{n,n+3\mu} - U_{-3\mu}^{eff,1}(n)\delta_{n,n-3\mu} \right) \right] \quad (753)$$

$$\begin{aligned} U_{\mu}^{eff,2}(n) &= S_{Asqtad}(\epsilon)U_{\mu}^{eff,1}(n) \\ U_{\mu}^{eff,1}(n) &= P_{U_3}S_{fat7}(\epsilon=0)U_{\mu}(n) \end{aligned} \quad (754)$$

$$\begin{aligned} S_{fat7}U &= \frac{1}{8}U + \frac{1}{16}fat3 + \frac{1}{64}fat5 + \frac{1}{384}fat7 \\ S_{Asqtad}(\epsilon)U &= \frac{1+\epsilon}{8}U + \frac{1}{16}fat3 + \frac{1}{64}fat5 + \frac{1}{384}fat7 - \frac{1}{8}Lepage \end{aligned} \quad (755)$$

Because different fermions have different ϵ , it seems easier to implement this,

$$\begin{aligned} D &= 2m\delta_{m,n} + \sum_{\mu} \eta_{\mu}(n) \left[U_{\mu}^{eff,2}(n)\delta_{n,n+\mu} - U_{-\mu}^{eff,2}(n)\delta_{n,n-\mu} \right. \\ &\left. + \frac{\epsilon}{8} \left(U_{\mu}^{eff,1}(n)\delta_{n,n+\mu} - U_{-\mu}^{eff,1}(n)\delta_{n,n-\mu} \right) + \left(-\frac{1+\epsilon}{24} \right) \left(U_{3\mu}^{eff,1}(n)\delta_{n,n+3\mu} - U_{-3\mu}^{eff,1}(n)\delta_{n,n-3\mu} \right) \right] \end{aligned} \quad (756)$$

$$\begin{aligned} U_{\mu}^{eff,2}(n) &= S_{Asqtad}(\epsilon=0)U_{\mu}^{eff,1}(n) \\ U_{\mu}^{eff,1}(n) &= P_{U_3}S_{fat7}U_{\mu}(n) \end{aligned} \quad (757)$$

10.2 Cayley-Hamilton method and some results

Cayley-Hamilton method is widely used in the calculation of smearing, here is some results.

10.2.1 Cayley-Hamilton method

Cayley-Hamilton theory demonstrated, for square matrices,

$$p(M) = 0, \quad (758)$$

where, $p(x) = 0$ is the characteristic equation, i.e., $p(x) = \det(M - xI)$ ('p' means polynomial).

It is used to calculate the function of a $N \times N$ matrix,

$$f(M) = \sum_{i=0}^{N-1} f_i M^i, \quad (759)$$

To determine the coefficients, note that, when λ_j is eigen-value of M , $f(\lambda_j)$ is eigen-value of $f(M)$. So, we can,

- (1) Calculate $f(\lambda_j)$ use f .
- (2) Calculate $f(\lambda_j)$ use $f(M) = \sum_{i=0}^{N-1} f_i M^i$ with TBD coefficients, f_i .
- (3) Solve the equation $f(\lambda_j) = \sum_{i=0}^{N-1} f_i \lambda_j^i$ to determine f_i .

10.2.2 3 x 3 matrix

$$f(Q) = f_0 + f_1 Q + f_2 Q^2, \quad (760)$$

with,

$$\begin{aligned} f(\lambda_0) &= f_0 + f_1 \lambda_0 + f_2 \lambda_0^2, \\ f(\lambda_1) &= f_0 + f_1 \lambda_1 + f_2 \lambda_1^2, \\ f(\lambda_2) &= f_0 + f_1 \lambda_2 + f_2 \lambda_2^2, \end{aligned} \quad (761)$$

where $\lambda_{0,1,2}$ are eigen-values of Q .

10.2.3 Exponential**10.2.4 Sqrt**

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